MATHEMATICS

VECTOR 3D

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VECTOR 3D

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BASIC PROPERTIES OF VECTORS

1. The vertices of a triangle are A (1, 1, 2), B(4, 3, 1) and C(2, 3, 5). A vector representing the internal bisector of the angle A is :

एक त्रिभुज के शीर्ष A (1, 1, 2), B(4, 3, 1) और C(2, 3, 5) हैं, तो कोण A के अन्तः समद्विभाजक को निरूपित करने वाला सदिश है:

(A) $\hat{i}+\hat{j}+2\hat{k}$ (B) $2\hat{i}-2\hat{j}+\hat{k}$ (C) $2\hat{i}+2\hat{j}-\hat{k}$ (D) $2\hat{i}+2\hat{j}+\hat{k}$

Ans. D

Sol. D is mid–point of AB

D (3,3,3)

$\overrightarrow{AD}=2\hat{i}+2\hat{j}+\hat{k}$

2. Consider points A, B, C and D with position vectors $7\hat{i}-4\hat{j}+7\hat{k}$, $\hat{i}-6\hat{j}+10\hat{k}$, $-\hat{i}-3\hat{j}+4\hat{k}$ and $5\hat{i}-\hat{j}+5\hat{k}$ respectively. Then ABCD is a :

(A) parallelogram but not a rhombus (B) square
(C) rhombus (D) None of these

माना बिन्दु A, B, C तथा D हैं जिनके स्थिति सदिश क्रमशः $7\hat{i}-4\hat{j}+7\hat{k}$, $\hat{i}-6\hat{j}+10\hat{k}$, $-\hat{i}-3\hat{j}+4\hat{k}$ तथा $5\hat{i}-\hat{j}+5\hat{k}$ हैं, तब ABCD होगा :

(A) समान्तर चतुर्भुज परन्तु समचतुर्भुज नहीं (B) वर्ग
(C) समचतुर्भुज (D) इनमें से कोई नहीं

Ans. D

Sol. AB = 7
BC = 7

$CD=\sqrt{41}$

$AD=\sqrt{17}$

3. If C is the mid point of AB and P is any point outside AB, then :

यदि C , AB का मध्य बिन्दु P है तथा कोई भी एक बिन्दु है, जो AB के बाहर है,तो :

(A) $\overrightarrow{PA}+\overrightarrow{PB}=2\overrightarrow{PC}$ (B) $\overrightarrow{PA}+\overrightarrow{PB}=\overrightarrow{PC}$
(C) $\overrightarrow{PA}+\overrightarrow{PB}+2\overrightarrow{PC}=\vec{0}$ (D) $\overrightarrow{PA}+\overrightarrow{PB}+\overrightarrow{PC}=\vec{0}$

Ans. A

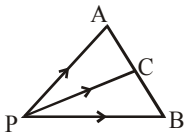
Sol. $\overrightarrow{PA}+\overrightarrow{PB}=\left(\overrightarrow{PA}+\overrightarrow{AC}\right)+\left(\overrightarrow{PB}+\overrightarrow{BC}\right)-\left(\overrightarrow{AC}+\overrightarrow{BC}\right)$

$=\overrightarrow{PC}+\overrightarrow{PC}-\left(\overrightarrow{AC}-\overrightarrow{CB}\right)$

$=2\overrightarrow{PC}-0$

$\left(\because \overrightarrow{AC}=\overrightarrow{CB}\right)$

$\therefore \overrightarrow{PA}+\overrightarrow{PB}=2\overrightarrow{PC}$



4. Let $\vec{a},\vec{b},\vec{c}$ be three non-zero vectors which are pairwise non-collinear. If $\vec{a}+3\vec{b}$ is collinear with $\vec{c}$ and $\vec{b}+2\vec{c}$ is collinear with $\vec{a}$, then $\vec{a}+3\vec{b}+6\vec{c}$ is : [AIEEE-2011]

माना $\vec{a},\vec{b},\vec{c}$ तीन शून्येतर सदिश है जो युग्मत सरेख नहीं है। यदि $\vec{a}+3\vec{b}$ तथा $\vec{c}$ सरेख हैं तथा $\vec{b}+2\vec{c}$ तथा $\vec{a}$ सरेख हैं, तो $\vec{a}+3\vec{b}+6\vec{c}$ बराबर हैं : [AIEEE-2011]

(A) $\vec{a}+\vec{c}$ (B) $\vec{a}$ (C) $\vec{c}$ (D) $\vec{0}$

Ans. D

Sol. Let Let

$a+3b=\lambda\vec{c}$ $\vec{b}+2\vec{c}=\mu\vec{a}$

add $6\vec{c}$ both side | $3b+6\vec{c}=3\mu\vec{a}$

$\vec{a}+3\vec{b}+6\vec{c}=(\lambda+6)\vec{c}$ add $\vec{a}$ both side $\vec{a}+3\vec{b}+6\vec{c}=(3\mu+1)\vec{a}$

Hence $(\lambda+6)\vec{c}=(3\mu+1)\vec{a}$

But given $\vec{a}$ and $\vec{c}$ are non coliner

Hence $\lambda+6=3\mu+1=0$

so $\vec{a}+3\vec{b}+6\vec{c}=\vec{0}$

DOT PRODUCT & ITS APPLICATION

5. If $|\vec{a}|=5, |\vec{a}-\vec{b}|=8$ and $|\vec{a}+\vec{b}|=10$, then $|\vec{b}|$ is equal to :

(A) 1 (B) $\sqrt{57}$ (C) 3 (D) none of these

यदि $|\vec{a}| = 5, |\vec{a} - \vec{b}| = 8$ और $|\vec{a} + \vec{b}| = 10$ हो, तो $|\vec{b}| =$

(A) 1 (B) $\sqrt{57}$ (C) 3 (D) इनमें से कोई नहीं

Ans. B

Sol. $\left|\vec{a}-\vec{b}\right|^2+\left|\vec{a}+\vec{b}\right|^2=164$

$\left|\vec{a}\right|^2+\left|\vec{b}\right|^2=82$

$25+\left|\vec{b}\right|^2=82$

$\left|\vec{b}\right|=\sqrt{57}$

6. Angle between diagonals of a parallelogram whose side are represented by $\vec{a}=2\hat{i}+\hat{j}+\hat{k}$ and $\vec{b}=\hat{i}-\hat{j}-\hat{k}$:

एक समान्तर चतुर्भुज जिसकी भुजाऐं $\vec{a}=2\hat{i}+\hat{j}+\hat{k}$ तथा $\vec{b}=\hat{i}-\hat{j}-\hat{k}$ से दी गई है, के विकर्णों के मध्य कोण होगा :

(A) $\cos^{-1}\left(\frac{1}{3}\right)$ (B) $\cos^{-1}\left(\frac{1}{2}\right)$ (C) $\cos^{-1}\left(\frac{4}{9}\right)$ (D) $\cos^{-1}\left(\frac{5}{9}\right)$

Ans. A

Sol. $\vec{\alpha}+(1,-1,-1)=(2,1,1)$

$\vec{\alpha}=(1,2,2)$

$\vec{\beta}=(3,0,0)$

$\vec{\alpha}\cdot\vec{\beta}=3\times3\cos\theta=3$

$\cos\theta=\frac{1}{3}$

7. Two adjacent sides of a parallelogram ABCD are given by $\overrightarrow{AB}=2\hat{i}+10\hat{j}+11\hat{k}$ and $\overrightarrow{AD}=-\hat{i}+2\hat{j}+2\hat{k}$. The side AD is rotated by an acute angle α in the plane of the parallelogram so that AD becomes AD′. If AD′ makes a right angle with the side AB, then the cosine of the angle α is given by :

समान्तर चतुर्भुज ABCD की दो आसन्न भुजाऐं $\overrightarrow{AB}=2\hat{i}+10\hat{j}+11\hat{k}$ तथा $\overrightarrow{AD}=-\hat{i}+2\hat{j}+2\hat{k}$ द्वारा दी गई है। यदि भुजा AD को न्यून कोण α द्वारा समान्तर चतुर्भुज के तल में घूर्णन किया जाता है। इस प्रकार कि यह AD′ हो जाती है एवं AD′ भुजा AB के साथ समकोण बनाती है तो कोण α की कोज्या होगी :

(A) $\frac{8}{9}$ (B) $\frac{\sqrt{17}}{9}$ (C) $\frac{1}{9}$ (D) $\frac{4\sqrt{5}}{9}$

Ans. B

Sol. $\cos\theta = -2 + 20 + 22 = 40$

$\cos\theta > 0 \Rightarrow \theta < 90$

$\theta + \alpha = 90$

$\cos\alpha = \cos(90-\theta) = \sin\theta$

$$= \frac{|\vec{a}\times\vec{b}|}{|\vec{a}||\vec{b}|}$$

$$\frac{\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 10 & 11\\ -1 & 2 & 2\end{vmatrix}}{15\times 3} = \frac{\hat{i}(-2)-\hat{j}(15)+\hat{k}(14)}{45}$$

$$= \frac{\sqrt{425}}{45} = \frac{\sqrt{17}}{9}$$

8. Let two non-collinear unit vector $\hat{a}$ and $\hat{b}$ form an acute angle. A point P moves so that at any time t the position vector $\overrightarrow{OP}$ (where O is the origin) is given by $\hat{a}\cos t + \hat{b}\sin t$. When P is farthest from origin O, let M be the length of $\overrightarrow{OP}$ and $\hat{u}$ be the unit vector along $\overrightarrow{OP}$. Then :

(A) $\hat{u} = \dfrac{\hat{a}+\hat{b}}{|\hat{a}+\hat{b}|}$ and $M = (1+\hat{a}\cdot\hat{b})^{\frac{1}{2}}$

(B) $\hat{u} = \dfrac{\hat{a}-\hat{b}}{|\hat{a}-\hat{b}|}$ and $M = (1+\hat{a}\cdot\hat{b})^{\frac{1}{2}}$

(C) $\hat{u} = \dfrac{\hat{a}+\hat{b}}{|\hat{a}+\hat{b}|}$ and $M = (1+2\hat{a}\cdot\hat{b})^{\frac{1}{2}}$

(D) $\hat{u} = \dfrac{\hat{a}-\hat{b}}{|\hat{a}-\hat{b}|}$ and $M = (1+2\hat{a}\cdot\hat{b})^{\frac{1}{2}}$

माना असंरेखीय इकाई सदिश $\hat{a}$ तथा $\hat{b}$ एक न्यूनकोण बनाते है। एक बिन्दु P ऐसे गमन करता है कि किसी समय t पर स्थिति सदिश $\overrightarrow{OP}$ (जहाँ O मूल बिन्दु है) $\hat{a}\cos t + \hat{b}\sin t$ द्वारा दिया जाता है। माना जब P मूल बिन्दु O से अधिकतम दूरी पर है, तब $\overrightarrow{OP}$ की लम्बाई M तथा $\overrightarrow{OP}$ के समान्तर इकाई सदिश $\hat{u}$ है, तब :

(A) $\hat{u} = \dfrac{\hat{a}+\hat{b}}{|\hat{a}+\hat{b}|}$ तथा $M = (1+\hat{a}\cdot\hat{b})^{\frac{1}{2}}$

(B) $\hat{u} = \dfrac{\hat{a}-\hat{b}}{|\hat{a}-\hat{b}|}$ तथा $M = (1+\hat{a}\cdot\hat{b})^{\frac{1}{2}}$

(C) $\hat{u} = \dfrac{\hat{a}+\hat{b}}{|\hat{a}+\hat{b}|}$ तथा $M = (1+2\hat{a}\cdot\hat{b})^{\frac{1}{2}}$

(D) $\hat{u} = \dfrac{\hat{a}-\hat{b}}{|\hat{a}-\hat{b}|}$ तथा $M = (1+2\hat{a}\cdot\hat{b})^{\frac{1}{2}}$

Ans. A

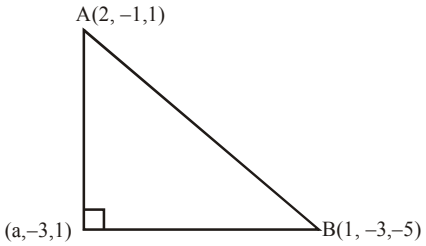
Sol. AC ⊥ BC

∴ dr's of AC and BC will be (2–a,2,0) and (1–a,0,–6)

So that (2 – a) (1 – a) + 2 × 0 + 0 × (–6) = 0

$\Rightarrow \quad a^2 - 3a + 2 = 0$

$\therefore \quad a = 1, 2$



9. Let $\hat{a}$ and $\hat{b}$ be two unit vectors. If the vectors $\vec{c} = \hat{a} + 2\hat{b}$ and $\vec{d} = 5\hat{a} - 4\hat{b}$ are perpendicular to each other, then the angle between $\hat{a}$ and $\hat{b}$ is : [AIEEE-2012]

माना $\hat{a}$ तथा $\hat{b}$ दो मात्रक सदिश है। यदि $\vec{c} = \hat{a} + 2\hat{b}$ तथा $\vec{d} = 5\hat{a} - 4\hat{b}$ परस्पर लम्बवत् हैं, तो $\hat{a}$ तथा $\hat{b}$ के बीच का कोण हैं : [AIEEE-2012]

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{3}$

Ans. D

Sol. $\vec{c}.\vec{d} = 0$

$\Rightarrow (\hat{a} + 2\hat{b}).(5\hat{a} - 4\hat{b}) = 0$

$\Rightarrow 5 - 8 + 6\,\hat{a} \,.\, \hat{b} = 0$

$\Rightarrow \hat{a} \,.\, \hat{b} = 1/2$

$\Rightarrow \cos\theta = 1/2$

$\Rightarrow \theta = \frac{\pi}{3}$

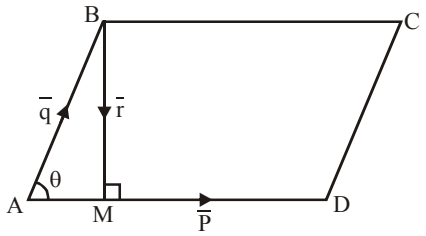
10. Let ABCD be a parallelogram such that $AB = \vec{q}, AD = \vec{p}$ and $\angle BAD$ be an acute angle. If $\vec{r}$ is the vector that coincides with the altitude directed from the vertex B to the side AD, then $\vec{r}$ is given by : [AIEEE-2012]

माना ABCD एक ऐसा समांतर चतुर्भुज है, कि $AB = \vec{q}, AD = \vec{p}$ तथा $\angle BAD$ एक न्यून कोण है। यदि सदिश $\vec{r}$ शीर्ष B से AD पर खींचे गए लंब के संपाती है, तो $\vec{r}$ निम्न द्वारा प्रदत हैं : [AIEEE-2012]

(A) $\vec{r} = -3\vec{q} + \frac{3(\vec{p}.\vec{q})}{(\vec{p}.\vec{p})}\vec{p}$ (B) $\vec{r} = 3\vec{q} - \frac{3(\vec{p}.\vec{q})}{(\vec{p}.\vec{p})}\vec{p}$ (C) $\vec{r} = -\vec{q} + \left(\frac{\vec{p}.\vec{q}}{\vec{p}.\vec{p}}\right)\vec{p}$ (D) $\vec{r} = \vec{q} - \left(\frac{\vec{p}.\vec{q}}{\vec{p}.\vec{p}}\right)\vec{p}$

Ans. C

Sol.



$$\bar{q}+\bar{r}=\overline{AM}$$

$$\Rightarrow \bar{r}=-\bar{q}+\overline{AM}$$

$$\Rightarrow \bar{r}=-\bar{q}+\frac{\bar{p}.\bar{q}}{|\bar{p}|^2}\vec{p}$$

$$\Rightarrow \bar{r}=-\bar{q}+\left(\frac{\bar{p}.\bar{q}}{\bar{p}.\bar{p}}\right)\bar{p}$$

11. Let $\vec{u}$ be a vector coplanar with the vectors $\vec{a}=2\hat{i}+3\hat{j}-\hat{k}$ and $\vec{b}=\hat{j}+\hat{k}$. If $\vec{u}$ is perpendicular to $\vec{a}$ and $\vec{u}.\vec{b}=24$, then $|\vec{u}|^2$ is equal to : [JEE(Main)-2018]

माना $\vec{u}$ एक ऐसा सदिश है जो सदिशों $\vec{a}=2\hat{i}+3\hat{j}-\hat{k}$ तथा $\vec{b}=\hat{j}+\hat{k}$ के साथ समतलीय है। यदि $\vec{u}$, $\vec{a}$ पर लंबवत् है तथा $\vec{u}.\vec{b}=24$ है, तो $|\vec{u}|^2$ बराबर है : [JEE(Main)-2018]

(A) 256 (B) 84 (C) 336 (D) 315

Ans. C

Sol. $\vec{u}=\lambda\left(\vec{a}\times\left(\vec{a}\times\vec{b}\right)\right)$

$$=\lambda\left(\left(\vec{a}.\vec{b}\right)\vec{a}-\left(\vec{a}.\vec{a}\right)\vec{b}\right)$$

$$=\lambda\left(2\vec{a}-14\vec{b}\right)$$

$$\vec{u}=\lambda\left(4\hat{i}-8\hat{j}-16\hat{k}\right)$$

$$\vec{u}.\vec{b}=24 \Rightarrow \lambda=-1$$

$$\Rightarrow \vec{u}=-4\hat{i}+8\hat{j}+16\hat{k}$$

$$\Rightarrow|\vec{u}|^2=16+64+256=336$$

CROSS PRODUCT AND ITS APPLICATION

12. Let $\vec{a} = \hat{i} + \hat{j}$ and $\vec{b} = 2\hat{i} - \hat{k}$. The point of intersection of the lines $\vec{r} \times \vec{a} = \vec{b} \times \vec{a}$ and $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$ is :

मानाकि $\vec{a} = \hat{i} + \hat{j}$ तथा $\vec{b} = 2\hat{i} - \hat{k}$ हैं, रेखाओं $\vec{r} \times \vec{a} = \vec{b} \times \vec{a}$ तथा $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$ का प्रतिच्छेद बिन्दु है :

(A) $-\hat{i} + \hat{j} + 2\hat{k}$ (B) $3\hat{i} - \hat{j} + \hat{k}$ (C) $3\hat{i} + \hat{j} - \hat{k}$ (D) $\hat{i} - \hat{j} - \hat{k}$

Ans. C

Sol. $\vec{a} = \hat{i} + \hat{j}$ and $\vec{b} = 2\hat{i} - \hat{k}$

$\vec{r} \times \vec{a} = \vec{b} \times \vec{a} \Rightarrow (\vec{r} - \vec{b}) \times \vec{a} = 0$

$\Rightarrow \vec{r} = \vec{b} + \lambda\vec{a}$ ...(i)

similarly $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$

$\Rightarrow \vec{r} = \vec{a} + \mu\vec{b}$ ...(ii)

Putting the vector $\vec{a}$ and $\vec{b}$ in (i) and (ii) and equating

we get $2\hat{i} - \hat{k} + \lambda(\hat{i} + \hat{j}) = \hat{i} + \hat{j} + \mu(2\hat{i} - \hat{k})$

$\Rightarrow \quad 2 + \lambda = 1 + 2\mu, \ \lambda = 1, \mu = 1$

$\therefore$ Point of intersection is $3\hat{i} + \hat{j} - \hat{k}$.

13. Unit vector perpendicular to the plane of the triangle ABC with position vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ of the vertices A, B, C is : (where Δ is the area of the triangle ABC).

(A) $\dfrac{\left(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}\right)}{\Delta}$ (B) $\dfrac{\left(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}\right)}{2\Delta}$

(C) $\dfrac{\left(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}\right)}{4\Delta}$ (D) None of these

त्रिभुज ABC के शीर्षो A, B, C के स्थिति सदिश क्रमशः $\vec{a}$, $\vec{b}$, $\vec{c}$ है, तो त्रिभुज के समतल के लम्बवत् इकाई सदिश है :
(जहाँ Δ त्रिभुज ABC का क्षेत्रफल है)

(A) $\dfrac{\left(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}\right)}{\Delta}$ (B) $\dfrac{\left(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}\right)}{2\Delta}$

(C) $\dfrac{\left(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}\right)}{4\Delta}$ (D) इनमें से कोई नहीं

Ans. B

Sol. $\vec{\Delta} = |\vec{\Delta}|\hat{\Delta} = \frac{1}{2}\left(\vec{a}\times\vec{b} + \vec{b}\times\vec{c} + \vec{c}\times\vec{a}\right)$

14. If $\vec{u}$ and $\vec{v}$ are unit vectors and θ is the acute angle between them, then $2\vec{u}\times 3\vec{v}$ is a unit vector for :

(A) Exactly two values of θ (B) More than two values of θ

(C) No value of θ (D) Exactly one value of θ

यदि $\vec{u}$ तथा $\vec{v}$ इकाई सदिश हैं तथा θ उनके मध्य न्यून कोण है, तब $2\vec{u}\times 3\vec{v}$ जिनके लिए इकाई सदिश है :

(A) θ के ठीक दो मानों के लिए (B) θ के दो मानों से ज्यादा के लिए

(C) θ के किसी मान के लिए नहीं (D) θ के ठीक एक मान के लिए

Ans. D

Sol. $|2\vec{u}\times 3\vec{v}| = 1$

$|2\vec{u}\times 3\vec{v}| = 6|\vec{u}||\vec{v}|\sin\theta = 1$

$= 6\sin\theta = 1$

$\sin\theta = \frac{1}{6}$

15. Let $\vec{a}, \vec{b}, \vec{c}$ be unit vectors such that $\vec{a}+\vec{b}+\vec{c} = \vec{0}$. Which one of the following is correct?

(A) $\vec{a}\times\vec{b} = \vec{b}\times\vec{c} = \vec{c}\times\vec{a} = \vec{0}$ (B) $\vec{a}\times\vec{b} = \vec{b}\times\vec{c} = \vec{c}\times\vec{a} \neq \vec{0}$

(C) $\vec{a}\times\vec{b} = \vec{b}\times\vec{c} = \vec{a}\times\vec{c} \neq \vec{0}$ (D) $\vec{a}\times\vec{b}, \vec{b}\times\vec{c}, \vec{c}\times\vec{a}$ are mutually perpendicular

मानाकि $\vec{a}, \vec{b}, \vec{c}$ इकाई सदिश इस प्रकार है कि $\vec{a}+\vec{b}+\vec{c} = \vec{0}$ निम्न में से कौनसा सत्य है?

(A) $\vec{a}\times\vec{b} = \vec{b}\times\vec{c} = \vec{c}\times\vec{a} = \vec{0}$ (B) $\vec{a}\times\vec{b} = \vec{b}\times\vec{c} = \vec{c}\times\vec{a} \neq \vec{0}$

(C) $\vec{a}\times\vec{b} = \vec{b}\times\vec{c} = \vec{a}\times\vec{c} \neq \vec{0}$ (D) $\vec{a}\times\vec{b}, \vec{b}\times\vec{c}, \vec{c}\times\vec{a}$ परस्पर लम्बवत् है

Ans. B

Sol. $|\vec{a}+\vec{b}+\vec{c}| = 0$

$\vec{b}\times\vec{a} + \vec{c}\times\vec{a} = 0$ $\qquad \vec{a}\times\vec{b} = \vec{c}\times\vec{a}$

$\vec{a}\times\vec{b} + \vec{c}\times\vec{b} = 0$

$\vec{a}\times\vec{c} + \vec{b}\times\vec{c} = 0$

$\vec{a}, \vec{b}, \vec{c}$ are co-planar.

$\vec{c} = -\left(\vec{a}+\vec{b}\right)$

16. If the vectors $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$, $\vec{b} = 2\hat{i} + 4\hat{j} + \hat{k}$ and $\vec{c} = \lambda\hat{i} + \hat{j} + \mu\hat{k}$ are mutually orthogonal, then $(\lambda, \mu) =$ [AIEEE-2010]

यदि सदिश $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$, $\vec{b} = 2\hat{i} + 4\hat{j} + \hat{k}$ तथा $\vec{c} = \lambda\hat{i} + \hat{j} + \mu\hat{k}$ परस्पर लम्बकोणीय हैं, तो (λ, μ) बराबर है : [AIEEE-2010]

(A) $(-3, 2)$ (B) $(2, -3)$ (C) $(-2, 3)$ (D) $(3, -2)$

Ans. A

Sol. Give $\vec{a} \perp \vec{b}$, $\vec{a} \perp \vec{c}$ and $\vec{b} \perp \vec{c}$

so $\vec{a}.\vec{c} = 0$ and $\vec{b}.\vec{c} = 0$

$\Rightarrow \lambda - 1 + 2\mu = 0$ and $2\lambda + 4 + \mu = 0$

$\Rightarrow \lambda = -3$ and $\mu = 2$

17. The vectors $\vec{a}$ and $\vec{b}$ are not perpendicular and $\vec{c}$ and $\vec{d}$ are two vectors satisfying : $\vec{b} \times \vec{c} = \vec{b} \times \vec{d}$ and $\vec{a}.\vec{d} = 0$. Then the vector $\vec{d}$ is equal to : [AIEEE-2011]

$\vec{a}$ तथा $\vec{b}$ लम्बवत् नहीं है तथा $\vec{c}$ तथा $\vec{d}$ इस प्रकार के है कि $\vec{b} \times \vec{c} = \vec{b} \times \vec{d}$ तथा $\vec{a}.\vec{d} = 0$ हैं, तो $\vec{d}$ बराबर है : [AIEEE-2011]

(A) $\vec{b} + \left(\frac{\vec{b}.\vec{c}}{\vec{a}.\vec{b}}\right)\vec{c}$ (B) $\vec{c} - \left(\frac{\vec{a}.\vec{c}}{\vec{a}.\vec{b}}\right)\vec{b}$ (C) $\vec{b} - \left(\frac{\vec{b}.\vec{c}}{\vec{a}.\vec{b}}\right)\vec{c}$ (D) $\vec{c} + \left(\frac{\vec{a}.\vec{c}}{\vec{a}.\vec{b}}\right)\vec{b}$

Ans. B

Sol. $a.b. \neq 0$

$a.d = 0$

$b \times c = b \times d$

$a \times (b \times c) = a \times (b \times d)$

$(a.c)b - (a.b)c - (a.c)b - (a.b)d$ {a.d=0}

$\Rightarrow (a.b)d = (a.b)c$ (a.c)b (divide by a.b)

$$\boxed{d = c - \frac{(a.c)}{(a.b)}b}$$

18. The value of $\left[\left(\vec{a} + 2\vec{b} - \vec{c}\right), \left(\vec{a} - \vec{b}\right), \left(\vec{a} - \vec{b} - \vec{c}\right)\right]$ is equal to the box product :

$\left[\left(\vec{a} + 2\vec{b} - \vec{c}\right), \left(\vec{a} - \vec{b}\right), \left(\vec{a} - \vec{b} - \vec{c}\right)\right]$ का मान निम्न में से किस बॉक्स गुणनफल (box product) के बराबर है:

(A) $\left[\vec{a}\ \vec{b}\ \vec{c}\right]$ (B) $2\left[\vec{a}\ \vec{b}\ \vec{c}\right]$ (C) $3\left[\vec{a}\ \vec{b}\ \vec{c}\right]$ (D) $4\left[\vec{a}\ \vec{b}\ \vec{c}\right]$

Ans. C

Sol. $\left(\vec{a}+2\vec{b}-\vec{c}\right),\left(\vec{a}-\vec{b}\right),\left(\vec{a}-\vec{b}-\vec{c}\right)$

$\left(\vec{a}-\vec{b}\right)\times\left(\vec{a}-\vec{b}-\vec{c}\right)$

$\vec{a}\times\vec{b}-\vec{a}\times\vec{c}-\vec{b}\times\vec{a}+\vec{b}\times\vec{c}$

$\left(\vec{a}+2\vec{b}-\vec{c}\right)\cdot\left(\vec{c}\times\vec{a}+\vec{b}\times\vec{a}\right)$

$2\vec{b}\cdot\left(\vec{c}\times\vec{a}\right)-\vec{c}\left(\vec{b}\times\vec{a}\right)$

2[a, b, c] + [a, b, c] = 3[a, b, c]

19. If $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{b}=\hat{i}-\hat{j}+\hat{k}$, $\vec{c}=\hat{i}+2\hat{j}-\hat{k}$, then the value of $\begin{vmatrix}\vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c}\\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c}\\ \vec{c}\cdot\vec{a} & \vec{c}\cdot\vec{b} & \vec{c}\cdot\vec{c}\end{vmatrix}$ is equal to :

यदि $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{b}=\hat{i}-\hat{j}+\hat{k}$, $\vec{c}=\hat{i}+2\hat{j}-\hat{k}$ हो, तो $\begin{vmatrix}\vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c}\\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c}\\ \vec{c}\cdot\vec{a} & \vec{c}\cdot\vec{b} & \vec{c}\cdot\vec{c}\end{vmatrix}$ का मान है :

(A) 2 (B) 4 (C) 16 (D) 64

Ans. C

Sol. $\begin{vmatrix}3 & 1 & 2\\ 1 & 3 & -2\\ 2 & -2 & 6\end{vmatrix}=3(18-4)-1(6+4)+2(-2-6)$

= 42 – 10 – 16

= 42 – 26

= 16

20. If $\vec{a},\vec{b},\vec{c}$ are non–coplanar vectors and λ is a real number, then the vectors $\vec{a}+2\vec{b}+3\vec{c}$, $\lambda\vec{b}+4\vec{c}$ and $(2\lambda-1)\vec{c}$ are non- coplanar for :

(A) all values of λ (B) all except one value of λ

(C) all except two values of λ (D) no value of λ

यदि $\vec{a},\vec{b},\vec{c}$ विषमतलीय सदिश हैं तथा λ एक वास्तविक संख्या है तो सदिश $\vec{a}+2\vec{b}+3\vec{c}$, $\lambda\vec{b}+4\vec{c}$ तथा $(2\lambda-1)\vec{c}$ विषमतलीय है :

(A) λ के सभी मानों के लिए (B) एक को छोड़ कर λ के सभी मानों के लिए

(C) दो को छोड़ कर λ के सभी मानों के लिए (D) λ के किसी भी मान के लिए नहीं

Ans. C

Sol. Let $\vec{\alpha}=\vec{a}+2\vec{b}+3\vec{c}$, $\beta=\lambda\vec{b}+4\vec{c}$ and $\vec{\gamma}=(2\lambda-1)\vec{c}$.

Then, $\left[\vec{\alpha}\ \vec{\beta}\ \vec{\gamma}\right]=\begin{vmatrix}1 & 2 & 3\\ 0 & \lambda & 4\\ 0 & 0 & (2\lambda-1)\end{vmatrix}\left[\vec{a}\,\vec{b}\,\vec{c}\right]$

$\Rightarrow \quad \left[\vec{\alpha}\ \vec{\beta}\ \vec{\gamma}\right]=\lambda(2\lambda-1)\left[\vec{a}\,\vec{b}\,\vec{c}\right]$

$\Rightarrow \quad \left[\alpha\ \vec{\beta}\ \vec{\gamma}\right]=0,\ \text{if}\ \lambda=0,\ \frac{1}{2} \qquad \left[\because\ \left[\vec{a}\,\vec{b}\,\vec{c}\right]\neq 0\right]$

Hence, $\vec{\alpha},\vec{\beta},\vec{\gamma}$ are non-coplanar for all values of λ except two values 0 and $\frac{1}{2}$.

21. The edges of a parallelopiped are of unit length and are parallel to non-coplanar unit vectors $\hat{a}$, $\hat{b}$, $\hat{c}$ such that $\hat{a}\cdot\hat{b}=\hat{b}\cdot\hat{c}=\hat{c}\cdot\hat{a}=\frac{1}{2}$. Then the volume of the parallelopiped is :

एक षट्फलक (parallelopiped) की भुजाऐं इकाई लम्बाई की है व विषमतलीय इकाई सदिशों $\hat{a}$, $\hat{b}$, $\hat{c}$ जहाँ $\hat{a}\cdot\hat{b}=\hat{b}\cdot\hat{c}=\hat{c}\cdot\hat{a}=\frac{1}{2}$ के समान्तर है। तो षट्फलक का आयतन निम्न है :

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2\sqrt{2}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{\sqrt{3}}$

Ans. A

Sol. $\hat{a}\cdot\hat{b}=\hat{b}\cdot\hat{c}=\hat{c}\cdot\hat{a}=\frac{1}{2}$

$V=\left(\vec{a}\times\vec{b}\right)\cdot\frac{\vec{a}\times\vec{b}}{\vec{a}\times\vec{b}}\vec{c}$ $\qquad h=\frac{\left(\hat{a}\times\hat{b}\right)}{\left|\hat{a}\times\hat{b}\right|}\cdot\vec{c}$

$V=\left|\left(\vec{a}\times\vec{b}\right)\cdot\vec{c}\right|$

$$\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]^2 = \begin{vmatrix} \vec{a}.\vec{a} & \vec{a}.\vec{b} & \vec{a}.\vec{c} \\ \vec{b}.\vec{a} & \vec{b}.\vec{b} & \vec{b}.\vec{c} \\ \vec{c}.\vec{a} & \vec{c}.\vec{b} & \vec{c}.\vec{c} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 \end{vmatrix}$$

$$= 1 + \frac{1}{8} + \frac{1}{8} - \frac{1}{4} - \frac{1}{4} - \frac{1}{4}$$

$$= 1 - \frac{1}{2} = \frac{1}{2}$$

$$\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right] = \frac{1}{\sqrt{2}}$$

22. The vector $\vec{a} = \alpha\hat{i} + 2\hat{j} + \beta\hat{k}$ lies in the plane of the vectors $\vec{b} = \hat{i} + \hat{j}$ and $\vec{c} = \hat{j} + \hat{k}$ and bisects the angle between $\vec{b}$ and $\vec{c}$. Then which one of the following gives possible values of α and β? [AIEEE-2008]

सदिश $\vec{a} = \alpha\hat{i} + 2\hat{j} + \beta\hat{k}$ सदिशों $\vec{b} = \hat{i} + \hat{j}$ तथा $\vec{c} = \hat{j} + \hat{k}$ के तल में स्थित है तथा $\vec{b}$ तथा $\vec{c}$ के बीच के कोण का समद्विभाजन करता है तो निम्न में से कौनसा α एवं β के संभावित मान प्रदान करता है ? [AIEEE-2008]

(A) $\alpha = 1, \beta = 2$ (B) $\alpha = 2, \beta = 1$ (C) $\alpha = 1, \beta = 1$ (D) $\alpha = 2, \beta = 2$

Ans. C

Sol. $\vec{a} = \lambda\left(\hat{b} + \hat{c}\right)$

$$\Rightarrow \alpha\hat{i} + 2\hat{j} + \beta\hat{k} = \lambda\left(\frac{\hat{i} + 2\hat{j} + \hat{k}}{\sqrt{2}}\right)$$

$\lambda = \sqrt{2}\alpha$ and $\lambda = \sqrt{2}$ and $\lambda = \sqrt{2}\beta$

Þ a = 1 and b = 1.

23. If $\vec{u}, \vec{v}, \vec{w}$ are non-coplanar vectors and p, q are real numbers, then the equality

$\left[3\vec{u}\ \ p\vec{v}\ \ p\vec{w}\right] - \left[p\vec{v}\ \ \vec{w}\ \ q\vec{u}\right] - \left[2\vec{w}\ \ q\vec{v}\ \ q\vec{u}\right] = 0$ holds for : [AIEEE-2009]

(A) exactly two values of (p,q) (B) more than two but not all values of (p, q)
(C) all values of (p, q) (D) exactly one value of (p, q)

यदि $\vec{u}, \vec{v}, \vec{w}$ असमतलीय सदिश है तथा p, q वास्तविक संख्याऐं है तो समिका $[3\vec{u}\ p\vec{v}\ p\vec{w}]-[p\vec{v}\ \vec{w}\ q\vec{u}]-[2\vec{w}\ q\vec{v}\ q\vec{u}]=0$ निम्न के लिए सत्य होगी : [AIEEE-2009]

(A) (p,q) के ठीक दो मानों के लिए (B) दो से अधिक परन्तु (p, q) के सभी मानों के लिए नहीं
(C) (p, q) के सभी मानों के लिए (D) (p, q) के ठीक एक मान के लिए

Ans. D

Sol. $[3\vec{u}\ p\vec{v}\ p\vec{w}]-[p\vec{v}\ \vec{w}\ q\vec{u}]-[2\vec{w}\ q\vec{v}\ q\vec{u}]=0$

$3p^2[\vec{u}\ \vec{v}\ \vec{w}] - pq[\vec{v}\ \vec{w}\ \vec{u}] - 2q^2[\vec{w}\ \vec{v}\ \vec{u}] = 0$

$(3p^2 - pq + 2q^2) . [\vec{u}\ \vec{v}\ \vec{w}] = 0$

$3p^2 - pq + 2q^2 = 0$

has exactly one solution

$p = q = 0$

24. If the vectors $p\hat{i}+\hat{j}+\hat{k}, \hat{i}+q\hat{j}+\hat{k}$ and $\hat{i}+\hat{j}+r\hat{k}$ $(p \neq q \neq r \neq 1)$ are coplanar, then the value of $pqr-(p+q+r)$ is : [AIEEE-2011]

यदि यदिश $p\hat{i}+\hat{j}+\hat{k}, \hat{i}+q\hat{j}+\hat{k}$ तथा $\hat{i}+\hat{j}+r\hat{k}$ $(p \neq q \neq r \neq 1)$ समतलीय हैं, तो $pqr-(p+q+r)$ का मान हैं : [AIEEE-2011]

(A) –2 (B) 2 (C) 0 (D) –1

Ans. A

Sol. $\begin{vmatrix} p & 1 & 1 \\ 1 & q & 1 \\ 1 & 1 & r \end{vmatrix} = 0$

$p(qr - 1) - (r - 1) + (1 - q) = 0$

$pqr - p - r + 1 + 1 - q = 0$

$pqr - (p + r + q) + 2 = 0$

$pqr - (p + r + q) = -2$

VTP AND ITS APPLICATION

25. If $\vec{b}$ and $\vec{c}$ are two non-collinear vectors such that $\vec{a} \parallel (\vec{b}\times\vec{c})$, then $(\vec{a}\times\vec{b}) . (\vec{a}\times\vec{c})$ is equal to :

(A) $\vec{a}^2 (\vec{b} . \vec{c})$ (B) $\vec{b}^2 (\vec{a} . \vec{c})$ (C) $\vec{c}^2 (\vec{a} . \vec{b})$ (D) None of these

यदि $\vec{b}$ तथा $\vec{c}$ दो असंरेखीय सदिश इस प्रकार है कि $\vec{a} \parallel (\vec{b} \times \vec{c})$, तो $(\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{c})$ बराबर होगा :

(A) $\vec{a}^2 (\vec{b} \cdot \vec{c})$ (B) $\vec{b}^2 (\vec{a} \cdot \vec{c})$ (C) $\vec{c}^2 (\vec{a} \cdot \vec{b})$ (D) इनमें से कोई नहीं

Ans. A

Sol. $\vec{a} \parallel \vec{b} \times \vec{c}$

$(\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{c}) = ?$

$\vec{a} \perp \vec{b}$

$\vec{a} \perp \vec{c}$

$|\vec{a}||\vec{b}|(\vec{a} \times \vec{b}) \cdot |\vec{a}||\vec{c}|(\vec{a} \times \vec{c})$

$(\vec{a})^2 \left(|\vec{b}||\vec{c}|\right)(\cos\theta)$

$\vec{a}^2 (\vec{b} \cdot \vec{c})$

26. Vector of length 3 unit which is perpendicular to $\hat{i} + \hat{j} + \hat{k}$ and lies in the plane of $\hat{i} + \hat{j} + \hat{k}$ and $2\hat{i} - 3\hat{j}$:

परिमाण 3 का एक सदिश ज्ञात कीजिये जो $\hat{i} + \hat{j} + \hat{k}$ के लम्बवत् है तथा $\hat{i} + \hat{j} + \hat{k}$ तथा $2\hat{i} - 3\hat{j}$ के समतलीय है :

(A) $\frac{3}{\sqrt{6}}(\hat{i} - 2\hat{j} + \hat{k})$ (B) $\frac{3}{\sqrt{6}}(2\hat{i} - \hat{j} - \hat{k})$ (C) $\frac{3}{\sqrt{114}}(8\hat{i} - 7\hat{j} - \hat{k})$ (D) $\frac{3}{\sqrt{114}}(-7\hat{i} + 8\hat{j} - \hat{k})$

Ans. D

Sol. $\vec{a} = \hat{i} + \hat{j} + \hat{k}$

$\vec{b} = 2\hat{i} - 3\hat{j}$

$\vec{v} = (\vec{a} \times \vec{b}) \times \vec{a}$

$= (\vec{a} \cdot \vec{a})\vec{b} - \vec{a}(\vec{a} \cdot \vec{b})$

$= 3 \cdot (2\hat{i} - 3\hat{j}) - (-1)(\hat{i} + \hat{j} + \hat{k})$

$7\hat{i} - 8\hat{j} + \hat{k}$

For magnitude 3 $= \pm 3 \frac{\vec{v}}{|\vec{v}|}$

$= \pm \frac{3}{\sqrt{114}}(-7\hat{i} + 8\hat{j} - \hat{k})$

27. If $\left[\vec{a}\,\vec{b}\,\vec{c}\right]=4$ then $\left[\vec{a}\times\vec{b}\;\;\vec{b}\times\vec{c}\;\;\vec{c}\times\vec{a}\right]=$

यदि $\left[\vec{a}\,\vec{b}\,\vec{c}\right]=4$ तब $\left[\vec{a}\times\vec{b}\;\;\vec{b}\times\vec{c}\;\;\vec{c}\times\vec{a}\right]=$

(A) 4 (B) 2 (C) 8 (D) 16

Ans. D

Sol. [a b c] = 4

$\left(\vec{a}\times\vec{b}\right)\times\left(\vec{b}\times\vec{c}\right)\cdot\left(\vec{c}\times\vec{a}\right)$

$\left[\vec{b}\left(\left(\vec{a}\times\vec{b}\right)\cdot\vec{c}\right)-\vec{c}\left(\left(\vec{a}\times\vec{b}\right)\cdot\vec{b}\right)\right]\cdot\left(\vec{c}\times\vec{a}\right)$

$4\vec{b}\cdot\left(\vec{c}\times\vec{a}\right)$

= 16

28. If $\vec{c}=2\lambda\left(\vec{a}\times\vec{b}\right)+3\mu\left(\vec{b}\times\vec{a}\right);\vec{a}\times\vec{b}\neq 0,\ \vec{c}.\left(\vec{a}\times\vec{b}\right)=0$ then :

(A) $\lambda = 3\mu$ (B) $2\lambda = 3\mu$ (C) $\lambda + \mu = 0$ (D) None of these

यदि $\vec{c}=2\lambda\left(\vec{a}\times\vec{b}\right)+3\mu\left(\vec{b}\times\vec{a}\right);\vec{a}\times\vec{b}\neq 0,\ \vec{c}.\left(\vec{a}\times\vec{b}\right)=0$ तो :

(A) $\lambda = 3\mu$ (B) $2\lambda = 3\mu$ (C) $\lambda + \mu = 0$ (D) इनमें से कोई नहीं

Ans. B

Sol. $\vec{c}=2\lambda\left(\vec{a}\times\vec{b}\right)+3\mu\left(\vec{b}\times\vec{a}\right)\vec{a}\times\vec{b}\neq 0$

dot with $\vec{c}$ $\qquad\vec{c}\cdot\left(\vec{a}\times\vec{b}\right)=0$

$|\vec{c}|^2=0+0$

$\vec{c}=0$

$-2\lambda+3\mu=0$

29. For any vector $\vec{a}$, $\left|\vec{a}\times\hat{i}\right|^2+\left|\vec{a}\times\hat{j}\right|^2+\left|\vec{a}\times\hat{k}\right|^2$ is equal to :

(A) $|\vec{a}|^2$ (B) $2|\vec{a}|^2$ (C) $3|\vec{a}|^2$ (D) None of these

किसी भी सदिश $\vec{a}$ के लिए, $\left|\vec{a}\times\hat{i}\right|^2+\left|\vec{a}\times\hat{j}\right|^2+\left|\vec{a}\times\hat{k}\right|^2$ का मान निम्न में से किस के बराबर है :

(A) $|\vec{a}|^2$ (B) $2|\vec{a}|^2$ (C) $3|\vec{a}|^2$ (D) इनमें से कोई नहीं

Ans. B

Sol. $\left|\vec{a}\times\hat{i}\right|^2+\left|\vec{a}\times\hat{j}\right|^2+\left|\vec{a}\times\hat{k}\right|^2$

$\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$

$\left(\vec{a}\times\hat{i}=-a_2\hat{k}+a_3\hat{j}\right)$

$=$ $a_2^2+a_3^2+a_1^2+a_3^2+a_1^2+a_2^2$

$=2\left(a_1^2+a_2^2+a_3^2\right)$

$=2\left|\vec{a}\right|^2$

30. If $\left(\vec{a}\times\vec{b}\right)\times\vec{c}=\vec{a}\times\left(\vec{b}\times\vec{c}\right)$, where $\vec{a}$, $\vec{b}$ and $\vec{c}$ are any three vectors such that $\vec{a}\cdot\vec{b}\neq0$, $\vec{b}\cdot\vec{c}\neq0$, then $\vec{a}$ and $\vec{c}$ are :

(A) inclined at an angle of $\pi/6$ between them (B) perpendicular

(C) parallel (D) inclined at an angle of $\pi/3$ between them

यदि $\left(\vec{a}\times\vec{b}\right)\times\vec{c}=\vec{a}\times\left(\vec{b}\times\vec{c}\right)$, जहाँ $\vec{a}$, $\vec{b}$ तथा $\vec{c}$ कोई तीन ऐसे ऐसे सदिश है कि $\vec{a}\cdot\vec{b}\neq0$, $\vec{b}\cdot\vec{c}\neq0$, तो $\vec{a}$ तथा $\vec{c}$ है :

(A) के बीच $\pi/6$ का कोण है (B) लम्बवत् है

(C) समान्तर है (D) के बीच $\pi/3$ का कोण है

Ans. C

Sol. $(\vec{a}\times\vec{b})\times\vec{c}=\vec{a}\times(\vec{b}\times\vec{c})$

$\Rightarrow(\vec{a}.\vec{c})\vec{b}-(\vec{b}.\vec{c})\vec{a}=(\vec{a}.\vec{c})\vec{b}-(\vec{a}.\vec{b})\vec{c}$

$\Rightarrow(\vec{b}.\vec{c})\vec{a}=(\vec{a}.\vec{b})\vec{c}$

So that $\vec{a}$ is parallel to $\vec{c}$

31. If $\vec{a}=\frac{1}{\sqrt{10}}(3\vec{i}+\vec{k})$ and $\vec{b}=\frac{1}{7}(2\vec{i}+3\vec{j}-6\vec{k})$, then the value of $(2\vec{a}-\vec{b}).[(\vec{a}\times\vec{b})\times(\vec{a}+2\vec{b})]$ is :

[AIEEE-2011]

यदि $\vec{a}=\frac{1}{\sqrt{10}}(3\vec{i}+\vec{k})$ तथा $\vec{b}=\frac{1}{7}(2\vec{i}+3\vec{j}-6\vec{k})$, तो $(2\vec{a}-\vec{b}).[(\vec{a}\times\vec{b})\times(\vec{a}+2\vec{b})]$ का मान होगा : [AIEEE-2011]

(A) 5 (B) 3 (C) –5 (D) –3

Ans. C

Sol. $\vec{a}.\vec{b}=0$ and $|a| = |b| = 1$

$(\vec{a}\times\vec{b})\times(\vec{a}+2\vec{b}) = (\vec{a}\times\vec{b})\times\vec{a}+(\vec{a}\times\vec{b})\times 2\vec{b}$

$= -\left[\vec{a}\times(\vec{a}\times\vec{b})+2\vec{b}\times(\vec{a}\times\vec{b})\right]$

$= -\left[(\vec{a}.\vec{b})\vec{a}-(\vec{a}\ .\ \vec{a})\vec{b}+2(\vec{b}\ .\vec{b})\vec{a}-2(\vec{b}\ .\ \vec{a})\vec{b}\right]$

$= -\left[0-\vec{b}+2\vec{a}+0\right]=\left[\vec{b}-2\vec{a}\right]$

$\therefore\ (2\vec{a}-\vec{b}).\left[(\vec{a}\times\vec{b})\times(\vec{a}+2\vec{b})\right]$

$= (2\vec{a}-\vec{b})\ .\ (\vec{b}-2\vec{a})$

$= -4a^2 - b^2 + 4\,\vec{a}\ .\ \vec{b} = -5$

32. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three non-zero vectors such that no two of them are collinear and $(\vec{a}\times\vec{b})\times\vec{c} = \frac{1}{3}|\vec{b}||\vec{c}|\vec{a}$. If θ is the angle between vectors $\vec{b}$ and $\vec{c}$, then a value of $\sin\theta$ is : [JEE(Main)-2015]

माना $\vec{a}$, $\vec{b}$ तथा $\vec{c}$ तीन शून्योतर ऐसे सदिश हैं कि उनमें से कोई दो संरेख नहीं हैं तथा $(\vec{a}\times\vec{b})\times\vec{c} = \frac{1}{3}|\vec{b}||\vec{c}|\vec{a}$ हैं। यदि सदिशों $\vec{b}$ तथा $\vec{c}$ के बीच का कोण θ है, तो $\sin\theta$ का एक मान है : [JEE(Main)-2015]

(A) $\frac{2\sqrt{2}}{3}$ (B) $\frac{-\sqrt{2}}{3}$ (C) $\frac{2}{3}$ (D) $\frac{-2\sqrt{3}}{3}$

Ans. A

Sol. $(\vec{a}.\vec{c})\vec{b}-(\vec{b}.\vec{c})\vec{a} = \frac{1}{3}|\vec{b}||\vec{c}|\vec{a}$

$-\vec{b}.\vec{c} = \frac{1}{3}(\vec{b})(\vec{c})$

$\text{Cos}\,\theta = -\frac{1}{3}$

$\sin\theta = \frac{2\sqrt{2}}{3}$

33. Let $\vec{a},\vec{b}$ and $\vec{c}$ be three unit vectors such that $\vec{a}\times\left(\vec{b}\times\vec{c}\right)=\frac{\sqrt{3}}{2}\left(\vec{b}+\vec{c}\right)$. If $\vec{b}$ is not parallel to $\vec{c}$, then the angle between $\vec{a}$ and $\vec{b}$ is : [JEE(Main)-2016]

माना $\vec{a},\vec{b}$ तथा $\vec{c}$ तीन ऐसे मात्रक सदिश हैं कि $\vec{a}\times\left(\vec{b}\times\vec{c}\right)=\frac{\sqrt{3}}{2}\left(\vec{b}+\vec{c}\right)$ है। यदि $\vec{b}$, $\vec{c}$ के समान्तर नहीं है, तो $\vec{a}$ तथा $\vec{b}$ के बीच का कोण है : [JEE(Main)-2016]

(A) $\frac{2\pi}{3}$ (B) $\frac{5\pi}{6}$ (C) $\frac{3\pi}{4}$ (D) $\frac{\pi}{2}$

Ans. B

Sol. $\left(\vec{a}.\vec{c}\right)\vec{b}-\left(\vec{a}.\vec{b}\right)\vec{c}=\frac{\sqrt{3}}{2}\vec{b}+\frac{\sqrt{3}}{2}\vec{c}$

$-\vec{a}.\vec{b}=\frac{\sqrt{3}}{2}$

$-\cos\theta=\frac{\sqrt{3}}{2}$

$\theta=\frac{5\pi}{6}$

34. Let $\vec{a}=2\hat{i}+\hat{j}-2\hat{k}$ and $\vec{b}=\hat{i}+\hat{j}$. Let $\vec{c}$ be a vector such that $|\vec{c}-\vec{a}|=3$, $\left|\left(\vec{a}\times\vec{b}\right)\times\vec{c}\right|=3$ and the angle between $\vec{c}$ and $\vec{a}\times\vec{b}$ be 30°. Then $\vec{a}.\vec{c}$ is equal to : [JEE(Main)-2017]

माना $\vec{a}=2\hat{i}+\hat{j}-2\hat{k}$ तथा $\vec{b}=\hat{i}+\hat{j}$ है। माना $\vec{c}$ एक ऐसा सदिश है कि $|\vec{c}-\vec{a}|=3$, $\left|\left(\vec{a}\times\vec{b}\right)\times\vec{c}\right|=3$ तथा $\vec{c}$ और $\vec{a}\times\vec{b}$ के बीच कोण 30° है, तो $\vec{a}.\vec{c}$ बराबर होगा : [JEE(Main)-2017]

(A) 5 (B) $\frac{1}{8}$ (C) $\frac{25}{8}$ (D) 2

Ans. D

Sol. $\vec{a}\times\vec{b}=2\hat{i}-2\hat{j}+\hat{k}\quad\Rightarrow\left|\vec{a}\times\vec{b}\right|=3\quad|\vec{c}-\vec{a}|=3$

$\left|\vec{a}\times\vec{b}\right||\vec{c}|\sin 30°=3$ $\qquad c^2+a^2-2\vec{a}.\vec{c}=9$

$|\vec{c}|=2$ $\qquad \vec{c}.\vec{a}=2$

35. A line with direction cosines proportional to 2, 1, 2 meets each of the lines $x = y + a = z$ and $x + a = 2y = 2z$. The coordinates of each of the points of intersection are given by :

एक रेखा जिसके दिक्कोज्या 2, 1, 2 के अनुपात में हैं, रेखाओं $x = y + a = z$ तथा $x + a = 2y = 2z$ को काटती हैं। इन काटे गए बिन्दुओं के निर्देशांक हैं :

(A) (3a, 3a, 3a), (a, a, a)
(B) (3a, 2a, 3a), (a, a, a)
(C) (3a, 2a, 3a), (a, a, 2a)
(D) (2a, 3a, 3a), (2a, a, a)

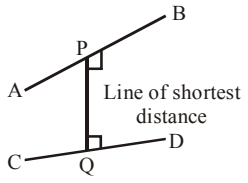
Ans. B

Sol. Let the two lines be AB and CD having equation $\frac{x}{1} = \frac{y+a}{1} = \frac{z}{1} = \lambda$ and $\frac{x+a}{2} = \frac{y}{1} = \frac{z}{1} = \mu$

then $P \equiv (\lambda, \lambda - a, \lambda)$ and $Q = (2\mu - a, \mu, \mu)$

So according to question,

$$\frac{\lambda - 2\mu + a}{2} = \frac{\lambda - a - \mu}{1} = \frac{\lambda - \mu}{2}$$



$\Rightarrow \quad \mu = a$ and $\lambda = 3a$

$\therefore \quad P \equiv (3a, 2a, 3a)$

and $\quad Q \equiv (a, a, 0)$

Trick : Put the option and check it

36. If a line makes an angle of $\pi/4$ with the positive directions of each of x-axis and y-axis, then the angle that the line makes with the positive direction of the z-axis is : [AIEEE-2007]

यदि एक रेखा प्रत्येक x- अक्ष तथा y-अक्ष की धनात्मक दिशा के साथ $\pi/4$ का कोण बनाती है, तब वह कोण, जो रेखा z-अक्ष की धनात्मक दिशा के साथ बनायेगी, होगा : [AIEEE-2007]

(A) $\pi/6$ (B) $\pi/3$ (C) $\pi/4$ (D) $\pi/2$

Ans. D

Sol. $\ell = \cos\frac{\pi}{4}, m = \cos\frac{\pi}{4}$

we know that $\ell^2 + m^2 + n^2 = 1$

$$\frac{1}{2} + \frac{1}{2} + n^2 = 1 \Rightarrow n = 0$$

Hence angle with positive direction of z-axis is $\frac{\pi}{2}$

37. The angle between the lines whose direction cosines satisfy the equations $l + m + n = 0$ and $l^2 = m^2 + n^2$ is : [JEE(Main)-2014]

दो रेखाएँ, जिनके दिक्–कोज्या, समीकरणों $l + m + n = 0$ तथा $l^2 = m^2 + n^2$ को संतुष्ट करते हैं, के बीच का कोण है :

[JEE(Main)-2014]

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$

Ans. C

Sol. Solving two equation

$l = -(m + n)$ then $(m + n)^2 = m^2 + n^2$

Þ $mn = 0$

If $m = 0$, If $n = 0$

$l = -n$ $l = -m$

hence dr are 1, 0, –1 and 1, –1, 0

$$\cos\theta = \frac{1}{\sqrt{2}\sqrt{2}} = \frac{1}{2}$$

$$\theta = \frac{\pi}{3}$$

38. The two lines $x = ay + b$, $z = cy + d$; and $x = a'y + b'$, $z = c'y + d'$ are perpendicular to each other if :

दो रेखाऐं $x = ay + b$, $z = cy + d$; तथा $x = a'y + b'$, $z = c'y + d'$ लम्बवत् होंगी, यदि और केवल यदि :

(A) $aa' + cc' = 1$ (B) $\frac{a}{a'} + \frac{c}{c'} = -1$ (C) $\frac{a}{a'} + \frac{c}{c'} = 1$ (D) $aa' + cc' = -1$

Ans. D

Sol. $\ell_1 - am_1 = 0$ and $cm_1 - n_1 = 0 \Rightarrow \frac{\ell_1}{a} = \frac{m_1}{1} = \frac{n_1}{c}$

Also $\ell_2 - a'm_2 = 0$ and $c'm_2 - n_2 = 0$

$\Rightarrow \frac{\ell_2}{a'} = \frac{m_2}{1} = \frac{n_2}{c'}$

$\therefore \ \ell_1\ell_2 + m_1m_2 + n_1n_2 = aa' + cc' + 1 = 0$

39. If the straight lines $\frac{x-1}{k} = \frac{y-2}{2} = \frac{z-3}{3}$ and $\frac{x-2}{3} = \frac{y-3}{k} = \frac{z-1}{2}$ intersect at a point, then the integer k is equal to :

[AIEEE-2008]

यदि सरल रेखाऐं $\frac{x-1}{k} = \frac{y-2}{2} = \frac{z-3}{3}$ तथा $\frac{x-2}{3} = \frac{y-3}{k} = \frac{z-1}{2}$ एक बिन्दु पर प्रतिच्छेद करती हैं, तो पूर्णांक k का मान है:

[AIEEE-2008]

(A) 5 (B) 2 (C) –2 (D) –5

Ans. D

Sol. $\frac{x-1}{k}=\frac{y-2}{2}=\frac{z-3}{3}$ and $\frac{x-2}{3}=\frac{y-3}{k}=\frac{z-1}{2}$

Since lines intersect in a point

$$\begin{vmatrix} k & 2 & 3 \\ 3 & k & 2 \\ 1 & 1 & -2 \end{vmatrix} = 0$$

\ 2k2 + 5k – 25 = 0

k = –5, 5/2

PLANE

40. Equation of the plane containing the straight line $\frac{x}{2}=\frac{y}{3}=\frac{z}{4}$ and perpendicular to the plane containing the straight lines $\frac{x}{3}=\frac{y}{4}=\frac{z}{2}$ and $\frac{x}{4}=\frac{y}{2}=\frac{z}{3}$, is :

रेखा T को अन्तर्विष्ट करने वाले तल का समीकरण ज्ञात कीजिये जो रेखा $\frac{x}{3}=\frac{y}{4}=\frac{z}{2}$ तथा $\frac{x}{4}=\frac{y}{2}=\frac{z}{3}$ को समाहित करने वाले तल के लम्बवत् है :

(A) x + 2y – 2z = 0 (B) 3x + 2y – 2z = 0 (C) x – 2y + z = 0 (D) 5x + 2y – 4z = 0

Ans. C

Sol. Let $l_1=\frac{x}{2}=\frac{y}{3}=\frac{z}{4}$, $l_2=\frac{x}{3}=\frac{y}{4}=\frac{z}{2}$, $l_3=\frac{x}{4}=\frac{y}{2}=\frac{z}{3}$

Equation of vector perpendicular to l_2 & l_3

$$NP_2=\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 4 & 2 \\ 4 & 2 & 3 \end{vmatrix}$$

$=8\hat{i}-\hat{j}-10\hat{k}$

Normal vector to the plane will be perpendicular to NP_2 & l_1

$$NP_1=\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 8 & -1 & 10 \end{vmatrix}$$

$= \hat{i}(-26) - \hat{j}(-52) + \hat{k}(-26)$

$\vec{n} = \hat{i} - 2\hat{j} + \hat{k}$

Plane $x - 2y + z = 0$

41. A line with positive direction cosines passes through the point P(2, – 1, 2) and makes equal angles with the coordinate axes. The line meets the plane $2x + y + z = 9$ at point Q. The length of the line segment PQ equals :

एक रेखा जो बिन्दु P(2, – 1, 2) से गुजरती है एवं धनात्मक दिक् कोज्याऐं तथा निर्देशी अक्षों के साथ समान कोण बनाती है। रेखा तल $2x + y + z = 9$ को बिन्दु Q पर मिलती है तो रेखाखण्ड PQ की लम्बाई होगी :

(A) 1 (B) $\sqrt{2}$ (C) $\sqrt{3}$ (D) 2

Ans. C

Sol. $(2, -1, 2) + \lambda (1, 1, 1)$

On plane $2(2 + \lambda) + (-1 + \lambda) + (2 + \lambda) = 9$

$5 + 4\lambda = 9$

$\lambda = 1$

Q (3, 0, 3) P(2, –1, 2)

$PQ = \sqrt{1+1+1} = \sqrt{3}$

42. Two systems of rectangular axes have the same origin. If a plane makes intercepts a, b, c and a′, b′, c′ on the two systems of axes respectively, then :

आयतीय अक्ष के दो निकायों के मूलबिन्दु समान हैं। यदि एक समतल अक्षों की दो निकायों पर क्रमशः a, b, c एवं a′, b′, c′ अन्तःखण्ड काटता है, तब :

(A) $a^2 + b^2 + c^2 = a'^2 + b'^2 + c'^2$ (B) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{1}{a'} + \frac{1}{b'} + \frac{1}{c'}$

(C) $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{a'^2} + \frac{1}{b'^2} + \frac{1}{c'^2}$ (D) $\frac{1}{a^2 - a'^2} + \frac{1}{b^2 - b'^2} + \frac{1}{c^2 - c'^2} = 0$

Ans. C

Sol. In system I In system II

$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ $\frac{x}{a'} + \frac{y}{b'} + \frac{z}{c'} = 1$

$\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$ = $\frac{1}{a'^2} + \frac{1}{b'^2} + \frac{1}{c'^2}$

43. Let P(3, 2, 6) be a point in space and Q be a point on the line $\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \mu(-3\hat{i} + \hat{j} + 5\hat{k})$. Then the value of μ for which the vector $\overrightarrow{PQ}$ is parallel to the plane $x - 4y + 3z = 1$ is :

माना P(3, 2, 6) समष्टि (space) में स्थित एक बिन्दु है तथा Q रेखा $\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \mu(-3\hat{i} + \hat{j} + 5\hat{k})$ पर स्थित एक बिन्दु है। तब μ का मान जिसके लिये सदिश $\overrightarrow{PQ}$ समतल $x - 4y + 3z = 1$ के समान्तर है, निम्न है :

(A) 1/4 (B) – 1/4 (C) 1/8 (D) – 1/8

Ans. A

Sol. Angle between line and normal to plane is

$$\cos\left(\frac{\pi}{2} - \theta\right) = \frac{1\times 2 - 2\times 1 + 2\sqrt{\lambda}}{3\times\sqrt{5+\lambda}}$$, where θ is the angle between line and plane

$$\Rightarrow \sin\theta = \frac{1\times 2 + 2\times(-1) + 2\sqrt{\lambda}}{3\times\sqrt{5+\lambda}} \Rightarrow \frac{1}{3} = \frac{2\sqrt{\lambda}}{3\sqrt{5+\lambda}}$$

$$\Rightarrow \frac{1}{3} = \frac{2\sqrt{\lambda}}{3\sqrt{5+\lambda}} \Rightarrow \lambda = \frac{5}{3}$$

44. If the distance of the point P(l, – 2, 1) from the plane $x + 2y - 2z = \alpha$, where $\alpha > 0$. is 5, then the foot of the perpendicular from P to the plane is :

यदि बिन्दु P(l, – 2, 1) की तल $x + 2y - 2z = \alpha$, जहाँ $\alpha > 0$ से दूरी 5 है तो बिन्दु P से तल पर डाला गया लम्ब पाद होगा :

(A) $\left(\frac{8}{3}, \frac{4}{3}, -\frac{7}{3}\right)$ (B) $\left(\frac{4}{3}, -\frac{4}{3}, \frac{1}{3}\right)$ (C) $\left(\frac{1}{3}, \frac{2}{3}, \frac{10}{3}\right)$ (D) $\left(\frac{2}{3}, -\frac{1}{3}, \frac{5}{2}\right)$

Ans. A

Sol. $\left|\frac{1-4-2-\alpha}{3}\right| = 5$

$|5 + \alpha| = 15$

$(1,-2,1) + \lambda(1,2,-2)$

$(1 + \lambda) + 2(-2 + 2\lambda) - 2(1 - 2\lambda) = 10$

$1 + \lambda - 4 + 4\lambda - 2 + 4\lambda = 10$

$9\lambda - 15 = 0$

$9\lambda = 15$

$\lambda = \frac{5}{3}$

$\left(\frac{8}{3}, \frac{4}{3}, \frac{7}{3}\right)$

45. The distance of the point (1, –5, 9) from the plane x – y + z = 5 measured along a straight line x = y = z is :

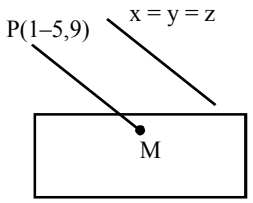
[AIEEE-2011]

बिन्दु (1, –5, 9) की समतल x – y + z = 5 से दूरी जो सरल रेखा x = y = z के अनुदिश मापी गयी है, हैं : [AIEEE-2011]

(A) $3\sqrt{5}$ (B) $10\sqrt{3}$ (C) $5\sqrt{3}$ (D) $3\sqrt{10}$

Ans. B

Sol.



eq^{n}. of a line || to x = y = z and
passing through (1, –5, 9) is

$$\frac{x-1}{1}=\frac{y+5}{1}=\frac{z-9}{1}=r$$

Let is meets plane at M(r+1, r–5, r+9)

Put in equation of plane

$x - y + z = 5$

$r + 1 - r + 5 + r + 9 = 5$

$r = -10$

Hence M (–9, –15, –1)

Distance PM = $\sqrt{100+100+100}=10\sqrt{3}$

46. The equation of the plane containing the line 2x – 5y + z = 3; x + y + 4z = 5, and parallel to the plane, x + 3y + 6z = 1, is : [JEE(Main)-2015]

रेखा 2x – 5y + z = 3; x + y + 4z = 5 को अंतर्विष्ट करने वाले समतल, जो समतल x + 3y + 6z = 1 के समांतर है, का समीकरण है : [JEE(Main)-2015]

(A) 2x + 6y+12z = 13 (B) x + 3y + 6z = – 7 (C) x + 3y + 6z = 7 (D) 2x + 6y + 12z = – 13

Ans. C

Sol. $\vec{u}=\lambda\left(\vec{a}\times\left(\vec{a}\times\vec{b}\right)\right)$

$$=\lambda\left(\left(\vec{a}.\vec{b}\right)\vec{a}-\left(\vec{a}.\vec{a}\right)\vec{b}\right)$$

$$=\lambda\left(2\vec{a}-14\vec{b}\right)$$

$$\vec{u}=\lambda\left(4\hat{i}-8\hat{j}-16\hat{k}\right)$$

$\vec{u}.\vec{b}=24 \Rightarrow \lambda=-1$

$\Rightarrow \vec{u}=-4\hat{i}+8\hat{j}+16\hat{k}$

$\Rightarrow|\vec{u}|^2=16+64+256=336$

47. The distance of the point (1, 3, –7) from the plane passing through the point (1, – 1, – 1) having normal perpendicular to both the lines $\frac{x-1}{1}=\frac{y+2}{-2}=\frac{z-4}{3}$ तथा $\frac{x-2}{2}=\frac{y+1}{-1}=\frac{z+7}{-1}$ is : [JEE(Main)-2017]

एक समतल जो बिन्दु (1, –1, –1) से होकर जाता है तथा जिसका अभिलम्ब दोनों रेखाओं $\frac{x-1}{1}=\frac{y+2}{-2}=\frac{z-4}{3}$ तथा $\frac{x-2}{2}=\frac{y+1}{-1}=\frac{z+7}{-1}$ पर लम्ब है, की बिन्दु (1, 3, –7) से दूरी है : [JEE(Main)-2017]

(A) $\frac{5}{\sqrt{83}}$ (B) $\frac{10}{\sqrt{74}}$ (C) $\frac{20}{\sqrt{74}}$ (D) $\frac{10}{\sqrt{83}}$

Ans. D

Sol. Normal vector of plane $=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & -2 & 3\\ 2 & -1 & -1\end{vmatrix}$

$=5\hat{i}+7\hat{j}+3\hat{k}$

equation of plane $5x+7y+3z=-5$

$\text{Distance}=\left|\frac{5(1)+7(3)+3(-7)+5}{\sqrt{83}}\right|$

$=\frac{10}{\sqrt{83}}$

48. If L_1 is the line of intersection of the planes $2x-2y+3z-2=0$, $x-y+z+1=0$ and L_2 is the line of intersection of the planes $x+2y-z-3=0$, $3x-y+2z-1=0$, then the distance of the origin from the plane containing the lines L_1 and L_2 is : [JEE(Main)-2018]

यदि समतलों $2x-2y+3z-2=0$, $x-y+z+1=0$ की परिच्छेदी रेखा L_1 है तथा समतलों $x+2y-z-3=0$, $3x-y+2z-1=0$ की परिच्छेदी रेखा L_2 है, तो मूल बिन्दु की दूरी उस सममतल से जो रेखाओं L_1 और L_2 का अंतर्विष्ट करता है, है : [JEE(Main)-2018]

(A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{1}{4\sqrt{2}}$ (D) $\frac{1}{3\sqrt{2}}$

Ans. D

Sol. required plane will pass though line of intersection of planes so equation of required plane will be

$x+2y-z-3+\lambda(3x-y+2z-1)=0$

take a fixed point on L_1 $(y=0)$

P(–5, 0, 4)

point P will lie on required plane

$\Rightarrow \lambda = -\frac{3}{2}$

So equation of required plane will be

$7x - 7y + 8z + 3 = 0$

distance from origin $= \left|\frac{3}{\sqrt{162}}\right| = \frac{1}{3\sqrt{2}}$

CONCEPT OF IMAGE

49. The image of the point (–1, 3, 4) in the plane $x - 2y = 0$ is :

(A) (15, 11, 4) (B) $\left(-\frac{17}{3}, -\frac{19}{3}, 1\right)$ (C) (8, 4, 4) (D) None of these

बिन्दु (–1, 3, 4) का तल $x - 2y = 0$ में प्रतिबिम्ब है :

(A) (15, 11, 4) (B) $\left(-\frac{17}{3}, -\frac{19}{3}, 1\right)$ (C) (8, 4, 4) (D) इनमें से कोई नहीं

Ans. D

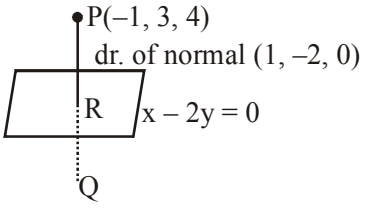
Sol. Equation of line PQ is

$\frac{x+1}{1} = \frac{y-3}{-2} = \frac{z-4}{0} = \lambda$

For some suitable value of λ, co-ordinates of point $Q(\lambda - 1, 3 - 2\lambda, 4)$

R is the mid point of P and Q.

$\therefore R \equiv \left(\frac{\lambda - 2}{2}, \frac{6 - 2\lambda}{2}, 4\right)$



$R \equiv \left(\frac{\lambda}{2} - 1, 3 - \lambda, 4\right)$

It satisfies $x - 2y = 0$

$\Rightarrow \lambda = \frac{14}{5}$

$\therefore \quad Q = \left(\frac{2}{5}, \frac{1}{5}, 4\right)$

50. If the image of the point P(1, – 2, 3) in the plane, 2x + 3y – 4z + 22 = 0 measured parallel to the line, $\frac{x}{1}=\frac{y}{4}=\frac{z}{5}$ is Q, then PQ is equal to : [JEE(Main)-2017]

यदि बिन्दु P(1, – 2, 3) का समतल 2x + 3y – 4z + 22 = 0 में वह प्रतिबिम्ब, जो रेखा $\frac{x}{1}=\frac{y}{4}=\frac{z}{5}$ के समांतर है, Q है, तो PQ बराबर है : [JEE(Main)-2017]

(A) $\sqrt{42}$ (B) $6\sqrt{5}$ (C) $3\sqrt{5}$ (D) $2\sqrt{42}$

Ans. D

Sol. Equation of PQ

$\vec{r} = (1,-2,3) + \lambda(1,4,5)$

N(1 + λ, 4λ – 2, 3 + 5λ)

2(1 + λ) + 3(4λ – 2) – 4(3 + 5λ) + 22 = 0

$6\lambda = 6$

N(2, 2, 8)

$\Rightarrow$ Q(3, 6, 13)

$\Rightarrow$ PQ = $2\sqrt{42}$

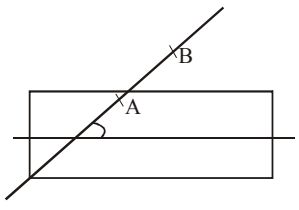
51. The length of the projection of the line segment joining the points (5, –1, 4) and (4, –1, 3) on the plane, x + y + z = 7 is : [JEE(Main)-2018]

बिन्दुओं (5, –1, 4) तथा (4, –1, 3) को मिलाने वाले रेखाखण्ड का समतल x + y + z = 7 पर डाले गए प्रक्षेप की लम्बाई है : [JEE(Main)-2018]

(A) $\frac{1}{3}$ (B) $\sqrt{\frac{2}{3}}$ (C) $\frac{2}{\sqrt{3}}$ (D) $\frac{2}{3}$

Ans. B

Sol.



A (5,–1,4)

B (4,–1,3)

θ = angle between plane & line

$$\cos(90-\theta)=\left|\frac{\overrightarrow{AB}.\left(\hat{i}+\hat{j}+\hat{k}\right)}{\left(\overrightarrow{AB}\right)\left|\hat{i}+\hat{j}+\hat{k}\right|}\right|=\frac{2}{\sqrt{6}}$$

Req. length $=|\overrightarrow{AB}|\cos\theta$

$$=\sqrt{2}\times\frac{\sqrt{2}}{\sqrt{6}}=\sqrt{\frac{2}{3}}$$

MISCELLANEOUS

52. Vector $\vec{x}$ satisfying the relation $\vec{A}\,.\,\vec{x}=c$ and $\vec{A}\times\vec{x}=\vec{B}$ is :

सदिश $\vec{x}$ जो समीकरण $\vec{A}\,.\,\vec{x}=c$ तथा $\vec{A}\times\vec{x}=\vec{B}$ को सन्तुष्ट करता है :

(A) $\frac{c\vec{A}-(\vec{A}\times\vec{B})}{|\vec{A}|}$ (B) $\frac{c\vec{A}-(\vec{A}\times\vec{B})}{|\vec{A}|^2}$ (C) $\frac{c\vec{A}+(\vec{A}\times\vec{B})}{|\vec{A}|^2}$ (D) $\frac{c\vec{A}-2(\vec{A}\times\vec{B})}{|\vec{A}|^2}$

Ans. B

Sol. $\vec{A}\cdot\vec{x}=c$ $\qquad\qquad$ $\left(\vec{A}\times\vec{x}\right)\times\vec{B}=0$

$\vec{A}\times\vec{x}=\vec{B}$

$\left(\vec{A}\times\vec{x}\right)\times\vec{A}=\vec{B}\times\vec{A}$

$\vec{x}\left(\vec{A}\cdot\vec{A}\right)-\vec{A}\left(\vec{A}\cdot\vec{x}\right)=\vec{B}\times\vec{A}$

$\vec{x}=\frac{c\vec{A}+\vec{B}\times\vec{A}}{|A|^2}$

$\frac{c\vec{A}-(\vec{A}\times\vec{B})}{|\vec{A}|^2}$

53. Let $\vec{a}=\hat{j}-\hat{k}$ and $\vec{c}=\hat{i}-\hat{j}-\hat{k}$. Then the vector $\vec{b}$ satisfying $\vec{a}\times\vec{b}+\vec{c}=\vec{0}$ and $\vec{a}.\vec{b}=3$ is : [AIEEE-2010]

माना $\vec{a}=\hat{j}-\hat{k}$ तथा $\vec{c}=\hat{i}-\hat{j}-\hat{k}$ तो सदिश $\vec{b}$ जो $\vec{a}\times\vec{b}+\vec{c}=\vec{0}$ तथा $\vec{a}.\vec{b}=3$ को संतुष्ट करता है : [AIEEE-2010]

(A) $-\hat{i}+\hat{j}-2\hat{k}$ (B) $2\hat{i}-\hat{j}+2\hat{k}$ (C) $\hat{i}-\hat{j}-2\hat{k}$ (D) $\hat{i}+\hat{j}-2\hat{k}$

Ans. A

Sol. $(\vec{a}\times\vec{b})+\vec{c}=0$

$(\vec{a}\times\vec{b}) = -\vec{c}$

$\Rightarrow \vec{a}\times(\vec{a}\times\vec{b}) = -\vec{a}\times\vec{c}$

$\Rightarrow (\vec{a}.\vec{b})\vec{a} - |\vec{a}|^2\ \vec{b} = -\vec{a}\times\vec{c}$

$\Rightarrow 3(\vec{j}-\vec{k}) - 2\vec{b} = -(-2i-j-k)\ (\vec{a}\times\vec{c} = -2i-j-k)$

$\Rightarrow 2\vec{b} = (-2i+2j-4k)$

$\Rightarrow \vec{b} = -i+j-2k$

54. If $\vec{a}$, $\vec{b}$, $\vec{c}$ are linearly independent vectors, then which one of the following set of vectors is linearly dependent?

(A) $\vec{a}+\vec{b}$, $\vec{b}+\vec{c}$, $\vec{c}+\vec{a}$ (B) $\vec{a}-\vec{b}$, $\vec{b}-\vec{c}$, $\vec{c}-\vec{a}$ (C) $\vec{a}\times\vec{b}$, $\vec{b}\times\vec{c}$, $\vec{c}\times\vec{a}$ (D) None

यदि $\vec{a}$, $\vec{b}$, $\vec{c}$ रेखीय रूप से स्वतंत्र सदिश हैं, तो निम्न में से सदिशों का कौनसा समुच्चय रेखीय रूप से परतंत्र है?

(A) $\vec{a}+\vec{b}$, $\vec{b}+\vec{c}$, $\vec{c}+\vec{a}$ (B) $\vec{a}-\vec{b}$, $\vec{b}-\vec{c}$, $\vec{c}-\vec{a}$ (C) $\vec{a}\times\vec{b}$, $\vec{b}\times\vec{c}$, $\vec{c}\times\vec{a}$ (D) कोई नहीं

Ans. (B)

Sol. $(\vec{a}\times\vec{b}) + \vec{c} = 0$

$(\vec{a}\times\vec{b}) = -\vec{c}$

$\Rightarrow \vec{a}\times(\vec{a}\times\vec{b}) = -\vec{a}\times\vec{c}$

$\Rightarrow (\vec{a}.\vec{b})\vec{a} - |\vec{a}|^2\ \vec{b} = -\vec{a}\times\vec{c}$

$\Rightarrow 3(\vec{j}-\vec{k}) - 2\vec{b} = -(-2i-j-k)\ (\vec{a}\times\vec{c} = -2i-j-k)$

$\Rightarrow 2\vec{b} = (-2i+2j-4k)$

$\Rightarrow \vec{b} = -i+j-2k$

Exercise J-2

BASIC PROPERTIES OF VECTORS

1. If $\vec{a}$ and $\vec{b}$ are non collinear vectors such that, $\vec{p} = (x + 4y)\vec{a} + (2x + y + 1)\vec{b}$ and $\vec{q} = (y - 2x + 2)\vec{a} + (2x - 3y - 1)\vec{b}$, find x and y such that $3\vec{p} = 2\vec{q}$.

यदि $\vec{a}$ तथा $\vec{b}$ असंरेखीय सदिश इस प्रकार है कि $\vec{p} = (x + 4y)\vec{a} + (2x + y + 1)\vec{b}$ तथा $\vec{q} = (y - 2x + 2)\vec{a} + (2x - 3y - 1)\vec{b}$, यदि $3\vec{p} = 2\vec{q}$ है, तो x एवं y का मान ज्ञात कीजिए।

Ans. $x = 2, y = -1$

Sol. $\vec{P} = (x + 4y)\vec{a} + (2x + y + 1)\vec{b}$

$\vec{q} = (y - 2x + 2)\vec{a} + (2x - 3y - 1)\vec{b}$

$3P = 2\vec{q}$

$3(x + 4y) = 2\,(y - 2x + 2)$

$$\left.\begin{array}{l} 7x + 10y = 4 \\ 6x + 3y + 3 = 4x - 6y - 2 \\ 2x + 9y = -5 \\ 5x + y = 9 \end{array}\right| \begin{array}{l} x = 2 \\ y = -1 \end{array}$$

2. Show that the points $\vec{a} - 2\vec{b} + 3\vec{c}$; $2\vec{a} + 3\vec{b} - 4\vec{c}$ and $-7\vec{b} + 10\vec{c}$ are collinear.

सिद्ध कीजिए कि बिन्दु $\vec{a} - 2\vec{b} + 3\vec{c}$; $2\vec{a} + 3\vec{b} - 4\vec{c}$ तथा $-7\vec{b} + 10\vec{c}$ संरेखीय है।

Ans.

Sol. $$\left.\begin{array}{l} \vec{p} = \vec{a} - 2\vec{b} + 3\vec{c} \\ \vec{q} = 2\vec{a} + 3\vec{b} - 4\vec{c} \\ \vec{r} = -7\vec{b} + 10\vec{c} \end{array}\right\} \text{Collinear}$$

$2\vec{p} - \vec{q} = \vec{r}$

$\overrightarrow{qp} = -\vec{a} - 5\vec{b} + 7\vec{c}$

$\overrightarrow{rq} = 2\vec{a} + 10\vec{b} + 14\vec{c}$

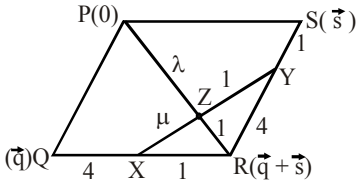
$\overrightarrow{qp} = \lambda\overrightarrow{rq}$

3. Points X and Y are taken on the sides QR and RS, respectively of a parallelogram PQRS, so that $\vec{QX} = 4\vec{XR}$ and $\vec{RY} = 4\vec{YS}$. The line XY cuts the line PR at Z. Prove that $\vec{PZ} = \left(\frac{21}{25}\right)\vec{PR}$.

समान्तर चतुर्भुज PQRS की भुजा QR तथा RS पर बिन्दु X एवं Y इस प्रकार से है कि $\vec{QX} = 4\vec{XR}$ तथा $\vec{RY} = 4\vec{YS}$. यदि रेखा XY रेखा PR को Z पर काटती है तो सिद्ध करो कि $\vec{PZ} = \left(\frac{21}{25}\right)\vec{PR}$.

Ans.

Sol. $\overrightarrow{QX} = 4\overrightarrow{XR}$



$\overrightarrow{RY} = 4\overrightarrow{YS}$

Let $\vec{p}$ be origin

and R $(\vec{q}+\vec{s})$

from figure

P.V. of X $= \dfrac{4(\vec{q}+\vec{s})+\vec{q}}{5} = \dfrac{5\vec{q}+4\vec{s}}{5}$

P.V. of Y $= \dfrac{4\vec{s}+\vec{q}+\vec{s}}{5} = \dfrac{5\vec{s}+\vec{q}}{5}$

Now Let Z divides PR in ratio λ :1

Now Let Z divides XY in ratio μ:1

P.V. of Z $= \dfrac{\lambda(\vec{q}+\vec{s})}{\lambda+1}$ (from PR)

P.V. of Z $= \dfrac{\frac{\mu(5\vec{s}+\vec{q})}{5}+\frac{5\vec{q}+4\vec{s}}{5}}{\mu+1}$ (from XY)

equating both Z then we get

$\dfrac{\lambda}{\lambda+1} = \dfrac{\mu+5}{5(\mu+1)}$ .....(i)

$\dfrac{\lambda}{\lambda+1} = \dfrac{5\mu+4}{5(\mu+1)}$ .....(ii)

from (i) & (ii), $\mu = \dfrac{1}{4}$ and $\lambda = \dfrac{21}{4}$

So P.V. of Z $= \dfrac{\frac{21}{4}}{\frac{21}{4}+1}(\vec{q}+\vec{s})$

$= \dfrac{21}{25}(\vec{q}+\vec{s}) = \dfrac{21}{25}\overrightarrow{PR}$

4. In a ΔABC, points E and F divide sides AC and AB respectively so that $\frac{AE}{EC}=4$ and $\frac{AF}{FB}=1$. Suppose D is a point on side BC. Let G be the intersection of EF and AD and suppose D is situated so that $\frac{AG}{GD}=\frac{3}{2}$. If the ratio $\frac{BD}{DC}=\frac{a}{b}$, where a and b are in their lowest form, find the value of (a + b).

त्रिभुज ΔABC में बिन्दु E एवं F ऐसा AC एवं AB को इस प्रकार से विभाजित करती है कि $\frac{AE}{EC}=4$ तथा $\frac{AF}{FB}=1$ है। रेखा BC पर D एक बिन्दु है। यदि EF तथा AD का प्रतिच्छेद बिन्दु G है तथा D इस प्रकार से है कि $\frac{AG}{GD}=\frac{3}{2}$ यदि $\frac{BD}{DC}=\frac{a}{b}$ जहाँ a एवं b का महत्तम समापवर्तक 1 है तो (a + b) का मान बताओं।

Ans. 9

Sol. $\frac{BD}{DC}=\frac{a}{b}\Rightarrow a+b$

$F\left(\frac{\vec{b}}{2}\right)\ E\left(\frac{4\vec{c}}{5}\right)\ G\left(\frac{3\vec{d}}{5}\right)$ $\qquad \vec{d}=\frac{\alpha\vec{c}+\vec{b}}{\alpha+1}$

$$\vec{G}=\left(\frac{\left(\frac{4\mu\vec{c}}{5}+\frac{\vec{b}}{2}\right)}{\mu+1}\right)=\frac{3}{5}\left(\frac{\alpha\vec{c}+\vec{b}}{\alpha+1}\right)$$

$$\frac{4\mu}{5(\mu+1)}=\frac{3\alpha}{5(\alpha+1)}$$

$$4\mu\alpha+4\mu=3\mu\alpha+3\alpha$$

$$\mu\alpha=3\alpha-4\mu$$

$$\frac{1}{2(\mu+1)}=\frac{3}{5(\alpha+1)}$$

$5\alpha + 5 = 6\mu + 6$
$5\alpha - 6\mu = 1$
$5\alpha = 1+ 6\mu$

$2\mu\ (5\alpha) = 3(5\alpha) - 20\mu$
$2\mu + 12\mu^2 = 3 + 18\mu - 20\mu$
$12\mu^2 + 4\mu - 3 = 0$

$$\mu=\frac{-4\pm\sqrt{\sqrt{16+12\times12}}}{2\times12}$$

$6\mu\alpha = 18\alpha - 4\cdot6\mu$
$5\alpha^2 - \alpha = 18\alpha - 20\alpha + 4$

$\alpha = 1$

$5\alpha^2 - \alpha - 4 = 0$

$\alpha=-\frac{4}{9}$

$5\alpha^2 - 5\alpha + 4\alpha - 4 = 0$

DOT PRODUCT AND ITS APPLICATION

5. Find a unit vector â which makes an angle ($\pi/4$) with axis of z and is such that $\hat{a}+\hat{i}+\hat{j}$ is a unit vector.

एक इकाई सदिश â ऐसा ज्ञात करो जो कि z अक्ष से ($\pi/4$) का कोण बनाता है तथा $\hat{a}+\hat{i}+\hat{j}$ एक इकाई सदिश है।

Ans. $\frac{-1}{2}i-\frac{1}{2}j+\frac{1}{\sqrt{2}}k$

Sol. Let $\hat{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$

$\hat{a}\cdot\hat{k}=\cos\frac{\pi}{4}$

$a_3=\frac{1}{\sqrt{2}}$

$a_1^2+a_2^2=\frac{1}{2}$

$\hat{a}+\hat{i}+\hat{j}=(a_1+1)\hat{i}+(a_2+1)\hat{j}+\frac{1}{\sqrt{2}}\hat{k}$

$(a_1+1)^2+(a_2+1)^2=\frac{1}{2}$

$2a_1+1+2a_2+1=0$

$a_1+a_2=-1$ ...(1)

$2a_1a_2=+\frac{1}{2}$

$a_1a_2=\frac{1}{4}$ ...(2)

from (1) & (2)

$a_1=-\frac{1}{2}$

and $a_2=-\frac{1}{2}$

6. Prove that $\left(\frac{\vec{a}}{\vec{a}^2}-\frac{\vec{b}}{\vec{b}^2}\right)^2=\left(\frac{\vec{a}-\vec{b}}{|\vec{a}|\,|\vec{b}|}\right)^2$

सिद्ध करो कि $\left(\frac{\vec{a}}{\vec{a}^2}-\frac{\vec{b}}{\vec{b}^2}\right)^2=\left(\frac{\vec{a}-\vec{b}}{|\vec{a}|\,|\vec{b}|}\right)^2$

Ans.

Sol. $\left(\frac{\vec{a}}{(\vec{a})^2}-\frac{\vec{b}}{(\vec{b})^2}\right)=\left(\frac{\vec{a}|\vec{b}|-\vec{b}|\vec{a}|^2}{|\vec{a}|^2\cdot|\vec{b}|^2}\right)^2$

$=\left(\frac{\vec{a}\cdot\vec{b}(\vec{b}-\vec{a})}{(\vec{a}\cdot\vec{b})^2}\right)^2$

$=\frac{\vec{b}-\vec{a}}{\vec{a}}$

7. If $\vec{a}$ and $\vec{b}$ are any two unit vectors, then find the range range of $\frac{3|\vec{a}+\vec{b}|}{2}+2|\vec{a}-\vec{b}|$.

यदि $\vec{a}$ एवं $\vec{b}$ दो सदिश है तो $\frac{3|\vec{a}+\vec{b}|}{2}+2|\vec{a}-\vec{b}|$ का परिसर बताओं।

Ans. Range: [3, 5]

Sol. $\frac{3}{2}|\vec{a}+\vec{b}|+2|\vec{a}-\vec{b}|$ $\qquad |\vec{a}+\vec{b}|=2\cos\frac{\alpha}{2}$

$|\vec{a}-\vec{b}|=2\sin\frac{\alpha}{2}$

$3\cos\frac{\alpha}{2}+2\sin\frac{\alpha}{2}$

[3, 5]

8. Let $\vec{V}_1$ and $\vec{V}_2$ are two vectors such that $\vec{V}_1=2(\sin\alpha+\cos\alpha)\hat{i}+\hat{j}$ and $\vec{V}_2=\sin\beta\hat{i}+\cos\beta\hat{j}$ where α and β satisfy the relation $2(\sin\alpha+\cos\alpha)\sin\beta=3-\cos\beta$, find the value of $(3\tan^2\alpha+4\tan^2\beta)$.

माना $\vec{V}_1$ तथा $\vec{V}_2$ दो सदिश इस प्रकार है कि $\vec{V}_1=2(\sin\alpha+\cos\alpha)\hat{i}+\hat{j}$ तथा $\vec{V}_2=\sin\beta\hat{i}+\cos\beta\hat{j}$ जहाँ α तथा β समीकरण $2(\sin\alpha+\cos\alpha)\sin\beta=3-\cos\beta$, को सन्तुष्ट करते है तो $(3\tan^2\alpha+4\tan^2\beta)$ का मान होगा :

Ans. 35

Sol. We have $\vec{V}_1\cdot\vec{V}_2=2(\sin\alpha+\cos\alpha)\sin\beta+\cos\beta$

$\therefore \quad \vec{V}_1\cdot\vec{V}_2=3 \qquad$ [given that $2(\sin\alpha+\cos\alpha)\sin\beta+\cos\beta-3$]

$|\vec{V}_1|^2\ |\vec{V}_2|^2\cos^2\theta=9 \quad (\theta=\vec{V}_1\wedge\vec{V}_2)$

$[4(1 + \sin 2\alpha) + 1]\,(1)\cos^2\theta = 9$

$(5 + 4\sin 2\alpha)\cos^2\theta = 9$

$$\cos^2\theta = \frac{9}{5+4\sin 2\alpha} \le 1 \;\Rightarrow\; 9 \le 5 + 4\sin 2\alpha \;\Rightarrow\; 4 \le 4\sin 2\alpha$$

$$\therefore \quad \sin 2\alpha = 1 \Rightarrow \alpha = \frac{\pi}{4}$$

$$\therefore \quad \cos^2\theta = \frac{9}{9} = 1 \;\Rightarrow\; \theta = 0$$

Hence $\theta = 0 \Rightarrow \vec{V}_1$ and $\vec{V}_2$ are collinear i.e. $\dfrac{2(\sin\alpha + \cos\alpha)}{\sin\beta} = \dfrac{1}{\cos\beta} \qquad \left(\alpha = \dfrac{\pi}{4}\right)$

$$\tan\beta = 2\left(\frac{2}{\sqrt{2}}\right) = 2\sqrt{2}$$

Hence $3\tan^2\alpha + 4\tan^2\beta = 3 + (4)(8) = 35$ Ans.

9. In a a unit cube. Find :

(a) The angle between the diagonal of the cube and a diagonal of a face skew to it.

(b) The angle between the diagonals of two faces of the cube through the same vertex.

(c) The angle between a diagonal of a cube and a diagonal of a face intersecting it.

एक इकाई घन में ज्ञात कीजिए :

(a) एक काय विकर्ण के तथा फलक विकर्ण, जो इसके साथ विषमतलीय (skew) है, के मध्य कोण बताओं।

(b) एक ही शीर्ष से जाने वाले दो समतल (face) के विकर्ण के बीच कोण बताओ।

(c) घन के काय विकर्ण तथा फलक विकर्ण, जो इसको प्रतिच्छेद करता है, के मध्य कोण बताओ।

Ans. (a) $\cot^{-1}(0)$ (b) $\cot^{-1}\dfrac{1}{\sqrt{3}}$ (c) $\cot^{-1}\sqrt{2}$

Sol. $\overrightarrow{BE} = (\vec{a}, \vec{a}, -\vec{a})$

$\overrightarrow{GD} = (\vec{a}, -\vec{a}, \vec{a})$

$\overrightarrow{BE}\cdot\overrightarrow{GD} = a^2 - a^2 - a^2$

$\sqrt{3}a\cdot\sqrt{3}a\cdot\cos\theta = a^2$

$\cos\theta = \dfrac{1}{3}$

10. Given that $\vec{u} = \hat{i} - 2\hat{j} + 3\hat{k};\ \vec{v} = 2\hat{i} + \hat{j} + 4\hat{k};\ \vec{w} = \hat{i} + 3\hat{j} + 3\hat{k}$ and $(\vec{u}\cdot\vec{R} - 10)\hat{i} + (\vec{v}\cdot\vec{R} - 20)\hat{j} + (\vec{w}\cdot\vec{R} - 20)\hat{k} = 0$.

Find the unknown vector $\vec{R}$.

दिया है कि $\vec{u}=\hat{i}-2\hat{j}+3\hat{k};\vec{v}=2\hat{i}+\hat{j}+4\hat{k};\vec{w}=\hat{i}+3\hat{j}+3\hat{k}$ तथा $(\vec{u}\cdot\vec{R}-10)\hat{i}+(\vec{v}\cdot\vec{R}-20)\hat{j}+(\vec{w}\cdot\vec{R}-20)\hat{k}=0$ तो सदिश $\vec{R}$ ज्ञात कीजिए।

Ans. $-\hat{i}+2\hat{j}+5\hat{k}$

Sol. $\vec{u}(1,-2,3)\quad \vec{R}(\alpha,\beta,\gamma)\quad \vec{v}(2,1,4)\quad \vec{w}(1,3,3)$

$\vec{u}\cdot\vec{R}=10,\vec{v}\cdot\vec{R}=20,\vec{w}\cdot\vec{R}=20$

$\alpha-2\beta+3\gamma=10$

$2\alpha+\beta+4\gamma=20$

$\alpha+3\beta+3\gamma-20$

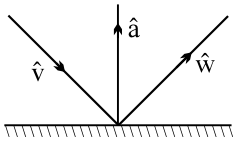
$\alpha=-1$

$\beta=2$

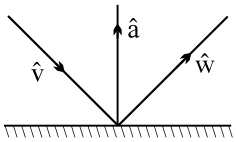
$\gamma=5$

LINEAR COMBINATION THEOREM IN A PLANE

11. Incident ray is along the unit vector $\hat{v}$ and the reflected ray is along the unit vector $\hat{w}$. The normal is along unit vector $\hat{a}$ outwards. Express $\hat{w}$ in terms of $\hat{a}$ and $\hat{v}$.

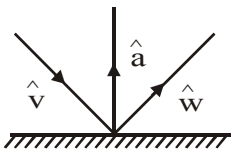


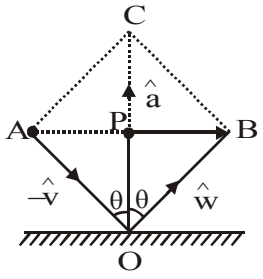
आवर्तित किरण इकाई सदिश $\hat{v}$ के अनुदिश तथा परावर्तित किरण इकाई सदिश $\hat{w}$ के अनुदिश है। अभिलम्ब इकाई सदिश $\hat{a}$ के अनुदिश बाहर की तरफ है। $\hat{w}$ को $\hat{a}$ और $\hat{v}$ के पदों में व्यक्त कीजिये।



Ans. $\hat{w}=\hat{v}-2(\hat{a}\cdot\hat{v})\hat{a}$

Sol. Given that incident ray is along $\hat{v}$, reflected ray is along $\hat{w}$ and normal is along $\hat{a}$, outwards. The given figure can be redrawn as shown.





We know that incident ray, reflected ray and normal lie in a plane, and angle of incidence = angle of reflection.

Therefore $\hat{a}$ will be along the angle bisector of $\hat{w}$ and $-\hat{v}$, i.e.,

$$\hat{a} = \frac{\hat{w} + (-\hat{v})}{|\hat{w} - \hat{v}|} \qquad ...(i)$$

[$\because$ angle bisector will along a vector dividing in same ratio as the ratio of the sides forming that angle.]

But $\hat{a}$ is a unit vector

where $|\hat{w} - \hat{v}| = OC = 2OP$

$= 2|\hat{w}|\cos\theta = 2\cos\theta$

Substituting this value in equation (i) we get

$$\hat{a} = \frac{\hat{w} - \hat{v}}{2\cos\theta}$$

$\therefore\ \hat{w} = \hat{v} + (2\cos\theta)\hat{a}$

$= \hat{v} - 2(\hat{a}\,.\,\hat{v})\hat{a}$ [$\because\ \hat{a}\,.\,\hat{v} = -\cos\theta$].

CROSS PRODUCT AND ITS APPLICATION

12. Find the minimum area of the triangle whose vertices are A(–1, 1, 2); B(1, 2, 3) and C(t, 1, 1) where t is a real number.

यदि किसी त्रिभुज ABC का शीर्ष यदि A(–1, 1, 2); B(1, 2, 3) एवं C(t, 1, 1) है जहाँ t एक वास्तविक संख्या है तो त्रिभुज का न्यूनतम क्षेत्रफल बताओं।

Ans. $\frac{\sqrt{3}}{2}$

Sol. $$Ar = \frac{1}{2}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ t-1 & -1 & -2 \\ -2 & -1 & -1 \end{vmatrix}$$

$$= \frac{1}{2}\hat{i}(-1) - \hat{j}(1-t-4) + \hat{k}(1-t-2)$$

$$= \frac{1}{2}(-1, +t+3, -t-1)$$

$$Ar = \frac{1}{2}\sqrt{1+(t+3)^2+(t+1)^2} = \frac{1}{2}\sqrt{2t^2+8t+11}$$

$4Ar^2 = 2t^2 + 8t + 11$

$$\frac{d(4Ar^2)}{dx} = 4t + 8$$

$4t + 8 = 0$

$\Rightarrow t = -2$

$$Ar = \frac{1}{2}\sqrt{2t^2 + 8t + 11}$$

$$Ar = \frac{1}{2}\sqrt{3}$$

13. Let $\overline{OA} = \vec{a}$; $\overline{OB} = 100\vec{a} + 2\vec{b}$ and $\overline{OC} = \vec{b}$ where O, A, and C are non collinear points. Let P denotes the area of the parallelogram with $\overline{OA}$ and $\overline{OC}$ as adjacent sides and Q denotes the area of the quadrilateral OABC. If Q = λP. Find the value of λ.

माना $\overline{OA} = \vec{a}$; $\overline{OB} = 100\vec{a} + 2\vec{b}$ तथा $\overline{OC} = \vec{b}$ जहाँ O, A, तथा C विषम रेखीय बिन्दु है। माना P एक समान्तर चतुर्भुज, जिसकी आसन्न भुजाऐं $\overline{OA}$ तथा $\overline{OC}$ हैं, का क्षेत्रफल है तथा Q चतुर्भुज OABC का क्षेत्रफल है यदि Q = λP तो λ का मान ज्ञात कीजिये।

Ans. 51

Sol. $P = \left|\left(\overline{OA}\right) \times \left(\overline{OC}\right)\right|$

$P = \left|\vec{a} \times \vec{b}\right|$

$$Q = \frac{1}{2}\left|\overline{OB} \times \overline{AC}\right| = \frac{1}{2}\left|\left(100\vec{a} + 2\vec{b}\right) \times \left(\vec{a} - \vec{b}\right)\right|$$

$$= \frac{1}{2}\left|-100\vec{a} \times \vec{b} + 2\vec{b} \times \vec{a}\right| = \frac{1}{2}\left|102\left(\vec{a} \times \vec{b}\right)\right| = 51\left|\vec{a} \times \vec{b}\right|$$

Now $Q = \lambda P$ [FOT]

$$51\left|\vec{a} \times \vec{b}\right| = \lambda\left|\vec{a} \times \vec{b}\right|$$

$\lambda = 51$ Ans.

14. Let O be the interior point of ΔABC such that $2\overrightarrow{OA} + 3\overrightarrow{OB} + 6\overrightarrow{OC} = 0$. If $\frac{\text{Ar. of } \Delta ABC}{\text{Ar. of } \Delta AOB} = \frac{m}{n}$ where m and n are relatively prime, then find (m + n).

माना O ΔABC में स्थित एक आन्तरिक बिन्दु है इस प्रकार कि $2\overrightarrow{OA} + 3\overrightarrow{OB} + 6\overrightarrow{OC} = 0$ यदि $\frac{\text{Ar. of } \Delta ABC}{\text{Ar. of } \Delta AOB} = \frac{m}{n}$ जहाँ m तथा n सापेक्षिक अभाज्य है तो (m + n) का मान ज्ञात कीजिये।

Ans. 17

Sol. Area of $\Delta ABC = \frac{1}{2}\left|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}\right|$

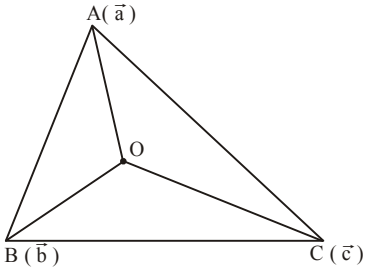
Now given $2\vec{a} + 3\vec{b} + 6\vec{b} = 0$

Cross with $\vec{a}$, $3\vec{a}\times\vec{b} + 6\vec{a}\times\vec{c} = 0$

or $\vec{a}\times\vec{b} = 2(\vec{c}\times\vec{a}) = 0$ ....(1)

again cross with $\vec{b}$, $2\vec{a}\times\vec{b} + 6\vec{c}\times\vec{b} = 0$

or $\vec{a}\times\vec{b} = 3(\vec{b}\times\vec{c}) = 0$ ....(2)



Area of $\Delta OAB = \frac{1}{2}\left|\vec{a}\times\vec{b}\right|$ [FOT]

$$\frac{\text{Ar. of } \Delta ABC}{\text{Ar. of } \Delta AOB} = \frac{\frac{1}{2}\left|\vec{a}\times\vec{b}\right|\left(1+\frac{1}{2}+\frac{1}{3}\right)}{\frac{1}{2}\left|\vec{a}\times\vec{b}\right|} = \frac{11}{6} \Rightarrow \quad m+n = 17 \text{ Ans.}$$

15. Determine vector of magnitude 9 which is perpendicular to both the vectors : $4\hat{i}-\hat{j}+3\hat{k}$ and $-2\hat{i}+\hat{j}-2\hat{k}$.

एक ऐसे सदिश जो बताओं जिसका परिमाण 9 है एवं वह दोनों सदिश $4\hat{i}-\hat{j}+3\hat{k}$ तथा $-2\hat{i}+\hat{j}-2\hat{k}$ के लम्बवत् है।

Ans. $\pm 3(\hat{i}-2\hat{j}-2\hat{k})$

Sol. $$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -1 & 3 \\ -2 & 1 & -2 \end{vmatrix} = \hat{i}(-1)-\hat{j}(-2)+\hat{k}(+2)$$

$= (-1, +2, 2)$

$\vec{a} = \pm 3(-1,2,2)$

16. A triangle has vertices (1, 1, 1) ; (2, 2, 2), (1, 1, y) and has the area equal to cosec $\left(\frac{\pi}{4}\right)$ sq. units. Find the value of y.

एक त्रिभुज का शीर्ष (1, 1, 1) ; (2, 2, 2) एवं (1, 1, y) है तथा उसका क्षेत्रफल cosec $\left(\frac{\pi}{4}\right)$ वर्ग मात्रक है तो y का मान बताओं।

Ans. y = 3 or y = – 1

Sol. Arg $= \sqrt{2}$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & 1 & 2y \end{vmatrix} = 2\sqrt{2}$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 0 & 0 & 1-y \end{vmatrix}$$

$$\left|(1-y)\left(\hat{i}-\hat{j}\right)\right| = 2\sqrt{2}$$

$$(1-y)\sqrt{2} = \pm 2\sqrt{2}$$

$y = 1 \pm 2$

$= 3, -1$

17. Given that $\vec{a}$ and $\vec{b}$ are two unit vectors such that angle between $\vec{a}$ and $\vec{b}$ is $\cos^{-1}\left(\frac{1}{4}\right)$. If $\vec{c}$ be a vector in the plane of $\vec{a}$ and $\vec{b}$, such that $|\vec{c}| = 4, \vec{c}\times\vec{b} = 2\vec{a}\times\vec{b}$ and $\vec{c} = \lambda\vec{a} + \mu\vec{b}$ then, find :

(a) the value of λ (b) the sum of values of μ and (c) the product of all possible values of μ

$\vec{a}$ एवं $\vec{b}$ इकाई सदिश है तथा इनके बीच का कोण $\cos^{-1}\left(\frac{1}{4}\right)$ है यदि उस तल में एक सदिश $\vec{c}$ इस प्रकार से है कि $|\vec{c}| = 4, \vec{c}\times\vec{b} = 2\vec{a}\times\vec{b}$ तथा $\vec{c} = \lambda\vec{a} + \mu\vec{b}$ तो :

(a) λ का मान बताओं (b) μ के सभी मानो का योग बताओं (c) μ के सारे मानों का गुणनफल बताओं

Ans. (a) 2 (b) –1 (c) –12

Sol. $\vec{c}\times\vec{b} = 2\vec{a}\times\vec{b}$

$(\vec{c} - 2\vec{a})\times\vec{b} = 0$

$\vec{c} = 2\vec{a} + \mu\vec{b}$

$\lambda = 2$

$|\vec{c}| = 4$

$c^2 = 4a^2 + \mu^2 b^2 + 4\mu \times \frac{1}{4}$

$\mu^2 + 4 + \mu = 16$

$\mu^2 + \mu - 12 = 0$

$\mu = -4, 3$

18. Let $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are four distinct vectors satisfying $\vec{a}\times\vec{b} = \vec{c}\times\vec{d}$ and $\vec{a}\times\vec{c} = \vec{b}\times\vec{d}$. Show that $\vec{a}\cdot\vec{b} + \vec{c}\cdot\vec{d} \neq \vec{a}\cdot\vec{c} + \vec{b}\cdot\vec{d}$.

माना $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ चार भिन्न सदिश हैं जो $\vec{a}\times\vec{b} = \vec{c}\times\vec{d}$ तथा $\vec{a}\times\vec{c} = \vec{b}\times\vec{d}$ को सन्तुष्ट करते हैं। सिद्ध कीजिए $\vec{a}\cdot\vec{b} + \vec{c}\cdot\vec{d} \neq \vec{a}\cdot\vec{c} + \vec{b}\cdot\vec{d}$

Ans.

Sol. Given that $\vec{a} \neq \vec{b} \neq \vec{c} \neq \vec{d}$

Such that $\vec{a} \times \vec{c} = \vec{b} \times \vec{d}$ ...(i)

$\vec{a} \times \vec{b} = \vec{c} \times \vec{d}$ ...(ii)

To prove that $(\vec{a} - \vec{d}).(\vec{b} - \vec{c}) \neq 0$

Subtracting equation (ii) from (i) we get

$\vec{a} \times (\vec{c} - \vec{b}) = (\vec{b} - \vec{c}) \times \vec{d}$

$\Rightarrow \vec{a} \times (\vec{c} - \vec{b}) = \vec{d} \times (\vec{c} - \vec{b})$

$\Rightarrow \vec{a} \times (\vec{c} - \vec{b}) - \vec{d} \times (\vec{c} - \vec{b}) = 0$

$\Rightarrow (\vec{a} - \vec{d}) \times (\vec{c} - \vec{b}) = 0 \Rightarrow \vec{a} - \vec{d} \| \vec{c} - \vec{b}$

$[\because \vec{a} - \vec{d} \neq 0, \vec{c} - \vec{b} \neq 0$ as all distinct]

$\Rightarrow$ Angle between $\vec{a} - \vec{d}$ and $\vec{c} - \vec{b}$ is either 0 or 180°.

$\Rightarrow (\vec{a} - \vec{d}) \cdot (\vec{c} - \vec{b}) = |\vec{a} - \vec{d}| |\vec{c} - \vec{b}|$

$[\cos 0° \text{ or } \cos 180°] \neq 0$ as $\vec{a}, \vec{d}, \vec{c}, \vec{b}$ all are different.

19. Let $\vec{a} = -\hat{i} - \hat{k}, \vec{b} = -\hat{i} + \hat{j}$ and $\vec{c} = \hat{i} + 2\hat{j} + 3\hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{r}.\vec{a} = 0$, then the value of $\vec{r}.\vec{b}$ is : [JEE-2011]

माना $\vec{a} = -\hat{i} - \hat{k}, \vec{b} = -\hat{i} + \hat{j}$ तथा $\vec{c} = \hat{i} + 2\hat{j} + 3\hat{k}$ तीन सदिश है यदि सदिश $\vec{r}$ इस प्रकार है कि $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ तथा $\vec{r}.\vec{a} = 0$ है, तो $\vec{r}.\vec{b}$ का मान होगा : [JEE-2011]

Ans. 9

Sol. $\vec{a} = -\hat{i} - \hat{k}$

$\vec{b} = -\hat{i} + \hat{j}$

$\vec{c} = \hat{i} + 2\hat{j} + 3\hat{k}$

$\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$

Taking cross product by $\vec{a}$

$(\vec{r} \times \vec{b}) \times \vec{a} = (\vec{c} \times \vec{b}) \times \vec{a}$

$\Rightarrow (\vec{r}.\vec{a})\vec{b} - (\vec{b}.\vec{a})\vec{r} = (\vec{c}.\vec{a})\vec{b}) - (\vec{b}.\vec{a})\vec{c}$

$\Rightarrow 0-\vec{r}=(-1-3)(-\hat{i}+\hat{j})-(1)(\hat{i}+2\hat{j}+3\hat{k})$

$\vec{r}=-3\hat{i}+6\hat{j}+3\hat{k}$

$\vec{r}.\vec{b}=3+6=9$

STP AND ITS APPLICATION

20. Let the volume of a tetrahedron having the vertices as $\vec{0},\vec{a},\vec{b}$ and $\vec{c}$ is λ_1 (unit cubes) and volume of another tetrahedron formed by the centroids of all faces of given tetrahedron is λ_2 (in unit cubes), then $\frac{\lambda_1}{3\lambda_2}$ is.

माना चतुष्फलक जिसके शीर्ष $\vec{0},\vec{a},\vec{b}$ तथा $\vec{c}$ है, का आयतन λ_1 (घन इकाई में) तथा एक दूसरा चतुष्फलक जो दिये गये चतुष्फलक के फलकों के केन्द्रकों से मिलकर बना है, का आयतन λ_2 (घन इकाई में) है तो $\frac{\lambda_1}{3\lambda_2}$ का मान होगा।

Ans. 9

Sol. Volume $\Rightarrow$

$\lambda_1=\frac{1}{6}\left[\vec{a}\,\vec{b}\,\vec{c}\right]$

$\lambda_2=\frac{1}{6}\left[\frac{\vec{a}}{3}\ \frac{\vec{b}}{3}\ \frac{\vec{c}}{3}\right]$

$=\frac{1}{6\times 27}\left[\vec{a}\,\vec{b}\,\vec{c}\right]$

$\frac{\lambda_1}{3\lambda_2}=\frac{\frac{1}{6}\left[\vec{a}\,\vec{b}\,\vec{c}\right]}{3\times\frac{1}{6\times 27}\left[\vec{a}\,\vec{b}\,\vec{c}\right]}=9$

21. Let the volume of tetrahedron ABCD is 81 cubic units and G_1,G_2,G_3 are centroids of triangular faces ABC, ABD and ACD respectively, then volume of tetrahedron $AG_1G_2G_3$, is. (in cubic units)

माना चतुष्फलक ABCD का आयतन 81 घन इकाई तथा G_1,G_2,G_3 क्रमशः ABC, ABD एवं ACD के त्रिकोणीय फलकों के केन्द्रक हो, तो चतुष्फलक $AG_1G_2G_3$ का आयतन होगा। (घन इकाई में)

Ans. 6

Sol. $\frac{1}{6}\left[\vec{a}\,\vec{b}\,\vec{c}\right]=81$

$AG_1G_2G_3$

$$= \frac{1}{6}\left[\frac{\vec{a}+\vec{b}}{3}\,\frac{\vec{b}+\vec{c}}{3}\,\frac{\vec{c}+\vec{a}}{3}\right]$$

$$= \frac{1}{27\times 6}\times 2\left[\vec{a}\,\vec{b}\,\vec{c}\right]$$

$= 6$

VTP AND ITS APPLICATION

22. If $\vec{a}$ and $\vec{b}$ are vectors in space given by $\vec{a}=\frac{\hat{i}-2\hat{j}}{\sqrt{5}}$ and $\vec{b}=\frac{2\hat{i}+\hat{j}+3\hat{k}}{\sqrt{14}}$, then the value of $\left(2\vec{a}+\vec{b}\right)\cdot\left[\left(\vec{a}\times\vec{b}\right)\times\left(\vec{a}-2\vec{b}\right)\right]$, is.

यदि $\vec{a}$ तथा $\vec{b}$ इस प्रकार है कि $\vec{a}=\frac{\hat{i}-2\hat{j}}{\sqrt{5}}$ तथा $\vec{b}=\frac{2\hat{i}+\hat{j}+3\hat{k}}{\sqrt{14}}$ है, तो $\left(2\vec{a}+\vec{b}\right)\cdot\left[\left(\vec{a}\times\vec{b}\right)\times\left(\vec{a}-2\vec{b}\right)\right]$ का मान होगा।

Ans. 5

Sol. $\vec{a}=\frac{1}{\sqrt{5}}(1,-2,0)$

$\vec{b}=\frac{1}{\sqrt{14}}(2,1,3)$

$\left(2\vec{a}+\vec{b}\right)\cdot\left[\left(\vec{a}\times\vec{b}\right)\times\left(\vec{a}-2\vec{b}\right)\right]$

$\left[2\vec{a}+\vec{b}\right]\cdot\left[\vec{b}\left(a^2-2\vec{a}\cdot\vec{b}\right)-\vec{a}\left(\vec{b}\cdot\vec{a}-2b^2\right)\right]$

$\left[2\vec{a}+\vec{b}\right]\cdot\left[\vec{b}\left(1-2\times 0\right)-\vec{a}\left(0-2\right)\right]$

$\left(2\vec{a}+\vec{b}\right)\cdot\left(\vec{b}+2\vec{a}\right)$

$4 + 1 + 0 = 5$

DIRECTION COSINES AND DIRECTION RATIOS

23. Find the angle between the two straight lines whose direction cosines l, m, n are given by $2l + 2m - n = 0$ and $mn + nl + lm = 0$.

उन दो रेखाओं के बीच का कोण बताओं जिनके दिक् कोज्याओं l, m, n में निम्न $2l + 2m - n = 0$ तथा $mn + nl + lm = 0$ है।

Ans. $\theta = 90^0$

Sol. $2l + 2m = n$

$lm + mn + nl = 0$

$lm + (m + l)(2m + 2l) = 0$

$lm + 2m^2 + 2l^2 + 4ml = 0$

$2m^2 + 2l^2 + 5ml = 0$

$2m^2 + 4ml + ml + 2l^2 = 0$

$(2m + l)(m + 2l) = 0$

Case–I

$2m + l = 0$

$n = -2m = l$

$$v_1 = \left\langle \mp\frac{2}{3}, \pm\frac{1}{3}, \mp\frac{2}{3} \right\rangle$$

Case–II

$m + 2l = 0$

$m = n = -2l$

$$v_2 = \left\langle \pm\frac{1}{3}, \mp\frac{2}{3}, \mp\frac{2}{3} \right\rangle$$

$$v_1 \cdot v_2 = -\frac{2}{3} - \frac{2}{3} + \frac{4}{3} = 0$$

EQUATION OF LINE

24. Find out whether the following pairs of lines are parallel, non-parallel and intersecting, or non-parallel and non-intersecting :

ज्ञात करो कि इनमें से कौन सा रेखा युग्म समान्तर, असमान्तर प्रतिच्छेदित एवं विषमतलीय होगा :

(i)
$$\vec{r}_1 = \hat{i} + \hat{j} + 2\hat{k} + \lambda\left(3\hat{i} - 2\hat{j} + 4\hat{k}\right)$$
$$\vec{r}_2 = 2\hat{i} + \hat{j} + 3\hat{k} + \mu\left(-6\hat{i} + 4\hat{j} - 8\hat{k}\right)$$

(ii)
$$\vec{r}_1 = \hat{i} - \hat{j} + 3\hat{k} + \lambda\left(\hat{i} - \hat{j} + \hat{k}\right)$$
$$\vec{r}_2 = 2\hat{i} + 4\hat{j} + 6\hat{k} + \mu\left(2\hat{i} + \hat{j} + 3\hat{k}\right)$$

(iii)
$$\vec{r}_1 = \hat{i} + \hat{k} + \lambda\left(\hat{i} + 3\hat{j} + 4\hat{k}\right)$$
$$\vec{r}_2 = 2\hat{i} + 3\hat{j} + \mu\left(4\hat{i} - \hat{j} + \hat{k}\right)$$

Ans. (i) parallel (ii) the lines intersect at the point p.v. $-2\hat{i} + 2\hat{j}$ (iii) lines are skew

Sol. (i) $(3,-2,4)$, $(-6,4,-8)$ } Parallel

(ii) $(1,-1,1)$, $(2,1,3)$ } Intersect

(iii) $\left.\begin{matrix}(1,3,4)\\(4,-1,1)\end{matrix}\right\}$ Skew

25. Find the equations of the straight line passing through the point (1, 2, 3) to intersect the straight line $x + 1 = 2(y - 2) = z + 4$ and parallel to the plane $x + 5y + 4z = 0$.

बिन्दु (1, 2, 3) से गुजरने वाली तथा रेखा $x + 1 = 2(y - 2) = z + 4$ को प्रतिच्छेदित करने वाली उस रेखा का समीकरण बताओं जो तल $x + 5y + 4z = 0$ के समान्तर हो।

Ans. $\frac{x-1}{2} = \frac{y-2}{2} = \frac{z-3}{-3}$

Sol. $\frac{x-1}{2} = \frac{y-2}{2} = \frac{z-3}{-3}$

$(2\lambda - 2, \lambda, -7 + 2\lambda)\cdot(1,5,4) = 0$

$2\lambda - 2 + 5\lambda - 28 + 8\lambda = 0$

$15\lambda - 30 = 0$

$\lambda = 2$

DR's = <2, 2, –3>

$\frac{x-1}{2} = \frac{y-2}{2} = \frac{z-3}{-3}$

26. Find the equations of the two lines through the origin which intersect the line $\frac{x-3}{2} = \frac{y-3}{1} = \frac{z}{1}$ at an angle of $\frac{\pi}{3}$.

मूल बिन्दु से गुजरने वाली उन दोनों रेखाओं का समीकरण बताओं जो रेखा $\frac{x-3}{2} = \frac{y-3}{1} = \frac{z}{1}$ को $\frac{\pi}{3}$ के कोण पर प्रतिच्छेदित करती है।

Ans. $\frac{x}{1} = \frac{y}{2} = \frac{z}{-1}$ or $\frac{x}{-1} = \frac{y}{1} = \frac{z}{-2}$

Sol. Let point on line $\frac{x-3}{2} = \frac{y-3}{1} = \frac{z}{1}$ .....(1)

are $(3 + 2\lambda, 3 + \lambda, \lambda)$

Equation of line which pass through origin is

$\frac{x-0}{3+2\lambda} = \frac{y-0}{3+\lambda} = \frac{z-0}{\lambda}$ .....(2)

Angle between (1) & (2)

$$\cos\frac{\pi}{3}=\frac{(3+2\lambda)2+(3+\lambda)1+\lambda\times1}{\sqrt{(3+2\lambda)^2+(3+\lambda)^2+\lambda^2}\sqrt{2^2+1^2+1}}$$

Solving we get

$\lambda^2+3\lambda+2=0$

$\Rightarrow \lambda=-1,-2$

Putting the value of λ in equation (2)

$$\frac{x}{1}=\frac{y}{2}=\frac{z}{-1} \text{ or } \frac{x}{-1}=\frac{y}{1}=\frac{z}{-2}$$

27. Find the equation to the line passing through the point (1, –2, –3) and parallel to the line $2x+3y-3z+2=0=3x-4y+2z-4$.

बिन्दु (1, –2, –3) से गुजरने वाली तथा रेखा $2x+3y-3z+2=0=3x-4y+2z-4$ के समानान्तर रेखा का समीकरण ज्ञात कीजिए।

Ans. $\frac{x-1}{6}=\frac{y+2}{13}=\frac{z+3}{17}$

Sol. $\text{DR's of line}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&3&-3\\3&-4&2\end{vmatrix}=\hat{i}(-6)-\hat{j}(13)+\hat{k}(-17)$

$=(6, 13, 17)$

Line passing from (1, –2, –3) in the direction = (6, 13, 17)

$$\frac{x-1}{6}=\frac{y+2}{13}=\frac{z+3}{17}$$

28. Find the equation of the line passing through the point (4, –14, 4) and intersecting the line of intersection of the planes : $3x+2y-z=5$ and $x-2y-2z=-1$ at right angles.

बिन्दु (4, –14, 4) से गुजरने वाली तथा तलों $3x+2y-z=5$ तथा $x-2y-2z=-1$ के प्रतिच्छेदी रेखा को समकोण पर काटने वाली रेखा का समीकरण ज्ञात कीजिए।

Ans. $\frac{x-4}{3}=\frac{y+14}{10}=\frac{z-4}{4}$

Sol. $\text{DR's of line of intersection of planes}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\3&2&-1\\1&-2&-2\end{vmatrix}=\hat{i}(-4-2)-\hat{j}(-6+1)+\hat{k}(-6-2)$

DR′s of line of intersection of planes = (–6, –5, –8)
For the point on the intersection line of planes let z = 0
$3x + 2y = 5$
$x - 2y = -1$
From these x = 1, y = 1
equation of line of intersection

$$\frac{x-1}{6} = \frac{y-1}{-5} = \frac{z}{8}$$

Variable point on the intersection line of planes $P(1 + 6\lambda, 1 - 5\lambda, 8\lambda)$
DR of line joining point (4, –14, 4) & $P(1 + 6\lambda, 1 - 5\lambda, 8\lambda) = (-3 + 6\lambda, 15 - 5\lambda, -4 + 8\lambda)$
$(-6, 5, -8) \cdot (-3 + 6\lambda, 15 - 5\lambda, -4 + 8\lambda)$
$18 - 36\lambda + 75 - 25\lambda + 32 - 64\lambda = 0$
$+125 - 125\lambda = 0$
$\lambda = 1$
DR of the line (3,10,4)

$$\frac{x-4}{3} = \frac{y+14}{10} = \frac{z-4}{4}$$

29. Find the equations to the line which can be drawn from the point (2, –1, 3) perpendicular to the lines

$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-4}{4} = \frac{y}{5} = \frac{z+3}{3}$ at right angles.

बिन्दु (2, –1, 3) से उन लम्बवत् रेखाओं का समीकरण ज्ञात कीजिए जो रेखाओं $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ तथा $\frac{x-4}{4} = \frac{y}{5} = \frac{z+3}{3}$ के लंबवत हो ।,

Ans. $\frac{x-2}{11} = \frac{y+1}{-10} = \frac{z-3}{2}$

Sol. DR′s for perpendicular line $= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 4 & 5 & 3 \end{vmatrix} = \hat{i}(-11) - \hat{j}(-10) + \hat{k}(-2)$

line $\frac{x-2}{11} = \frac{y+1}{-10} = \frac{z-3}{2}$

30. Find the value of p so that the lines $\frac{x+1}{-3} = \frac{y-p}{2} = \frac{z+2}{1}$ and $\frac{x}{1} = \frac{y-7}{-3} = \frac{z+7}{2}$ are in the same plane. For this value of p, find the coordinates of their point of intersection and the equation of the plane containing them.

p का मान ज्ञात करो यदि रेखाएं $\frac{x+1}{-3}=\frac{y-p}{2}=\frac{z+2}{1}$ तथा $\frac{x}{1}=\frac{y-7}{-3}=\frac{z+7}{2}$ एक तल में हो । p के इस मान के लिये इनका प्रतिच्छेद बिन्दु तथा इनको समातित करने वाले तल का समीकरण ज्ञात कीजिए।

Ans. $p = 3, (2, 1, -3)$; $x + y + z = 0$

Sol. $\begin{vmatrix} -3 & 2 & 1 \\ 1 & -3 & 2 \\ +1 & 7-P & -5 \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} -3 & 2 & 1 \\ 1 & -3 & 2 \\ 0 & 10-P & -7 \end{vmatrix} = 0$

$+3(21 - 20 + 2P) = 0$
$+1(-14 - 10 + P)$
$3 + 6P - 24 + P = 0$
$P = 3$

$$\text{Equation of plane} = \begin{vmatrix} x & y-7 & z+7 \\ -3 & 2 & 1 \\ 1 & -3 & 2 \end{vmatrix} = 0$$

$p = 3, (2, 1, -3)$; $x + y + z = 0$

PLANE

31. Let P = (1, 0, – 1) ; Q = (1, 1, 1) and R = (2, 1, 3) are three points :
(a) Find the area of the triangle having P, Q and R as its vertices.
(b) Give the equation of the plane through P, Q and R in the form ax + by + cz = 1.
(c) Where does the plane in part (b) intersect the y-axis.
(d) Give parametric equations for the line through R that is perpendicular to the plane in part (b).

यदि P = (1, 0, – 1) ; Q = (1, 1, 1) तथा R = (2, 1, 3) तीन बिन्दु है तो :
(a) त्रिभुज PQR का क्षेत्रफल ज्ञात कीजिए।
(b) P, Q तथा R से जाने वाले तल का समीकरण ax + by + cz = 1 के रूप में ज्ञात कीजिए।
(c) खण्ड (b) में तल y-अक्ष पर किस बिन्दु पर प्रतिच्छेदित करेगा।
(d) R से गुजरने वाली उस रेखा का प्राचल समीकरण ज्ञात करो जो खण्ड (b) में दिये गये तल के लम्बवत् हो।

Ans. (a) $\frac{3}{2}$

(b) $\frac{2x}{3}+\frac{2y}{3}-\frac{z}{3}=1$

(c) $\left(0, \frac{3}{2}, 0\right)$

(d) $x = 2t + 2$; $y = 2t + 1$ and $z = -t + 3$

Sol. (i) $\frac{1}{2}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -1 & -2 \\ 1 & 0 & 2 \end{vmatrix}$

$\frac{1}{2}\left|\hat{i}(-2)-\hat{j}(+2)+\hat{k}(+1)\right|$

$=\frac{1}{2}3=\frac{3}{2}$

(ii) $\vec{n}=(-2,-2,1)$

$2x+2y-z=3$

(iii) y–axis $\quad 0,\frac{3}{2}$

(iv) $\frac{x-2}{2}=\frac{y-1}{2}=\frac{z-3}{-1}=\lambda$

$x=2\lambda+2 \qquad y=2\lambda+1 \qquad z=-t+3$

32. Find the point where the line of intersection of the planes $x-2y+z=1$ and $x+2y-2z=5$, intersects the plane $2x+2y+z+6=0$.

उस बिन्दु को ज्ञात करो जहाँ पर तलों $x-2y+z=1$ तथा $x+2y-2z=5$ की प्रतिच्छेदी रेखा तल $2x+2y+z+6=0$ को प्रतिच्छेदित करती है।

Ans. $(1,-2,-4)$

Sol. Planes are $x-2y+z=1$ ....(i)

$x+2y-2z=5$ ....(ii)

$2x+2y+z=-6$ ....(iii)

Add (i) + (ii) + (iii)

$4x+2y=0 \Rightarrow y=-2x$ ....(iv)

From equations (iii) – (i)

$x+4y=-7$ ....(v)

from (iv) and (v) we get

$x=1,\ y=-2$

Put in (i) we get $z=-4$

So point of intersection is $(1,-2,-4)$

33. Feet of the perpendicular drawn from the point P (2, 3, –5) on the axes of coordinates are A, B and C. Find the equation of the plane passing through their feet and the area of ΔABC.

बिन्दु P (2, 3, –5) से निर्देशी अक्षों पर डाले गये लम्बो के पाद A, B तथा C है। इन पादों से गुजरने तल का समीकरण तथा त्रिभुज ΔABC का क्षेत्रफल ज्ञात कीजिए।

Ans. $\frac{x}{2}+\frac{y}{3}+\frac{z}{-5}=1$, Area $=\frac{19}{2}$ sq. units

Sol. $\frac{x}{2}+\frac{y}{3}+\frac{z}{-5}=0$

$$\frac{1}{2}\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & -3 & 0\\ 0 & 3 & 5\end{vmatrix}=\frac{1}{2}\left|\hat{i}(-15)-\hat{j}(10)+\hat{k}(6)\right|$$

$$=\frac{1}{2}\left|\sqrt{225+100+36}\right|$$

$$=\frac{19}{2}$$

34. Find the equation of the plane containing the straight line $\frac{x-1}{2}=\frac{y+2}{-3}=\frac{z}{5}$ and perpendicular to the plane $x-y+z+2=0$.

रेखा $\frac{x-1}{2}=\frac{y+2}{-3}=\frac{z}{5}$ को समाहित करने वाले उस तल का समीकरण ज्ञात करो जो तल $x-y+z+2=0$ के लम्बवत् हो।

Ans. $2x+3y+z+4=0$

Sol. Line : $\frac{x-1}{2}=\frac{y+2}{-3}=\frac{z}{5}$

Plane : $x-y+z+2=0$

The vector perpendicular to required plane is

$$\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & -3 & 5\\ 1 & -1 & 2\end{vmatrix}=2\hat{i}+3\hat{j}+\hat{k}$$

Now equation of plane passing through $(1,-2,0)$

and perpendicular to $2\hat{i}+3\hat{j}+\hat{k}$

$(x-1)\,2+(y+2)\,3+(z-0)1=0$

$\Rightarrow \quad 2x+3y+z+4=0$

35. Let P_1 be a plane passing through two points A(1, 2, 0) and B(2, – 1, 1) and perpendicular to $\bar{r}\cdot(i+j-3k)=1$. Let L be a line through (1, 1, 1) and intersect line of intersection of planes P_1 and P_2 perpendicularly where equation of plane P_2 is $\bar{r}\cdot(i+j+k)=5$. If the equation of line L is $\frac{x+1}{-1}=\frac{y-y_1}{b}=\frac{z-z_1}{c}$, then find the value of $\left(\frac{2y_1+z_1}{bc}\right)$.

माना P_1 एक तल है जो A(1, 2, 0) तथा B(2, – 1, 1) से गुजरता है एवं तल $\bar{r}\cdot(i+j-3k)=1$ के लम्बवत् है। L एक रेखा है जो (1, 1, 1) से गुजरती है तथा समतल P_1 तथा P_2 की प्रतिछेदी रेखा को लम्बवत् प्रतिच्छेद करती है जहाँ तल P_2 का समीकरण $\bar{r}\cdot(i+j+k)=5$ है। यदि रेखा L का समीकरण $\frac{x+1}{-1}=\frac{y-y_1}{b}=\frac{z-z_1}{c}$ है तो $\left(\frac{2y_1+z_1}{bc}\right)$ का मान ज्ञात कीजिये।

Ans. 9

Sol. Equation of plane P

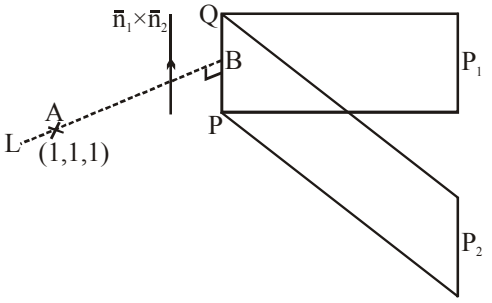
$\bar{n}_1=\bar{n}\times\overline{AB}=(i+j-3k)\times(i-3j+k)=2i+j+k$

$\therefore$ Equation of plane $P_1: 2(x-1)+1(y-2)+1(z-0)=0 \Rightarrow P_1: 2x+y+z=4$

Also, $P_2: x+y+z=5.$

Line PQ is parallel to vector

$$\bar{n}_1\times\bar{n}_2=\begin{vmatrix} i & j & k\\ 2 & 1 & 1\\ 1 & 1 & 1\end{vmatrix}=\begin{vmatrix} i & j & k\\ 1 & 0 & 0\\ 1 & 1 & 1\end{vmatrix}=k-j$$



$\therefore$ Equatin PQ: $\frac{x+1}{0}=\frac{y-6}{-1}=\frac{z-0}{1}=\lambda$

Let B(– 1, 6 – λ, λ) be any point on line PQ. So, $\overrightarrow{AB}=-2\hat{i}+(5-\lambda)\hat{j}+(\lambda-1)\hat{k}$

Now, $\overrightarrow{AB}\cdot\vec{n}_1\times\vec{n}_2=0 \Rightarrow 0-5+\lambda+\lambda-1=0 \Rightarrow \lambda=3$

$\therefore$ B ≡ (– 1, 3, 3)

So, equation of line L is $\frac{x+1}{1+1}=\frac{y-3}{1-3}=\frac{z-3}{1-3}$

i.e., $\frac{x+1}{-1}=\frac{y-3}{1}=\frac{z-3}{1}$

Hence $\left(\frac{2y_1+z_1}{bc}=\frac{6+3}{1}\right)=9$. Ans.

36. Let L be the line given by $\vec{r} = \begin{bmatrix} 2 \\ -2 \\ -1 \end{bmatrix} + \lambda \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ and let P be the point (2, – 1, 1). Also suppose that E be the plane containing three non collinear points A = (0, 1, 1); B(1, 2, 2) and C = (1, 0, 1)

Find

(a) Distance between the point P and the line L.

(b) Equation of the plane E.

(c) Equation the plane F containing the line L and the point P.

(d) Acute between the plane E and F.

(e) Volume of the parallelopiped by A, B, C and the point D(– 3, 0, 1).

रेखा L का समीकरण है $\vec{r} = \begin{bmatrix} 2 \\ -2 \\ -1 \end{bmatrix} + \lambda \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ तथा बिन्दु P के निर्देशांक (2, – 1, 1) है। यदि E एक ऐसा तल है जिस पर तीन असंरेखीय बिन्दु A = (0, 1, 1); B(1, 2, 2) तथा C = (1, 0, 1) हो तो

ज्ञात कीजिए

(a) P तथा L के बीच दूरी बताइये।

(b) तल E का समीकरण ज्ञात कीजिए।

(c) तल F का समीकरण, जिसमें रेखा L हो तथा बिन्दु P हो, ज्ञात कीजिए।

(d) तलों E तथा F के बीच न्यून कोण ज्ञात कीजिए।

(e) A, B, C तथा D(– 3, 0, 1) से बने हुए समान्तर षट्फलक का आयतन ज्ञात कीजिये।

Ans. (a) $\sqrt{3}$

(b) $x + y - 2z + 1 = 0$

(c) $x - 2y + z = 5$

(d) $\frac{\pi}{3}$

(e) 4

Sol. A(0,1,1) B(1,2,2) C(1,0,1)

$$\vec{n} = \begin{vmatrix} i & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & -1 & 0 \end{vmatrix} = \hat{i} - \hat{j}(-1) + \hat{k}(-2)$$

$(1,1,-2)$

(i) $(-\lambda, -1, -2 + \lambda) \cdot (-1,0,1) = 0$

$\lambda - 2 + \lambda = 0$

$\lambda = 1$

$\sqrt{1+1+1} = \sqrt{3}$

(ii) $x + y - 2z = -1$

(iii) $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 0 & 1 \\ 0 & -1 & -2 \end{vmatrix} = \hat{i} - \hat{j}(2) + \hat{k}$

$x - 2y + z = 5$

(iv) $= \dfrac{(1,1,-2)\cdot(1,-2,1)}{6}$

$= \dfrac{1-2-2}{6} = \dfrac{1}{2}$ $\qquad\qquad \theta = 60°$

(v) $\dfrac{1}{6}\begin{vmatrix} 0 & 2 & 1 \\ -4 & 0 & 0 \\ -1 & 1 & 0 \end{vmatrix}$

$(+4\ (-1)) = 4$

37. Find the foot and hence the length of the perpendicular from the point (5, 7, 3) to the line $\dfrac{x-15}{3} = \dfrac{y-29}{8} = \dfrac{5-z}{5}$.

Also find the equation of the plane in which the perpendicular and the given straight line lie.

बिन्दु (5, 7, 3) से रेखा $\dfrac{x-15}{3} = \dfrac{y-29}{8} = \dfrac{5-z}{5}$ पर लम्बपाद तथा लम्ब की लम्बाई ज्ञात करो। उस तल का समीकरण ज्ञात करो जिसमें दिया गया लम्ब तथा दी गयी रेखा आती है।

Ans. (9, 13, 15) ; 14 ; $9x - 4y - z = 14$

Sol. $\dfrac{x-15}{3} = \dfrac{y-29}{8} = \dfrac{z-5}{-5}$ $\qquad$ L $\qquad$ 1

DR′s of perpendicular line from (5,7,3)

$(3\lambda + 10, 8\lambda + 22, -5\lambda + 2)$

Now $9\lambda + 30 + 64\lambda + 176 + 25\lambda - 10 = 0$

$98\lambda + 196 = 0 \Rightarrow \lambda = -2$

foot perpendicular (x,y,z) = (9,13,15)

perpendicular line $\dfrac{x-5}{4} = \dfrac{y-7}{6} = \dfrac{z-3}{12}$ $\qquad$ L_2

Equation of plane having line L_1 & L_2

$\begin{vmatrix} x-5 & y-7 & z-3 \\ 2 & 3 & 6 \\ 3 & 8 & -5 \end{vmatrix} = 0$

$-9(x-5)+4(y-7)+z-3=0$
$\Rightarrow -9x+4y+z+14=0$

38. Let the equation of the plane containing the line $x-y-z-4=0=x+y+2z-4$ and is parallel to the line of intersection of the planes $2x+3y+z=1$ and $x+3y+2z=2$ be $x+Ay+Bz+C=0$. Compute the value of $|A+B+C|$.

माना रेखा $x-y-z-4=0=x+y+2z-4$ को समाहित करने वाला तल जो $2x+3y+z=1$ तथा $x+3y+2z=2$ की प्रतिछेदी रेखा के समान्तर है, का समीकरण $x+Ay+Bz+C=0$ है तो $|A+B+C|$ का मान ज्ञात कीजिये।

Ans. 11

Sol. $(x-y-z-4)+\lambda(x+y+2z-4)=0$
$(1+\lambda)x-(1-\lambda)y-(1-2\lambda)z-4(1+\lambda)=0$

$$\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 3 & 1\\ 1 & 3 & 2\end{vmatrix}=\hat{i}(3)-\hat{j}(3)+\hat{k}(3)$$

$(1,-1,1)$
$1+\lambda+1-\lambda-1+2\lambda=0$
$1+2\lambda=0$

$$\lambda=-\frac{1}{2}$$

$$\frac{1}{2}x-\frac{3}{2}y-2z-z=0$$
$x-3y-4z-4=0$
$x+Ay+Bz+C=0$
From this $A=-3$, $B=-4$, $C=-4$

39. Find the equation of the plane containing the line $\frac{x-1}{2}=\frac{y}{3}=\frac{z}{2}$ and parallel to the line $\frac{x-3}{2}=\frac{y}{5}=\frac{z-2}{4}$. Find also the S.D. between the two lines.

रेखा $\frac{x-1}{2}=\frac{y}{3}=\frac{z}{2}$ को समाहित करने वाली, तथा $\frac{x-3}{2}=\frac{y}{5}=\frac{z-2}{4}$ के समानान्तर तल का समीकरण ज्ञात करो। इन दोनों रेखाओं के बीच न्यूनतम दूरी भी ज्ञात कीजिए।

Ans. $x-2y+2z-1=0$; 2 units

Sol. DR′s of vector perpendicular to both lines $=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 3 & 2\\ 2 & 5 & 4\end{vmatrix}=\hat{i}(2)-\hat{j}(+4)+\hat{k}(4)$

DR′s of vector perpendicular to both lines = $\lambda(1,-2,2)$
Equation of plane
$x-2y+2z=1$

Shortest distance between lines = $(2,0,2)\cdot\frac{(1,-2,2)}{3}=\frac{2+4}{3}=2$

40. Consider the plane

$$E:\ \vec{r}=\begin{bmatrix}-1\\1\\1\end{bmatrix}+\lambda\begin{bmatrix}1\\2\\0\end{bmatrix}+\mu\begin{bmatrix}1\\0\\1\end{bmatrix}$$

Let F be the plane containing the point A (– 4, 2, 2) and parallel to E.

Suppose the point B is on the plane E such that B has a minimum distance from the point A. If C (– 3, 0, 4) lies in the plane F. Find the area of the triangle ABC.

माना समतल

$$E:\ \vec{r}=\begin{bmatrix}-1\\1\\1\end{bmatrix}+\lambda\begin{bmatrix}1\\2\\0\end{bmatrix}+\mu\begin{bmatrix}1\\0\\1\end{bmatrix}$$

F बिन्दु A (– 4, 2, 2) से गुजरने वाला E के समान्तर तल है। यदि बिन्दु B तल E में स्थित हो तथा A से न्यूनतम दूरी पर तो तथा C (– 3, 0, 4) तल F में हो तो त्रिभुज ABC का क्षेत्रफल ज्ञात कीजिए।

Ans. $\frac{9}{2}$

Sol. $E:\vec{r}=(-1,1,1)+\lambda(1,2,0)+\mu(1,0,1)\to B$

$F\to A(-4,2,2)$ $\qquad$ $BA\Rightarrow \min$

C(–3, 0, 4)

$$E=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&2&0\\1&0&1\end{vmatrix}=\hat{i}(2)-\hat{j}(1)+\hat{k}(-2)\Rightarrow E=2x-y-2z=-5$$

= (2, –1, –2)
A→FOP
(–4 + 2λ, 2 – λ, 2 – 2λ)
B(–2, 1, 0)
– 8 + 4λ – 2 + λ – 4 + 4λ = –5
9λ = 9
λ = 1

$$\frac{1}{2}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 1 & 2 \\ -1 & 2 & -2 \end{vmatrix}$$

$$= \frac{1}{2}\left|-6\hat{i} + 6\hat{j} - 3\hat{k}\right|$$

$$= \frac{9}{2}$$

41. Let P be the plane passing through (1, 1, 1) and parallel to the lines L_1 and L_2 having direction ratios 1, 0, –1 and –1, 1, 0 respectively. If A, B and C are the points at which P intersects the coordinate axes, find the volume of the tetrahedron whose vertices are A, B, C and the origin.

माना P एक तल है जो (1, 1, 1) से गुजरता है तथा रेखाऐं L_1 तथा L_2 जिनके दिक् अनुपात 1, 0, –1 तथा –1, 1, 0 हैं, के समान्तर है। यदि A, B तथा C वह बिन्दु है जहाँ P निर्देशी अक्षों को काटता है तो चतुष्फलक का आयतन ज्ञात कीजिये जिसके शीर्ष A, B, C तथा मूल बिन्दु है।

Ans. $\frac{9}{2}$

Sol. $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{vmatrix} = \hat{i} + \hat{j} + \hat{k}(.)$ $\qquad x + y + z = 3$

$$\frac{1}{6}\begin{vmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{vmatrix} = \frac{1}{6} \times 27$$

$$= \frac{9}{2}$$

FAMILY OF PLANES

42. Let (x, y, z) be points with integer coordinates satisfying the system of homogeneous equations

$$3x - y - z = 0 \; ; -3x + z = 0 \; ; -3x + 2y + z = 0$$

Then the number of such points for which $x^2 + y^2 + z^2 \le 100$ is.

माना (x, y, z) बिन्दु है जिनके निर्देशांक पूर्णांक है तथा निम्न समघातीय समीकरणों को सन्तुष्ट करते है

$$3x - y - z = 0 \; ; -3x + z = 0 \; ; -3x + 2y + z = 0$$

तो ऐसे बिन्दुओं की संख्या होगी जिनके लिये $x^2 + y^2 + z^2 \le 100$ हो।

Ans. 7

Sol. $3x - y - z = 0$
$-3x + z = 0$
$-3x + 2y + z = 0$
$y = 0$
$z = 3x$
$x^2 + 0 + 9x^2 \le 100$
$10x^2 \le 100$
$x^2 \le 10$
$x = 0, \pm1, \pm2, \pm3$
$= 7$

43. If the distance between the plane $Ax - 2y + z = d$ and the plane containing the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ is $\sqrt{6}$, then $|d|$ is.

यदि तल $Ax - 2y + z = d$ तथा रेखाऐं $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ तथा $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ को समाहित करने वाले तल के मध्य दूरी $\sqrt{6}$ है तो $|d|$ का मान होगा।

Ans. 6

Sol. $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 1 & 1 & 1 \end{vmatrix} = \hat{i}(-1) - \hat{j}(-2) + \hat{k}(-1)$

$(1, -2, 1)$
$x - 2y + z = d$
$x - 2y + z = 0$
$\left|\frac{d}{\sqrt{6}}\right| = \sqrt{6}$
$|d| = 6$

44. The plane denoted by $\Pi_1 : 4x + 7y + 4z + 81 = 0$ is rotated through a right angle about its line of intersection with the plane $\Pi_2 : 5x + 3y + 10z = 25$. If the plane in its new position be denoted by Π, and the distance of this plane from the origin is $\sqrt{k}$ where $k \in N$, then find k.

तल $\Pi_1 : 4x + 7y + 4z + 81 = 0$ को इसके तथा तल $\Pi_2 : 5x + 3y + 10z = 25$ के प्रतिच्छेदी रेखा के सापेक्ष समकोण पर घुमाया जाता है। यदि तल का नयी स्थिति Π हो तथा इस तल का मूल बिन्दु से दूरी $\sqrt{k}$, $k \in N$ हो तो k का मान ज्ञात कीजिए।

Ans. 212

Sol. $\pi_1 : 4x + 7y + 4z + 81 = 0$
$\pi_2 : 5x + 3y + 10z = 25$
$\pi_1 + \lambda\pi_2 = 0$
$(4 + 5\lambda)x + (7 + 3\lambda)y + (4 + 10\lambda)z + (81 - 25\lambda) = 0$
$(4,7,4) = 0$
$16 + 20\lambda + 49 + 21\lambda + 16 + 40\lambda = 0$
$81\lambda + 81 = 0$
$\lambda = -1$
$-x + 4y - 6z + 106 = 0$

$$\frac{106}{\sqrt{16+1+36}} = \frac{106}{\sqrt{53}} = 2\sqrt{53} = \sqrt{212}$$

45. Find the equation of the plane containing the line $2x - y + z - 3 = 0, 3x + y + z = 5$ and at a distance of $\frac{1}{\sqrt{6}}$ from the point $(2, 1, -1)$.

उस समतल का समीकरण ज्ञात कीजिये, जिस पर रेखा $2x - y + z - 3 = 0, 3x + y + z = 5$ स्थित है तथा जिसकी बिन्दु $(2, 1, -1)$ से दूरी $\frac{1}{\sqrt{6}}$ है।

Ans. $2x - y + z - 3 = 0$ and $62x + 29y + 19z - 105 = 0$

Sol. Equation of plane containing the line,
$2x - y + z - 3 = 0$ and $3x + y + z = 5$ is
$(2x - y + z - 3) + \lambda(3x + y + z - 5) = 0$
$\Rightarrow (2+3\lambda)x + (\lambda - 1)y + (\lambda + 1)z - 3 - 5\lambda = 0$

Since distance of plane from $(2, 1, -1)$ to above plane is $\frac{1}{\sqrt{6}}$

$$\therefore \quad \left|\frac{6\lambda + 4 + \lambda - 1 - \lambda - 1 - 3 - 5\lambda}{\sqrt{(3\lambda+2)^2 + (\lambda-1)^2 + (\lambda+1)^2}}\right| = \frac{1}{\sqrt{6}}$$

$\Rightarrow \quad 6(\lambda - 1)^2 = 11\lambda^2 + 12\lambda + 6$

$\Rightarrow \quad \lambda = 0, -\frac{24}{5}$

$\therefore$ Equation of planes are,
$2x - y + z - 3 = 0$ and $62x + 29y + 19z - 105 = 0$

CONCEPT OF IMAGE

46. Find the equation of the line which is reflection of the line $\frac{x-1}{9}=\frac{y-2}{-1}=\frac{z+3}{-3}$ in the plane $3x-3y+10z=26$.

तल $3x-3y+10z=26$ में रेखा $\frac{x-1}{9}=\frac{y-2}{-1}=\frac{z+3}{-3}$ का प्रतिबिम्ब ज्ञात कीजिए।

Ans. $\frac{x-4}{9}=\frac{y+1}{-1}=\frac{z-7}{-3}$

Sol. Line parallel to plane
Reflection of line will be parallel to given line
mirror image of point P(1,2,–3)
FOP from P in the direction (3, –3, 10) will be $(1+3\lambda, 2-3\lambda, -3+10\lambda)$
FOP on plane
$3+9\lambda-6+9\lambda-30+100\lambda=26$

$\lambda=\frac{1}{2}$

FOP $\left(\frac{5}{7},\frac{1}{2},2\right)$

Mirror image of point P (4, –1, 7)

Equation of reflected line $\frac{x-4}{9}=\frac{y+1}{-1}=\frac{z-7}{-3}$

MISCELLANEOUS

47. Find the equations to the line of greatest slope through the point (7, 2 , –1) in the plane $x-2y+3z=0$ assuming that the axes are so placed that the plane $2x+3y-4z=0$ is horizontal.

तल $x-2y+3z=0$ में (7, 2 , –1) में महत्तम ढाल वाली रेखा का समीकरण ज्ञात करो। अक्षो को इस तरह रखा गया है कि तल $2x+3y-4z=0$ क्षैतिज है।

Ans. $\frac{x-7}{22}=\frac{y-2}{5}=\frac{z+1}{-4}$

Sol. Line is $\perp$ to intersection line of planes

$$\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 3 & -4\\ 1 & -2 & 3\end{vmatrix}=\hat{i}-\hat{j}10+\hat{k}(-7)=(1,-10,-7)$$

direction of line

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -10 & -7 \\ 1 & -2 & 3 \end{vmatrix} = \hat{i}(-44) - \hat{j}(10) + \hat{k}(8)$$

$< 22, 5, -4 >$

Equation of line $\frac{x-7}{22} = \frac{y-2}{5} = \frac{z+1}{-4}$

Paragraph for Question Nos. 48 to 50

Let A(1, 2, 3), B(0, 0, 1), C(–1, 1, 1) are the vertices of a ΔABC.

माना A(1, 2, 3), B(0, 0, 1), C(–1, 1, 1) त्रिभुज ABC के शीर्ष है

48. The equation of internal angle bisector through A to side BC is :

कोण A के आन्तरिक कोणार्धक का समीकरण है :

(A) $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \mu(3\hat{i} + 2\hat{j} + 3\hat{k})$ (B) $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \mu(3\hat{i} + 4\hat{j} + 3\hat{k})$

(C) $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \mu(3\hat{i} + 3\hat{j} + 2\hat{k})$ (D) $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \mu(3\hat{i} + 3\hat{j} + 4\hat{k})$

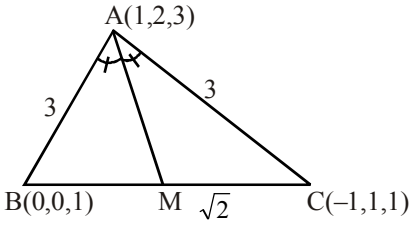
Ans. (D)

Sol. Here ΔABC is isosceles with AB = AC,

So internal bisector of A is perpendicular to BC.

As ΔAMB ≅ ΔAMC (R.H.S. rule)

⇒ M is midpoint of BC.



So, $M \equiv \left(\frac{-1}{2}, \frac{1}{2}, 1\right)$

∴ Equation of internal bisector through A to side BC is

$\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \mu\left(\frac{3}{2}\hat{i} + \frac{3}{2}\hat{j} + 2\hat{k}\right)$ OR $\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \mu\left(3\hat{i} + 3\hat{j} + 4\hat{k}\right)$

Alternative for (i): Equation of BC is

$\vec{r} = \hat{k} + \lambda(\hat{i} - \hat{j})$

Let p.v. of M on BC be $\vec{r}$

Now, $\vec{AM}$ = p.v. of M – p.v. of A = $(\lambda\hat{i} - \lambda\hat{j} + \hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k}) = (\lambda - 1)\hat{i} - (\lambda + 2)\hat{j} - 2\hat{k}$

Since $\vec{AM} \cdot (\hat{i} - \hat{j}) = 0$ ⇒ $\lambda = -1/2$

∴ p.v. of point $M = \frac{-1}{2}\hat{i} + \frac{1}{2}\hat{j} + \hat{k}$

⇒ Equation of internal bisector through A to the side BC is

$\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \mu\left(\frac{3}{2}\hat{i} + \frac{3}{2}\hat{j} + 2\hat{k}\right)$ OR $\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \mu(3\hat{i} + 3\hat{j} + 4\hat{k})$

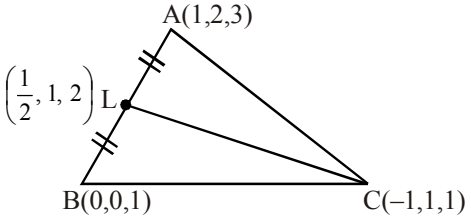
49. The equation of median through C to side AB is :
C से मध्यिका का समीकरण होगा :

(A) $\vec{r} = -\hat{i} + \hat{j} + \hat{k} + p(3\hat{i} - 2\hat{k})$ (B) $\vec{r} = -\hat{i} + \hat{j} + \hat{k} + p(3\hat{i} + 2\hat{k})$

(C) $\vec{r} = -\hat{i} + \hat{j} + \hat{k} + p(-3\hat{i} + 2\hat{k})$ (D) $\vec{r} = -\hat{i} + \hat{j} + \hat{k} + p(3\hat{i} + 2\hat{j})$

Ans. (B)

Sol. Clearly midpoint L of AB is $\left(\frac{1}{2}, 1, 2\right)$



$\Rightarrow$ Equation of median through C to AB is

$\vec{r} = (-\hat{i} + \hat{j} + \hat{k}) + p\left(\frac{3}{2}\hat{i} + \hat{k}\right)$ OR $\vec{r} = (-\hat{i} + \hat{j} + \hat{k}) + p(3\hat{i} + 2\hat{k})$

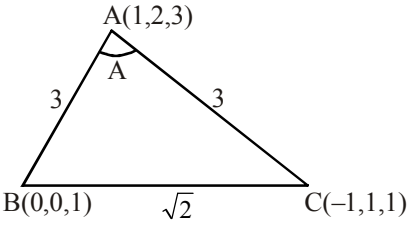
50. The area (ΔABC) is equal to :
ΔABC का क्षेत्रफल है :

(A) $\frac{9}{2}$ (B) $\frac{\sqrt{17}}{2}$ (C) $\frac{17}{2}$ (D) $\frac{7}{2}$

Ans. (B)

Sol. We have $\cos A = \frac{3^2 + 3^2 - \left(\sqrt{2}\right)^2}{2(3)(3)}$

$\therefore \cos A = \frac{16}{18} = \frac{8}{9}$



Now area (ΔABC) $= \frac{1}{2}(3)(3)\sin A = \frac{9}{2}\sqrt{1 - \frac{64}{81}} = \frac{9}{2} \times \frac{\sqrt{17}}{9} = \frac{\sqrt{17}}{2}$ sq. unit.

Paragraph for Question Nos. 51 to 52

Consider a plane $x + y - z = 1$ and the point A(1, 2, –3) है। A line has the equation $x = 1 + 3r$, $y = 2 - r$, $z = 3 + 4r$.

माना एक तल $x + y - z = 1$ तथा बिन्दु A(1, 2, –3) है। एक रेखा L का समीकरण $x = 1 + 3r$, $y = 2 - r$, $z = 3 + 4r$

51. The co-ordinate of a point B of line L, such that AB is parallel to the plane, is :

बिन्दु B निर्देशांक ज्ञात करें जो रेखा L पर स्थित है तथा AB समतल के समानान्तर है :

(A) 10, –1, 15 (B) –5, 4, –5 (C) 4, 1, 7 (D) –8, 5, –9

Ans. D

Sol. line $\frac{x-1}{3} = \frac{y-2}{-1} = \frac{z-3}{4} = r$; A(1, 2, – 3)

Any point say $B \equiv 3r + 1, 2 - r, 3 + 4r$ (on the line L)

$\overline{AB} = 3r, -r, 4r + 6$

Hence, $\overline{AB}$ is parallel to $x + y - z = 1$

hence, $3r - r - 4r - 6 = 0$

$2r = -6 \quad \Rightarrow \quad r = -3$

hence B is –8, 5, –9

52. Equation of the plane containing the line L and the point A has the equation :

रेखा L को समाहित करने वाले तथा बिन्दु A से गुजरने वाले समतल का समीकरण है :

(A) $x - 3y + 5 = 0$ (B) $x + 3y - 7 = 0$ (C) $3x - y - 1 = 0$ (D) $3x + y - 5 = 0$

Ans. B

Sol. Equation of plane containing the line L is

$A(x - 1) + B(y - 2) + C(z - 3) = 0$, where $3A - B + 4C = 0$ ....(1)

$\therefore$ (1) also contains the point A(1, 2, –3)

hence $C = 0$; $3A = B$

equation of plane $x - 1 + 3(y - 2) = 0$

$x + 3y - 7 = 0$ Ans.

Paragraph for Question Nos. 53 to 55

Consider the lines $L_1 : \frac{x+1}{3} = \frac{y+2}{1} = \frac{z+1}{2}$; $L_2 : \frac{x-2}{1} = \frac{y+2}{2} = \frac{z-3}{3}$

माना रेखायें $L_1 : \frac{x+1}{3} = \frac{y+2}{1} = \frac{z+1}{2}$; $L_2 : \frac{x-2}{1} = \frac{y+2}{2} = \frac{z-3}{3}$

53. The unit vector perpendicular to both L_1 and L_2 is :

दोनों रेखाओं L_1 व L_2 के लम्बवत् इकाई सदिश है :

(A) $\frac{-\hat{i}+7\hat{j}+7\hat{k}}{\sqrt{99}}$ (B) $\frac{-\hat{i}-7\hat{j}+5\hat{k}}{5\sqrt{3}}$ (C) $\frac{-\hat{i}+7\hat{j}+5\hat{k}}{5\sqrt{3}}$ (D) $\frac{7\hat{i}-7\hat{j}-\hat{k}}{\sqrt{99}}$

Ans. B

Sol. $\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 3 & 1 & 2\\ 1 & 2 & 3\end{vmatrix} = \hat{i}(-1)-\hat{j}(7)+\hat{k}(5)$

$= \frac{<-1,-7,5>}{\sqrt{75}}$

54. The shortest distance between L_1 and L_2 is :

L_1 तथा L_2 के बीच की न्यूनतम दूरी है :

(A) 0 (B) $\frac{17}{\sqrt{3}}$ (C) $\frac{41}{5\sqrt{3}}$ (D) $\frac{17}{5\sqrt{3}}$

Ans. D

Sol. $(\vec{a}_2-\vec{a}_1)\cdot\frac{(\vec{b}_1\times\vec{b}_2)}{|\vec{b}_1\times\vec{b}_2|}$

$=<-3,+0,-4>\cdot\frac{<-1,-7,5>}{\sqrt{75}}$

$=\frac{|3-20|}{5\sqrt{3}}=\frac{17}{5\sqrt{3}}$

55. The distance of the point (1, 1, 1) from the plane passing through the point (–1, –2, –1) and whose normal is perpendicular to both the lines L_1 and L_2 is :

बिन्दु (–1, –2, –1) से होकर जाने वाले तल जिसका अभिलम्ब दोनों रेखाओं L_1 व L_2 के लम्बवत् है, से बिन्दु (1, 1,1) की दूरी है :

(A) $\frac{2}{\sqrt{75}}$ (B) $\frac{7}{\sqrt{75}}$ (C) $\frac{13}{\sqrt{75}}$ (D) $\frac{23}{\sqrt{75}}$

Ans. C

Sol. Plane $\vec{n}=<-1,-7,5>$ Point $<-1,-2,-1>$

$-x-7y+5z=10$

distance from $(1,1,1)$

$\left|\frac{-13}{\sqrt{75}}\right|=\frac{13}{\sqrt{75}}$

ONE OR MORE THAN ONE CORRECT TYPE

56. A vector of magnitude 10 along the normal to the curve $3x^2 + 8xy + 2y^2 - 3 = 0$ at its point P(1, 0) can be :

वक्र $3x^2 + 8xy + 2y^2 - 3 = 0$ के लम्बवत् इसके बिन्दु P (1, 0) पर दस मापांक का सदिश है :

(A) $6\hat{i} + 8\hat{j}$ (B) $-6\hat{i} + 8\hat{j}$ (C) $6\hat{i} - 8\hat{j}$ (D) $-6\hat{i} - 8\hat{j}$

Ans. AD

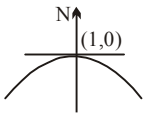
Sol. differentiate the curve

$6x + 8(xy_1 + y) + 4yy_1 = 0$

m_T at (1, 0) is $6 + 8(y_1 (1)) = 0$

$y_1(0) = -\frac{3}{4}$

$m_N = \frac{4}{3} \Rightarrow$ vector along normal $3\hat{i} + 4\hat{j}$



unit vector $= \pm \frac{(3\hat{i} + 4\hat{j})}{5}$ along the normal

again normal vector of magnitude $10 = \pm (6\hat{i} + 8\hat{j})$ Ans.

57. Consider a plane P passing through A(λ , 3, μ), B(– 1, 3, 2) and C(7, 5, 10) and a straight line L with positive direction cosines passing through A, bisecting BC and makes equal angles with the coordinate axes. Let L_1 be a line parallel to L and passing through origin. Which of the following is(are) correct?

(A) The value of $(\lambda + \mu)$ is equal to 5

(B) Equation of straight line L_1 is $\frac{x-1}{1} = \frac{y-1}{1} = \frac{z-1}{1}$

(C) Equation of the plane perpendicular to the plane P and containing line L_1 is $x - 2y + z = 0$

(D) Area of triangle ABC is equal to $3\sqrt{2}$

एक समतल P जो बिन्दुओं A(λ , 3, μ), B(– 1, 3, 2) तथा C(7, 5, 10) से गुजरता है। एक रेखा L जिसकी दिक्कोज्याऐं धनात्मक है एवं बिन्दु A से गुजरती है तथा BC को समभाजित करती है एवं समतल निर्देशांक अक्षों के साथ बराबर कोण बनाती है। माना L_1 रेखा L के समानान्तर अन्य रेखा है जो मूल बिन्दु से होकर गुजरती है। निम्न में से कौनसे कथन सही है?

(A) $(\lambda + \mu) = 5$

(B) रेखा L_1 का समीकरण $\frac{x-1}{1} = \frac{y-1}{1} = \frac{z-1}{1}$ है

(C) समतल P के लम्बवत् तल जो रेखा L_1 को समाहित करता है, का समीकरण $x - 2y + z = 0$ है

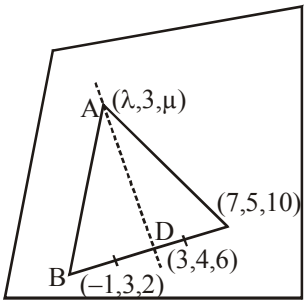
(D) त्रिभुज ABC का क्षेत्रफल $3\sqrt{2}$

Ans. BCD

Sol. (A) Dr's of Straight L which is passing through A and bisecting BC are $\lambda-3, -1, \mu-6$.
Straight line L is equally inclined with co-ordinate axes

$$\therefore \quad \frac{\lambda-3}{1}=\frac{-1}{1}=\frac{\mu-6}{1} \Rightarrow \lambda=2 \text{ and } \mu=5$$

$\lambda+\mu=7$. Ans.



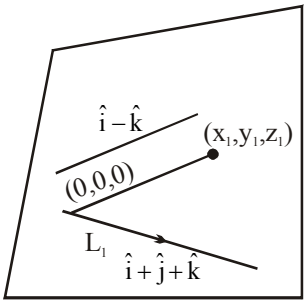
(B) Dr's of line L are 1, 1, 1

$\therefore$ Option 'c' is correct

$\frac{x-1}{1}=\frac{y-1}{1}=\frac{z-1}{1}$ it satisfied (0, 0, 0).

(C) Normal vector of plane P is [12th, 17-10-2010, vector]

$$\vec{n}=\overrightarrow{AB}\times\overrightarrow{AC}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 3 & 0 & 3\\ -5 & -2 & -5\end{vmatrix}=6\hat{i}-\hat{j}\times 0+\hat{k}(-6)=6(\hat{i}-\hat{k}).$$



Plane perpendicular to plane P and containing line L_1 is

$$=\begin{vmatrix}x & y & z\\ 1 & 0 & -1\\ 1 & 1 & 1\end{vmatrix}=0$$

$\Rightarrow \quad x(0+1)-y(1+1)+z(1)=0$

$\Rightarrow \quad x-2y+z=0$. Ans.

58. A line L passing through the point P(1, 4, 3), is perpendicular to both the lines

$$\frac{x-1}{2}=\frac{y+3}{1}=\frac{z-2}{4} \text{ and } \frac{x+2}{3}=\frac{y-4}{2}=\frac{z+1}{-2}$$

If the position vector of point Q on L is (a_1, a_2, a_3) such that $(PQ)^2 = 357$, then $(a_1 + a_2 + a_3)$ can be :

एक रेखा जो बिन्दु P(1, 4, 3) से गुजरती है तथा रेखाओं

$$\frac{x-1}{2}=\frac{y+3}{1}=\frac{z-2}{4} \text{ तथा } \frac{x+2}{3}=\frac{y-4}{2}=\frac{z+1}{-2}$$

के लम्बवत् है। यदि Q (a_1, a_2, a_3) रेखा पर स्थित है, $(PQ)^2 = 357$, तो $(a_1 + a_2 + a_3)$ हो सकता है :

(A) 16 (B) 15 (C) 2 (D) 1

Ans. BD

Sol. Equation of the line passing through P(1, 4, 3) is

$$\frac{x-1}{a}=\frac{y-4}{b}=\frac{z-3}{c} \quad ....(1)$$

where $a\hat{i}+b\hat{j}+c\hat{k}$ is along the common perpendicular to both the given lines.

Since (1) is perpendicular to $\frac{x-1}{2}=\frac{y+3}{1}=\frac{z-2}{4}$ and $\frac{x+2}{3}=\frac{y-4}{2}=\frac{z+1}{-2}$

Hence $2a + b + 4c = 0$

and $3a + 2b - 2c = 0$

$\therefore \quad \frac{a}{-2-8}=\frac{b}{12+4}=\frac{c}{4-3} \quad \Rightarrow \quad \frac{a}{-10}=\frac{b}{16}=\frac{c}{1}$

Hence the equation of the lines is $\frac{x-1}{-10}=\frac{y-4}{16}=\frac{z-3}{1}$ ....(2) Ans.

Now any point Q on (2) can be taken as $(1 - 10\lambda , 16\lambda + 4 , \lambda + 3)$

$\therefore$ Distance of Q from P (1, 4, 3) = $(10\lambda)^2 + (16\lambda)^2 + \lambda^2 = 357$

$\Rightarrow \quad (100 + 256 + 1)\lambda^2 = 357$

$\Rightarrow \quad \lambda = 1 \text{ or } -1$

$\therefore \quad$ Q is (–9, 20, 4) or (11, – 12, 2)

Hence $a_1 + a_2 + a_3 = 15$ or 1 Ans.

ONE OR MORE THAN ONE IS/ARE CORRECT

59. If $\vec{a}$, $\vec{b}$, $\vec{c}$ be three non zero vectors satisfying the condition $\vec{a} \times \vec{b} = \vec{c}$ and $\vec{b} \times \vec{c} = \vec{a}$ then which of the following always hold(s) good?

(A) $\vec{a}$, $\vec{b}$, $\vec{c}$ are orthogonal in pairs (B) $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = \left|\vec{b}\right|$

(C) $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = \left[\vec{a}\ \vec{b}\ \vec{c}\right] = |\vec{c}|^2$ (D) $\left|\vec{b}\right| = |\vec{c}|$

यदि $\vec{a}$, $\vec{b}$, $\vec{c}$ तीन अशून्य सदिश इस प्रकार है कि जो शर्त $\vec{a} \times \vec{b} = \vec{c}$ तथा $\vec{b} \times \vec{c} = \vec{a}$ को संतुष्ट करते है तो इनमें से कौनसे कथन सदैव सही होंगे?

(A) $\vec{a}$, $\vec{b}$, $\vec{c}$ युग्म के तौर पर लंबवत् है (B) $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = \left|\vec{b}\right|$

(C) $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = \left[\vec{a}\ \vec{b}\ \vec{c}\right] = |\vec{c}|^2$ (D) $\left|\vec{b}\right| = |\vec{c}|$

Ans. AC

Sol. Clearly $\vec{a} \cdot \vec{c} = 0$ and $\vec{b} \cdot \vec{c} = 0$ Also $\vec{a} \cdot \vec{b} = 0 \Rightarrow$ A

$\vec{a} \times \vec{b} = \vec{c}$

dot with $\vec{b} \Rightarrow \vec{b} \cdot \vec{c} = 0$; |||ly $\vec{b} \times \vec{c} = \vec{a}$

dot with $\vec{b} \Rightarrow \vec{a} \cdot \vec{b} = 0$; dot with $\vec{c} \Rightarrow \vec{a} \cdot \vec{c} = 0$

$\Rightarrow \quad \vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$

Again $\begin{matrix}|\vec{a}|\,|\vec{b}|=|\vec{c}| \\ |\vec{b}|\,|\vec{c}|=|\vec{a}|\end{matrix} \Rightarrow \frac{|\vec{a}|}{|\vec{c}|}=\frac{|\vec{c}|}{|\vec{a}|} \quad \Rightarrow \quad |\vec{a}|=|\vec{c}| \ \& \ |\vec{b}|=1$

$\Rightarrow \ \vec{a}\times\vec{b}\,.\,\vec{c} = |\vec{a}|\,|\vec{b}|\,|\vec{c}| = |\vec{a}|^2 = |\vec{c}|^2$

(children will assume $\vec{a}=\hat{i};\ \vec{b}=\hat{j}$ and $\vec{c}=\hat{k}$ but in this case all the four will be correct which will be wrong)

60. Given the following information about the non zero vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$

(i) $(\vec{A}\times\vec{B})\times\vec{A}=\vec{0}$ (ii) $\vec{B}\cdot\vec{B}=4$

(iii) $\vec{A}\cdot\vec{B}=-6$ (iv) $\vec{B}\cdot\vec{C}=6$

Which one of the following holds good?

अशून्य सदिशों $\vec{A}$, $\vec{B}$ तथा $\vec{C}$ के लिये निम्न जानकारी दी गई है :

(i) $(\vec{A}\times\vec{B})\times\vec{A}=\vec{0}$ (ii) $\vec{B}\cdot\vec{B}=4$

(iii) $\vec{A}\cdot\vec{B}=-6$ (iv) $\vec{B}\cdot\vec{C}=6$

निम्न में से कौनसा सही है?

(A) $\vec{A}\times\vec{B}=\vec{0}$ (B) $\vec{A}\cdot(\vec{B}\times\vec{C})=0$ (C) $\vec{A}\cdot\vec{A}=8$ (D) $\vec{A}\cdot\vec{C}=-9$

Ans. ABD

Sol. Given $|\vec{A}||\vec{B}|\cos\theta=-6$; $|\vec{B}|=2$ (given)

$\vec{B}\cdot\vec{C}=|\vec{B}||\vec{C}|\cos\phi=6$

and $(\vec{A}\times\vec{B})\times\vec{A}=0$

$(\vec{A}\cdot\vec{A})\vec{B}-(\vec{B}\cdot\vec{A})\vec{A}=0$

$(\vec{A}\cdot\vec{A})\vec{B}=-6\vec{A}$ .....(1)

$\therefore$ $\vec{A}$ and $\vec{B}$ are collinear and θ between $\vec{A}$ and $\vec{B}$ is π $\Rightarrow$ $\vec{A}\times\vec{B}=0$ $\Rightarrow$ A is correct

$\Rightarrow$ $\vec{A}\cdot(\vec{B}\times\vec{C})=(\vec{A}\times\vec{B})\cdot\vec{C}=0$ $\Rightarrow$ B is correct

also $\vec{A}\cdot\vec{B}=-6$ and $|\vec{B}|=2$

$\therefore$ $|\vec{A}||\vec{B}|\cos\pi=-6$

$|\vec{A}|\cdot(2)=6$ $\Rightarrow$ $|\vec{A}|=3$ $\Rightarrow$ $\vec{A}\cdot\vec{A}=9$ $\Rightarrow$ C is not correct

again $\vec{A}\cdot\vec{C}=?$

dot with $\vec{C}$ the equation (1)

$9(\vec{B}\cdot\vec{C}) = -6\,\vec{A}\cdot\vec{C}$

$9)(6) = -6(\vec{A}\cdot\vec{C}) \quad \Rightarrow \quad \vec{A}\cdot\vec{C} = -9 \quad \Rightarrow \quad$ (D) is correct

61. If $\vec{A}$, $\vec{B}$, $\vec{C}$ and $\vec{D}$ are four non zero vectors in the same plane no two of which are collinear then which of the following hold(s) good?

यदि $\vec{A}$, $\vec{B}$, $\vec{C}$ तथा $\vec{D}$ चार अशून्य सदिश में स्थित है जिनमें कोई दो समरेखीय नही है तो निम्न में से कौनसे कथन सत्य होंगे?

(A) $(\vec{A}\times\vec{B})\cdot(\vec{C}\times\vec{D}) = 0$ (B) $(\vec{A}\times\vec{C})\cdot(\vec{B}\times\vec{D}) \neq 0$

(C) $(\vec{A}\times\vec{B})\times(\vec{C}\times\vec{D}) = \vec{0}$ (D) $(\vec{A}\times\vec{C})\times(\vec{B}\times\vec{D}) \neq \vec{0}$

Ans. BC

Sol. Obviously (B) and (C)

62. If $\vec{a}\times\vec{b} = \vec{c}\times\vec{d}$ and $\vec{a}\times\vec{c} = \vec{b}\times\vec{d}$, then the vectors $\vec{a}-\vec{d}$ and $\vec{b}-\vec{c}$ are :

(A) collinear (B) linearly independent

(C) perpendicular (D) parallel

यदि $\vec{a}\times\vec{b} = \vec{c}\times\vec{d}$ और $\vec{a}\times\vec{c} = \vec{b}\times\vec{d}$ हो, तो सदिश $\vec{a}-\vec{d}$ और $\vec{b}-\vec{c}$ है :

(A) संरेखीय (B) रेखीय रूप से स्वतंत्र

(C) लम्बवत् (D) समान्तर

Ans. AD

Sol. $\vec{a}\times\vec{b} = \vec{c}\times\vec{d}$ ...(i)

$\vec{a}\times\vec{c} = \vec{b}\times\vec{d}$ $\qquad \vec{a}-\vec{d}$ & $\vec{b}-\vec{c}$

$\vec{d}\times\vec{b} = \vec{c}\times\vec{a}$ ...(ii)

(i) – (ii)

$\left(\vec{a}-\vec{d}\right)\times\vec{b} = \vec{c}\times\left(\vec{d}-\vec{a}\right)$

$\left(\vec{a}-\vec{d}\right)\times\left(\vec{b}-\vec{c}\right) = 0$

$\vec{a}-\vec{d} \parallel \vec{b}-\vec{c}$

63. If $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ are linearly independent set of vectors and $K_1\vec{a} + K_2\vec{b} + K_3\vec{c} + K_4\vec{d} = 0$, then :

(A) $K_1 + K_2 + K_3 + K_4 = 0$ (B) $K_1 + K_3 = K_2 + K_4 = 0$

(C) $K_1 + K_4 = K_2 + K_3 = 0$ (D) None of these

यदि $\vec{a}$, $\vec{b}$, $\vec{c}$ एवं $\vec{d}$ रेखीय रूप से स्वतंत्र सदिशों का समुच्चय है और $K_1\vec{a} + K_2\vec{b} + K_3\vec{c} + K_4\vec{d} = 0$ हो, तो :

(A) $K_1 + K_2 + K_3 + K_4 = 0$ (B) $K_1 + K_3 = K_2 + K_4 = 0$

(C) $K_1 + K_4 = K_2 + K_3 = 0$ (D) इनमें से कोई नहीं

Ans. ABC

Sol. $K_1 = K_2 = K_3 = K_4 = 0$

64. If $A(\bar{a}); B(\bar{b}); C(\bar{c})$ and $D(\bar{d})$ are four points such that $\bar{a} = -2\hat{i} + 4\hat{j} + 3\hat{k}$; $\bar{b} = 2\hat{i} - 8\hat{j}$; $\bar{c} = \hat{i} - 3\hat{j} + 5\hat{k}$; $\bar{d} = 4\hat{i} + \hat{j} - 7\hat{k}$, d is the shortest distance between the lines AB and CD, then which of the following is True?

(A) $d = 0$, hence AB and CD intersect (B) $d = \dfrac{[\overrightarrow{AB}\ \overrightarrow{CD}\ \overrightarrow{BD}]}{|\overrightarrow{AB} \times \overrightarrow{CD}|}$

(C) AB and CD are skew lines and $d = \dfrac{23}{13}$ (D) $d = \dfrac{[\overrightarrow{AB}\ \overrightarrow{CD}\ \overrightarrow{AC}]}{|\overrightarrow{AB} \times \overrightarrow{CD}|}$

यदि $A(\bar{a}); B(\bar{b}); C(\bar{c})$ तथा $D(\bar{d})$ चार बिन्दु इस प्रकार है कि $\bar{a} = -2\hat{i} + 4\hat{j} + 3\hat{k}$; $\bar{b} = 2\hat{i} - 8\hat{j}$; $\bar{c} = \hat{i} - 3\hat{j} + 5\hat{k}$; $\bar{d} = 4\hat{i} + \hat{j} - 7\hat{k}$, d रेखाओं AB तथा CD के मध्य न्यूनतम दूरी है, तो निम्न में से कौनसा कथन सत्य है?

(A) $d = 0$ तथा AB तथा CD प्रतिच्छेद करती है (B) $d = \dfrac{[\overrightarrow{AB}\ \overrightarrow{CD}\ \overrightarrow{BD}]}{|\overrightarrow{AB} \times \overrightarrow{CD}|}$

(C) AB तथा CD विसमतलीय रेखाऐं है तथा $d = \dfrac{23}{13}$ (D) $d = \dfrac{[\overrightarrow{AB}\ \overrightarrow{CD}\ \overrightarrow{AC}]}{|\overrightarrow{AB} \times \overrightarrow{CD}|}$

Ans. BCD

Sol. $\overrightarrow{AB} = \bar{b} - \bar{a} = 4\hat{i} - 12\hat{j} - 3\hat{k}$

$\overrightarrow{CD} = \bar{d} - \bar{c} = 3\hat{i} + 4\hat{j} - 12\hat{k}$

$\overrightarrow{AC} = \bar{c} - \bar{a} = 3\hat{i} - 7\hat{j} + 2\hat{k}$

$\overrightarrow{BD} = \bar{d} - \bar{b} = 2\hat{i} + 9\hat{j} - 7\hat{k}$

By definition $d = \dfrac{(\overrightarrow{AB} \times \overrightarrow{CD}) \cdot \overrightarrow{AC}}{|\overrightarrow{AB} \times \overrightarrow{CD}|}$ ... (i)

$= \dfrac{(\overrightarrow{AB} \times \overrightarrow{CD}) \cdot \overrightarrow{BD}}{|\overrightarrow{AB} \times \overrightarrow{CD}|}$ ... (ii)

$\overrightarrow{AB} \times \overrightarrow{CD} = 13(12\hat{i} + 3\hat{j} + 4\hat{k})$ $\therefore$ $|\overrightarrow{AB} \times \overrightarrow{CD}| = 169$

$\therefore \quad d = \dfrac{13(12\hat{i} + 3\hat{j} + 4\hat{k})}{169} \cdot (3\hat{i} - 7\hat{j} + 2\hat{k}) = \dfrac{23}{13}$ using (i)

also $d = \dfrac{13(12\hat{i} + 3\hat{j} + 4\hat{k})}{169} \cdot (2\hat{i} - 9\hat{j} + 7\hat{k}) = \dfrac{23}{13}$ using (ii)

65. Given the equations of the line $3x - y + z + 1 = 0, 5x + y + 3z = 0$. Then which of the following is correct ?

(A) symmetical form of the equations of line is $\dfrac{x}{2} = \dfrac{y - \frac{1}{8}}{-1} = \dfrac{z + \frac{5}{8}}{1}$

(B) symmetrical form of the equations of line is $\dfrac{x + \frac{1}{8}}{1} = \dfrac{y - \frac{5}{8}}{1} = \dfrac{z}{-2}$

(C) equation of the plane through (2, 1, 4) and prependicular to the given lines is $2x - y + z - 7 = 0$

(D) equation of the plane through (2, 1, 4) and prependicular to the given lines is $x + y - 2z + 5 = 0$

एक रेखा का समीकरण $3x - y + z + 1 = 0, 5x + y + 3z = 0$ है। तो निम्न में से कौन सही है?

(A) रेखा का सम्मित रूप है $\dfrac{x}{2} = \dfrac{y - \frac{1}{8}}{-1} = \dfrac{z + \frac{5}{8}}{1}$

(B) रेखा का सम्मित रूप है $\dfrac{x + \frac{1}{8}}{1} = \dfrac{y - \frac{5}{8}}{1} = \dfrac{z}{-2}$

(C) (2, 1, 4) से गुजरने वाला तथा दिये गये रेखा की लम्बवत् समतल का समीकरण $2x - y + z - 7 = 0$ है

(D) (2, 1, 4) से गुजरने वाला तथा दिये गये रेखा की लम्बवत् समतल का समीकरण $x + y - 2z + 5 = 0$ है

Ans. BD

Sol. Let a, b, c be the d.r's. of the given line. Then

we have $3a - b + c = 0$

$5a + b + 3c = 0$

Solving we get $\dfrac{a}{1} = \dfrac{b}{1} = \dfrac{c}{-2}$

Again, suppose the given line intersect the plane $z = 0$ at $(x_1, y_1, 0)$ then $3x_1 - y_1 + 1 = 0$ and $5x_1 + y_1 = 0$

Solving we get $x_1 = -\dfrac{1}{8}, y_1 = \dfrac{5}{8}$

Hence the symmetrical form of the line is $\dfrac{x + \frac{1}{8}}{1} = \dfrac{y - \frac{5}{8}}{1} = \dfrac{z}{-2}$

Equation of plane through (2, 1, 4) is

$A(x-2)+B(y-1)+c(z-4)=0$

when $A=l,\ b=l$ and $c=-2$

$\Rightarrow \quad x-2+y-1-2(z-4)=0 \quad \Rightarrow \quad x+y-2z+5=0$

66. Given three vectors

$\vec{U}=2\hat{i}+3\hat{j}-6\hat{k}\,; \qquad \vec{V}=6\hat{i}+2\hat{j}+3\hat{k}\,; \qquad \vec{W}=3\hat{i}-6\hat{j}-2\hat{k}$

Which of the following hold good for the vectors $\vec{U}$, $\vec{V}$ and $\vec{W}$?

(A) $\vec{U}$, $\vec{V}$, $\vec{W}$ are linearly depedent

(B) $(\vec{U}\times\vec{V})\times\vec{W}=\vec{0}$

(C) $\vec{U}$, $\vec{V}$ and $\vec{W}$ form a triplet of mutually perpendicular vectors

(D) $\vec{U}\times(\vec{V}\times\vec{W})=\vec{0}$

दिया गया है कि तीन सदिश

$\vec{U}=2\hat{i}+3\hat{j}-6\hat{k}\,; \qquad \vec{V}=6\hat{i}+2\hat{j}+3\hat{k}\,; \qquad \vec{W}=3\hat{i}-6\hat{j}-2\hat{k}$

निम्न में से कौनसे कथन सदिशों $\vec{U}$, $\vec{V}$ तथा $\vec{W}$ के लिए सही है?

(A) $\vec{U}$, $\vec{V}$, $\vec{W}$ रेखिक स्वतंत्र नहीं है

(B) $(\vec{U}\times\vec{V})\times\vec{W}=\vec{0}$

(C) $\vec{U}$, $\vec{V}$ तथा $\vec{W}$ परस्पर लम्बवत् है

(D) $\vec{U}\times(\vec{V}\times\vec{W})=\vec{0}$

Ans. BCD

Sol. If may be observed that $[\vec{U}\,\vec{V}\,\vec{W}]=\begin{vmatrix}2 & 3 & -6\\ 6 & 2 & 3\\ 3 & -6 & -2\end{vmatrix}=343\neq 0$

$\Rightarrow \quad \vec{U},\vec{V},\vec{W}$ are non coplanar hence linearly independent

Further $\vec{U}\times\vec{V}=\vec{W}$ and $\vec{V}\times\vec{W}=\vec{U}$

$\Rightarrow$ They form a right handed triplet of mutually perpendicular vectors and of course!

$(\vec{U}\times\vec{V})\times\vec{W}=\vec{0}=\vec{U}\times(\vec{V}\times\vec{W})$

67. Consider the following four pairs of lines in column-I and match them with one or more entries in column-II :

Column-I	**Column-II**
(A) $L_1 : x = 1 + t, y = t, z = 2 - 5t$ $L_2 : \vec{r} = (2,1,-3) + \lambda(2, 2, -10)$	(P) non coplanar lines
(B) $L_1 : \frac{x-1}{2} = \frac{y-3}{2} = \frac{z-2}{-1}$ $L_2 : \frac{x-2}{1} = \frac{y-6}{-1} = \frac{z+2}{3}$	(Q) lines lie in a unique plane
(C) $L_1 : x = -6t, y = 1 + 9t, z = -3t$ $L_2 : x = 1 + 2s, y = 4 - 3s, z = s$	(R) infinite planes containing both the lines
(D) $L_1 : \frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ $L_2 : \frac{x-3}{-4} = \frac{y-2}{-3} = \frac{z-1}{2}$	(S) lines do not intersect

माना Column–I में चार रेखा युग्म है एवं Column–II उनसे सम्बन्धित एक या एक से अधिक प्रविष्टि पंजिकायें है :

Column-I	**Column-II**
(A) $L_1 : x = 1 + t, y = t, z = 2 - 5t$ $L_2 : \vec{r} = (2,1,-3) + \lambda(2, 2, -10)$	(P) विसमतलीय रेखायें
(B) $L_1 : \frac{x-1}{2} = \frac{y-3}{2} = \frac{z-2}{-1}$ $L_2 : \frac{x-2}{1} = \frac{y-6}{-1} = \frac{z+2}{3}$	(Q) अद्वितीय तल पर स्थित रेखायें
(C) $L_1 : x = -6t, y = 1 + 9t, z = -3t$ $L_2 : x = 1 + 2s, y = 4 - 3s, z = s$	(R) दोनों रेखाओं से अनन्त तल गुजरते है
(D) $L_1 : \frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ $L_2 : \frac{x-3}{-4} = \frac{y-2}{-3} = \frac{z-1}{2}$	(S) रेखा प्रतिच्छेद नही करती है

Ans. (A) R (B) Q (C) QS (D) PS

Sol. (A) $L_1 : \frac{x-1}{1} = \frac{y-0}{1} = \frac{z-2}{-5}$; $\vec{V}_1 = \hat{i} + \hat{j} - 5\hat{k}$

$L_2 : \frac{x-2}{2} = \frac{y-1}{2} = \frac{z+3}{-10}$; $\vec{V}_2 = 2(\hat{i} + \hat{j} - 5\hat{k})$

Hence lines are parallel and both contains the points (1, 0, 2) and (2, 1, – 3) $\Rightarrow$ coincident line

both L_1 and L_2 may lie in an infinite number of planes hence $\Rightarrow$ (R)

(B) $\left.\begin{array}{l}\vec{V}_1 = 2\hat{i}+2\hat{j}-\hat{k} \\ \vec{V}_2 = \hat{i}-2\hat{j}+3\hat{k}\end{array}\right] \Rightarrow$ lines not parallel

Also both intersect at (3, 5, 1)

Hence lines are intersecting hence they lie on a unique plane $\Rightarrow$ (P)

(C) $L_1 : \frac{x-0}{-6} = \frac{y-1}{9} = \frac{z-0}{-3} = t$

$L_2 : \frac{x-1}{2} = \frac{y-4}{-3} = \frac{z-0}{1} = s$

$\left.\begin{array}{l}L_1 \text{ is parallel to } -6\hat{i}+9\hat{j}-3\hat{k} \\ L_2 \text{ is parallel to } 2\hat{i}-3\hat{j}+\hat{k}\end{array}\right| \Rightarrow$ lines parallel but not conicident

as (0, 1, 0) does not line on L_2, not intersecting

Hence L_1, L_2 lies in a unique planes $\Rightarrow$ (Q), (S)

(D) Lines are skew can be verified $\Rightarrow$ (P), (S)

68. P(0, 3, – 2); Q(3, 7, – 1) and R(1, – 3, – 1) are 3 given points. Let L_1 be the line passing through P and Q and L_2 be the line through R and parallel to the vector $\vec{V} = \hat{i}+\hat{k}$:

Column-I	**Column-II**
(A) perpendicular distance of P from L_2	(P) $7\sqrt{3}$
(B) shortest distance between L_1 and L_2	(Q) 2
(C) area of the triangle PQR	(R) 6
(D) distance from (0, 0, 0) to the plane PQR	(S) $\frac{19}{\sqrt{147}}$

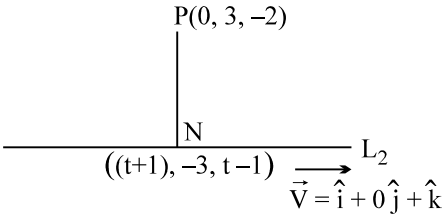
P(0, 3, – 2); Q(3, 7, – 1) तथा R(1, – 3, – 1) तीन दिये हुये बिन्दु है। यदि L_1 रेखा बिन्दु P तथा Q को मिलाती है तथा L_2 रेखा R से गुजरती है एवं सदिश $\vec{V} = \hat{i}+\hat{k}$ के समानान्तर है :

Column-I	**Column-II**
(A) P की L_2 से लम्बवत् दूरी है	(P) $7\sqrt{3}$
(B) L_1 तथा L_2 के मध्य न्यूनतम दूरी है	(Q) 2
(C) ΔPQR का क्षेत्रफल	(R) 6
(D) (0, 0, 0) की PQR समतल से दूरी है	(S) $\frac{19}{\sqrt{147}}$

Ans. (A) R (B) Q (C) P (D) S

Sol. $L_1 : \dfrac{x}{3} = \dfrac{y-3}{4} = \dfrac{z+2}{1}$ ....(1)

(passing through P and Q)



$L_2 : \dfrac{x-1}{1} = \dfrac{y-3}{0} = \dfrac{z+1}{1}$ ....(2)

(passing through R and parallel to $\vec{v} = \hat{i} + \hat{k}$)

(A) <u>distance of P(0, 3, – 2) from L_2</u>

$\overrightarrow{PN} = (t+1)\hat{i} - 6\hat{j} + 2(t-1)\hat{k}$

now $\overrightarrow{PN} \cdot \vec{V} = 0 \Rightarrow [(t+1)\hat{i} - 6\hat{j} + (t+1)\hat{k}] \cdot (\hat{i} + \hat{k}) = 0$

$(t + 1) + (t + 1) = 0 \Rightarrow t = -1$

hence $\overrightarrow{PN} = -6\hat{j}$

$\left|\overrightarrow{PN}\right| = \left|-6\hat{j}\right| = 6$ Ans. $\Rightarrow$ (R)

<u>Distance between L_1 and L_2</u>

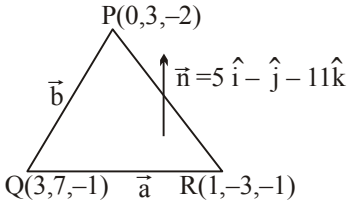
Equation of plane containing L_1 and parallel to L_2

$Ax + B(y - 3) + C(z + 2) = 0$

where $3A + 4B + C = 0$

and $A + 0B + C = 0 \Rightarrow A + C = 0$

$C = \lambda,\ A = -\lambda,\ B = +\lambda/2$



$\therefore$ equation of plane

$$-\lambda x + \frac{\lambda}{2}(y-3) + \lambda(z+2) = 0$$

$2x - y + 3 - 2z - 4 = 0$

$2x - y - 2z = 1$ ....(1)

now distance of the point $(1, -3, -1)$ lying on the line L_2 from the plane (1)

$$d = \left|\frac{2+3+2-1}{3}\right| = 2 \text{ Ans.} \Rightarrow \text{(Q)}$$

Area of ΔPQR

$\overrightarrow{QR} = \vec{a} = 2\hat{i} + 10\hat{j} + 0\hat{k}$

$\overrightarrow{QP} = \vec{b} = 3\hat{i} + 4\hat{j} + \hat{k}$

$$\vec{a} \times \vec{b} = 2\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 5 & 0 \\ 3 & 4 & 1 \end{vmatrix} = 2[\hat{i}(5) - \hat{j}(1) + \hat{k}(4-15)] = 2[5\hat{i} - \hat{j} - 11\hat{k}]$$

$$\frac{|\vec{a} \times \vec{b}|}{2} = \sqrt{25+1+121} = \sqrt{147} = \sqrt{3 \cdot 49} = 7\sqrt{3} \text{ Ans.} \Rightarrow \text{(P)}$$

69. Match the following Column :

Column-I	**Column-II**
(A) Let O be an interior point of ΔABC such that $\overrightarrow{OA}+2\overrightarrow{OB}+3\overrightarrow{OC}=\vec{0}$, then the ratio of the area of ΔABC to the area of ΔAOC, is with O is the origin	(P) 0
(B) Let ABC be a triangle whose centroid is G, orthocentre is H and circumcentre is the origin 'O'. If D is any point in the plane of the triangle such that no three of O, A, B, C and D are collinear satisfying the relation $\overrightarrow{AD}+\overrightarrow{BD}+\overrightarrow{CH}+3\overrightarrow{HG}=\lambda\overrightarrow{HD}$ then the value of the scalar 'λ' is	(Q) 1 (R) 2
(C) If $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ are non zero vectors such that no three of them are in the same plane and no two are orthogonal then the value of the scalar $\frac{(\vec{b}\times\vec{c})\cdot(\vec{a}\times\vec{d})+(\vec{c}\times\vec{a})\cdot(\vec{b}\times\vec{d})}{(\vec{a}\times\vec{b})\cdot(\vec{d}\times\vec{c})}$ is	(S) 3

निम्न Column को सुमेलित कीजिए :

Column-I	**Column-II**
(A) माना O त्रिभुज ΔABC के अन्तः में स्थित एक बिन्दु इसप्रकार है कि $\overrightarrow{OA}+2\overrightarrow{OB}+3\overrightarrow{OC}=\vec{0}$ तो Δ ABC का क्षेत्रफल का Δ AOC के क्षेत्रफल के साथ अनुपात है, जहाँ O मूल बिन्दु है	(P) 0
(B) माना ABC एक त्रिभुज जिसका केन्द्रक G, लम्बकेन्द्र H तथा परिकेन्द्र मूल बिन्दु 'O' है। यदि D एक ऐसा बिन्दु त्रिभुज के तल में इसप्रकार है कि जिसके लिए O, A, B, C तथा D में से कोई भी तीन समरेखीय नहीं है। एवं $\overrightarrow{AD}+\overrightarrow{BD}+\overrightarrow{CH}+3\overrightarrow{HG}=\lambda\overrightarrow{HD}$ को सन्तुष्ट करते है, तो अदिश 'λ'=	(Q) 1 (R) 2
(C) यदि $\vec{a}$, $\vec{b}$, $\vec{c}$ तथा $\vec{d}$ चार अशून्य सदिश है। इनमें से कोई भी तीन समतलीय नहीं है। तथा कोई भी दो लम्बवत् नहीं है, तो अदिश $\frac{(\vec{b}\times\vec{c})\cdot(\vec{a}\times\vec{d})+(\vec{c}\times\vec{a})\cdot(\vec{b}\times\vec{d})}{(\vec{a}\times\vec{b})\cdot(\vec{d}\times\vec{c})}$	(S) 3

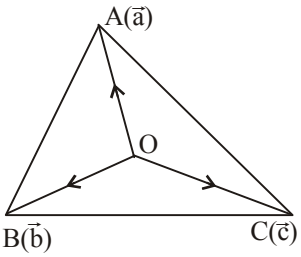
Ans. (A) S (B) R (C) Q

Sol. (A) $\frac{\text{Area of }\Delta ABC}{\text{Area of }\Delta AOC}=\frac{\frac{1}{2}\left|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\right|}{\frac{1}{2}\left|\vec{a}\times\vec{c}\right|}$

now $\vec{a}+2\vec{b}+3\vec{c}=0$

cross with $\vec{b}$, $\vec{a}\times\vec{b}+3\vec{c}\times\vec{b}=0 \quad\Rightarrow\quad \vec{a}\times\vec{b}=3(\vec{b}\times\vec{c})$

cross with $\vec{a}$, $2\vec{a}\times\vec{b}+3\vec{a}\times\vec{c}=0 \quad\Rightarrow\quad \vec{a}\times\vec{b}=\frac{3}{2}(\vec{c}\times\vec{a})$



$\therefore \quad \vec{a}\times\vec{b}=\frac{3}{2}(\vec{c}\times\vec{a})=3(\vec{b}\times\vec{c})$

Let $(\vec{c}\times\vec{a})=\vec{p}$

$\vec{a}\times\vec{b}=\frac{3\vec{p}}{2}; \quad \vec{b}\times\vec{c}=\frac{\vec{p}}{2}$

$\therefore \quad \text{ratio}=\frac{\left|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\right|}{\left|\vec{c}\times\vec{a}\right|}=\frac{\left|\frac{3\vec{p}}{2}+\frac{\vec{p}}{2}+\vec{p}\right|}{|\vec{p}|}=\frac{3|\vec{p}|}{|\vec{p}|}=3$ Ans. $\Rightarrow$ (S)

(B) $\text{LHS}=\vec{d}-\vec{a}+\vec{d}-\vec{b}+\vec{h}-\vec{c}+3(\vec{g}-\vec{h})$

$=2\vec{d}-(\vec{a}+\vec{b}+\vec{c})+3\frac{(\vec{a}+\vec{b}+\vec{c})}{3}-2\vec{h}$

$=2\vec{d}-2\vec{h}=2(\vec{d}-\vec{h})=2\overrightarrow{HD} \quad\Rightarrow\quad \lambda=2$ Ans. $\Rightarrow$ (R)

(C) $(\vec{b}\times\vec{c})\cdot(\vec{a}\times\vec{d})=\begin{vmatrix}\vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{d}\\ \vec{c}\cdot\vec{a} & \vec{c}\cdot\vec{d}\end{vmatrix}$

|||ly compute others which gives (1) $\Rightarrow$ (Q)

70. Consider the following linear equations $ax + by + cz = 0$, $bx + cy + az = 0$, and $cx + ay + bz = 0$
Match the conditions/ expressions in Column I with statements in Column II.

Column I	**Column II**
(A) $a + b + c \neq 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$	(P) the equation represent planes meeting only at a single point
(B) $a + b + c = 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$	(Q) the equation represent the line $x = y = z$
(C) $a + b + c \neq 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$	(R) the equation represent identical planes
(D) $a + b + c = 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$	(S) the equation represent the whole of the three dimensional space

माना रैखिक समीकरणें $ax + by + cz = 0$, $bx + cy + az = 0$, तथा $cx + ay + bz = 0$ है।
Column–I में दिये गये प्रतिबन्धों/व्यंजको का Column–II में दिये गये सुमेल कीजिये :

Column I	**Column II**
(A) $a + b + c \neq 0$ तथा $a^2 + b^2 + c^2 = ab + bc + ca$	(P) समीकरणें समतलों को प्रदर्शित करती है जो केवल एक बिन्दु पर मिलते है।
(B) $a + b + c = 0$ तथा $a^2 + b^2 + c^2 \neq ab + bc + ca$	(Q) समीकरणें रेखा $x = y = z$ को प्रदर्शित करती है
(C) $a + b + c \neq 0$ क $a^2 + b^2 + c^2 \neq ab + bc + ca$	(R) समीकरणें समान तलों को प्रदर्शित करती है

(D) $a + b + c = 0$ तथा
$a^2 + b^2 + c^2 = ab + bc + ca$

(S) समीकरणें सम्पूर्ण त्रिविमीय समष्टि को प्रदर्शित करती है

Ans. (A) R (B) Q (C) P (D) S

Sol. Let $\Delta = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$

$= \frac{-1}{2}(a + b + c)[(a - b)^2 + (b - c)^2 + (c - a)^2]$

(A) If $a + b + c \neq 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$

$\Rightarrow (a - b)^2 + (b - c)^2 + (c - a)^2 = 0$

$\Rightarrow \Delta = 0$ and $a = b = c \neq 0$

$\Rightarrow$ the equation represent identical planes.

(B) $a + b + c = 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$

$\Rightarrow \Delta = 0$

Since all the three planes pass through (1,1,1)

So equation of the line of intersection of these plane will be $\frac{x - 0}{1 - 0} = \frac{y - 0}{1 - 0} = \frac{z - 0}{1 - 0}$

(C) $a + b + c \neq 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$

$\Rightarrow \Delta \neq 0$

$\Rightarrow$ the equations represent planes meeting at only one point i.e. (0,0,0)

(D) $a + b + c = 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$

$\Rightarrow a = b = c = 0$

$\Rightarrow$ the equations represent whole of the three dimensional space.

Exercise J-M

1. Let $\vec{a}=2\hat{i}+\hat{j}-2\hat{k}$ and $\vec{b}=\hat{i}+\hat{j}$. If $\vec{c}$ is a vector such that $\vec{a}.\vec{c}=|\vec{c}|,|\vec{c}-\vec{a}|=2\sqrt{2}$ and the angle between $(\vec{a}\times\vec{b})$ and $\vec{c}$ is $\frac{\pi}{6}$, then the value of $|(\vec{a}\times\vec{b})\times\vec{c})|$ is : **[JEE(Main)July2021]**

माना $\vec{a}=2\hat{i}+\hat{j}-2\hat{k}$ तथा $\vec{b}=\hat{i}+\hat{j}$ है। यदि एक सदिश $\vec{c}$ के लिए $\vec{a}.\vec{c}=|\vec{c}|,|\vec{c}-\vec{a}|=2\sqrt{2}$ हैं तथा $(\vec{a}\times\vec{b})$ और $\vec{c}$ के बीच का कोण $\frac{\pi}{6}$ है, तो $|(\vec{a}\times\vec{b})\times\vec{c})|$ का मान है : **[JEE(Main)July2021]**

(A) $\frac{3}{2}$ (B) 4 (C) 3 (D) $\frac{2}{3}$

Ans. A

Sol. $\vec{a}=2\hat{i}+\hat{j}-2\hat{k}$ $\quad\vec{b}=\hat{i}+\hat{j}$

$$\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 1 & -2\\ 1 & 1 & 0\end{vmatrix}=2\hat{i}-2\hat{j}+\hat{k}$$

$|\vec{a}\times\vec{b}|=3$

$|\vec{c}-\vec{a}|=2\sqrt{2}$

$\vec{c}^{2}+a^{2}-2\vec{a}.\vec{c}=8$

$c^2 + a^2 - 2c = 8$

$[\because a^2 = 9]$

$c^2 - 2c + 1 = 0$

$(c - 1)^2 = 0$

$(c - 1)^2 = 0$

$c = 1$

$\Rightarrow|\vec{c}|=1$

$|(\vec{a}\times\vec{b})\times\vec{c}|=|\vec{a}\times\vec{b}||\vec{c}|\sin(\pi/6)$

$$=(3)(1)\left(\frac{1}{2}\right)$$

$$=\frac{3}{2}$$

2. In a triangle ABC, if $\left|\overrightarrow{BC}\right|=3,\left|\overrightarrow{CA}\right|=5$ and $\left|\overrightarrow{BA}\right|=7,$ then the projection of the vector $\overrightarrow{BA}$ on $\overrightarrow{BC}$ is equal to : **[JEE(Main)July2021]**

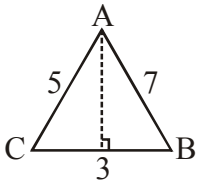
त्रिभुज ABC में यदि $\left|\overrightarrow{BC}\right|=3,\left|\overrightarrow{CA}\right|=5$ तथा $\left|\overrightarrow{BA}\right|=7,$ तो सदिश $\overrightarrow{BA}$ का $\overrightarrow{BC}$ के अनुदिश प्रक्षेप होगा :

[JEE(Main)July2021]

(A) $\frac{11}{2}$ (B) $\frac{19}{2}$ (C) $\frac{13}{2}$ (D) $\frac{15}{2}$

Ans. A

Sol. $\left|\overrightarrow{BC}\right|=3,\left|\overrightarrow{CA}\right|=5,\left|\overrightarrow{BA}\right|=7$



Projection of $\overrightarrow{BA}$ on $\overrightarrow{BC}=\left|\overrightarrow{BA}\right|$ Cos$\angle$ABC

$=7\left(\frac{3^2+7^2-5^2}{2\times3\times7}\right)=\frac{11}{2}$

3. Let a vector $\vec{a}$ be coplanar with vectors $\vec{b}=2\hat{i}+\hat{j}+\hat{k}$ and $\vec{c}=\hat{i}-\hat{j}+\hat{k}$. If $\vec{a}$ is perpendicular to $\vec{d}=3\hat{i}+2\hat{j}+6\hat{k}$, and $|\vec{a}|=\sqrt{10}$. Then possible value of $\left[\vec{a}\ \vec{b}\ \vec{c}\right]+\left[\vec{a}\ \vec{b}\ \vec{d}\right]+\left[\vec{a}\ \vec{c}\ \vec{d}\right]$ is equal to :

[JEE(Main)July2021]

माना एक सदिश $\vec{a}$, सदिशों $\vec{b}=2\hat{i}+\hat{j}+\hat{k}$ तथा $\vec{c}=\hat{i}-\hat{j}+\hat{k}$ के सहतलीय है। यदि $\vec{a}$, सदिश $\vec{d}=3\hat{i}+2\hat{j}+6\hat{k}$ पर लम्बवत् है और $|\vec{a}|=\sqrt{10}$ है, तो $\left[\vec{a}\ \vec{b}\ \vec{c}\right]+\left[\vec{a}\ \vec{b}\ \vec{d}\right]+\left[\vec{a}\ \vec{c}\ \vec{d}\right]$ का एक संभावित मान है : **[JEE(Main)July2021]**

(A) – 40 (B) – 38 (C) – 42 (D) –29

Ans. C

Sol. $\vec{a}=\lambda\vec{b}+\mu\vec{c}$

$\vec{a}=\lambda\left(2i+j+k\right)+\mu\left(i-j+k\right)$

$\vec{a}=i\left(2\lambda+\mu\right)+j\left(\lambda-\mu\right)+k\left(\lambda+\mu\right)$

Now $\vec{a}.\vec{d}=0$

$3\left(2\lambda+\mu\right)+2\left(\lambda-\mu\right)+6\left(\lambda+\mu\right)=0$

$14\lambda+7\mu=0$

$2\lambda+\mu=0$ ....(1)

$|\vec{a}|=\sqrt{10}$

$\sqrt{\left(2\lambda+\mu\right)^2+\left(\lambda-\mu\right)^2+\left(\lambda+\mu\right)^2}=\sqrt{10}$

$4\lambda^2+4\lambda\mu+\mu^2+2\lambda^2+2\mu^2=10$

$6\lambda^2+3\mu^2+4\lambda\mu=10$ ....(2)

by (1) $\mu=-2\lambda$

$6\lambda^2+12\lambda^2-8\lambda^2=10$

$10\lambda^2=10$

$\lambda=\pm1$

if $\lambda=1$ then $\mu=-2$

if $\lambda=-1$ then $\mu=2$

Now $\vec{a}=3j-k$

or

$\vec{a}=-3j+k$

if we choose $\vec{a}=3j-k$

$\left[\vec{a}\,\vec{b}\,\vec{c}\right]+\left[\vec{a}\,\vec{b}\,\vec{d}\right]+\left[\vec{a}\,\vec{c}\,\vec{d}\right]=0-28-14=-42$

4. Let three vector $\vec{a}$, $\vec{b}$ and $\vec{c}$ be such that $\vec{a}\times\vec{b}=\vec{c}$, $\vec{b}\times\vec{c}=\vec{a}$ and $|\vec{a}|=2$. Then which one of the following is not true ? **[JEE(Main)July2021]**

(A) Projection of $\vec{a}$ on $\left(\vec{b}\times\vec{c}\right)$ is 2 (B) $\left|3\vec{a}+\vec{b}-2\vec{c}\right|^2=51$

(C) $\vec{a}\times\left(\left(\vec{b}+\vec{c}\right)\times\left(\vec{b}-\vec{c}\right)\right)=\vec{0}$ (D) $\left[\vec{a}\ \vec{b}\ \vec{c}\right]+\left[\vec{c}\ \ \vec{a}\ \ \vec{b}\right]=8$

माना तीन सदिश $\vec{a}$, $\vec{b}$ तथा $\vec{c}$, $\vec{a}\times\vec{b}=\vec{c}$, $\vec{b}\times\vec{c}=\vec{a}$ तथा $|\vec{a}|=2$ को संतुष्ट करते हैं। तो निम्न में से कौनसा कथन असत्य है? **[JEE(Main)July2021]**

(A) $\left(\vec{b}\times\vec{c}\right)$ पर $\vec{a}$ का प्रक्षेप 2 है (B) $\left|3\vec{a}+\vec{b}-2\vec{c}\right|^2=51$

(C) $\vec{a}\times\left(\left(\vec{b}+\vec{c}\right)\times\left(\vec{b}-\vec{c}\right)\right)=\vec{0}$ (D) $\left[\vec{a}\ \vec{b}\ \vec{c}\right]+\left[\vec{c}\ \ \vec{a}\ \ \vec{b}\right]=8$

Ans. B

Sol. $\vec{a}\times\vec{b}=\vec{c}$ .....(1)

$\vec{b}\times\vec{c}=\vec{a}$ .....(2)

and $|\vec{a}|=2$.

By (1) and (2)

$\vec{a}.\vec{c}=0$

$\vec{b}.\vec{c}=0$

$\vec{a}.\vec{b}=0$

Hence $\vec{a}$, $\vec{b}$ and $\vec{c}$ are mutually perpendicular

$\vec{a}\times\vec{b}=\vec{c}$

$|\vec{a}\times\vec{b}|=|\vec{c}|$

$a^2b^2\sin^2 90°=c^2$

$4b^2=c^2$ ....(3)

$\vec{b}\times\vec{c}=\vec{a}$

$|\vec{b}\times\vec{c}=\vec{a}|$

$b^2c^2\sin^2 90°=a^2$

$b^2c^2=4$ ....(4)

by (3) and (4)

$|\vec{b}|=1$

$|\vec{c}|=2$

Now

(A) option (A)

$$\left|\frac{\vec{a}.(\vec{b}\times\vec{c})}{\vec{b}\times\vec{c}}\right|=2$$

$$\frac{|\vec{a}||\vec{b}||\vec{c}|\sin\theta\cos\theta}{|\vec{b}||\vec{c}|\sin\theta}=2$$

2 = 2 which is correct.

(B) $\left|3\vec{a}+\vec{b}-2\vec{c}\right|^2=9a^2+b^2+4c^2+0+0+0$

$=36+1+16$

Which is wrong

hence option (2) is correct

5. Let the vectors

$(2+a+b)\hat{i}+(a+2b+c)\hat{j}-(b+c)\hat{k}$,

$(1+b)\hat{i}+2b\hat{j}-b\hat{k}$ and $(2+b)\hat{i}+2b\hat{j}+(1-b)\hat{k}$, $a, b, c \in R$

be co-planar. Then which of the following is true? **[JEE(Main)July2021]**

माना

$(2+a+b)\hat{i}+(a+2b+c)\hat{j}-(b+c)\hat{k}$,

$(1+b)\hat{i}+2b\hat{j}-b\hat{k}$ तथा $(2+b)\hat{i}+2b\hat{j}+(1-b)\hat{k}$, $a, b, c \in R$

सहतलीय सदिश हैं। तो निम्न में से कौन–सा कथन सही है ? **[JEE(Main)July2021]**

(A) $2b = a + c$ (B) $a = b + 2c$ (C) $3c = a + b$ (D) $2a = b + c$

Ans. A

Sol. vectors are co-planer then S.T.P = 0

$$\begin{vmatrix} 2+a+b & a+2b+c & -b-c \\ 1+b & 2b & -b \\ 2+b & 2b & 1-b \end{vmatrix} = 0$$

$R_3 \to R_3 - R_2$

$$\begin{vmatrix} 2+a+b & a+2b+c & -b-c \\ 1+b & 2b & -b \\ 1 & 0 & 1 \end{vmatrix} = 0$$

$C_1 \to C_1 - C_3$

$$\begin{vmatrix} 2+a+2b+c & a+2b+c & -b-c \\ 1+2b & 2b & -b \\ 0 & 0 & 1 \end{vmatrix} = 0$$

$C_1 \to C_1 - C_2$

$$\begin{vmatrix} 2 & a+2b+c & -b-c \\ 1 & 2b & -b \\ 0 & 0 & 1 \end{vmatrix} = 0$$

$4b - a - 2b - c = 0$

$2b = a + c$

6. Let a, b and c be distinct positive numbers. If the vectors $a\hat{i}+a\hat{j}+c\hat{k}, \hat{i}+\hat{k}$ and $c\hat{i}+c\hat{j}+b\hat{k}$ are co-planar, then c is equal to : **[JEE(Main)July2021]**

माना a, b तथा c तीन भिन्न धनात्मक संख्याएँ हैं। यदि सदिश $a\hat{i}+a\hat{j}+c\hat{k}, \hat{i}+\hat{k}$ तथा $c\hat{i}+c\hat{j}+b\hat{k}$ समतलीय हैं, तो c बराबर हैं: **[JEE(Main)July2021]**

(A) $\dfrac{2}{\frac{1}{a}+\frac{1}{b}}$ (B) $\sqrt{ab}$ (C) $\dfrac{a+b}{2}$ (D) $\dfrac{1}{a}+\dfrac{1}{b}$

Ans. B

Sol. $\begin{vmatrix} a & a & c \\ 1 & 0 & 1 \\ c & c & b \end{vmatrix} = 0$

$a(-c) - a(b-c) + c(c) = 0$

$c^2 = ab$

$c = \sqrt{ab}$

7. If $|\vec{a}| = 2, |\vec{b}| = 5$ and $|\vec{a} \times \vec{b}| = 8$, then $|\vec{a}.\vec{b}|$ is equal to : **[JEE(Main)July2021]**

यदि $|\vec{a}| = 2, |\vec{b}| = 5$ तथा $|\vec{a} \times \vec{b}| = 8$ हैं, तो $|\vec{a}.\vec{b}|$ बराबर है : **[JEE(Main)July2021]**

(A) 6 (B) 4 (C) 3 (D) 5

Ans. A

Sol. $\left|\vec{a} \times \vec{b}\right| = |\vec{a}||\vec{b}| \sin\theta$

$8 = 2 \times 5 \times \sin\theta$

$\sin\theta = 4/5 = \dfrac{P}{H} \quad B = 3$

$\cos\theta = 3/5$

$\left|\vec{a} \cdot \vec{b}\right| = |\vec{a}||\vec{b}| \cos\theta$

$= 2 \times 5 \times \dfrac{3}{5} = 6$

8. Let $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = -\hat{i} + 2\hat{j} + 3\hat{k}$. Then the vector product $\left(\vec{a} + \vec{b}\right) \times \left(\left(a \times \left(\left(\vec{a} - \vec{b}\right) \times \vec{b}\right)\right) \times \vec{b}\right)$ is equal to.

माना $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$ तथा $\vec{b} = -\hat{i} + 2\hat{j} + 3\hat{k}$ हैं। तो सदिश गुणनफल $\left(\vec{a} + \vec{b}\right) \times \left(\left(a \times \left(\left(\vec{a} - \vec{b}\right) \times \vec{b}\right)\right) \times \vec{b}\right)$ बराबर है :

(A) $7\left(30\hat{i} - 5\hat{j} + 7\hat{k}\right)$ (B) $7\left(34\hat{i} - 5\hat{j} + 3\hat{k}\right)$ (C) $5\left(30\hat{i} - 5\hat{j} + 7\hat{k}\right)$ (D) $5\left(34\hat{i} - 5\hat{j} + 3\hat{k}\right)$

Ans. B

Sol. $\left(\vec{a} + \vec{b}\right) \times \left(\left(\vec{a} \times \left(\left(\vec{a} - \vec{b}\right) \times \vec{b}\right)\right) \times \vec{b}\right)$

$\Rightarrow \left(\vec{a} + \vec{b}\right) \times \left(\left(\vec{a} \times \left(\vec{a} \times b\right)\right) \times \vec{b}\right)$

$\Rightarrow \left(\vec{a} + \vec{b}\right) \times \left(\left(\left(\vec{a}.\vec{b}\right)\vec{a} - \left(\vec{a}.\vec{a}\right)\vec{b}\right) \times \vec{b}\right)$

$\Rightarrow \left(\vec{a} + \vec{b}\right) \times \left(\left(7\vec{a} - 6\vec{b}\right) \times \vec{b}\right)$

$\Rightarrow \left(\vec{a} + \vec{b}\right) \times \left(7\left(\vec{a} \times \vec{b}\right)\right)$

$\Rightarrow 7\left\{\vec{a} \times \left(\vec{a} \times \vec{b}\right) + \vec{b} \times \left(\vec{a} \times \vec{b}\right)\right\}$

$\Rightarrow 7\left\{\left(\vec{a}.\vec{b}\right)\vec{a} - \left(\vec{a}.\vec{a}\right)\vec{b} + \left(\vec{b}.\vec{b}\right)\vec{a} - \left(\vec{b}.\vec{a}\right)\vec{b}\right\}$

$\Rightarrow 7\left\{7\vec{a}-6\vec{b}+14\vec{a}-7\vec{b}\right\}$

$\Rightarrow 7\left\{21\vec{a}-13\vec{b}\right\}$

$\Rightarrow 7\left\{21(i+j+2k)-13(-i+2j+3k)\right\}$

$\Rightarrow 7\ (34i-5j+3k)$

9. Let $\vec{a},\vec{b}$ and $\vec{c}$ be three vectors such that $\vec{a}=\vec{b}\times\left(\vec{b}\times\vec{c}\right)$. If magnitudes of the vectors $\vec{a},\vec{b}$ and $\vec{c}$ are $\sqrt{2},1$ and 2 respectively and the angle between $\vec{b}$ and $\vec{c}$ is $\theta\left(0<\theta<\frac{\pi}{2}\right)$, then the value of 1 + tan θ is equal to : **[JEE(Main)July2021]**

माना तीन सदिशों $\vec{a},\vec{b}$ तथा $\vec{c}$ के लिए $\vec{a}=\vec{b}\times\left(\vec{b}\times\vec{c}\right)$ है। यदि सदिशों $\vec{a},\vec{b}$ तथा $\vec{c}$ के परिमाण क्रमशः $\sqrt{2},1$ तथा 2 हैं और $\vec{b}$ तथा $\vec{c}$ के बीच कोण $\theta\left(0<\theta<\frac{\pi}{2}\right)$ है, तो 1 + tan θ का मान बराबर है : **[JEE(Main)July2021]**

(A) $\frac{\sqrt{3}+1}{\sqrt{3}}$ (B) $\sqrt{3}+1$ (C) 1 (D) 2

Ans. D

Sol. $\vec{a}=\vec{b}\times(\vec{b}\times\vec{c})$ ..... (i)

$\vec{a}=(\vec{b}.\vec{c})\vec{b}-(\vec{b}.\vec{b})\vec{c}$

Take dot producet with $\vec{b}$ in (i)

$\because \vec{a}.\vec{b}=0$ ..... (ii)

$\vec{a}=(2\cos\theta\vec{b}-\vec{c}$ ..... (iii)

Dot product with $\vec{a}$ in

$\vec{a}.\vec{b}=(2\cos\theta)\vec{b}.\vec{a}-\vec{c}.\vec{a}$

$\Rightarrow 2=0-\vec{c}.\vec{a}$

$\therefore \vec{c}\ .\vec{a}=-2$ .... (iv)

Take dot product with $\vec{c}$ in (iii)

$\vec{a}.\vec{c}=(2\cos\theta)\vec{b}.\vec{c}-\vec{c}.\vec{c}$

$2=4\cos^2\theta$

$\cos\theta=\pm\frac{1}{\sqrt{2}}$

$\cos^2\pm=\frac{1}{\sqrt{2}}$

$\theta \in \left(0, \frac{\pi}{2}\right)$

$\therefore \quad \cos\theta = \frac{1}{\sqrt{2}}$

$\theta = 45^\circ$

$1 + \tan\theta = 1 + \tan 45^\circ = 2$

10. If the shortest distance between the lines $\vec{r_1} = \alpha\hat{i} + 2\hat{j} + 2\hat{k} + \lambda(\hat{i} - 2\hat{j} + 2\hat{k}), \lambda \in R$, $\alpha > 0$ and $\vec{r_2} = -4\hat{i} - \hat{k} + \mu(3\hat{i} - 2\hat{j} - 2\hat{k}), \mu \in R$ is 9, then α is equal to _____. **[JEE(Main)July2021]**

यदि रेखाओं $\vec{r_1} = \alpha\hat{i} + 2\hat{j} + 2\hat{k} + \lambda(\hat{i} - 2\hat{j} + 2\hat{k}), \lambda \in R$, $\alpha > 0$ तथा $\vec{r_2} = -4\hat{i} - \hat{k} + \mu(3\hat{i} - 2\hat{j} - 2\hat{k}), \mu \in R$ के मध्य न्यूनतम दूरी 9 है, तो α बराबर है _____ **[JEE(Main)July2021]**

Ans. 6

Sol. $L_1 \ \vec{r} = \vec{a} + \lambda\vec{p} \qquad \vec{a} = (\alpha, 2, 2)$

$L_2 \ \vec{r} = \vec{b} + \mu\vec{q} \qquad \vec{b} = (-4, 0, -1)$

$\vec{p} \times \vec{q} = 8\hat{i} + 8\hat{j} + 4\hat{k}$

Shortest distance $= \left|\frac{(\vec{a} - \vec{b}).(\vec{p} \times \vec{q})}{|\vec{p} \times \vec{q}|}\right| = 9$

$$\left|\frac{((\alpha + 4)\hat{i} + 2\hat{j} + 3\hat{k}).(2\hat{i} + 2\hat{j} + \hat{k})}{3}\right| = 9$$

$|2\alpha + 15| = 27$

$2\alpha + 15 = 27$ or $2\alpha + 15 = -27$

$\alpha = 6$ or $\alpha = -21$ (Rejected)

11. Let $\vec{a}, \vec{b}, \vec{c}$ be three mutually perpendicular vectors of the same magnitude and equally inclined at an angle θ, with the vector $\vec{a} + \vec{b} + \vec{c}$. Then $36\cos^2 2\theta$ is equal to _____. **[JEE(Main)July2021]**

माना बराबर परिमाण के तीन सदिश $\vec{a}, \vec{b}, \vec{c}$ परस्पर लंबवत है तथा सदिश $\vec{a} + \vec{b} + \vec{c}$ के साथ एक समान कोण θ बनाते हैं। तो $36\cos^2 2\theta$ बराबर है _____ **[JEE(Main)July2021]**

Ans. 4

Sol. $\cos\theta = \frac{(\vec{a} + \vec{b} + \vec{c}).\vec{a}}{|\vec{a} + \vec{b} + \vec{c}||\vec{a}|}$

$= \frac{r^2}{(r\sqrt{3})(r)} = \frac{1}{\sqrt{3}}$

$$\cos(2\theta) = 2\cos^2\theta - 1 = \frac{-1}{3}$$

$$36\cos^2(2\theta) = 36\left(\frac{1}{9}\right) = 4$$

12. For p > 0, a vector $\overrightarrow{v_2} = 2\hat{i} + (p+1)\hat{j}$ is obtained by rotating the vector $\overrightarrow{v_1} = \sqrt{3}\,p\hat{i} + \hat{j}$ by an angle θ about origin in counter clockwise direction. If $\tan\theta = \frac{(\alpha\sqrt{3}-2)}{4\sqrt{3}+3}$, then the value of α is equal to _____.

[JEE(Main)July2021]

p > 0 के लिए, सदिश $\overrightarrow{v_1} = \sqrt{3}\,p\hat{i} + \hat{j}$ को केन्द्रबिन्दु के गिर्द वामावृत्त दिशा में कोण θ से आवर्तित करने पर सदिश $\overrightarrow{v_2} = 2\hat{i} + (p+1)\hat{j}$ प्राप्त होता है। यदि $\tan\theta = \frac{(\alpha\sqrt{3}-2)}{4\sqrt{3}+3}$ है, तो α का मान बराबर है _____

[JEE(Main)July2021]

Ans. 6

Sol. $\left|\vec{V}_1\right| = \left|\vec{V}_2\right|$

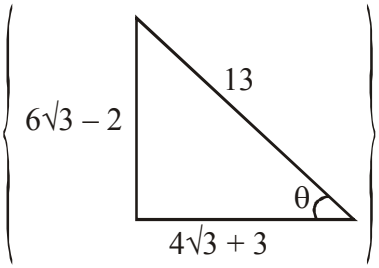
$3P^2 + 1 = 4 + (P+1)^2$

$P = 2$ or $P = -1$ (rejected)

$\vec{V}_1 = 2\sqrt{3}\,\hat{i} + \hat{j}$

$\vec{V}_2 = 2\hat{i} + 3\hat{j}$

$$\cos\theta = \frac{\vec{V}_1.\vec{V}_2}{\left|\vec{V}_1\right|\left|\vec{V}_2\right|} = \frac{4\sqrt{3}+3}{13}$$



$$\tan\theta = \frac{6\sqrt{3}-2}{4\sqrt{3}+3}$$

$\alpha = 6$

13. Let $\vec{p} = 2\hat{i} + 3\hat{j} + \hat{k}$ and $\vec{q} = \hat{i} + 2\hat{j} + \hat{k}$ be two vectors. If a vector $\vec{r} = (\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k})$ is perpendicular to each of the vectors $(\vec{p}+\vec{q})$ and $(\vec{p}-\vec{q})$, and $|\vec{r}| = \sqrt{3}$, then $|\alpha| + |\beta| + |\gamma|$ is equal to _____.

[JEE(Main)July2021]

माना $\vec{p}=2\hat{i}+3\hat{j}+\hat{k}$ तथा $\vec{q}=\hat{i}+2\hat{j}+\hat{k}$ दो सदिश हैं। यदि सदिश $\vec{r}=(\alpha\hat{i}+\beta\hat{j}+\gamma\hat{k})$, दोनों सदिशों $(\vec{p}+\vec{q})$ तथा $(\vec{p}-\vec{q})$ के लम्बवत् है तथा $|\vec{r}|=\sqrt{3}$ है, तो $|\alpha|+|\beta|+|\gamma|$ बराबर है _____। **[JEE(Main)July2021]**

Ans. 3

Sol. $\vec{p}+\vec{q}=(3,5,2)$

$\vec{p}-\vec{q}=(1,1,0)$

$\vec{r}=(\alpha,\beta,\gamma)$

$\vec{r}.(\vec{p}+\vec{q})=0$

$3\alpha+5\beta+2\gamma=0$

$\vec{r}.(\vec{p}-\vec{q})=0$

$\alpha+\beta=0$

$\beta=-\alpha$

$\gamma=\alpha$

$\vec{r}=\alpha\hat{i}-\alpha\hat{j}+\alpha\hat{k}$

$|\vec{r}|=\sqrt{3\alpha^2}=\sqrt{3}$

$\alpha^2=1\Rightarrow\alpha=\pm1$

$|\alpha|+|\beta|+|\gamma|=3$

14. If $\left(\vec{a}+3\vec{b}\right)$ is perpendicular to $\left(7\vec{a}-5\vec{b}\right)$ and $\left(\vec{a}-4\vec{b}\right)$ is perpendicular to $\left(7\vec{a}-2\vec{b}\right)$, then the angle between $\vec{a}$ and $\vec{b}$ (in degrees) is _____. **[JEE(Main)July2021]**

यदि $\left(\vec{a}+3\vec{b}\right)$, $\left(7\vec{a}-5\vec{b}\right)$ के लंबवत है तथा $\left(\vec{a}-4\vec{b}\right)$, $\left(7\vec{a}-2\vec{b}\right)$ के लंबवत् है, तो $\vec{a}$ तथा $\vec{b}$ के बीच का कोण (डिग्री में) है _____ **[JEE(Main)July2021]**

Ans. 60

Sol. $\left(\vec{a}+3\vec{b}\right).\left(7\vec{a}-5\vec{b}\right)=0$

$7|\vec{a}|^2-5\vec{a}\cdot\vec{b}+21\vec{a}\cdot\vec{b}-15|\vec{b}|^2=0$

$7|\vec{a}|^2+16\vec{a}\cdot\vec{b}-15|\vec{b}|^2=0---\{1\}$

$(\vec{a}-4\vec{b}).(7\vec{a}-2\vec{b})=0$

$7|\vec{a}|^2-2\vec{a}\cdot\vec{b}-28\,\vec{a}\cdot\vec{b}+8|\vec{b}|^2=0$

$7|\vec{a}|^2-30\,\vec{a}\cdot\vec{b}+8|\vec{b}|^2=0----(2)$

$46\vec{a}\cdot\vec{b}-23|\vec{b}|^2=0$

$$\Rightarrow \vec{a}\cdot\vec{b}=\frac{|\vec{b}|^2}{2}$$

$$7|\vec{a}|^2+16\vec{a}\cdot\vec{b}-30\ \vec{a}\cdot\vec{b}=0$$

$$7|\vec{a}|^2=14\vec{a}\cdot\vec{b}$$

$$|\vec{a}|^2=2\vec{a}\cdot\vec{b} \quad \Rightarrow \quad \vec{a}\cdot\vec{b}=\frac{|\vec{a}|^2}{2}$$

$$\frac{|\vec{b}|^2}{2}=\frac{|\vec{a}|^2}{2}\Rightarrow|\vec{a}|=|\vec{b}|$$

$$2\vec{a}\cdot\vec{b}=|\vec{b}|^2$$

$$2|\vec{a}||\vec{b}|\cos\theta=|\vec{b}|^2$$

$$\cos\theta=1/2\Rightarrow\ \theta=\frac{\pi}{3}$$

15. Let $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{b}$ and $\vec{c}=\hat{j}-\hat{k}$ be three vectors such that $\vec{a}\times\vec{b}=\vec{c}$ and $\vec{a}\,.\,\vec{b}=1$. If the length of projection vector of the vector $\vec{b}$ on the vector $\vec{a}\times\vec{c}$ is l, then the value of $3l^2$ is equal to _____.

[JEE(Main)July2021]

माना $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{b}$ तथा $\vec{c}=\hat{j}-\hat{k}$ तीन सदिश हैं जिनके लिए $\vec{a}\times\vec{b}=\vec{c}$ तथा $\vec{a}\,.\,\vec{b}=1$ हैं। यदि सदिश $\vec{b}$ के, सदिश $\vec{a}\times\vec{c}$ पर प्रक्षेप सदिश की लम्बाई l है, तो $3l^2$ का मान बराबर है _____ **[JEE(Main)July2021]**

Ans. 2

Sol. $$l=\left|\frac{\vec{b}.(\vec{a}\times\vec{c})}{|\vec{a}\times\vec{c}|}\right|=\left|\frac{\left[\vec{b}\,\vec{a}\,\vec{c}\right]}{|\vec{a}\times\vec{c}|}\right|=\left|\frac{\left[\vec{a}\ \vec{b}\ \vec{c}\right]}{|\vec{a}\times\vec{c}|}\right|$$

$$\vec{a}\times\vec{c}=\begin{vmatrix}i & j & k\\ 1 & 1 & 1\\ 0 & 1 & -1\end{vmatrix}=-2i+j+k$$

$$\vec{a}\times\vec{b}=\vec{c}$$

$$\left(\vec{a}\times\vec{b}\right).\vec{c}=\vec{c}.\vec{c}\ \Rightarrow\ \left[\vec{a}\ \ \vec{b}\ \vec{c}\right]=2$$

$$l=\frac{2}{\sqrt{6}}=\sqrt{\frac{2}{3}}$$

$$3l^2=2$$

16. Let $\vec{a}=\hat{i}-\alpha\hat{j}+\beta\hat{k}$, $\vec{b}=3\hat{i}+\beta\hat{j}-\alpha\hat{k}$ and $\vec{c}=-\alpha\hat{i}-2\hat{j}+\hat{k}$, where α and β are integers. If $\vec{a}\cdot\vec{b}=-1$ and $\vec{b}\cdot\vec{c}=10$ then $\left(\vec{a}\times\vec{b}\right).\vec{c}$ is equal to _____. **[JEE(Main)July2021]**

माना $\vec{a}=\hat{i}-\alpha\hat{j}+\beta\hat{k}$, $\vec{b}=3\hat{i}+\beta\hat{j}-\alpha\hat{k}$ तथा $\vec{c}=-\alpha\hat{i}-2\hat{j}+\hat{k}$ हैं, जहाँ α तथा β पूर्णांक हैं। यदि $\vec{a}\cdot\vec{b}=-1$ तथा $\vec{b}\cdot\vec{c}=10$ हैं, तो $\left(\vec{a}\times\vec{b}\right).\vec{c}$ बराबर है _____ **[JEE(Main)July2021]**

Ans. 9

Sol. $\vec{a}=\hat{i}-\alpha\hat{j}+\beta\hat{k}$

$\vec{b}=3\hat{i}+\beta\hat{j}-\alpha\hat{k}$

$\vec{c}=-\alpha j-2\hat{j}+\hat{k}$

$\therefore\ \vec{a}.\vec{b}=1$

$\Rightarrow\ 3-\alpha\beta-\alpha\beta=-1 \qquad \therefore\ \ \alpha\beta=2 \qquad$ .....(1)

$\therefore\ \vec{b}.\vec{c}=10$

$\therefore\ 3\alpha-2\beta-\alpha=10$

$4\alpha+2\beta=-10$

$\therefore\ 2\alpha+\beta=-5 \qquad$ .....(2)

$\therefore\ \alpha, \beta \in$ integers and satisfying above conditions

$\therefore\ \alpha=-2$ and $\beta=-1$

$$\left(\vec{a}\times\vec{b}\right).\vec{c}=\begin{vmatrix}1 & -\alpha & \beta\\ 3 & \beta & -\alpha\\ -\alpha & -2 & 1\end{vmatrix}$$

$$=\begin{vmatrix}1 & 2 & -1\\ 3 & -1 & 2\\ 2 & -2 & 1\end{vmatrix}$$

$= 1(-1+4)-2(3-4)-1(-6+2)$

$= 3+2+4=9$

3D Geometry

17. The lines $x = ay - 1 = z - 2$ and $x = 3y - 2 = bz - 2$, $(ab \neq 0)$ are coplanar, if : **[JEE(Main)July2021]**

रेखाएँ $x = ay - 1 = z - 2$ तथा $x = 3y - 2 = bz - 2$, $(ab \neq 0)$ समतलीय हैं, यदि : **[JEE(Main)July2021]**

(A) $b = 1, a \in R - \{0\}$ (B) $a = 2, b = 2$ (C) $a = 2, b = 3$ (D) $a = 1, b \in R - \{0\}$

Ans. A

Sol. $\dfrac{x-0}{1}=\dfrac{y-\frac{1}{a}}{1/a}=\dfrac{z-2}{1}$

$$\frac{x}{1}=\frac{y-\frac{2}{3}}{\frac{1}{3}}=\frac{z-\frac{2}{b}}{\frac{1}{b}}$$

if lines are coplaner then

$$\begin{vmatrix} 0 & \frac{2}{3}-\frac{1}{a} & \frac{2}{b}-2 \\ 1 & \frac{1}{a} & 1 \\ 1 & \frac{1}{3} & \frac{1}{b} \end{vmatrix}=0$$

$3b - 4a + 4ab = 3$

from option (1)

$b = 1\ a \in R - \{0\}$

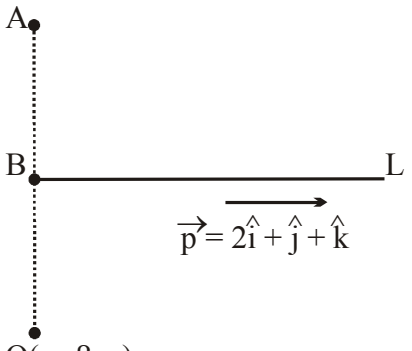
18. Consider the line L given by the equation $\frac{x-3}{2}=\frac{y-1}{1}=\frac{z-2}{1}$. Let Q be the mirror image of the point (2,3, –1) with respect to L. Let a plane P be such that it passes through Q, and the line L is perpendicular to P. Then which of the following points is on the plane P ? **[JEE(Main)July2021]**

माना Q, बिन्दु (2, 3, – 1) का रेखा L : $\frac{x-3}{2}=\frac{y-1}{1}=\frac{z-2}{1}$ में दपर्ण प्रतिबिम्ब है। माना एक समतल P बिन्दु Q से होकर जाता है तथा रेखा L, P पर लम्बवत् है। तो निम्न में से कौनसा बिन्दु समतल P है ? **[JEE(Main)July2021]**

(A) (1, 1, 2) (B) (1, 2, 2) (C) (1, 1, 2) (D) (–1, 1, 2)

Ans. B

Sol.



$Q(\alpha, \beta, \gamma)$

$B\ (2\lambda + 3, \lambda + 1, \lambda + 2)$

$\overrightarrow{AB}=(2\lambda+1)\hat{i}+(\lambda-2)\hat{j}+(\lambda+3)\hat{k}$

$\overrightarrow{AB}\ .\ \vec{P}=0$

$2(2\lambda + 1) + \lambda - 2 + \lambda + 3 = 0$

$6\lambda = -3$

$\lambda = \frac{-1}{2}$

$B\left(2, \frac{1}{2}, \frac{3}{2}\right)$

$\frac{\alpha+2}{2} = 2, \qquad \frac{\beta+3}{2} = \frac{1}{2}, \qquad \frac{\gamma-1}{2} = \frac{3}{2}$

$\alpha=2, \qquad \beta = 2, \qquad \gamma = 2$

equation of plane $2x + y + z =$ of passing through $(2, -2, 4)$

$d = 6$

$2x + y + z = 6$

19. Let L be the line of intersection of planes $\vec{r}.(\hat{i}-\hat{j}+2\hat{k})=2$ and $\vec{r}.(2\hat{i}+\hat{j}-\hat{k})=2$. If P (α, β, γ) is the foot of perpendicular on L from the point (1,2, 0), then the value of $35(\alpha + \beta + \gamma)$ is equal to : **[JEE(Main)July2021]**

माना L, समतलों $\vec{r}.(\hat{i}-\hat{j}+2\hat{k})=2$ तथा $\vec{r}.(2\hat{i}+\hat{j}-\hat{k})=2$ की प्रतिच्छेदन रेखा है। यदि बिन्दु (1, 2, 0) से रेखा L पर डाले गए लम्ब का पाद P (α, β, γ) है, तो $35(\alpha + \beta + \gamma)$ का मान बराबर है : **[JEE(Main)July2021]**

(A) 134 (B) 143 (C) 119 (D) 101

Ans. C

Sol. Direction of line of inter section

$$\vec{P} = \vec{n}_2 \times \vec{n}_2$$

$$= \begin{vmatrix} i & j & k \\ 1 & -1 & 2 \\ 2 & 1 & -1 \end{vmatrix}$$

$= i(1 - 2) - j(-1 - 4) + k(1 + 2)$

$= -1 + 5j + 3k$

Equation of plane in cartesicen form

$x - y + 2z = 2$ .....(1)

$2x + y - z = 2$ .....(2)

Now we will get one point on Line if intersection

then put $z = 0$ in (1) and (2)

$x = \frac{4}{3}, \quad y = -\frac{2}{3}$

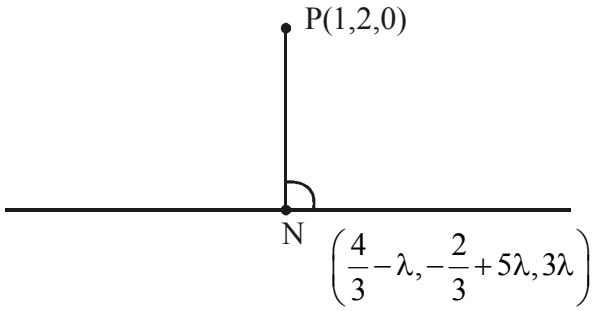
point $\left(\frac{4}{3}, -\frac{2}{3}, 0\right)$

Now equation of L

$\vec{r} = \vec{a} + \lambda\vec{P}$

$\vec{r} = \left(\frac{4}{3}i - \frac{2}{3}j + 0k\right) + \lambda\left(-i + 5j + 3k\right)$

We calcualte foot of perpendicular



$P\vec{N} = \left(\frac{1}{3} - \lambda\right)i + \left(5\lambda - \frac{8}{3}\right)j + \left(3\lambda\right)k$

$P\vec{N}.\vec{P} = 0$

$-1\left(\frac{1}{3} - \lambda\right) + 5\left(5\lambda - \frac{8}{3}\right) + 9\lambda = 0$

$\lambda = \frac{41}{105}$

Now $\alpha = \frac{4}{3} - \lambda$

$\beta = 5\lambda - \frac{2}{3}$

$\gamma = 3\lambda$

$\alpha + \beta + \gamma = 7\lambda + \frac{2}{3}$

$= 7\left(\frac{41}{105}\right) + \frac{2}{5} = \frac{51}{15}$

Now $35\left(\alpha + \beta + \gamma\right) = 35 \times \frac{51}{15} = 119$

20. If the shortest distance between the straight lines $3(x-1) = 6(y-2) = 2(z-1)$ and $4(x-2) = 2(y-\lambda) = (z-3)$, $\lambda \in R$ is $\frac{1}{\sqrt{38}}$, then the integral value of λ is equal to : **[JEE(Main)July2021]**

यदि सरल रेखाओं $3(x-1)=6(y-2)=2(z-1)$ तथा $4(x-2)=2(y-\lambda)=(z-3)$, $\lambda\in R$ के बीच की न्यूनतम दूरी $\frac{1}{\sqrt{38}}$ है, तो λ का पूर्णांक मान बराबर है : **[JEE(Main)July2021]**

(A) 5 (B) – 1 (C) 2 (D) 3

Ans. D

Sol. Given lines

$3(x-1)=6(y-2)=2(z-1)$

$$\frac{x-1}{2}=\frac{y-2}{1}=\frac{z-1}{3}---(1)$$

$4(x-2)=2(y-\lambda)=1(z-3)$

$$\frac{x-2}{1}=\frac{y-\lambda}{2}=\frac{z-3}{4}---(2)$$

Minimum distance between two skew lines

$$d=\left|\frac{\overrightarrow{AB}\cdot(\vec{p}\times\vec{q})}{|\vec{p}\times\vec{q}|}\right| \qquad \vec{p}=2i+j+3k \quad \vec{q}=i+2j+4k$$

$$\vec{AB}=i+(\lambda-2)j+(2)\hat{k}$$

$$\vec{p}\times\vec{q}=\begin{vmatrix} i & j & k \\ 2 & 1 & 3 \\ 1 & 2 & 4 \end{vmatrix}=i(-2)-j(5)+k(3)$$

$$d=\left|\frac{-2+5(2-\lambda)+6}{\sqrt{4+25+9}}\right|$$

$$d=\left|\frac{-2+10-5\lambda+6}{\sqrt{38}}\right|=\frac{1}{\sqrt{38}}$$

$$d=\left|\frac{14-5\lambda}{\sqrt{38}}\right|=\frac{1}{\sqrt{38}} \qquad =-2i-5j+3k$$

$|5\lambda-14|=1$

$5\lambda-14=\pm1$

$5\lambda=14\pm1$

$\lambda=3$ (integral value)

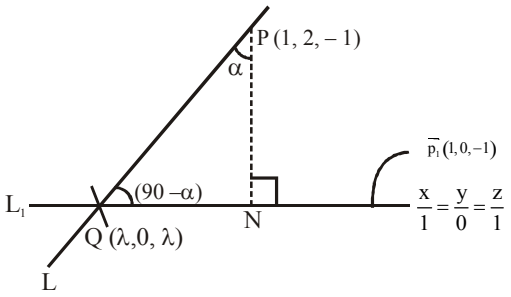
21. Let the foot of perpendicular from a point P(1, 2, –1) to the straight line L : $\frac{x}{1}=\frac{y}{0}=\frac{z}{-1}$ be N. Let a line be drawn from P parallel to the plane $x+y+2z=0$ which meets L at point Q. If α is the acute angle between the lines PN and PQ, then $\cos\alpha$ is equal to _____ . **[JEE(Main)July2021]**

माना सरल रेखा L : $\frac{x}{1} = \frac{y}{0} = \frac{z}{-1}$ पर बिन्दु P(1, 2, – 1) से डाले गए लम्ब का पाद N है। माना P से समतल, x + y + 2z = 0 के समान्तर खींची गई एक रेखा, L को बिन्दु Q पर मिलती है। यदि रेखाओं PN तथा PQ के बीच का न्यूनकोण α है, तो cosα बराबर है: **[JEE(Main)July2021]**

(A) $\frac{1}{\sqrt{3}}$ (B) $\frac{1}{\sqrt{5}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{2\sqrt{3}}$

Ans. A

Sol.



$\overrightarrow{PQ} \parallel$ Plane

$\overrightarrow{PQ}(\lambda-1,-2,-\lambda+1)$

$\overrightarrow{PQ}.\vec{n}=0 \quad [\vec{n}=(1,1,2)]$

$\lambda-1-2-2\lambda+2=0$

$-\lambda-1=0$

$Q(-1,0,-1)$

$\overrightarrow{PQ}=(-2,-2,2)$

$$\cos(90-\alpha)=\frac{\overrightarrow{PQ}.\overrightarrow{P_1}}{|\overrightarrow{PQ}|.|\overrightarrow{P_1}|}$$

$$\sin\alpha\left|\frac{-2+0-2}{\sqrt{4+4+4\sqrt{2}}}\right|=\frac{4}{2\sqrt{3}\sqrt{2}}=\frac{\sqrt{2}}{\sqrt{3}}$$

$$\cos\alpha=\frac{1}{\sqrt{3}}$$

22. Let the plane passing through the point (–1, 0, – 2) and perpendicular to each of the planes 2x + y – z = 2 and x – y – z = 3 be ax + by + cz + 8 = 0. Then the value of a + b + c is equal to : **[JEE(Main)July2021]**

माना बिन्दु (–1, 0, – 2) से होकर जाने वाले तथा समतलों 2x + y – z = 2 और x – y – z = 3 पर लम्बवत् समतल का समीकरण ax + by + cz + 8 = 0 है, तो a + b + c का मान बराबर है : **[JEE(Main)July2021]**

(A) 5 (B) 8 (C) 3 (D) 4

Ans. D

Sol. A (–1, 0, – 2)

P_1; $2x + y - z = 2$; $\vec{n}_1 = 2i + j - k$

P_2; $x - y - z = 3$; $\vec{n}_2 = i - j - k$

$\vec{n} = \vec{n}_1 \times \vec{n}_2$

$$\vec{n} = \begin{vmatrix} i & j & k \\ 2 & 1 & -1 \\ 1 & -1 & -1 \end{vmatrix} = -2i + j - 3k$$

DR's of normal = 2, –1, 3

Equation of plane $\Rightarrow 2(x + 1) - 1(y - 0) + 3(z + 2) = 0$

$\Rightarrow 2x - y + 3z + 8 = 0$

$a + b + c = 2 - 1 + 3 = 4$

23. For real numbers α and $\beta \neq 0$, if the point of intersection of the straight lines

$$\frac{x-\alpha}{1} = \frac{y-1}{2} = \frac{z-1}{3} \text{ and } \frac{x-4}{\beta} = \frac{y-6}{3} = \frac{z-7}{3},$$

lies on the plane $x + 2y - z = 8$ then $\alpha - \beta$ is equal to : **[JEE(Main)July2021]**

वास्तविक संख्याओं α तथा $\beta \neq 0$ के लिए, यदि सरल रेखाओं

$$\frac{x-\alpha}{1} = \frac{y-1}{2} = \frac{z-1}{3} \text{ तथा } \frac{x-4}{\beta} = \frac{y-6}{3} = \frac{z-7}{3},$$

का प्रतिच्छेदन बिन्दु, समतल $x + 2y - z = 8$ पर है, तो $\alpha - \beta$ बराबर है : **[JEE(Main)July2021]**

(A) 5 (B) 7 (C) 3 (D) 9

Ans. B

Sol. Given lines :

$$L_1 : \frac{x-\alpha}{1} = \frac{y-1}{2} = \frac{z-1}{3}$$

$$L_2 : \frac{x-4}{\beta} = \frac{y-6}{3} = \frac{z-7}{3}$$

General point on the line L_1 :

$= (\alpha + t_1, 1 + 2t_1, 1 + 3t_1)$

General point on the line L_2 :

$= (4 + \beta + t_2, 6 + 3t_2, 7 + 3t_2)$

for point of intersection of 4 L_2

$\alpha + t_1 = 4 + \beta t_2$ .....(i)

$1 + 2t_2 = 6 + 3t_2$ .....(ii)

$1 + 3t_1 = 7 + 3t_2$ .....(iii)

On solving (ii) and (iii)

$t_1 = 1, t_2 = -1$

Using (i) $\alpha + 1 = 4 - \beta$

$\alpha + \beta = 3$ .....(iv)

Point of intersection of L_1 and L_2 in term of α

$= (\alpha + 1, 1+ 2, 1 +3)$

$= (1 +\alpha , 3, 4)$ lies on the plane $x + 2y - z = 8$

$1 + \alpha + 6 - 4 = 8 \quad \alpha = 5, \beta = 2$

$\alpha - \beta = 7$

24. Let P be a plane passing through the pionts (1,0,1), (1, –2, 1) and (0, 1, –2). Let a vector $\vec{a} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ be such that $\vec{a}$ is parallel to the plane P, perpendicular to $(\hat{i} + 2\hat{j} + 3\hat{k})$ and $\vec{a}.(\hat{i} + \hat{j} + 2\hat{k}) = 2$, then $(\alpha - \beta + \gamma)^2$ equals _____. **[JEE(Main)July2021]**

माना समतल P बिंदुओं (1,0,1), (1, –2, 1) तथा (0, 1, –2) से होकर जाता है। माना एक सदिश $\vec{a} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$, समतल P के समांतर है, सदिश $(\hat{i} + 2\hat{j} + 3\hat{k})$ के लंबवत है तथा $\vec{a}.(\hat{i} + \hat{j} + 2\hat{k}) = 2$ को संतुष्ट करता है। तो $(\alpha - \beta + \gamma)^2$ बराबर है _____ **[JEE(Main)July2021]**

Ans. 81

Sol. Plane P A(1, 0, 1), B(1, –2, 1), C(0, 1, –2)

$\vec{P} = \overrightarrow{AB} = 2\hat{j}$

$\vec{q} = \overrightarrow{AC} = -\hat{i} + \hat{j} - 3\hat{k}$

let $\hat{i} + 2\hat{j} + 3\hat{k} = \vec{r}$

$\vec{a}$ is lying in plane of $\vec{p}$ & $\vec{q}$ and perpendicular or $\vec{r}$

$\vec{a} = \lambda(\vec{p} \times \vec{q}) \times \vec{r}$

$= \lambda((\vec{p}.\vec{r})\vec{q} - (\vec{q}.\vec{r})\vec{p})$

$= \lambda(4\vec{q} + 8\vec{p})$

$\vec{a} = \lambda(-4\hat{i} + 10\hat{j} - 12\hat{k})$

$\vec{a}.(\hat{i} + \hat{j} + 2\hat{k}) = 2$

$-4\lambda + 20\lambda - 24\lambda = 2$

$-8\lambda = 2$

$\lambda = -1/4$

$\alpha - \beta + \gamma = -4\lambda - 20\lambda - 12\lambda$

$= -36\lambda$

$\alpha - \beta + \gamma = 9$

$(\alpha - \beta + \gamma)^2 = 81$

25. If the line $\frac{x-k}{1}=\frac{y-2}{2}=\frac{z-3}{3}$ and $\frac{x+1}{3}=\frac{y+2}{2}=\frac{z+3}{1}$ are co-planar, then the value of k is _____.

[JEE(Main)July2021]

यदि रेखाएँ $\frac{x-k}{1}=\frac{y-2}{2}=\frac{z-3}{3}$ तथा $\frac{x+1}{3}=\frac{y+2}{2}=\frac{z+3}{1}$ समतलीय हैं, तो k का मान है _____

[JEE(Main)July2021]

Ans. 1

Sol. $\vec{r}=(k,2,3)+\lambda(1,2,3)$

$\frac{1}{2}$

$\vec{d}=(k+1)\hat{i}+4\hat{j}+6\hat{k}$

$$\begin{vmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ k+1 & 4 & 6 \end{vmatrix}=0$$

$1(8) - 2(18 - k - 1) + 3\,(12 - 2k - 2) = 0$

$2k + 10 - 6K - 6 = 0$

$4 = 4k$

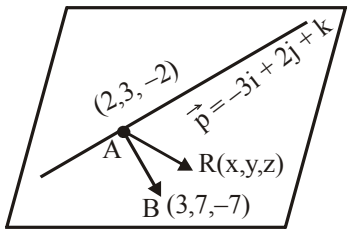
$k = 1$

26. Let a plane P pass through the point (3, 7, –7) and contain the line $\frac{x-2}{-3}=\frac{y-3}{2}=\frac{z+2}{1}$. If distance of the plan P from the originis d, then d^2 is equal to. **[JEE(Main)July2021]**

माना समतल P बिन्दु (3, 7, –7) से होकर जाता है। तथा रेखा $\frac{x-2}{-3}=\frac{y-3}{2}=\frac{z+2}{1}$ इसमें स्थित है। यदि समतल P की मूलबिन्दु से दूरी d है, तो d^2 बराबर है ––––––– **[JEE(Main)July2021]**

Ans. 3

Sol.



$\overrightarrow{AR}, \overrightarrow{AB}, \vec{P}$ are collinear vectors

$$\begin{vmatrix} x-2 & y-3 & z+2 \\ 1 & 4 & -5 \\ -3 & 2 & 1 \end{vmatrix}=0$$

$\Rightarrow (x-2)\,(14) - (y-3)\,(-14) + (z+2)\,(14) = 0$

$\Rightarrow x + y + 7 - 3 = 0$

$$d = \left|\frac{0+0+0-3}{\sqrt{3}}\right| = \sqrt{3}$$

$d^2 = 3$

27. The distance of the point P(3, 4, 4) from the point of intersection of the line joining the points Q(3, –4, –5) and R(2, –3, 1) and the plane 2x + y + z = 7, is equal to _____. **[JEE(Main)July2021]**

बिन्दुओं Q(3, – 4, – 5) तथा R(2, – 3, 1) को मिलाने वाली रेखा तथा समतल 2x + y + z = 7 के प्रतिच्छेदन बिन्दु से बिन्दु P (3,4, 4) की दूरी है _____ **[JEE(Main)July2021]**

Ans. 7

Sol. D.R. of the line QR

$= < -1, 1, 6 >$

General point on the line QR

$= (3 - t, -4 + t, -5 + 6t)$

Let It lies on the plane 2x + y + z = 7

$\Rightarrow 6 - 2t - 9 + 7t = 7$

$\Rightarrow 5t = 10$

$t = 2$

$\therefore$ Point of intersection of line QR and the plane = (3 – 2, – 4 + 2, –5 + 12)

$= (1, -2, 7)$

Required distance = $\sqrt{(3-1)^2 + (4+2)^2 + (4-7)^2}$

$= \sqrt{4+36+9}$

$= 7$

28. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{j} - \hat{k}$. If $\vec{c}$ is a vector such that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a} \cdot \vec{c} = 3$, then $\vec{a} \cdot (\vec{b} \times \vec{c})$ is equal to : **[JEE(Main)Aug.2021]**

माना $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ तथा $\vec{b} = \hat{j} - \hat{k}$ है। यदि एक सदिश $\vec{c}$ इस प्रकार है कि $\vec{a} \times \vec{c} = \vec{b}$ तथा $\vec{a} \cdot \vec{c} = 3$ है, तो $\vec{a} \cdot (\vec{b} \times \vec{c})$ बराबर है: **[JEE(Main)Aug.2021]**

(A) – 6 (B) 6 (C) 2 (D) – 2

Ans. D

Sol. $\vec{a} \times \vec{c} = \vec{b}$

cross with $\vec{a}$ $\qquad \vec{a} \times \vec{b} = -2\hat{i} + \hat{j} + \hat{k}$

$\vec{a} \times (\vec{a} \times \vec{c}) = \vec{a} \times \vec{b}$ $\qquad \left|\vec{a} \times \vec{b}\right| = \sqrt{6}$

$3\vec{a} - 3\vec{c} = \vec{a} \times \vec{b}$

$3\vec{c} = 3\vec{a} - \vec{a} \times \vec{b}$

dot with $\vec{a} \times \vec{b}$

$3\left(\vec{a}\times\vec{b}\right).\vec{c}=0-\left(\vec{a}\times\vec{b}\right)^2$

$3\left(\vec{a}\times\vec{b}\right).\vec{c}=-\ 6$

$\left(\vec{a}\times\vec{b}\right).\vec{c}=-2=\vec{a}.\left(\vec{b}\times\vec{c}\right)$

29. Let $\vec{a}$ and $\vec{b}$ be two vectors such that $\left|2\vec{a}+3\vec{b}\right| = \left|3\vec{a}+\vec{b}\right|$ and the angle between $\vec{a}$ and $\vec{b}$ is 60º. If $\frac{1}{8}\ \vec{a}$ is a unit vector, then $|\vec{b}|$ is equal to : **[JEE(Main)Aug.2021]**

माना $\vec{a}$ तथा $\vec{b}$ दो सदिशों हैं जिनके लिए $\left|2\vec{a}+3\vec{b}\right| = \left|3\vec{a}+\vec{b}\right|$ हैं और $\vec{a}$ तथा $\vec{b}$ के बीच का कोण 60º है। यदि $\frac{1}{8}\ \vec{a}$ एक इकाई सदिश हैं, तो $|\vec{b}|$ बराबर है : **[JEE(Main)Aug.2021]**

(A) 6 (B) 4 (C) 8 (D) 5

Ans. D

Sol. $\vec{a}\hat{}\vec{b}=60^o$

$|\vec{a}|=8$

$|2\vec{a}+3\vec{b}|^2=|3\vec{a}+\vec{b}|^2$

$4a^2+12ab\cos 60^o+9b^2=9a^2+6ab\cos 60^o+b^2$

$\Rightarrow b^2+3b-40=0$

$\Rightarrow b=5(b>0)$

30. Let $\vec{a},\vec{b},\vec{c}$ be three vectors mutually perpendicular to each other and have same magnitude. If a vector $\vec{r}$ satisfies $\vec{a}\times\{(\vec{r}-\vec{b})\times\vec{a}\}+\vec{b}\times\{(\vec{r}-\vec{c})\times\vec{b}\}+\vec{c}\times\{(\vec{r}-\vec{a})\times\vec{c}\}=\vec{0}$ then $\vec{r}$ is equal to : **[JEE(Main)Aug.2021]**

माना तीन सदिश $\vec{a},\vec{b},\vec{c}$ परस्पर लम्बवत हैं तथा इनके परिमाण बराबर है। यदि एक सदिश $\vec{r}$, $\vec{a}\times\{(\vec{r}-\vec{b})\times\vec{a}\}+\vec{b}\times\{(\vec{r}-\vec{c})\times\vec{b}\}+\vec{c}\times\{(\vec{r}-\vec{a})\times\vec{c}\}=\vec{0}$ को सन्तुष्ट करता है, तो $\vec{r}$ बराबर है : **[JEE(Main)Aug.2021]**

(A) $\frac{1}{2}(\vec{a}+\vec{b}+2\vec{c})$ (B) $\frac{1}{3}(2\vec{a}+\vec{b}-\vec{c})$ (C) $\frac{1}{3}(\vec{a}+\vec{b}+\vec{c})$ (D) $\frac{1}{2}(\vec{a}+\vec{b}+\vec{c})$

Ans. D

Sol. $\vec{r}=x\vec{a}+y\vec{b}+z\vec{c}$

$|\vec{a}|=|\vec{b}|=|\vec{c}|=k$

$\vec{a}\times\{(\vec{r}-\vec{b})\times\vec{a}\}+\vec{b}\times\{(\vec{r}-\vec{c})\times\vec{b}\}+\vec{c}\times\{(\vec{r}-\vec{a})\times\vec{c}\}=0$

$\vec{a}\times\left((\vec{r}-\vec{b})\times\vec{a}\right)$

$= |\vec{a}|^2 (\vec{r} - \vec{b}) - \left(\vec{a}.(\vec{r} - \vec{b})\right)\vec{a}$

$= k^2(\vec{r} - \vec{b}) - \left(\vec{a}.\vec{r}\right)\vec{a}$

$= k^2(\vec{r} - \vec{b}) - \left(\vec{a}.(x\vec{a} + y\vec{b} + z\vec{c})\right)\vec{a}$

$= k^2(\vec{r} - \vec{b}) - \left(k^2 x\right)\vec{a}$

$= k^2(\vec{r} - \vec{b}) - k^2 x\vec{a} + k^2(\vec{r} - \vec{c}) - k^2 y\vec{b} + k^2(\vec{r} - \vec{a}) - k^2 z\vec{c} = 0$

$\vec{r} - \vec{b} - x\vec{a} + \vec{r} - \vec{c} - y\vec{b} + \vec{r} - \vec{a} - z\vec{c} = 0$

$3\vec{r} - \vec{a} - \vec{b} - \vec{c} - x\vec{a} - y\vec{b} - z\vec{c} = 0$

$3\vec{r} - (\vec{a} + \vec{b} + \vec{c}) - (x\vec{a} + y\vec{b} + z\vec{c}) = 0$

$2\vec{r} = \vec{a} + \vec{b} + \vec{c}$

$\vec{r} = \dfrac{\vec{a} + \vec{b} + \vec{c}}{2}$

31. If the proejction of the vector $\hat{i} + 2\hat{j} + \hat{k}$ on the sum of the two vectors $2\hat{i} + 4\hat{j} - 5\hat{k}$ and $-\lambda\hat{i} + 2\hat{j} + 3\hat{k}$ is 1, then λ is equal to _____. **[JEE(Main)Aug.2021]**

यदि दो सदिशों $2\hat{i} + 4\hat{j} - 5\hat{k}$ तथा $-\lambda\hat{i} + 2\hat{j} + 3\hat{k}$ के योग पर सदिश $\hat{i} + 2\hat{j} + \hat{k}$ का प्रक्षेप 1 है, तो λ बराबर है_____ **[JEE(Main)Aug.2021]**

Ans. 5

Sol. $\overline{v}_1 = \hat{i} + 2\hat{j} + \hat{k}, \overline{v}_2 = 2\hat{i} + 4\hat{j} - 5\hat{k}, \overline{v}_3 = -\lambda\hat{i} + 2\hat{j} + 3\hat{k}$

$\overline{v}_2 + \overline{v}_3 = (2 - \lambda)\hat{i} + 6\hat{j} - 2\hat{k} = \overline{v}_4$

Projection of $\overline{v}_1$ on $\overline{v}_4 = \overline{v}_1 \cdot \dfrac{\overline{v}_4}{|\overline{v}_4|}$

$\Rightarrow \dfrac{1\times(2-\lambda) + 2\times 6 + 1\times(-2)}{\sqrt{(2-\lambda)^2 + 6^2 + (-2)^2}} = 1$

$\Rightarrow (12 - \lambda)^2 = (2 - \lambda)^2 + 40$

On solving

$\lambda = 5$

32. Let $\vec{a} = \hat{i} + 5\hat{j} + a\hat{k}, \vec{b} = \hat{i} + 3\hat{j} + \beta\hat{k}$ and $\vec{c} = -\hat{i} + 2\hat{j} - 3\hat{k}$ be three vectors such that, $\left|\vec{b} \times \vec{c}\right| = 5\sqrt{3}$ and $\vec{a}$ is perpendicular to $\vec{b}$. Then the greatest amongst the values of $|\vec{a}|^2$ is _____.

[JEE(Main)Aug.2021]

माना तीन सदिशों $\vec{a}=\hat{i}+5\hat{j}+a\hat{k},\vec{b}=\hat{i}+3\hat{j}+\beta\hat{k}$ तथा $\vec{c}=-\hat{i}+2\hat{j}-3\hat{k}$ के लिए $\left|\vec{b}\times\vec{c}\right|=5\sqrt{3}$ है तथा सदिश $\vec{a}$ सदिश $\vec{b}$ के लम्बवत् है तो $|\vec{a}|^2$ के मानों में अधिकतम मान है _____ **[JEE(Main)Aug.2021]**

Ans. 90

Sol. $\vec{a}=i+5j+\alpha k$

$\vec{b}=i+3j+\beta k$

$\vec{c}=-i+2j-3k$

$|\vec{b}\times\vec{c}|^2=75$

$\Rightarrow b^2c^2-(\vec{b}.\vec{c})^2=75$

$\Rightarrow (10+\beta^2).14-(5-3\beta)^2=75$

$\Rightarrow \beta=-2,-4$

$\vec{a}.\vec{b}=0$

$\Rightarrow 1+15+\alpha\beta=0$

$\Rightarrow \alpha=8,4$

$|\vec{a}|^2=1+25+\alpha^2$

$|\vec{a}|^2_{max}=1+25+64$

$=90$

33. Let $\vec{a}=2\hat{i}-\hat{j}+2\hat{k}$ and $\vec{b}=\hat{i}+2\hat{j}-\hat{k}$. Let a vector $\vec{\upsilon}$ be in the plane containing $\vec{a}$ and $\vec{b}$. If $\vec{\upsilon}$ is perpendicular to the vector $3\hat{i}+2\hat{j}-\hat{k}$ and its projection on $\vec{a}$ is 19 units, then $|2\vec{\upsilon}|^2$ is equal to ____ .

[JEE(Main)Aug.2021]

माना $\vec{a}=2\hat{i}-\hat{j}+2\hat{k}$ तथा $\vec{b}=\hat{i}+2\hat{j}-\hat{k}$ है। माना $\vec{a}$ तथा $\vec{b}$ को रखने वाले समतल में एक सदिश $\vec{\upsilon}$ है। यदि $\vec{\upsilon}$ सदिश $3\hat{i}+2\hat{j}-\hat{k}$ के अभिलम्ब है तथा $\vec{a}$ पर उसका प्रक्षप 19 इकाई है, तब $|2\vec{\upsilon}|^2$ बराबर है ____

[JEE(Main)Aug.2021]

Ans. 1494

Sol. $\vec{a}\times\vec{b}=\begin{vmatrix} i & j & k \\ 2 & -1 & 2 \\ 1 & 2 & -1 \end{vmatrix}$

$\vec{a}\times\vec{b}=-3\hat{i}+4\hat{j}+5\hat{k}$

$\therefore$ $\vec{v}$ be in the plane containing $\vec{a}$ and $\vec{b}$ and it is perpendicular to $\vec{c}=(3\hat{i}+2\hat{j}-\hat{k})$

$\therefore \vec{v}=\lambda(\vec{c}\times(\vec{a}\times\vec{b}))$

$\vec{v} = \lambda((\vec{c}.\vec{b})\vec{a} - (\vec{c}.\vec{a})\vec{b})$

$\vec{v} = \lambda(8\vec{a} - 2\vec{b})$

$\vec{v} = \lambda(14\hat{i} - 12\hat{j} + 18\hat{k})$

Given,

$\vec{v}.\hat{a} = 19$

$\lambda(14\hat{i} - 12\hat{j} + 18\hat{k}).\frac{(2\hat{i} - \hat{j} + 2\hat{k})}{3} = 19$

$\Rightarrow \lambda\frac{(28 + 12 + 36)}{3} = 19$

$\lambda \times \frac{76}{3} = 19$

$\therefore \lambda = \frac{3}{4}$

$\vec{v} = \frac{3}{4}(14\hat{i} - 12\hat{j} + 18k) = \frac{3}{2}(7\hat{i} - 6\hat{j} + 9\hat{k})$

$2\vec{v} = 3(7\hat{i} - 6\hat{j} + 9\hat{k})$

$|2\vec{v}|^2 = \left(\sqrt{21^2 + 18^2 + 27^2}\right)^2 = 1494$

34. A plane P contains the line $x + 2y + 3z + 1 = 0 = x - y - z - 6$, and is perpendicular to the plane $-2x + y + z + 8 = 0$. Then which of the following points lies on P ? **[JEE(Main)Aug.2021]**

एक समतल P में रेखा $x + 2y + 3z + 1 = 0 = x - y - z - 6$ स्थित है तथा P, समतल $-2x + y + z + 8 = 0$ के लम्बवत् है तो निम्न में से कौनसा बिन्दु समतल P पर है ? **[JEE(Main)Aug.2021]**

(A) (2, – 1, 1) (B) (1, 0, 1) (C) (–1, 1, 2) (D) (0, 1, 1)

Ans. D

Sol. Equation of required plane

$x + 2y + 3z + 1 + \lambda(x - y - z - 6) = 0$

It is perpendicular to $-2x + y + z + 8 = 0$

$(1 + \lambda, 2 - \lambda, 3 - \lambda).(-2, 1, 1) = 0$

$-2 - 2\lambda + 2 - \lambda + 3 - \lambda = 0$

$\lambda = \frac{3}{4}$

$x + 2y + 3z + 1 + \frac{3}{4}(x - y - z - 6) = 0$

$7x + 5y + 9z - 14 = 0$

(0, 1, 1) lies in this plane

35. Let P be the plane passing through the point (1, 2, 3) and the line of intersection of the planes $\vec{r}.\left(\hat{i}+\hat{j}+4\hat{k}\right)=$ 16 and $\vec{r}.\left(-\hat{i}+\hat{j}+\hat{k}\right)=6$. Then which of the following points does NOT lie on P ?

[JEE(Main)Aug.2021]

माना P एक समतल है जो बिन्दु (1, 2, 3) तथा समतलों $\vec{r}.\left(\hat{i}+\hat{j}+4\hat{k}\right)=16$ और $\vec{r}.\left(-\hat{i}+\hat{j}+\hat{k}\right)=6$ की प्रतिच्छेदन रेखा से होकर जाता है तो निम्न में से कौनसा बिन्दु P पर स्थित नहीं है? **[JEE(Main)Aug.2021]**

(A) (3, 3, 2) (B) (6, – 6, 2) (C) (– 8, 8, 6) (D) (4, 2, 2)

Ans. D

Sol. Equation of plane

$$(x+y+4x-16)+\lambda(-x+y+z-6)=0$$

$$(1,2,3)$$

$$\lambda=\frac{-1}{2}$$

equation of Req. plane $3x+y+7z=26$

36. Equation of a plane at a distance $\sqrt{\frac{2}{21}}$ from the origin, which contains the line of intersection of the planes $x-y-z-1=0$ and $2x+y-3z+4=0$, is : **[JEE(Main)Aug.2021]**

मूलबिन्दु से $\sqrt{\frac{2}{21}}$ की दूरी पर एक समतल, जिसमें समतलों $x-y-z-1=0$ तथा $2x+y-3z+4=0$ की प्रतिच्छेदन रेखा स्थित है, का समीकरण है : **[JEE(Main)Aug.2021]**

(A) $3x-4z+3=0$ (B) $-x+2y+2z-3=0$

(C) $4x-y-5z+2=0$ (D) $3x-y-5z+2=0$

Ans. C

Sol. $P_1: x-y-z-1=0$

$P_2: 2x-y-3z+4=0$

$P: P_1+\lambda P_2=0$

$(x-y-z-1)+\lambda(2x+y-3z+4)=0$

$\Rightarrow (1+2\lambda)x+(-1+\lambda)y+(-1-3\lambda)z-1+4\lambda=0$

Distance from $(0,0)=\sqrt{\frac{2}{21}}$

$$\left|\frac{0+0+0-1+4\lambda}{\sqrt{(1+2\lambda)^2+(-1+\lambda)^2+(-1-3\lambda)^2}}\right|=\sqrt{\frac{2}{21}}$$

$\Rightarrow 308\lambda^2-184\lambda+15=0$

$(2\lambda-1)(154\lambda-15)=0$

$\lambda=\frac{1}{2},\frac{15}{154}$

for $\lambda=\frac{1}{2}$

$P:(x-y-z-1)+\frac{1}{2}(2x+y-3z+4)=0$

$P:4x-y-5z+2=0$

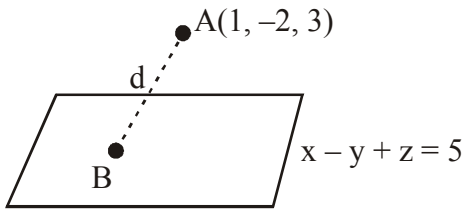
37. The distance of the point (1, –2, 3) from the plane x – y + z = 5 measured parallel to a line, whose direction ratios are 2, 3, –6 is : **[JEE(Main)Aug.2021]**

बिन्दु (1, –2, 3) की, एक रेखा जिसके दिक् अनुपात 2, 3, –6 हैं, के समान्तर समतल x – y + z = 5 से दूरी है : **[JEE(Main)Aug.2021]**

(A) 3 (B) 2 (C) 5 (D) 1

Ans. D

Sol.



DR's of AB = 2, 3, – 6

Equation of AB $=\frac{x-1}{2}=\frac{y+2}{3}=\frac{z-3}{-6}=\lambda$

$B(1+2\lambda,-2+3\lambda,3-6\lambda)$

Lies on given plane

$1+2\lambda+2-3\lambda+3-6\lambda=5$

$\lambda=\frac{1}{7}$

$\overrightarrow{AB}=2\lambda i+3\lambda j-6\lambda k$

$d=|\overrightarrow{AB}|=\lambda.7=\frac{1}{7}.7=1$

38. The equation of the plane passing through the line of intersection of the planes $\vec{r}.\left(\hat{i}+\hat{j}+\hat{k}\right)=1$ and $\vec{r}.\left(2\hat{i}+3\hat{j}-\hat{k}\right)+4=0$ and parallel to the x-axis is : **[JEE(Main)Aug.2021]**

समतलों $\vec{r}.\left(\hat{i}+\hat{j}+\hat{k}\right)=1$ तथा $\vec{r}.\left(2\hat{i}+3\hat{j}-\hat{k}\right)+4=0$ की प्रतिच्छेदन रेखा से होकर जाने वाले तथा x-अक्ष के समान्तर समतल का समीकरण है : **[JEE(Main)Aug.2021]**

(A) $\vec{r}\left(\hat{j}-3\hat{k}\right)+6=0$

(B) $\vec{r}\left(\hat{j}-3\hat{k}\right)-6=0$

(C) $\vec{r}\left(\hat{i}-3\hat{k}\right)+6=0$

(D) $\vec{r}\left(\hat{i}+3\hat{k}\right)+6=0$

Ans. A

Sol. Given plane in cartesian form

$x + y + z = 1$ _____(1)

$2x + 3y - z + 4 = 0$ _____(2)

Required plane

$(x + y + z - 1) + \lambda\ (2x + 3y - z + 4) = 0$

$(1 + 2\lambda)x + (1 + 3\lambda)y + (1 - \lambda)z + 4\lambda - 1 = 0$

Here, normal is perpendicular to x-axis.

Hence $(1+2\lambda)+(1+3\lambda)(0)+(1-\lambda)(0)=0$

$d = -\frac{1}{2}$

Now required plane

$y - 3z + 6 = 0$

39. The angle between teh straight lines, whose directioncosines are given by the equations $2l + 2m - n = 0$ and $mn + nl + lm = 0$, is : **[JEE(Main)Aug.2021]**

सरल रेखाओं, जिनके दिक्–कोसाइन समीकरणों $2l + 2m - n = 0$ तथा $mn + nl + lm = 0$ द्वारा दिए गए हैं, के बीच का कोण है : **[JEE(Main)Aug.2021]**

(A) $\frac{\pi}{3}$ (B) $\cos^{-1}\left(\frac{8}{9}\right)$ (C) $\frac{\pi}{2}$ (D) $\pi-\cos^{-1}\left(\frac{4}{9}\right)$

Ans. C

Sol. $n = 2(l + m)$ _____ (1)

$n(m + l) + lm = 0$ _____ (2)

$2(l + m)^2 + lm = 0$

$2l^2 + 5lm + 2m^2 = 0$

$(m + 2l)(2m + l) = 0$

$l = -2m$	$l = -\frac{m}{2}$
Case - I	
$<-2m, m, -2m,>$	$<-\frac{m}{2}, m, m>$
$<-2, 1, -2>$	$<-1, 2, 2>$

Hence, angle between these lines

$$\cos\theta = \frac{2+2-4}{\sqrt{9}\sqrt{9}}$$

$$\cos\theta = 0$$

$$\theta = \frac{\pi}{2}$$

40. Let the equation of the plane, that passes through the point (1, 4, –3) and contains the line of intersection of the planes $3x - 2y + 4z - 7 = 0$ and $x + 5y - 2z + 9 = 0$, be $\alpha x + \beta y + \gamma z + 3 = 0$, then $\alpha + \beta + \gamma$ is equal to : **[JEE(Main)Aug.2021]**

माना समतल, जो बिन्दु (1, 4, –3) से होकर जाता है तथा जिसमें समतलों $3x - 2y + 4z - 7 = 0$ तथा $x + 5y - 2z + 9 = 0$ की प्रतिच्छेदन रेखा स्थित है, का समीकरण $\alpha x + \beta y + \gamma z + 3 = 0$ है, तो $\alpha + \beta + \gamma$ बराबर है : **[JEE(Main)Aug.2021]**

(A) 15 (B) 23 (C) – 23 (D) – 15

Ans. C

Sol. $P_1 : 3x - 2y + 4z - 7 = 0$

$P_2 : x + 5y - 2z + 9 = 0$

$P : P_1 + \lambda P_2 = 0$

$(3x - 2y + 4z - 7) + \lambda(x + 5y - 2z + 9) = 0$

$\alpha + \beta + \gamma = -11 - 4 - 8 = -23$

$P : (3x - 2y + 4z - 7) + \frac{2}{3}(x + 5y - 2z + 9) = 0$

$P : -11x - 4y - 8z + 3 = 0$

$\alpha + \beta + \gamma = -11 - 4 - 8 = -23$

41. The distance of the point (–1, 2, –2) from the line of intersection of the planes $2x + 3y + 2z = 0$ and $x - 2y + z = 0$ is : **[JEE(Main)Aug.2021]**

बिन्दु (–1, 2, –2) की समतलों $2x + 3y + 2z = 0$ और $x - 2y + z = 0$ की प्रतिच्छेदन रेखा से दूरी है : **[JEE(Main)Aug.2021]**

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{\sqrt{34}}{2}$ (C) $\frac{5}{2}$ (D) $\frac{\sqrt{42}}{2}$

Ans. B

Sol. $P_1 : 2x + 3y + 2z = 0$

$\vec{n}_1 = 2\hat{i} + 3\hat{j} + 2\hat{k}$

$P_2 : x - 2y + z = 0$

$\vec{n}_2 = \hat{i} - 2\hat{j} + \hat{k}$

$\vec{b} = \vec{n}_1 \times \vec{n}_2$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 2 \\ 1 & -2 & 1 \end{vmatrix}$$

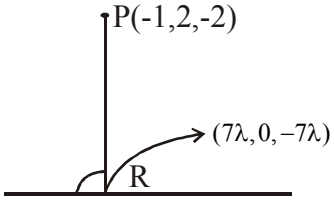
$= \hat{i}(7) - \hat{j}(0) + \hat{k}(-7)$

$\vec{b} = 7\hat{i} - 7\hat{k}$

$\vec{r} = \vec{a} + \lambda\vec{b}$

$= (0,0,0) + \lambda(7\hat{i} - 7\hat{k})$

$= 7\lambda(\hat{i} - \hat{k})$



$\overrightarrow{PR} = (7\lambda + 1)\hat{i} - 2\hat{j} + (2 - 7\lambda)\hat{k}$

$\overrightarrow{PR}.\vec{b} = 0$

$7(7\lambda + 1) - 7(2 - 7\lambda) = 0$

$\lambda = \frac{1}{14}$

$R = \left(-\frac{1}{2}, 0, -\frac{1}{2}\right)$

$PR = \sqrt{\left(\frac{3}{2}\right)^2 + 4 + \left(\frac{3}{2}\right)^2}$

$= \sqrt{\frac{9}{2} + 4} = \sqrt{\frac{17}{2}}$

42. The distance of line $3y - 2z - 1 = 0 = 3x - z + 4$ from the point $(2, -1, 6)$ is :

[JEE(Main)Aug.2021]

रेखा $3y - 2z - 1 = 0 = 3x - z + 4$ की बिन्दु $(2, -1, 6)$ से दूरी है : **[JEE(Main)Aug.2021]**

(A) $\sqrt{26}$ (B) $4\sqrt{2}$ (C) $2\sqrt{5}$ (D) $2\sqrt{6}$

Ans. D

Sol. Let direction ratio of the line :

$< a, b, c >$

Hence

$0.a+3b-2c=0$

$3.a+0.b-1.c=0$

Using cross multiplication rule

$$\frac{a}{-3-0}=\frac{b}{-6-0}=\frac{c}{0-9}$$

$$\Rightarrow \frac{a}{-3}=\frac{b}{-6}=\frac{c}{-9}$$

$$\frac{a}{1}=\frac{b}{2}=\frac{c}{3}$$

for point of the line

when x = 0, z = 4, y = 3

Equation of the line :

$$\frac{x-0}{1}=\frac{y-3}{2}=\frac{z-4}{3} \quad \text{.....(1)}$$

General point on the line

$P=(t,3+2t,4+3t)$

Let given point be $Q=(2,-1,6)$

D.R, of PQ = $(2-t,\ -4-2t,\ 2-3t)$

PQ $\perp^r$ to the line L

$1(2-t)+2(-4-2t)+3(2-3t)=0$

$\Rightarrow 2-t-8-4t+6-9t=0$

$\Rightarrow\ -14t=0$

$t=0$

$\therefore P\equiv(0,3,4)$

$PQ=\sqrt{2^2+4^2+2^2}$

$=\sqrt{4+16+4}=\sqrt{24}$

$=2\sqrt{6}$

43. Let the acute angle bisector of the two planes x – 2y – 2z + 1 = 0 and 2x – 3y – 6z + 1 = 0 be the plane P. Then which of the following points lies on P? **[JEE(Main)Aug.2021]**

माना दो समतलों x – 2y – 2z + 1 = 0 तथा 2x – 3y – 6z + 1 = 0 के न्यून कोण का समद्विभाजक समतल P है। तब इनमें से कौनसा बिन्दु P पर स्थित है? **[JEE(Main)Aug.2021]**

(A) $\left(-2,0,-\frac{1}{2}\right)$ (B) (0, 2, –4) (C) $\left(3,1,-\frac{1}{2}\right)$ (D) (4, 0, –2)

Ans. A

Sol. $P_1 : x - 2y - 2z + 1 = 0$

$P_2 : 2x - 3y - 6z + 1 = 0$

Equation of angle bisectors

$$\frac{x-2y+2z+1}{3} = \pm\frac{2x-3y+6z+1}{7}$$

since $a_1a_2 + b_1b_2 + c_1c_2 > 0$

$\therefore$ Negative sign will give acute angle bisector

$$\Rightarrow 7x-14y-14z+7 = -6x+9y+18z-3$$

$$13x-23y-32z+10 = 0$$

$\left(-2,0,-\frac{1}{2}\right)$ satisfy it.

44. Suppose the line $\frac{x-2}{\alpha} = \frac{y-2}{-5} = \frac{z+2}{2}$ lies on the plane $x + 3y - 2z + \beta = 0$. The $(\alpha + \beta)$ is equal to_____.

[JEE(Main)Aug.2021]

माना रेखा $\frac{x-2}{\alpha} = \frac{y-2}{-5} = \frac{z+2}{2}$ समतल $x + 3y - 2z + \beta = 0$ में स्थित है तो $(\alpha + \beta)$ बराबर है _____

[JEE(Main)Aug.2021]

Ans. 7

Sol. $\frac{x-2}{\alpha} = \frac{y-2}{-5} = \frac{z+2}{2}$

$\vec{b} = \alpha\hat{i} - 5\hat{j} + 2\hat{k}$

$\vec{n} = \hat{i} + 3\hat{j} - 2\hat{k}$

$\vec{b}.\vec{n} = 0$

$\alpha - 15 - 4 = 0$

$\alpha = 19$

$2 + 6 + 4 + \beta = 0$

$\beta = -12$

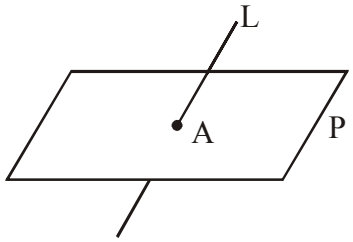
$\alpha + \beta = 19 - 12 = 7$

45. The square of the distance of the point of intersection of the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+1}{6}$ and the plane $2x - y + z = 6$ from the point (–1, – 1, 2) is _____. **[JEE(Main)Aug.2021]**

बिन्दु (–1, – 1, 2) से रेखा $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z+1}{6}$ तथा समतल 2x – y + z = 6 के प्रतिच्छेदन बिन्दु की दूरी का वर्ग बराबर है _____ **[JEE(Main)Aug.2021]**

Ans. 61

Sol.



$P: 2x-y+z=6$

$L: \frac{x-1}{2}=\frac{y-2}{3}=\frac{z+1}{6}=\lambda$

$A(1+2\lambda, 2+3\lambda, -1+6\lambda)$

$\Rightarrow 2+4\lambda-2-3\lambda-1+6\lambda=6$

$\Rightarrow \lambda=1$

$A(3,5,5)$

$B(-1,-1,2)$

$AB^2=61$

46. Let S be the mirror image of the point Q(1, 3, 4) with respect to the plane 2x–y+z+3=0 and let R (3, 5, λ) be a point of this plane. Then the square of the length of the line segment SR is _____. **[JEE(Main)Aug.2021]**

माना समतल 2x–y+z+3=0 के सापेक्ष बिन्दु Q(1, 3, 4) का दर्पण प्रतिबिम्ब S है तथा माना इस समतल पर एक बिन्दु R (3, 5, λ) है तो रेखाखण्ड SR की लम्बाई का वर्ग है_____ **[JEE(Main)Aug.2021]**

Ans. 72

Sol. R (3, 5, r) on the plane

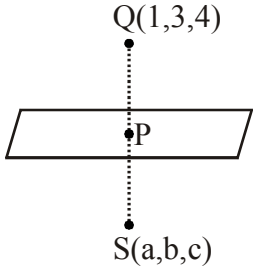
2x – y + z + 3 = 0

then 6 – 5 + r + 3 = 0

r = –4

R(3, 5, –4)

Equation of line QS



$\vec{r}=(1,3,4)+\lambda(2,-1,1)$

$P(1+2\lambda, 3-\lambda, 4+\lambda)$

P is one plane

$2(1+2\lambda)+\lambda-3+4+\lambda+3=0$

$2+4\lambda+\lambda-3+4+\lambda+3=0$

$6\lambda+6=0$

$\lambda = -1$

$(-1, 4, 3)$

P is mid-point of QS then $\frac{a+1}{2} = -1$ $\quad a = -3$

$\frac{b+3}{2} = 4$ $\quad b = 5$

$\frac{c+4}{2} = 3$ $\quad c = 2$

S(–3, 5, 2)

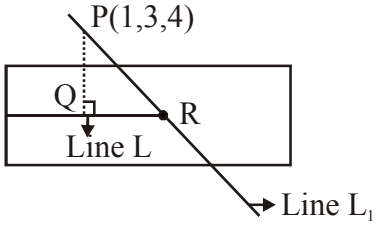
$RS^2 = (\sqrt{36+36})^2 = 72$

47. Let the line L be the projection of the line $\frac{x-1}{2} = \frac{y-3}{1} = \frac{z-4}{2}$ in the plane $x - 2y - z = 3$. If d is the distance of the point (0, 0, 6) from L, then d^2 is equal to _____. **[JEE(Main)Aug.2021]**

माना रेखा $\frac{x-1}{2} = \frac{y-3}{1} = \frac{z-4}{2}$ का समतल $x - 2y - z = 3$ में प्रक्षेप रेखा L है। यदि बिन्दु (0, 0, 6) की L से दूरी d है, तो d^2 बराबर है_____ **[JEE(Main)Aug.2021]**

Ans. 26

Sol.



R is POI of line L_1 and Plane

$R(1+2\gamma, 3+\gamma, 4+2\gamma)$

R will lie on $x - 2y - z = 3$

$1 + 2\gamma - 2(3+\gamma) - (4+2\gamma) = 3$

$2\gamma = 1 - 6 - 4 - 3 \Rightarrow \gamma = -6$

$R(-11, -3, -8)$

$Q \equiv$ Foot of perpendicular from P on P line

$Q(1+\mu, 3-2\mu, 4-\mu)$

$1 + \mu - 2(3-2\mu) - (4-\mu) = 3$

$6\mu = 6 + 4 + 3 - 1$

$\mu = 2$ $\quad \overrightarrow{QR} = (-14, -2, -10)$

$Q(3,-1,2)$

Equation of line L

$\vec{r} = (3,-1,2) + t(7,1,5)$

$S(0,0,6)$

distance of S from line $= \dfrac{|\overrightarrow{SQ} \times \overrightarrow{QR}|}{|\overrightarrow{QR}|}$

$$= \frac{|\hat{i} + 43\hat{j} - 10\hat{k}|}{|7\hat{i} + \hat{j} + 5\hat{k}|}$$

$d = \sqrt{26}$

$\Rightarrow d^2 = 26$

48. Let Q be the foot of the perpendicular from the point P(7, –2, 13) on the plane containing the lines $\dfrac{x+1}{6} = \dfrac{y-1}{7} = \dfrac{z-3}{8}$ and $\dfrac{x-1}{3} = \dfrac{y-2}{5} = \dfrac{z-3}{7}$ .Then $(PQ)^2$, is equal to _____.

[JEE(Main)Aug.2021]

माना बिन्दु P(7, –2, 13) से समतल, जिसमें रेखाएँ $\dfrac{x+1}{6} = \dfrac{y-1}{7} = \dfrac{z-3}{8}$ तथा $\dfrac{x-1}{3} = \dfrac{y-2}{5} = \dfrac{z-3}{7}$ स्थित है, पर डाले गये लम्ब का पाद Q है तो $(PQ)^2$ बराबर _____ है। **[JEE(Main)Aug.2021]**

Ans. 96

Sol. Equation of plane $\begin{vmatrix} x-1 & y-2 & z-3 \\ 6 & 7 & 8 \\ 3 & 5 & 7 \end{vmatrix} = 0$

$\Rightarrow x - 2y + z = 0$

$d^2(7,-2,13) = \dfrac{|7+4+13|^2}{6} = 96$

49. Let $\hat{a}$, $\hat{b}$ be unit vectors. If $\vec{c}$ be a vectors such that the angle between $\hat{a}$ and $\hat{c}$ is $\dfrac{\pi}{12}$, and $\hat{b} = \vec{c} + 2(\vec{c} \times \hat{a})$, then $|6\vec{c}|^2$ is equal to : **[JEE(Main)June2022]**

माना Let $\hat{a}$, $\hat{b}$ इकाई सदिश हैं। यदि एक सदिश $\vec{c}$ इस प्रकार है कि $\hat{a}$ तथा $\hat{c}$ के बीच का कोण $\dfrac{\pi}{12}$ है तथा $\hat{b} = \vec{c} + 2(\vec{c} \times \hat{a})$ है, तो $|6\vec{c}|^2$ बराबर है : **[JEE(Main)June2022]**

(A) $6(3-\sqrt{3})$ (B) $3 + \sqrt{3}$ (C) $6(3+\sqrt{3})$ (D) $6(\sqrt{3}+1)$

Ans. C

Sol. Let $|\vec{c}| = x$

$\hat{b} = \vec{c} + (\vec{c} \times \hat{a})$

$|\hat{b}| = 1$

$|\vec{c} + 2(\vec{c} \times \hat{a})| = 1$

$|\vec{c}|^2 + 4(\vec{c} \times \hat{a})^2 + 4\vec{c}.(\vec{c} \times \hat{a}) = 1$

$x^2 + 4(x^2)(1)\sin^2(15°) + 0 = 1$

$x^2 + 2x^2\left(1 - \frac{\sqrt{3}}{2}\right) = 1$

$x^2 + x^2(2 - \sqrt{3}) = 1$

$x^2(3 - \sqrt{3}) = 1$

$x^2 = \frac{3 + \sqrt{3}}{6}$

Ans = $(6x)^2 = 36x^2 = 6(3 + \sqrt{3})$

50. Let $\hat{a}$ and $\hat{b}$ be two unit vectors such that $\left|(\hat{a} + \hat{b}) + 2(\hat{a} \times \hat{b})\right| = 2$. If $\theta \in (0, \pi)$ is the angle between $\hat{a}$ and $\hat{b}$, then among the statements :

(S1) : $2|\hat{a} \times \hat{b}| = |\hat{a} - \hat{b}|$

(S2) : The projection of $\hat{a}$ on $(\hat{a} + \hat{b})$ is $\frac{1}{2}$ **[JEE(Main)June2022]**

(A) only (S1) is true (B) only (S2) is true

(C) Both (S1) and (S2) are true (D) Both (S1) and (S2) are false

माना दो इकाई सदिशों $\hat{a}$ तथा $\hat{b}$ के लिए $\left|(\hat{a} + \hat{b}) + 2(\hat{a} \times \hat{b})\right| = 2$ है। यदि $\hat{a}$ तथा $\hat{b}$ के बीच का कोण $\theta \in (0, \pi)$ है, है, तो कथनों :

(S1) : $2|\hat{a} \times \hat{b}| = |\hat{a} - \hat{b}|$

(S2) : $\hat{a}$ का $(\hat{a} + \hat{b})$ पर प्रक्षेप $\frac{1}{2}$ है **[JEE(Main)June2022]**

(A) केवल (S1) सत्य है (B) केवल (S2) सत्य है

(C) (S1) तथा (S2) दोनों सत्य हैं (D) (S1) तथा (S2) दोनों असत्य हैं

Ans. C

Sol. $\left|\left(\hat{a}+\hat{b}\right)+2\left(\hat{a}\times\hat{b}\right)\right|=2$

$\left(\hat{a}+\hat{b}\right)^2+4\left(\hat{a}+\hat{b}\right)^2+4\left(\vec{a}+\vec{b}\right)\left(\vec{a}\times\vec{b}\right)=4$

$1+1+2\cos\theta+4\sin^2\theta=4$

$1+\cos\theta+2\left(1-\cos^2\theta\right)=2$

$\cos\theta=\dfrac{-1}{2}$ is correct in $\theta\in(0,\pi)$

$\theta=\dfrac{2\pi}{3}$

(S1) $2\left|\hat{a}\times\hat{b}\right|=\left|\hat{a}-\hat{b}\right|$

$2\times\dfrac{\sqrt{3}}{2}=\sqrt{1+1+2\times\dfrac{1}{2}}$

$\sqrt{3}=\sqrt{3}$

which is correct

(S2) $\left|\dfrac{\hat{a}.\left(\hat{a}+\hat{b}\right)}{\left|\hat{a}+\hat{b}\right|}\right|=\dfrac{1}{2}$

$\left|\dfrac{1-\dfrac{1}{2}}{1}\right|=\dfrac{1}{2}$

which is also correct

51. Let $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$ $a_i>0$, i = 1, 2, 3 be vector which makes equal angles with the coordinate axes OX, OY and OZ. Also, Let the projection of $\vec{a}$ on the vector $3\hat{i}+4\hat{j}$ be 7. Let $\vec{b}$ be a vector obtained by rotating $\vec{a}$ with 90°. If $\vec{a},\vec{b}$ and x-axis are coplanar, then projection of a vector $\vec{b}$ on $3\hat{i}+4\hat{j}$ is equal to :

[JEE(Main)June2022]

माना सदिश $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$ $a_i>0$, i = 1, 2, 3 निर्देशांक अक्षों OX, OY तथा OZ से बराबर कोण बनाता है। यह भी मान लीजिए कि $\vec{a}$ का सदिश $3\hat{i}+4\hat{j}$ पर प्रक्षेप 7 है। माना सदिश $\vec{a}$ को 90° घुमाने पर सदिश $\vec{b}$ प्राप्त होता है। यदि $\vec{a},\vec{b}$ तथा x-अक्ष सह–तलीय हैं, तो एक सदिश $\vec{b}$ का $3\hat{i}+4\hat{j}$ पर प्रक्षेप बराबर है : **[JEE(Main)June2022]**

(A) $\sqrt{7}$ (B) $\sqrt{2}$ (C) 2 (D) 7

Ans. B

Sol. $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

$\vec{a} = \lambda\left(\frac{1}{\sqrt{3}}\hat{i} + \frac{1}{\sqrt{3}}\hat{j} + \frac{1}{\sqrt{3}}\hat{k}\right) = \frac{\lambda}{\sqrt{3}}\left(\hat{i} + \hat{j} + \hat{k}\right)$

Now projection of $\vec{a}$ on $\vec{c} \equiv 3\hat{i} + 4\hat{j}$ is 7

$\Rightarrow \frac{\vec{a}\cdot\vec{c}}{|\vec{c}|} = 7$

$\frac{\lambda}{\sqrt{3}}\frac{\left(\hat{i} + \hat{j} + \hat{k}\right)\cdot\left(3\hat{i} + 4\hat{j}\right)}{5} = 7$

$\lambda = 5\sqrt{3}$

$\vec{a} = 5(\hat{i} + \hat{j} + \hat{k})$

now $\vec{b} = 5\alpha\left(\hat{i} + \hat{j} + \hat{k}\right) + \beta\left(\hat{i}\right)$

$\vec{a}\cdot\vec{b} = 0$

$\Rightarrow 25\alpha(3) + 5\beta = 0$

$\Rightarrow 15\alpha + \beta = 0 \Rightarrow \beta = -15\alpha$

$\vec{b} = 5\alpha(-2\hat{i} + \hat{j} + \hat{k})$

$|\vec{b}| = 5\sqrt{3} = |\vec{a}|$

$\Rightarrow \alpha = \pm\frac{1}{\sqrt{2}}$

$\vec{b} = \pm\frac{5}{\sqrt{2}}(-2\hat{i} + \hat{j} + \hat{k})$

Projection of $\vec{b}$ on $3\hat{i} + 4\hat{j}$ is

$\frac{\vec{b}\cdot(3\hat{i} + 4\hat{j})}{5} = \pm\frac{5}{\sqrt{2}}\left(\frac{-6+4}{5}\right) = \pm\sqrt{2}$

52. If $\vec{a}\,.\,\vec{b} = 1, \vec{b}\,.\,\vec{c} = 2$ and $\vec{c}\,.\,\vec{a} = 3$, then the value of $\left[\vec{a}\times\left(\vec{b}\times\vec{c}\right), \vec{b}\times\left(\vec{c}\times\vec{a}\right), \vec{c}\times\left(\vec{b}\times\vec{a}\right)\right]$ is :

[JEE(Main)June2022]

यदि $\vec{a}\,.\,\vec{b} = 1, \vec{b}\,.\,\vec{c} = 2$ तथा $\vec{c}\,.\,\vec{a} = 3$ हैं, तो $\left[\vec{a}\times\left(\vec{b}\times\vec{c}\right), \vec{b}\times\left(\vec{c}\times\vec{a}\right), \vec{c}\times\left(\vec{b}\times\vec{a}\right)\right]$ का मान बराबर है :

[JEE(Main)June2022]

(A) 0 (B) $-6\vec{a}.\left(\vec{b}\times\vec{c}\right)$ (C) $12\vec{c}.\left(\vec{a}\times\vec{b}\right)$ (D) $-12\vec{b}.\left(\vec{c}\times\vec{a}\right)$

Ans. A

Sol. $\vec{a}\times\left(\vec{b}\times\vec{c}\right)=\left(\vec{a}.\vec{c}\right)\vec{b}-\left(\vec{a}.\vec{b}\right)\vec{c}=3\vec{b}-\vec{c}$

$\vec{b}\times\left(\vec{c}\times\vec{a}\right)=\left(\vec{b}.\vec{a}\right)\vec{c}-\left(\vec{b}.\vec{c}\right)\vec{a}=\vec{c}-2\vec{a}$

$\vec{c}\times\left(\vec{b}\times\vec{a}\right)=\left(\vec{c}.\vec{a}\right)\vec{b}-\left(\vec{c}.\vec{b}\right)\vec{a}=3\vec{b}-2\vec{a}$

$\left[3\vec{b}-\vec{c},\vec{c}-2\vec{a},3\vec{b}-2\vec{a}\right]$

$\left(3\vec{b}-\vec{c}\right).\left[\left(\vec{c}-2\vec{a}\right)\times\left(3\vec{b}-2\vec{a}\right)\right]$

$\left(3\vec{b}-\vec{c}\right).\left[3\left(\vec{c}\times\vec{b}\right)-2\left(\vec{c}\times\vec{a}\right)-6\left(\vec{a}\times\vec{b}\right)\right]$

$-6\left[\vec{b}\vec{c}\vec{a}\right]+6\left[\vec{c}\vec{a}\vec{b}\right]$

$=0$

53. If the lines $\vec{r}=\left(\hat{i}-\hat{j}+\hat{k}\right)+\lambda\left(3\hat{j}-\hat{k}\right)$ and $\vec{r}=\left(\alpha\,\hat{i}-\hat{j}\right)+\mu\left(2\hat{i}-3\hat{k}\right)$ are coplanar, then the distance of the plane containing these two lines from the point $(\alpha, 0,0)$ is : **[JEE(Main)June2022]**

यदि रेखाएँ $\vec{r}=\left(\hat{i}-\hat{j}+\hat{k}\right)+\lambda\left(3\hat{j}-\hat{k}\right)$ तथा $\vec{r}=\left(\alpha\,\hat{i}-\hat{j}\right)+\mu\left(2\hat{i}-3\hat{k}\right)$ सह–तलीय हैं, तो इन रेखाओं से होकर जाने वाले समतल की बिन्दु $(\alpha, 0,0)$ से दूरी है : **[JEE(Main)June2022]**

(A) $\frac{2}{9}$ (B) $\frac{2}{11}$ (C) $\frac{4}{11}$ (D) 2

Ans. B

Sol. $\vec{r}=(1,3\lambda-1,1-\lambda)$

$\vec{r}=(\alpha+2\mu,-1,-3\mu)$

$$\begin{bmatrix}\alpha-1 & 0 & -1\\ 0 & 3 & -1\\ 2 & 0 & -3\end{bmatrix}=0$$

$\Rightarrow\alpha=\frac{5}{3}$

$$\vec{n}=\begin{bmatrix}\hat{i} & \hat{j} & \hat{k}\\ 0 & 3 & -1\\ 2 & 0 & -3\end{bmatrix}=\hat{i}(-9)-\hat{j}(2)+\hat{k}(-6)$$

$=-9\hat{i}-2\hat{j}-6\hat{k}$

equation of plane

$9(x-1)+2(y+1)+6(z-1)=0$

$9x+2y+6z=13$

Perpendicular distance from (5/3, 0, 0)

$$=\left|\frac{9\frac{5}{3}+0+0-13}{\sqrt{81+4+36}}\right|=\left|\frac{2}{\sqrt{121}}\right|=\frac{2}{11}$$

54. Let $\vec{a}=\hat{i}+\hat{j}+2\hat{k},\vec{b}=2\hat{i}-3\hat{j}+\hat{k}$ and $\vec{c}=\hat{i}-\hat{j}+\hat{k}$ be three given vectors. Let $\vec{v}$ be a vector in the plane of $\vec{a}$ and $\vec{b}$ whose projection on $\vec{c}$ is $\frac{2}{\sqrt{3}}$. If $\vec{v}.\hat{j}=7$, then $\vec{v}.\left(\hat{i}+\hat{k}\right)$ is equal to : **[JEE(Main)June2022]**

माना तीन सदिश $\vec{a}=\hat{i}+\hat{j}+2\hat{k},\vec{b}=2\hat{i}-3\hat{j}+\hat{k}$ तथा $\vec{c}=\hat{i}-\hat{j}+\hat{k}$ है। माना $\vec{a}$ तथा $\vec{b}$ के समतल में एक सदिश $\vec{v}$ का $\vec{c}$ पर प्रक्षेप $\frac{2}{\sqrt{3}}$ है। यदि $\vec{v}.\hat{j}=7$ है, तो $\vec{v}.\left(\hat{i}+\hat{k}\right)$ बराबर है : **[JEE(Main)June2022]**

(A) 6 (B) 7 (C) 8 (D) 9

Ans. D

Sol. $\vec{v}=x\vec{a}+y\vec{b}$

$\vec{v}=x\left(\hat{i}+\hat{j}+2\hat{k}\right)+y\left(2\hat{i}-3\hat{j}+\hat{k}\right)$

$\frac{\vec{v}.\vec{c}}{|\vec{c}|}=2\sqrt{3}$

$x+2y-(x-3y)+2x+y=2$

$2x+6y=2$

$x+3y=1$ ——(1)

$\vec{v}.\hat{j}=7$

$x-3y=7$ ——(2)

$2y=8 \Rightarrow x=4$

$3y=-3 \Rightarrow y=-1$

$\vec{v}=2\hat{i}+7\hat{j}+7\hat{k}$

$\vec{v}.\left(\hat{i}+k\right)$

$2+7=9$

55. Let $\vec{a}=\hat{i}+\hat{j}-\hat{k}$ and $\vec{c}=2\hat{i}-3\hat{j}+2\hat{k}$ .Then the number of vectors $\vec{b}$ such that $\vec{b}\times\vec{c}=\vec{a}$ and $|\vec{b}|\in\{1,2,.....,10\}$ is: **[JEE(Main)June2022]**

माना $\vec{a}=\hat{i}+\hat{j}-\hat{k}$ तथा $\vec{c}=2\hat{i}-3\hat{j}+2\hat{k}$ हैं। तो सदिशों $\vec{b}$ की संख्या, जिनके लिए $\vec{b}\times\vec{c}=\vec{a}$ तथा $|\vec{b}|\in\{1,2,.....,10\}$ हैं, है: **[JEE(Main)June2022]**

(A) 0 (B) 1 (C) 2 (D) 3

Ans. A

Sol. $\vec{a}=\hat{i}+\hat{j}-\hat{k}$

$\vec{c}=2\hat{i}-3\hat{j}+2\hat{k}$

$\Rightarrow \vec{b}\times\vec{c}\ ||\ \vec{a}$

$\Rightarrow c\perp\vec{a}$

$\Rightarrow \vec{a}.\vec{c}=0$

$\left(\hat{i}+\hat{j}-\hat{k}\right).\left(2\hat{i}-3\hat{j}+2\hat{k}\right)$

$2-3-2\neq 0$

which is coutradiction

So no of vectors $\vec{b}=0$

56. Let $\vec{a}$ and $\vec{b}$ be the vectors along the diagonals of a parallelogram having area $2\sqrt{2}$. Let the angle between $\vec{a}$ and $\vec{b}$ be acute $|\vec{a}|=1$, and $|\vec{a}.\vec{b}|=|\vec{a}\times\vec{b}|$. If $c=2\sqrt{2}\left(\vec{a}\times\vec{b}\right)-2\vec{b}$, then an angle between $\vec{b}$ and $\vec{c}$ is : **[JEE(Main)June2022]**

माना एक समांतर चतुर्भुज, जिसका क्षेत्रफल $2\sqrt{2}$ है, के विकर्णों के अनुदिश सदिश $\vec{a}$ तथा $\vec{b}$ हैं। माना $\vec{a}$ तथा $\vec{b}$ के बीच कोण, न्यून कोण है, $|\vec{a}|=1$ तथा $|\vec{a}.\vec{b}|=|\vec{a}\times\vec{b}|$ हैं। यदि $c=2\sqrt{2}\left(\vec{a}\times\vec{b}\right)-2\vec{b}$ है, तो $\vec{b}$ तथा $\vec{c}$ के बीच एक कोण है : **[JEE(Main)June2022]**

(A) $\frac{\pi}{4}$ (B) $-\frac{\pi}{4}$ (C) $\frac{5\pi}{6}$ (D) $\frac{3\pi}{4}$

Ans. D

Sol. $\cos\theta=\sin\theta$

$\tan\theta=1$

$\theta=\frac{\pi}{4}$

$\frac{1}{2}|\vec{a}\times\vec{b}|=2\sqrt{2}$

$|\vec{a}||\vec{b}|\sin\theta=4\sqrt{2}$

$$\left|\vec{b}\right|.\frac{1}{\sqrt{2}}=4\sqrt{2}$$

$$\left|\vec{b}\right|=8$$

$$\left|\vec{c}\right|=\sqrt{8\left|\vec{a}\times\vec{b}\right|^2+4\left|\vec{b}\right|^2}$$

$$\sqrt{8\left(4\sqrt{2}\right)^2+4(8)^2}$$

$$=\sqrt{32\times8+32\times8}$$

$$=16\sqrt{2}$$

$$\vec{c}=2\sqrt{2}\left(\vec{a}\times\vec{b}\right)-2\vec{b}$$

$$\vec{b}.\vec{c}=0-2\vec{b}.\vec{b}$$

$$\left|\vec{c}\right|\cos\theta=-2\left|\vec{b}\right|$$

$$\cos\theta=\frac{-2(8)}{16\sqrt{2}}$$

$$\cos\theta=\frac{-1}{\sqrt{2}}$$

$$\Rightarrow\theta=\frac{3\pi}{4}$$

57. Let the plane P : $\vec{r}.\vec{a}=d$ contain the line of intersection of two planes $\vec{r}.\left(\hat{i}+3\hat{j}-\hat{k}\right)=6$ and $\vec{r}.\left(-6\hat{i}+5\hat{j}-\hat{k}\right)=7$. If the plane P passes through the point $\left(2,3,\frac{1}{2}\right)$, then the value of $\frac{\left|13\vec{a}\right|^2}{d^2}$ is equal to :

[JEE(Main)June2022]

माना दो समतलतों $\vec{r}.\left(\hat{i}+3\hat{j}-\hat{k}\right)=6$ तथा $\vec{r}.\left(-6\hat{i}+5\hat{j}-\hat{k}\right)=7$ की प्रतिच्छेदन रेखा समतल P : $\vec{r}.\vec{a}=d$ में है। यदि समतल P, बिन्दु $\left(2,3,\frac{1}{2}\right)$ से होकर जाता है, तो $\frac{\left|13\vec{a}\right|^2}{d^2}$ का मान है : **[JEE(Main)June2022]**

(A) 90 (B) 93 (C) 95 (D) 97

Ans. B

Sol. $P_1 : x + 3y - z = 6$

$P_2 : -6x + 5y - z = 7$

Let the required plane is $P_1 + \lambda P_2 = 0$

$\Rightarrow (1 - 6\lambda)x + (3 + 5\lambda)y - z(1 + \lambda) = 6 + 7\lambda$

$\left(2,3,\frac{1}{2}\right)$ will satisfy it

$$\lambda = 1$$

So the plane : $-5x + 8y - 2z - 13 = 0$

$$\Rightarrow \vec{r}.\left(-5\hat{i}+8\hat{j}-2\hat{k}\right)=13$$

$$\frac{|13\vec{a}|^2}{d^2}=\frac{13^2}{13^2}.(25+64+4)=93$$

58. Let $\vec{a}=\alpha\hat{i}+2\hat{j}-\hat{k}$ and $\vec{b}=-2\hat{i}+\alpha\hat{j}+\hat{k}$, where $\alpha \in R$. If the area of the parallelogram whose adjacent sides are represented by the vectors $\vec{a}$ and $\vec{b}$ is $\sqrt{15\left(\alpha^2+4\right)}$, then the value of $2|\vec{a}|^2+\left(\vec{a}.\vec{b}\right)|\vec{b}|^2$ is equal to :

[JEE(Main)June2022]

माना $\vec{a}=\alpha\hat{i}+2\hat{j}-\hat{k}$ तथा $\vec{b}=-2\hat{i}+\alpha\hat{j}+\hat{k}$ हैं, जहाँ $\alpha \in R$ है। यदि समान्तर चतुर्भुज, जिसकी संलग्न भुजाएँ $\vec{a}$ तथा $\vec{b}$ हैं, क्षेत्रफल $\sqrt{15\left(\alpha^2+4\right)}$ हैं, तो $2|\vec{a}|^2+\left(\vec{a}.\vec{b}\right)|\vec{b}|^2$ का मान बराबर है : **[JEE(Main)June2022]**

(A) 10 (B) 7 (C) 9 (D) 14

Ans. D

Sol. Given $\vec{a}=\alpha\hat{i}+2\hat{j}-\hat{k}$

$\vec{b}=-2\hat{i}+\alpha\hat{j}+\hat{k}$

area of parallelogram is $\sqrt{15\left(\alpha^2+4\right)}$

$$\Rightarrow |\vec{a}\times\vec{b}|=\sqrt{15\left(\alpha^2+4\right)}$$

$$\left\|\begin{matrix}\hat{i} & \hat{j} & \hat{k}\\ \alpha & 2 & -1\\ -2 & \alpha & 1\end{matrix}\right\|=\sqrt{15\left(\alpha^2+4\right)}$$

$$\Rightarrow \sqrt{(2+\alpha)^2+(\alpha-2)^2+\left(\alpha^2+4\right)^2}=\sqrt{15\alpha^2+60}$$

Squaring both sides and solving it.

$\alpha^4 - 5\alpha^2 - 36 = 0$

$\alpha^2 = 9$ and $\alpha^2 = -4$ (not possible)

$\Rightarrow \alpha = \pm 3$

Therefore

$$2|\vec{a}|^2+\left(\vec{a}.\vec{b}\right).|\vec{b}|^2$$

$= 2 \times 14 - 14 = 28 - 14 = 14$

59. Let $\vec{a}$ be a vector which is perpendicular to the vector $3\hat{i}+\frac{1}{2}\hat{j}+2\hat{k}$. If $\vec{a}\times\left(2\hat{i}+\hat{k}\right)=2\hat{i}-13\hat{j}-4\hat{k}$, then the projection of the vector $\vec{a}$ on the vector $2\hat{i}+2\hat{j}+\hat{k}$ is : **[JEE(Main)June2022]**

माना सदिश $3\hat{i}+\frac{1}{2}\hat{j}+2\hat{k}$ के लंबवत एक सदिश $\vec{a}$ है। यदि $\vec{a}\times\left(2\hat{i}+\hat{k}\right)=2\hat{i}-13\hat{j}-4\hat{k}$ है, तो सदिश $\vec{a}$ का सदिश $2\hat{i}+2\hat{j}+\hat{k}$ पर प्रक्षेप है : **[JEE(Main)June2022]**

(A) $\frac{1}{3}$ (B) 1 (C) $\frac{5}{3}$ (D) $\frac{7}{3}$

Ans. C

Sol. Let $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$ and $\vec{a}.\left(3\hat{i}-\frac{1}{2}\hat{j}+2\hat{k}\right)=0$

$\Rightarrow 3a_1-\frac{a_2}{2}+2a_3=0$

$\Rightarrow 6a_1-a_2+4a_3=0$ ———(1)

Also

$\vec{a}\times\left(2\hat{i}+\hat{k}\right)=2\hat{i}-13\hat{j}-4\hat{k}$

$-2a_2\hat{k}+2a_3\hat{j}-a_1\hat{j}+a_2\hat{i}=2\hat{i}-13\hat{j}-4\hat{k}$

By comparison

$a_2=2$ and $a_1-2a_3=13$ ———(2)

from eq. (1) and (2)

$a_1=3$ and $a_3=-5$

$\vec{a}=3\hat{i}+2\hat{j}-5\hat{k}$

Projection of $\vec{a}$ on $2\hat{i}+2\hat{j}+\hat{k}=\frac{6+4-5}{\sqrt{2^2+2^2+1^2}}$

$=\frac{5}{3}$ units

60. Let $\vec{a}=\alpha\hat{i}+3\hat{j}-\hat{k}$, $\vec{b}=3\hat{i}-\beta\hat{j}+4\hat{k}$ and $\vec{c}=\hat{i}+2\hat{j}-2\hat{k}$ where $\alpha, \beta \in R$, be three vectors. If the projection of $\vec{a}$ on $\vec{c}$ is $\frac{10}{3}$ and $\vec{b}\times\vec{c}=-6\hat{i}+10\hat{j}+7\hat{k}$, then the value of $\alpha+\beta$ is equal to : **[JEE(Main)June2022]**

माना तीन सदिश $\vec{a}=\alpha\hat{i}+3\hat{j}-\hat{k}$, $\vec{b}=3\hat{i}-\beta\hat{j}+4\hat{k}$ तथा $\vec{c}=\hat{i}+2\hat{j}-2\hat{k}$ हैं, जहाँ $\alpha, \beta \in R$ हैं। यदि $\vec{a}$ का $\vec{c}$ पर प्रक्षेप $\frac{10}{3}$ है तथा $\vec{b}\times\vec{c}=-6\hat{i}+10\hat{j}+7\hat{k}$ है, तो $\alpha+\beta$ का मान बराबर है : **[JEE(Main)June2022]**

(A) 3 (B) 4 (C) 5 (D) 6

Ans. A

Sol. $\vec{a}=\alpha\hat{i}+3\hat{j}-\hat{k}$, $\vec{b}=3\hat{i}-\beta\hat{j}+4\hat{k}$ $\alpha, \beta \in R$

$\vec{c}=\hat{i}+2\hat{j}-2\hat{k}$

Projection of $\vec{a}$ on $\vec{c}=\dfrac{\vec{a}.\vec{c}}{|\vec{c}|}=\dfrac{10}{3}$

$\alpha+6+2=10$

$$\vec{b}\times\vec{c}=\begin{vmatrix} i & j & k \\ 3 & -\beta & 4 \\ 1 & 2 & -2 \end{vmatrix}=i(2\beta-8)+j(4+6)+k(6+\beta)=-6i+10j+7k$$

$2\beta-8=-6$

$\beta=1$

$\alpha+\beta=3$

61. Let A, B, C be three points whose position vectors respectively are

$\vec{a}=\hat{i}+4\hat{j}+3\hat{k}$

$\vec{b}=2\hat{i}+\alpha\hat{j}+4\hat{k}$, $\alpha \in R$

$\vec{c}=3\hat{i}-2\hat{j}+5\hat{k}$

If α is the smallest positive integer for which $\vec{a}, \vec{b}, \vec{c}$ are noncollinear, then the length of the median, in ΔABC, through A is : **[JEE(Main)June2022]**

माना तीन बिन्दुओं A, B, C के स्थिति सदिश क्रमशः

$\vec{a}=\hat{i}+4\hat{j}+3\hat{k}$

$\vec{b}=2\hat{i}+\alpha\hat{j}+4\hat{k}$, $\alpha \in R$

$\vec{c}=3\hat{i}-2\hat{j}+5\hat{k}$

है। यदि α न्यूनतम धन पूर्णांक है जिसके लिए $\vec{a}, \vec{b}, \vec{c}$ असंरेख है, तो ΔABC की A से माध्यिका की लंबाई है :

[JEE(Main)June2022]

(A) $\dfrac{\sqrt{82}}{2}$ (B) $\dfrac{\sqrt{62}}{2}$ (C) $\dfrac{\sqrt{69}}{2}$ (D) $\dfrac{\sqrt{66}}{2}$

Ans. A

Sol. Given $\overrightarrow{AB} \| \overrightarrow{AC}$ if $\dfrac{1}{2}=\dfrac{\alpha-4}{-6}=\dfrac{1}{2}$

$\Rightarrow \alpha=1$

$\therefore \vec{a}, \vec{b}, \vec{c}$ are non-collineal for $\alpha = 2$ (smallest positive integel)

Mid point of $BC = M\left(\frac{5}{2}, 0, \frac{9}{2}\right)$

$AM = \sqrt{\frac{9}{4}+16+\frac{9}{4}} = \frac{\sqrt{82}}{2}$

62. Let θ be the angle between the vectors $\vec{a}$ and $\vec{b}$, where $|\vec{a}|=4, |\vec{b}|=3$ and $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{3}\right)$. Then $\left|\left(\vec{a}-\vec{b}\right)\times\left(\vec{a}+\vec{b}\right)\right|^2 + 4\left(\vec{a}.\vec{b}\right)^2$ is equal to _____. **[JEE(Main)June2022]**

माना सदिशों $\vec{a}$ तथा $\vec{b}$ के बीच कोण θ है, जहाँ $|\vec{a}|=4, |\vec{b}|=3$ तथा $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{3}\right)$ है। तो $\left|\left(\vec{a}-\vec{b}\right)\times\left(\vec{a}+\vec{b}\right)\right|^2 + 4\left(\vec{a}.\vec{b}\right)^2$ बराबर है _____ **[JEE(Main)June2022]**

Ans. 576

Sol. $|\vec{a}|=4, |\vec{b}|=3 \quad \theta \in \left(\frac{\pi}{4}, \frac{\pi}{3}\right)$

$|(\vec{a}-\vec{b})\times(\vec{a}+\vec{b})|^2 + 4(\vec{a}\cdot\vec{b})^2$

$|\vec{a}\times\vec{b}-\vec{b}\times\vec{a}|^2 + 4a^2b^2\cos^2\theta$

$4|\vec{a}\times\vec{b}|^2 + 4a^2b^2\cos^2\theta$

$4a^2b^2\sin^2\theta + 4a^2b^2\cos^2\theta$

$4a^2b^2 = 4\times16\times9 = 576$

63. Let the lines

$L_1\ \vec{r} = \lambda\left(\hat{i}+2\hat{j}+3\hat{k}\right), \lambda \in R$

$L_2\ \vec{r} = \left(\hat{i}+3\hat{j}+\hat{k}\right)+\mu\left(\hat{i}+\hat{j}+5\hat{k}\right); \mu \in R$, intersects at the point S. If a plane $ax + by - z + d = 0$ passes through S and is parallel to both the lines L_1 and L_2, then the value of $a + b + d$ is equal to _____. **[JEE(Main)June2022]**

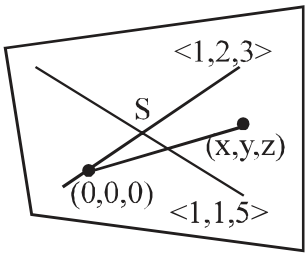
माना रेखाएँ

$L_1\ \vec{r} = \lambda\left(\hat{i}+2\hat{j}+3\hat{k}\right), \lambda \in R$

$L_2\ \vec{r} = \left(\hat{i}+3\hat{j}+\hat{k}\right)+\mu\left(\hat{i}+\hat{j}+5\hat{k}\right); \mu \in R$ बिन्दु S पर काटती हैं। यदि एक समतल $ax + by - z + d = 0$, बिन्दु S से होकर जाता है तथा दोनों रेखाओं L_1 तथा L_2 के समांत है, तो $a + b + d$ का मान बराबर है _____ **[JEE(Main)June2022]**

Ans. 5

Sol. Both the lines lie in the same plane



$\therefore$ equation of the plane

$$\begin{vmatrix} x & y & z \\ 1 & 2 & 3 \\ 1 & 1 & 5 \end{vmatrix} = 0$$

$\Rightarrow 7x - 2y - z = 0$

$\therefore \quad a + b + d = 5$

64. Let $\vec{b} = \hat{i} + \hat{j} + \lambda\hat{k}, \lambda \in R$. If $\vec{a}$ is a vector such that $\vec{a} \times \vec{b} = 13\hat{i} - \hat{j} - 4\hat{k}$ and $\vec{a} \cdot \vec{b} + 21 = 0$, then $(\vec{b} - \vec{a}) \cdot (\hat{k} - \hat{j}) + (\vec{b} + \vec{a}) \cdot (\hat{i} - \hat{k})$ is equal to _____. **[JEE(Main)June2022]**

माना $\vec{b} = \hat{i} + \hat{j} + \lambda\hat{k}, \lambda \in R$ है। यदि एक सदिश $\vec{a}$ के लिए $\vec{a} \times \vec{b} = 13\hat{i} - \hat{j} - 4\hat{k}$ तथा $\vec{a} \cdot \vec{b} + 21 = 0$ हैं, तो, $(\vec{b} - \vec{a}) \cdot (\hat{k} - \hat{j}) + (\vec{b} + \vec{a}) \cdot (\hat{i} - \hat{k})$ बराबर है _____ **[JEE(Main)June2022]**

Ans. 14

Sol. Let $\vec{a} = xi + yj + zk$

$\vec{b} = i + j + \lambda k$

$\vec{a} \cdot \vec{b} = x + y + \lambda z = -21$

$$\vec{a} \times \vec{b} = \begin{vmatrix} i & j & k \\ x & y & z \\ 1 & 1 & \lambda \end{vmatrix}$$

$= (\lambda y - z)\, i - (\lambda x - z)j + (x - y)k = 13i - j - 4k$

$$\Rightarrow \left.\begin{aligned} \lambda y - z &= 13 \\ \lambda x - z &= 1 \\ x - y &= -4 \end{aligned}\right\} \Rightarrow \begin{aligned} \lambda &= 3 \\ x &= -2 \\ y &= 2 \\ z &= -7 \end{aligned}$$

$\vec{a} = -2i + 2j - 7k$

$\vec{b} = i + j + 3k$

$\vec{b}-\vec{a}=3i-j+10k$

$\vec{b}+\vec{a}=-i+3j-4k$

$\left(\vec{b}.\vec{a}\right).(k-j)+\left(\vec{b}+\vec{a}\right).(i-k)=(10+1)+(-1+4)$

$=14$

65. If $\vec{a}=2\hat{i}+\hat{j}+3\hat{k}$, $\vec{b}=3\hat{i}+3\hat{j}+\hat{k}$ and $\vec{c}=c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$ are coplanar vectors and $\vec{a}.\vec{c}=5$, $\vec{b}\perp\vec{c}$ then $122(c_1+c_2+c_3)$ is equal to _____. **[JEE(Main)June2022]**

यदि $\vec{a}=2\hat{i}+\hat{j}+3\hat{k}$, $\vec{b}=3\hat{i}+3\hat{j}+\hat{k}$ तथा $\vec{c}=c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$ के सह–तलीय है $\vec{a}.\vec{c}=5$ है तथा $\vec{b}\perp\vec{c}$ के लंबवत है तब $122(c_1+c_2+c_3)$ का मान है _____ **[JEE(Main)June2022]**

Ans. 150

Sol. $\bar{a}\cdot\bar{c}=5\Rightarrow 2c_1+c_2+3c_3=5$ _____(1)

$\bar{b}\cdot\bar{c}=0\Rightarrow 3c_1+3c_2+c_3=0$ _____(2)

And $\left[\bar{a}\ \bar{b}\ \bar{c}\right]=0\Rightarrow\begin{vmatrix}c_1 & c_2 & c_3\\ 2 & 1 & 3\\ 3 & 3 & 1\end{vmatrix}=0$

$\Rightarrow 8c_1-7c_2-3c_3=0$ _____(3)

By solving (1), (2), (3) we get

$c_1=\frac{10}{122}, c_2=\frac{-85}{122}, c_3=\frac{225}{122}$

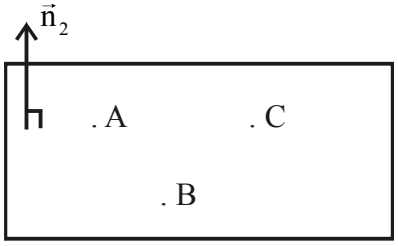
$\therefore\ 122(c_1+c_2+c_3)=150$

66. Let $P_1:\vec{r}.\left(2\hat{i}+\hat{j}-3\hat{k}\right)=4$ be a plane. Let P_2 be another plane which passes through the points (2, –3, 2), (2, –2, –3) and (1, –4, 2). If the direction ratios of the line of intersection of P_1 and P_2 be 16, α, β, then the value of $\alpha+\beta$ is equal to _____. **[JEE(Main)June2022]**

माना एक समतल $P_1:\vec{r}.\left(2\hat{i}+\hat{j}-3\hat{k}\right)=4$ है। माना P_2 एक अन्य समतल है जो बिन्दुओं (2, –3, 2), (2, –2, –3) तथा (1, –4, 2) से होकर जाता है। यदि P_1 तथा P_2 की प्रतिच्छेदन रेखा के दिक् अनुपात 16, α, β हैं, तो $\alpha+\beta$ का मान बराबर है _____ **[JEE(Main)June2022]**

Ans. 28

Sol. $P_1:\vec{r}.\left(2\hat{i}+\hat{j}-3\hat{k}\right)=4\ \Rightarrow\vec{n}_1=2i+j-3k$



P_2 : A (2, –3, 2), B (2, –2, –3), C (1, –4, 2)

$$\Rightarrow \vec{n}_2 = \overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} i & j & k \\ 0 & 1 & -5 \\ -1 & -1 & 0 \end{vmatrix} = -5i + 5j + k(1)$$

direction vector of line of intersection of planes P_1 and P_2

$$\vec{n}_1 \times \vec{n}_2 \begin{vmatrix} i & j & k \\ 2 & 1 & -3 \\ -5 & 5 & 1 \end{vmatrix} = +16i + 13j + 15k$$

$drs \equiv 16, \alpha, \beta \equiv 16, 13, 15 \Rightarrow \alpha + \beta = 28$ Ans.

67. Let $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{b} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{c}$ be a vector such that $\vec{a} + (\vec{b} \times \vec{c}) = \vec{0}$ and $\vec{b}.\vec{c} = 5$. Then the value of $3(\vec{c}.\vec{a})$ is equal to _____. (*Data error in question) **[JEE(Main)June2022]**

माना $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{b} = \hat{i} + \hat{j} + \hat{k}$ है तथा एक सदिश $\vec{c}$ इस प्रकार है कि $\vec{a} + (\vec{b} \times \vec{c}) = \vec{0}$ तथा $\vec{b}.\vec{c} = 5$ हैं। तो $3(\vec{c}.\vec{a})$ का मान बराबर है _____. (*Data error in question) **[JEE(Main)June2022]**

Ans. 0

Sol. $\vec{a} \times (\vec{b} \times \vec{c}) = 0$

$\vec{c}.\vec{a} + \vec{c}.(\vec{a} \times \vec{c}) = 0$

$\vec{c}.\vec{a} + 0 = 0$

$\therefore 3(\vec{c}.\vec{a}) = 0$

3D Geometry

68. If the shortest distance between the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{\lambda}$ and $\frac{x-2}{1} = \frac{y-4}{4} = \frac{z-5}{5}$ is $\frac{1}{\sqrt{3}}$, then the sum of all possible values of λ is : **[JEE(Main)June2022]**

यदि रेखाओं $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{\lambda}$ तथा $\frac{x-2}{1} = \frac{y-4}{4} = \frac{z-5}{5}$ के बीच न्यूनतम दूरी $\frac{1}{\sqrt{3}}$ है, तो λ के सभी संभव मानों का योगफल है : **[JEE(Main)June2022]**

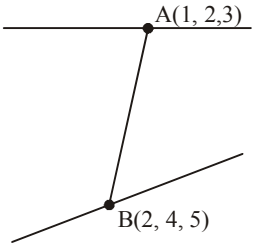
(A) 16 (B) 6 (C) 12 (D) 15

Ans. A

Sol. $\vec{p} = 2i + 3j + \lambda k$

$\vec{q} = i + 4j + 5k$

$\overrightarrow{AB} = i + 2j + 2k$



minimum distance

$$\left|\frac{\overrightarrow{AB}.(\vec{p}\times\vec{q})}{\vec{p}\times\vec{q}}\right| = \frac{1}{\sqrt{3}}$$

$$\frac{\left|\overrightarrow{AB}.(\vec{p}\times\vec{q})\right|}{|\vec{p}\times\vec{q}|} = \frac{1}{\sqrt{3}}$$

$$\overrightarrow{AB}.(\vec{p}\times\vec{q}) = \begin{vmatrix} 1 & 2 & 2 \\ 2 & 3 & \lambda \\ 1 & 4 & 5 \end{vmatrix} = 5 - 2\lambda$$

$$|\vec{p}\times\vec{q}| = |i(15-4\lambda) - j(10-\lambda) + k(5)|$$

$= 17\lambda^2 - 140\lambda + 350$

Now

$$\frac{5-2\lambda}{\sqrt{17\lambda^2 - 140\lambda + 350}} = \frac{1}{\sqrt{3}}$$

$\lambda^2 - 16\lambda + 55 = 0$

Sum of roots = 16

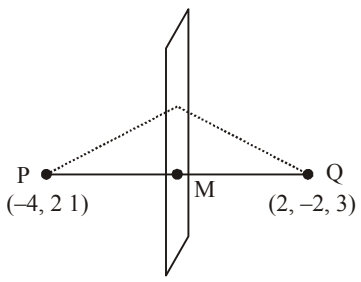
69. Let the points on the plane P be equidistant from the points (–4, 2, 1) and (2, –2, 3). Then the acute angle between the plane P and the plane $2x + y + 3z = 1$ is : **[JEE(Main)June2022]**

माना समतल P के बिन्दु, बिन्दुओं (–4, 2, 1) तथा (2, –2, 3) के बराबर दूरी पर है। तो समतल P तथा समतल $2x + y + 3z = 1$ के बीच न्यूनकोण है : **[JEE(Main)June2022]**

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{5\pi}{12}$

Ans. C

Sol.



M (–1, 0, 2) is point on plane

Normal $\overrightarrow{PQ} = \vec{n} = 6i - 4j + 2k$

equation of required plane

$\vec{r}.\vec{n} = \vec{a}.\vec{n}$

$\left(x\hat{i} + y\hat{j} + z\hat{k}\right).\left(6\hat{i} - 4\hat{j} + 2\hat{k}\right) = -6 + 4$

$6x - 4y + 2z + 2 = 0$

angle between two plane

$$\cos\theta = \left|\frac{12 - 4 + 6}{\sqrt{36 + 16 + 4} \times \sqrt{4 + 1 + 9}}\right|$$

$$= \left|\frac{14}{\sqrt{56} \times \sqrt{14}}\right|$$

$$\cos\theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{3}$$

70. Let Q be the mirror image of the point P(1, 0, 1) with respect to the plane S : x + y + z = 5. If a line L passing through (1, –1, –1), parallel to the line PQ meets the plane S at R, then QR^2 is equal to :

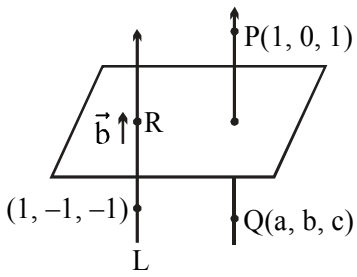
[JEE(Main)June2022]

माना समतल S : x + y + z = 5 के सापेक्ष बिन्दु P(1, 0, 1) का दर्पण प्रतिबिंब Q है। यदि बिन्दु (1, –1, –1) से होकर जाने वाली तथा रेखा PQ के समांतर रेखा L, समतल S को बिन्दु R पर मिलती है, तो QR^2 बराबर है : **[JEE(Main)June2022]**

(A) 2 (B) 5 (C) 7 (D) 11

Ans. B

Sol.



Let parallel vector of $L = \vec{b}$

mirror image of Q on given plane x + y + z = 5

$$\frac{a-1}{1} = \frac{b-0}{1} = \frac{c-1}{1} = \frac{-2(2-5)}{3}$$

$a = 3, b = 2, c = 3$

$Q \equiv (3,2,3)$

$\because \quad \vec{b} \parallel \overrightarrow{PQ}$

so, $\vec{b} = (1,1,1)$

Equation of line $L : \frac{x-1}{1} = \frac{y+1}{1} = \frac{z+1}{1}$

Let point R, $(\lambda+1, \lambda-1, \lambda-1)$

lying on plane $x + y + z = 5$

so, $3\lambda - 1 = 5$

$\Rightarrow \quad \lambda = 2$

Point R is (3, 1, 1)

$QR^2 = 5$

71. Let P be the plane passing through the intersection of the planes $\vec{r}.\left(\hat{i}+3\hat{j}-\hat{k}\right)=5$ and $\vec{r}.\left(2\hat{i}-\hat{j}+\hat{k}\right)=3$ and the point (2, 1, –2). Let the position vectors of the points X and Y be $\hat{i}-2\hat{j}+4\hat{k}$ and $5\hat{i}-\hat{j}+2\hat{k}$ respectively. Then the points : **[JEE(Main)June2022]**

(A) X and X + Y are on the same side of P

(B) Y and Y – X are on the opposite side of P

(C) X and Y are on the opposite side of P

(D) X + Y and X – Y are on the same side of P

माना समतलों $\vec{r}.\left(\hat{i}+3\hat{j}-\hat{k}\right)=5$ तथा $\vec{r}.\left(2\hat{i}-\hat{j}+\hat{k}\right)=3$ के प्रतिच्छेदन तथा बिन्दु (2, 1, –2) से होकर जाने वाला समतल P है। माना बिन्दुओं X तथा Y के स्थिति सदिश क्रमशः $\hat{i}-2\hat{j}+4\hat{k}$ तथा $5\hat{i}-\hat{j}+2\hat{k}$ हैं। तो बिन्दु : **[JEE(Main)June2022]**

(A) X तथा X + Y समतल P के एक ही ओर हैं

(B) Y तथा Y – X, समतल P के भिन्न ओर हैं

(C) X तथा Y, समतल P के भिन्न ओर हैं

(D) X + Y तथा X – Y, समतल P के एक ही ओर हैं

Ans. C

Sol. $P_1 : x + 3y - z - 5 = 0$

$P_2 : 2x - y + z - 3 = 0$

$P : P_1 + \lambda P_2 = O$

$(x + 3y - z - 5) + \lambda(2x - y + z - 3) = 0$

Passes through (2, 1, –2)

$\Rightarrow \lambda = 1$

P : 3x + 2y – 8 = O

X= (1, –2, 4)

Y = (5, –1, 2)

Putting value in plane P

X $\Rightarrow$ 3 – 4 – 8 < 0

Y $\Rightarrow$ 15 – 2 – 8 > 0

X and Y are on the opposite side of P.

72. If the two lines $l_1 : \frac{x-2}{3} = \frac{y+1}{-2}$, z = 2 and $l_2 : \frac{x-1}{1} = \frac{2y+3}{\alpha} = \frac{z+5}{2}$ are perpendicular, then an angle between the lines l_2 and $l_3 : \frac{1-x}{3} = \frac{2y-1}{-4} = \frac{z}{4}$ is : **[JEE(Main)June2022]**

यदि दो रेखाएँ $l_1 : \frac{x-2}{3} = \frac{y+1}{-2}, z = 2$ तथा $l_2 : \frac{x-1}{1} = \frac{2y+3}{\alpha} = \frac{z+5}{2}$ लंबवत है, तो रेखा l_2 तथा $l_3 : \frac{1-x}{3} = \frac{2y-1}{-4} = \frac{z}{4}$ के बीच एक कोण है : **[JEE(Main)June2022]**

(A) $\cos^{-1}\left(\frac{29}{4}\right)$ (B) $\sec^{-1}\left(\frac{29}{4}\right)$

(C) $\cos^{-1}\left(\frac{2}{29}\right)$ (D) $\cos^{-1}\left(\frac{2}{\sqrt{29}}\right)$

Ans. B

Sol. $l_1 : \frac{x-2}{3} = \frac{y+1}{-2} = \frac{z-2}{0}$

$l_2 : \frac{x-1}{1} = \frac{y+\frac{3}{2}}{\frac{\alpha}{2}} = \frac{z+5}{2}$

$l_3 : \frac{x-1}{-3} = \frac{y-\frac{1}{2}}{-2} = \frac{z-0}{4}$

$l_1 \perp l_2 \Rightarrow \frac{|3-\alpha+0|}{\sqrt{13}\sqrt{1+\frac{\alpha^2}{4}+4}} = 0 \Rightarrow \alpha = 3$

angle between l_2 & l_3

$$\cos\theta = \frac{\left|1\times(-3)+(-2)\left(\frac{\alpha}{2}\right)+2\times 4\right|}{\sqrt{1+4+\frac{\alpha^2}{4}}\sqrt{9+16+4}}$$

$$\cos\theta = \frac{\left|-3-\alpha+8\right|}{\sqrt{5+\frac{\alpha^2}{4}}\sqrt{29}}$$

put $\alpha = 3$

$$\cos\theta = \frac{2}{\sqrt{\frac{29}{4}}\sqrt{29}} = \frac{4}{29}$$

$$\theta = \cos^{-1}\left(\frac{4}{29}\right) \Rightarrow \theta = \sec^{-1}\left(\frac{29}{4}\right)$$

73. Let the plane $2x + 3y + z + 20 = 0$ be rotated through a right angle about its line of intersection with the plane $x - 3y + 5z = 8$. If the mirror image of the point $\left(2, -\frac{1}{2}, 2\right)$ in the rotated plane is B(a, b, c), then :

[JEE(Main)June2022]

माना समतल $2x + 3y + z + 20 = 0$ को, इसकी समतल $x - 3y + 5z = 8$ के साथ प्रतिच्छेदन रेखा के सापेक्ष एक समकोण तक घुमाया जाता है। यदि बिंदु $\left(2, -\frac{1}{2}, 2\right)$ का घुमे हुए समतल में दर्पण प्रतिबिंब B(a, b, c) है, तो :

[JEE(Main)June2022]

(A) $\frac{a}{8} = \frac{b}{5} = \frac{c}{-4}$ (B) $\frac{a}{4} = \frac{b}{5} = \frac{c}{-2}$

(C) $\frac{a}{8} = \frac{b}{-5} = \frac{c}{4}$ (D) $\frac{a}{4} = \frac{b}{5} = \frac{c}{2}$

Ans. A

Sol. Let equation of rotated plane be :

$(2x + 3y + z + 20) + \lambda(x - 3y + 5z - 8) = 0$

$(2 + \lambda 1)x + (3 - 3\lambda)y + (1 + 5\lambda)z + 20 - 8\lambda = 0$

Above plane is perpendicular to $2x + 3y + z + 20 = 0$

So, $(2 + \lambda).2 + (3 - 3\lambda).3 + (1 + 5\lambda).1 = 0 \Rightarrow \lambda = 7$

$\Rightarrow$ Equation of rotated plane : $x - 2y + 4z - 4 = 0$

Mirror image of $A\left(2, \frac{-1}{2}, 2\right)$ in rotated plane is B(a, b, c)

Equation of AB : $\frac{x-2}{1} = \frac{y+\frac{1}{2}}{-2} = \frac{z-2}{4} = k$

Let coordinate of B be $\left(2+k, \frac{-1}{2}-2k, 2+4k\right)$

midpoint of AB is $\left(2+\frac{k}{2}, \frac{-1}{2}-k, 2+2k\right)$ which willm lie on the plane $x - 2y + 4z - 4 = 0$

Hence $k = \frac{-2}{3}$

Therefore B is $\left(\frac{4}{3}, \frac{5}{6}, \frac{-2}{3}\right) \equiv \left(\frac{8}{6}, \frac{5}{6}, \frac{-4}{6}\right)$

So, $\frac{a}{8} = \frac{b}{5} = \frac{c}{-4}$

74. If the plane $2x + y - 5z = 0$ is rotated about its line of intersection with the plane $3x - y + 4z - 7 = 0$ by an angle of $\frac{\pi}{2}$, then the plane after the rotation passes throught the point : **[JEE(Main)June2022]**

समतल $2x + y - 5z = 0$ को, इसकी समतल $3x - y + 4z - 7 = 0$ से प्रतिच्छेदन रेखा के सापेक्ष $\frac{\pi}{2}$ कोण तक घुमाने पर प्राप्त समतल किस बिन्दु से होकर जाता है : **[JEE(Main)June2022]**

(A) (2, –2, 0) (B) (–2, 2, 0) (C) (1, 0, 2) (D) (–1, 0, –2)

Ans. C

Sol. $(2x + y - 5z) + \lambda(3x - y + 4z - 7) = 0$

Rotated by $\frac{\pi}{2}$

$(2+3\lambda)x + (1-\lambda)y + (-5+4\lambda)z - 7\lambda = 0$

& $2x + y - 5z = 0$

$\Rightarrow 2(2+3\lambda) + 1 - (1-\lambda) - 5(-5+4\lambda) - 7\lambda = 0$

$30 = 15\lambda$

$\lambda = 2$

Required plane : $8x - y + 32 - 14 = 0$

75. If two straight lines whose direction cosines are given by the relations $l+m-n=0, 3l^2+m^2+cnl=0$ are parallel, then the positive value of c is : **[JEE(Main)June2022]**

यदि दो सरल रेखाएँ, जिनकी दिक्कोज्याएँ $l+m-n=0, 3l^2+m^2+cnl=0$ द्वारा दी गई हैं, समांतर हैं, तो c का धनात्मक मान है: **[JEE(Main)June2022]**

(A) 6 (B) 4 (C) 3 (D) 2

Ans. A

Sol. $l + m - n = 0$

$3l^2 + m^2 + nlc = 0$ ——(1)

$m = n - l$

$m^2 = n^2 + l^2 - 2nl$ ——(2)

From (1)

$3\ell^2 + \left(n^2 + \ell^2 - 2n\ell\right) + n\ell c = 0$

divide by n^2

$3\frac{\ell^2}{n^2} + 1 + \frac{\ell^2}{n^2} - \frac{2\ell}{n} + \frac{\ell c}{n} = 0$

$4\frac{\ell^2}{n^2} + \frac{\ell}{n}(c-2) + 1 = 0$

lines are parallel so

$\frac{\ell_1}{\ell_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2}$

$\Rightarrow \frac{\ell_1}{n_1} = \frac{\ell_2}{n_2} \Rightarrow$ equal roots

$\Rightarrow D = 0$

$(c - 2)^2 - 4(4)(1) = 0$

$c = 6$ OR -2

$c = 6$

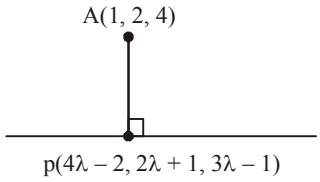
76. Let the foot of the perpendicular from the point (1, 2, 4) on the line $\frac{x+2}{4} = \frac{y-1}{2} = \frac{z+1}{3}$ be P. Then the distance of P from the plane $3x + 4y + 12z + 23 = 0$ is : **[JEE(Main)June2022]**

माना बिन्दु (1, 2, 4) से रेखा $\frac{x+2}{4} = \frac{y-1}{2} = \frac{z+1}{3}$ पर लंब का पाद P है। तो P की समतल $3x + 4y + 12z + 23 = 0$ से दूरी है : **[JEE(Main)June2022]**

(A) 5 (B) 50/13 (C) 4 (D) 63/13

Ans. A

Sol.



$\overrightarrow{AP}\cdot\left(4\hat{i}+2\hat{j}+3\hat{k}\right)=0$

$4(4\lambda-3)+2(2\lambda-1)+3(3\lambda-5)=0$

$29\lambda = 29$

$\lambda = 1$

P(2, 3, 2)

$d=\left|\frac{6+12+24+23}{\sqrt{3^2+4^2+12^2}}\right|=\frac{65}{13}=5$

77. The shortest distance between the line $\frac{x-3}{2}=\frac{y-2}{3}=\frac{z-1}{-1}$ and $\frac{x+3}{2}=\frac{y-6}{1}=\frac{z-5}{3}$, is :

[JEE(Main)June2022]

रेखाओं $\frac{x-3}{2}=\frac{y-2}{3}=\frac{z-1}{-1}$ तथा $\frac{x+3}{2}=\frac{y-6}{1}=\frac{z-5}{3}$ के बीच न्यूनतम दूरी है : **[JEE(Main)June2022]**

(A) $\frac{18}{\sqrt{5}}$ (B) $\frac{22}{3\sqrt{5}}$ (C) $\frac{46}{3\sqrt{5}}$ (D) $6\sqrt{3}$

Ans. A

Sol. $SD=\left|\frac{\left(6\hat{i}-4\hat{j}-4\hat{k}\right).\left(10\hat{i}-8\hat{j}-4\hat{k}\right)}{\sqrt{10^2+8^2+4^2}}\right|$

$\vec{n}_1\times\vec{n}_2=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&3&-1\\2&1&3\end{vmatrix}$

$=\frac{60+32+16}{2\sqrt{45}}$

$=10\hat{i}-8\hat{j}-4\hat{k}$

$=\frac{54}{3\sqrt{5}}=\frac{18}{\sqrt{5}}$

78. If two distinct points Q, R lie on the line of intersection of the planes –x + 2y – z = 0 and 3x – 5y + 2z = 0 and PQ = PR = $\sqrt{18}$ where the point P is (1, –2, 3), then the area of the triangle PQR is equal to : **[JEE(Main)June2022]**

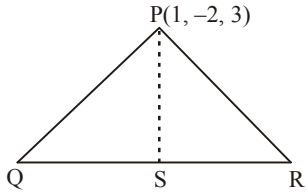
यदि समतलों –x + 2y – z = 0 तथा 3x – 5y + 2z = 0 की प्रतिच्छेदन रेखा पर दो भिन्न बिन्दु Q, R हैं तथा PQ = PR = $\sqrt{18}$ है, जहां P बिन्दु (1, –2, 3) है, तो त्रिभुज PQR का क्षेत्रफल बराबर है : **[JEE(Main)June2022]**

(A) $\frac{2}{3}\sqrt{38}$ (B) $\frac{4}{3}\sqrt{38}$ (C) $\frac{8}{3}\sqrt{38}$ (D) $\sqrt{\frac{152}{3}}$

Ans. B

Sol. Plane are $-x + 2y - z = 0$

$3x - 5y + 2z = 0$



Let R (k, k, k) Now PQ = $\sqrt{18}$

$\Rightarrow (k-2)^2 + (k+2)^2 + (k-3)^2 = 18$

$\Rightarrow 3k^2 - 4k - 4 = 0$

$\Rightarrow k = -\frac{2}{3}, 2$

So $Q\left(-\frac{2}{3}, -\frac{2}{3}, -\frac{2}{3}\right)$ and R(2, 2, 2)

mid point $S \equiv \left(\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right)$

base of $\Delta PQR = QR = \sqrt{3}.\frac{8}{3}$, height $= \frac{\sqrt{114}}{3} = PS$

area $= \frac{1}{2} \times \sqrt{3} \times \frac{8}{3} \times \frac{\sqrt{114}}{3} = \frac{4}{3}\sqrt{38}$

79. The acute angle between the planes P_1 and P_2, when P_1 and P_2 are the planes passing through the intersection of the planes $5x + 8y + 13z - 29 = 0$ and $8x - 7y + z - 20 = 0$ and the points (2, 1, 3) and (0, 1, 2), respectively, is : **[JEE(Main)June2022]**

मान समतलों $5x + 8y + 13z - 29 = 0$, $8x - 7y + z - 20 = 0$ के प्रतिच्छेदन और बिन्दुओं (2, 1, 3) तथा (0, 1, 2) से होकर जाने वाले दो समतल क्रमश: P_1 तथा P_2 हैं। तो समतलों P_1 तथा P_2 के बीच न्यून कोण है : **[JEE(Main)June2022]**

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{12}$

Ans. A

Sol. The required plane $P_1 + \lambda P_2 = 0$

$\Rightarrow (5 + 8\lambda)x + (8 - 7\lambda)y + (13 + \lambda)z = 29 + 20\lambda$

(2, 1, 3) will satisfy if $\lambda = \frac{7}{2}$

and plane : $2x - y + z = 6$ $(0, 1, 2)$ will satisfy if $\lambda = \frac{1}{5}$

So angle between then and plane : $x + y + 2z = 5$

$$\cos\theta = \left|\frac{2-1+2}{\sqrt{6}\sqrt{6}}\right| = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{3}$$

80. Let the plane $ax + by + cz = d$ pass through $(2, 3, -5)$ and is perpendicular to the planes $2x + y - 5z = 10$ and $3x + 5y - 7z = 12$. If a, b, c, d are integers $d > 0$ and gcd $(|a|, |b|, |c|, d) = 1$, then the value of $a + 7b + c + 20d$ is equal to : **[JEE(Main)June2022]**

माना समतल $ax + by + cz = d$ बिन्दु $(2, 3, -5)$ से होकर जाता है तथा समतलों $2x + y - 5z = 10$ और $3x + 5y - 7z = 12$ के लंबवत है। यदि a, b, c, d पूर्णांक हैं, $d > 0$ है तथा gcd $(|a|, |b|, |c|, d) = 1$ है, तो $a + 7b + c + 20d$ का मान बराबर है :

[JEE(Main)June2022]

(A) 18 (B) 20 (C) 24 (D) 22

Ans. D

Sol. Equation of plane passing through $(2, 3, -5)$ and perpendicular to the palnes $2x + y - 5z = 10$ and $3x + 5y - 7z = 12$ is

$$\left(\vec{r} - \vec{a}\right).\vec{N} = 0 \Rightarrow \begin{vmatrix} x-2 & y-3 & z+5 \\ 2 & 1 & -5 \\ 3 & 5 & -7 \end{vmatrix} = 0$$

$-18x + y - 7z = 2$
$a = -18, b = 1, c = -7, d = 2$
$\because$ gcd of $(|a|, |b|, |c|, d) = 1$
$\therefore a + 7b + c + 20d = 22$

81. If the mirror image of the point $(2, 4, 7)$ in the plane $3x - y + 4z = 2$ is (a, b, c), then $2a + b + 2c$ is equal to: **[JEE(Main)June2022]**

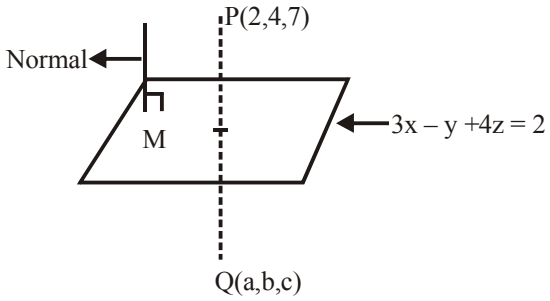
यदि बिन्दु $(2, 4, 7)$ का समतल $3x - y + 4z = 2$ में दर्पण प्रतिबिंब (a, b, c) है, तो $2a + b + 2c$ बराबर है :

[JEE(Main)June2022]

(A) 54 (B) 50 (C) –6 (D) –42

Ans. C

Sol.



$M\left(\frac{a+2}{2},\frac{b+4}{2},\frac{c+7}{2}\right)$

lies on plane

$3a - b + 4c = -26$ .....(1)

drs of PQ $a-2, b-4, c-7$

drs of normal plane $3, -1, 4$

$\frac{a-2}{3}=\frac{b-4}{-1}=\frac{c-7}{4}=k$ (say)

$a = 3k+2, b = 4-k, c = 4k+7$

Put in (1)

$13k = -28$ .....(2)

Now $2a + b + 2c = 13k + 22 = -28 + 22 = -6$ Ans.

82. Let $\frac{x-2}{3}=\frac{y+1}{-2}=\frac{z+3}{-1}$ lie on the plane $px - qy + z = 5$, for some $p, q \in R$. The shortest distance of the plane from the origin is : **[JEE(Main)June2022]**

माना $\frac{x-2}{3}=\frac{y+1}{-2}=\frac{z+3}{-1}$ किसी $p, q \in R$ के लिए समतल $px - qy + z = 5$ पर स्थित है। मूल बिन्दु से इस समतल की न्यूनतम दूरी है : **[JEE(Main)June2022]**

(A) $\sqrt{\frac{3}{109}}$ (B) $\sqrt{\frac{5}{142}}$ (C) $\frac{5}{\sqrt{71}}$ (D) $\frac{1}{\sqrt{142}}$

Ans. B

Sol. Given equation of line : $\frac{x-2}{3}=\frac{y+1}{-2}=\frac{z+3}{-1}=\lambda$

$(3\lambda + 2, -2\lambda - 1, -\lambda - 3)$ lies on plane $px - qy + z = 5$

$p(3\lambda + 2) - q(-2\lambda - 1) + (-\lambda - 3) = 5$

$\lambda(3p + 2q - 1) + (2p + q - 8) = 0$

$\therefore\ 3p + 2q - 1 = 0$ _____(1)

$2p + q - 8 = 0$ _____(2)

Solve (1) and (2) we get
$p = 15, \quad q = -22$
equation of plane : $15x + 22y + z - 5 = 0$

$\therefore$ Shortest distance from origin $= \left|\frac{0+0+0-5}{\sqrt{15^2+22^2+1}}\right|$

$= \frac{5}{\sqrt{710}} = \sqrt{\frac{5}{142}}$

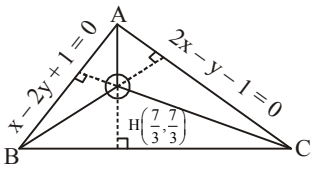
83. The distance of the origin from the centroid of the triangle whose two sides have the equations $x - 2y + 1 = 0$ and $2x - y - 1 = 0$ and whose orthocenter is $\left(\frac{7}{3}, \frac{7}{3}\right)$ is : **[JEE(Main)June2022]**

एक त्रिभुज की दो भुजाओं के समीकरण $x - 2y + 1 = 0$ तथा $2x - y - 1 = 0$ इसका लंबकेन्द्र $\left(\frac{7}{3}, \frac{7}{3}\right)$ है। इस त्रिभुज के केन्द्र से मूल बिन्दु की दूरी है : **[JEE(Main)June2022]**

(A) $\sqrt{2}$ (B) 2 (C) $2\sqrt{2}$ (D) 4

Ans. C

Sol. Given equation of sides of ΔABC
AB : $x - 2y + 1 = 0$
AC : $2x - y - 1 = 0 \Rightarrow A(1,1)$
Altitude from B is $BH = x + 2y - 7 = 0 \Rightarrow B(3, 2)$
Altitude from C is $CH = 2x + y - 7 = 0 \Rightarrow C(2, 3)$
Centroid of $\Delta ABC = C(2, 2)$



then, distance between origin and centroid of triangle $= \sqrt{(2-0)^2 + (2-0)^2} = 2\sqrt{2}$

84. Let Q be the mirror image of the point P(1, 2, 1) with respect to the plane $x + 2y + 2z = 16$. Let T be a plane passing through the point Q and contains the line $\vec{r} = -\hat{k} + \lambda\left(\hat{i} + \hat{j} + 2\hat{k}\right)$, $\lambda \in R$. Then, which of the following points lies on T? **[JEE(Main)June2022]**

माना समतल $x + 2y + 2z = 16$. के सापेक्ष बिन्दु P(1, 2, 1) का दर्पण प्रतिबिंब Q है। माना T एक समतल है जो बिन्दु Q से होकर जाता है तथा रेखा $\vec{r} = -\hat{k} + \lambda\left(\hat{i} + \hat{j} + 2\hat{k}\right)$, $\lambda \in R$ समतल T में स्थित है। तो निम्न में से कौनसा बिन्दु T पर स्थित है? **[JEE(Main)June2022]**

(A) (2, 1, 0) (B) (1, 2, 1) (C) (1, 2, 2) (D) (1, 3, 2)

Ans. B

Sol. Image of P(1, 2, 1) in plane $x + 2y + 2z = 16$

$$\frac{x-1}{1}=\frac{y-2}{2}=\frac{z-1}{2}=\frac{-2\left[1+2\times2+2\times1-16\right]}{1^2+2^2+2^2}$$

$$\frac{x-1}{1}=\frac{y-2}{2}=\frac{z-1}{2}=2$$

Q(3, 6, 5) is image of point P(1, 2, 1) in the plane x + 2y + 2z = 16.

Given $\vec{r}=-\hat{k}+\lambda\left(\hat{i}+\hat{j}+2\hat{k}\right)$ plane passing through Q.

$\dot{a}\left(3,8,5\right)$

$\dot{A}\left(0,0,-1\right)$

$\overrightarrow{AQ}=3\hat{i}+6\hat{j}+6\hat{k}=3\left(\hat{i}+2\hat{j}+2\hat{k}\right)$

$\vec{n}=\left(\hat{i}\times2\hat{j}\times2\hat{k}\right)\times\left(\hat{i}+\hat{j}+2\hat{k}\right)$

$$=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 2 & 2\\ 1 & 1 & 2\end{vmatrix}=2\hat{i}-o\hat{j}-\hat{k}$$

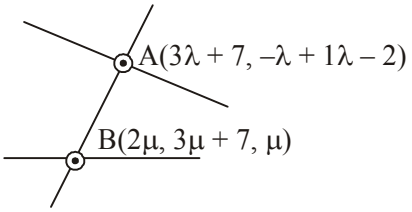
Equation of plane = 2(x – 0)+ O(y – 0) – 1(z + 1) = 0
2x – 2 = 0
So, point lying on the plane is (1, 2, 1)

85. Let a line having direction ratios 1, –4, 2 intersect the lines $\frac{x-7}{3}=\frac{y-1}{-1}=\frac{z+2}{1}$ and $\frac{x}{2}=\frac{y-7}{3}=\frac{z}{1}$ at the points. A and B. Then $(AB)^2$ is equal to _____. **[JEE(Main)June2022]**

माना एक रेखा जिसके दिक अनुपात 1, –4, 2 हैं रेखाओं $\frac{x-7}{3}=\frac{y-1}{-1}=\frac{z+2}{1}$ तथा $\frac{x}{2}=\frac{y-7}{3}=\frac{z}{1}$ को बिन्दुओं A तथा B पर काटती है। तो $(AB)^2$ बराबर है _____ **[JEE(Main)June2022]**

Ans. 84

Sol.



DR's of AB
$(3\lambda-2\mu+7,-\lambda-3\mu-6,\lambda-\mu-2)$

$$\frac{3\lambda-2\mu+7}{1}=\frac{-\lambda-3\mu-6}{-4}=\frac{\lambda-\mu-2}{2}$$

Taking first (2) $-12\lambda+8\mu-28=-\lambda-3\mu-6$
$\lambda-\mu+2=0$
Taking second & third $-2\lambda-6\mu-12=-4\lambda+4\mu+8$

$\lambda - 5\mu - 10 = 0$

After solving above two equation $\lambda = -5$, $\mu = -3$

$A = (-8, 6, -7)$

$B = (-6, -2, -3)$

$(AB)^2 = 4 + 64 + 16 = 84$

86. If the shortest distance between the lines $\vec{r} = \left(-\hat{i} + 3\hat{k}\right) + \lambda\left(\hat{i} - a\hat{j}\right)$ and $\vec{r} = \left(-\hat{j} + 2\hat{k}\right) + \mu\left(\hat{i} - \hat{j} + \hat{k}\right)$ is $\sqrt{\frac{2}{3}}$, then the integral value of a is equal to _____. **[JEE(Main)June2022]**

यदि रेखाओं $\vec{r} = \left(-\hat{i} + 3\hat{k}\right) + \lambda\left(\hat{i} - a\hat{j}\right)$ तथा $\vec{r} = \left(-\hat{j} + 2\hat{k}\right) + \mu\left(\hat{i} - \hat{j} + \hat{k}\right)$ के बीच न्यूनतम दूरी $\sqrt{\frac{2}{3}}$ है, तो a का पूर्णांकीय मान बराबर है _____ **[JEE(Main)June2022]**

Ans. 2

Sol. $a_1 = (-1, 0, 3)$

$a_2 = (0, -1, 2)$

$b_1 = (1, -a, 0)$ dr's of line (1)

$b_2 = (1, -1, 1)$ dr's of line (2)

$$\bar{a}_2 - \bar{a}_1 = (1, -1, -1)$$

$$\bar{b}_1 \times \bar{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & k \\ 1 & -a & 0 \\ 1 & -1 & 1 \end{vmatrix}$$

$$\bar{b}_1 \times \bar{b}_2 = \hat{i}(-a) - \hat{j} + k(a-1)$$

$$\left|\bar{b}_1 \times \bar{b}_2\right| = \sqrt{a^2 + 1 + (a-1)^2}$$

$$a_2 - a_1 . \bar{b}_1 \times \bar{b}_2 = 2 - 2a$$

$$\frac{2(1-a)}{\sqrt{a^2 + 1 + (a-1)^2}} = \sqrt{\frac{2}{3}}$$

Squaring an both the side

After solving $a = 2, \frac{1}{2}$

87. Let l_1 be the line in xy-plane with x and y intercepts $\frac{1}{8}$ and $\frac{1}{4\sqrt{2}}$ respectively and l_2 be the line in zx-plane with x and z intercepts $-\frac{1}{8}$ and $-\frac{1}{6\sqrt{3}}$ respectively. If d is the shortest distance between the line l_1 and l_2, then d^{-2} is equal to _____. **[JEE(Main)June2022]**

माना रेखा l_1 xy-समतल में है तथा इसके x तथा y अंतःखंड $\frac{1}{8}$ और $\frac{1}{4\sqrt{2}}$ है तथा रेखा l_2 zx-समतल में है तथा इसके x तथा z अंतःखंड क्रमशः $-\frac{1}{8}$ और $-\frac{1}{6\sqrt{3}}$ हैं। यदि रेखाओं l_1 तथा l_2 के बीच न्यूनतम दूरी d है, तो d^{-2} बराबर है _____

[JEE(Main)June2022]

Ans. 51

Sol. $l_1 : \frac{x}{\frac{1}{8}} + \frac{y}{\frac{1}{4\sqrt{2}}} = 1$ and $z = 0$

$$l_1 : \frac{x-\frac{1}{8}}{\sqrt{2}} = \frac{y-0}{-2} = \frac{z-0}{0},\ A = \left(\frac{1}{8}, 0, 0\right), \vec{p} = \sqrt{2}i - 2j$$

$$l_2 : \frac{x}{-\frac{1}{8}} + \frac{z}{\frac{-1}{6\sqrt{3}}} = 1 \text{ and } y = 0$$

$$l_2 : \frac{x+\frac{1}{8}}{3\sqrt{3}} = \frac{y-0}{0} = \frac{z-0}{4},\ B = \left(\frac{-1}{8}, 0, 0\right),\ \vec{q} = 3\sqrt{3}i + 4k$$

d = Projection of $\overrightarrow{AB}$ on $\vec{p} \times \vec{q}$

$$\overrightarrow{AB} = \frac{-1}{4}i$$

$$\vec{p} \times \vec{q} = \begin{vmatrix} i & j & k \\ \sqrt{2} & -2 & 0 \\ 3\sqrt{3} & 0 & 4 \end{vmatrix}$$

$$\vec{p} \times \vec{q} == -2\left(4i + 2\sqrt{2}j - 3\sqrt{3}\right)$$

$$d = \left|\frac{\overrightarrow{AB}.(\vec{p} \times \vec{q})}{|\vec{p} \times \vec{q}|}\right|$$

$$d = \frac{1}{\sqrt{51}}$$

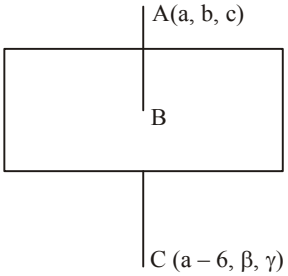
$$d^{-2} = 51$$

88. Let the mirror image of the point (a, b, c) with respect to the plane $3x - 4y + 12z + 19 = 0$ be $(a-6, \beta, \gamma)$. If $a + b + c = 5$, then $7\beta - 9\gamma$ is equal to _____. **[JEE(Main)June2022]**

माना बिन्दु (a, b, c) का समतल $3x - 4y + 12z + 19 = 0$ के सापेक्ष दर्पण प्रतिबिंब $(a-6, \beta, \gamma)$ है। यदि $a + b + c = 5$, है, तो $7\beta - 9\gamma$ बराबर है _____ **[JEE(Main)June2022]**

Ans. 137

Sol. $3x - 4y + 12z + 19 = 0$



Direction ratio of $\overrightarrow{AC} = (-6, \beta - b, \gamma - c)$

D. R. of plane = (3, –4, 12)

So $\dfrac{-6}{3} = \dfrac{\beta - b}{-4} = \dfrac{\gamma - c}{12}$

$\beta - b = 8$ $\qquad$ $\gamma - c = -24$

$\beta = 8 + b$ $\qquad$ $\gamma = c - 24$

Now mid point of AC will lie on given plane

So $B\left(\dfrac{a + a - 6}{2}, \dfrac{b + \beta}{2}, \dfrac{c + \gamma}{2}\right)$

$= \left(a - 3, \dfrac{b + \beta}{2}, \dfrac{c + \gamma}{2}\right)$

$3(a - 3) - 4\left(\dfrac{b + \beta}{2}\right) + 12\left(\dfrac{C + \gamma}{2}\right) + 19 = 0$

$3a - 2b + 6c - 2\beta + 6\gamma + 10 = 0$

$3a - 2b + 6c - 2(8 + b) + 6(c - 24) + 10 = 0$

$3(5 - b - c) - 2b + 6c - 2b + 6c - 150 = 0$

$-7b + 9c - 135 = 0$

$-7(\beta - 8) + 9(\gamma + 24) - 135 = 0$

$7\beta - 9\gamma = 137$

89. Let the image of the point P(1, 2, 3) in the line L : $\dfrac{x - 6}{3} = \dfrac{y - 1}{2} = \dfrac{z - 2}{3}$ be Q. Let R(α, β, γ) be a point that divides internally the line segment PQ in the ratio 1 : 3. Then the value of $22(\alpha + \beta + \gamma)$ is equal to _____.

[JEE(Main)June2022]

माना रेखा L : $\dfrac{x - 6}{3} = \dfrac{y - 1}{2} = \dfrac{z - 2}{3}$ में बिन्दु P(1, 2, 3) का प्रतिबिंब Q है। माना बिन्दु R(α, β, γ), रेखा खण्ड PQ को अंत 1 : 3 के अनुपात में विभाजित करता है। तो $22(\alpha + \beta + \gamma)$ का मान बराबर है _____.

[JEE(Main)June2022]

Ans. 125

Sol. From point P(1, 2, 3), P′ is foot of perpendicular on line $\frac{x-6}{3}=\frac{y-1}{2}=\frac{z-2}{3}=\lambda$

$\therefore$ P′ = $(3\lambda + 6, 2\lambda + 1, 3\lambda + 2)$

d.r's of PP′ $\equiv 3\lambda + 5, 2\lambda - 1, 3\lambda - 1$

as PP′ and line l is perpendicular

$\therefore (3\lambda + 5)3 + 2(2\lambda - 1) + 3(3\lambda - 1) = 0$

$$\Rightarrow \lambda = \frac{-5}{11}$$

$$P' = \left(\frac{51}{11}, \frac{1}{11}, \frac{7}{11}\right)$$

The point dividing PQ in the ratio 1 : 3 will be mid point of P and foot of perpendicular from P on line $\Rightarrow$ R will be mid point of PP′

$$\Rightarrow R = (\alpha, \beta, \gamma) = \left(\frac{\frac{51}{11}+1}{2}, \frac{\frac{1}{11}+2}{2}, \frac{\frac{7}{11}+3}{2}\right)$$

$$(\alpha, \beta, \gamma) = \left(\frac{62}{22}, \frac{23}{22}, \frac{40}{22}\right)$$

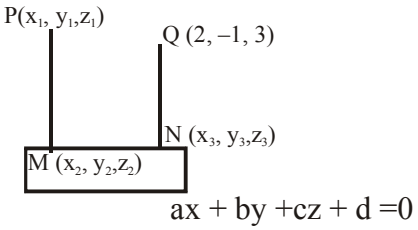
$$22(\alpha, \beta, \gamma) = 22\left(\frac{62+23+40}{22}\right) = 125$$

90. Let d be the distance between the foot of perpendiculars of the points P(1, 2, –1) and Q(2, –1, 3) on the plane $-x + y + z = 1$. Then d^2 is equal to _____. **[JEE(Main)June2022]**

माना बिन्दुओं P(1, 2, –1) तथा Q(2, –1, 3) से समतल $-x + y + z = 1$ पर डाले गए लंबों के पादों के बीच की दूरी d है, तो d^2 बराबर है _____ **[JEE(Main)June2022]**

Ans. 26

Sol.



for $M \equiv \frac{x_2 - x_1}{a} = \frac{y_2 - y_1}{b} = \frac{z_2 - z_1}{c} = -\frac{ax_1 + by_1 + cz_1 + d}{a^2 + b^2 + c^2}$

P(1, 2, –1)

$$M \equiv \frac{x_2 - 1}{1} = \frac{y_2 - 2}{-1} = -\frac{z_2 + 1}{-1} = -\frac{1 - 2 + 1 + 1}{3} = \frac{-1}{3}$$

plane x –y – z + 1 = 0

$M\ (x_2\, y_2\, z_2) \equiv \left(\frac{2}{3}, \frac{7}{3}, \frac{-2}{3}\right)$

Q (2, – 1, 3)

$N \to$ foot of $\perp N(x_3, y_3, z_3)$ when $\frac{x_3-2}{1} = \frac{y_3+1}{-1} = \frac{z_3-3}{-1} = -\frac{1}{3}$

$N\left(\frac{5}{3}, \frac{-2}{3}, \frac{10}{3}\right)$

$MN = d = \sqrt{1^2+3^2+4^2} = \sqrt{26}$

$d^2 = 26$

91. Let ABC be a triangle such that $\overrightarrow{BC} = \vec{a}$, $\overrightarrow{CA} = \vec{b}$, $\overrightarrow{AB} = \vec{c}$, $|\vec{a}| = 6\sqrt{2}$, $|\vec{b}| = 2\sqrt{3}$ and $\vec{b}.\vec{c} = 12$. Consider the statements :

(S1) : $\left|\left(\vec{a}\times\vec{b}\right)+\left(\vec{c}\times\vec{b}\right)\right| - |\vec{c}| = 6\left(2\sqrt{2}-1\right)$

(S2) : $\angle ACB = \cos^{-1}\left(\sqrt{\frac{2}{3}}\right)$

Then : **[JEE(Main)July2022]**

(A) Both (S1) and (S2) are true (B) Only (S1) is true

(C) Only (S2) is true (D) Both (S1) and (S2) are false

माना एक त्रिभुज ABC के लिए $\overrightarrow{BC} = \vec{a}$, $\overrightarrow{CA} = \vec{b}$, $\overrightarrow{AB} = \vec{c}$, $|\vec{a}| = 6\sqrt{2}$, $|\vec{b}| = 2\sqrt{3}$ तथा $\vec{b}.\vec{c} = 12$ हैं :

कथनों :

(S1) : $\left|\left(\vec{a}\times\vec{b}\right)+\left(\vec{c}\times\vec{b}\right)\right| - |\vec{c}| = 6\left(2\sqrt{2}-1\right)$

(S2) : $\angle ACB = \cos^{-1}\left(\sqrt{\frac{2}{3}}\right)$

में : **[JEE(Main)July2022]**

(A) (S1) तथा (S2) दोनों सत्य हैं (B) केवल (S1) सत्य हैं

(C) केवल (S2) सत्य है (D) दोनों (S1) तथा(S2) असत्य हैं

Ans. D

Sol. $\vec{a}+\vec{b}+\vec{c} = 0$

$\vec{b}+\vec{c} = -\vec{a}$

$|\vec{b}|^2+|\vec{c}|^2+2\vec{b}\cdot\vec{c}=|\vec{a}|^2$

$|\vec{c}|^2=36$

$|\vec{c}|=6$

$S1:|\vec{a}\times\vec{b}+\vec{c}\times\vec{b}|-|\vec{c}|$

$|(\vec{a}+\vec{c})\times\vec{b}|-|\vec{c}|$

$|-\vec{b}\times\vec{b}|-|\vec{c}|$

$0-6=-6$

$S2:\vec{a}+\vec{b}+\vec{c}=0$

$\vec{b}+\vec{c}=-\vec{a}$

$|\vec{a}|^2+|\vec{b}|^2-2|\vec{a}||\vec{b}|\cos(\angle ACB)=|\vec{c}|^2$

$\cos(\angle ACB)=\sqrt{\frac{2}{3}}$

92. Let $\vec{a}=\hat{i}-\hat{j}+2\hat{k}$ and let $\vec{b}$ be a vector such that $\vec{a}\times\vec{b}=2\hat{i}-\hat{k}$ and $\vec{a}\,.\,\vec{b}=3$. Then the projection of $\vec{b}$ on the vector $\vec{a}-\vec{b}$ is : **[JEE(Main)July2022]**

माना $\vec{a}=\hat{i}-\hat{j}+2\hat{k}$ तथा एक सदिश $\vec{b}$ के लिए $\vec{a}\times\vec{b}=2\hat{i}-\hat{k}$ तथा $\vec{a}\,.\,\vec{b}=3$ हैं, तो सदिश $\vec{b}$ का सदिश $\vec{a}-\vec{b}$ पर प्रक्षेप है: **[JEE(Main)July2022]**

(A) $\frac{2}{\sqrt{21}}$ (B) $2\sqrt{\frac{3}{7}}$ (C) $\frac{2}{3}\sqrt{\frac{7}{3}}$ (D) $\frac{2}{3}$

Ans. A

Sol. $\vec{a}=\hat{i}-\hat{j}+2\hat{k}$

$\vec{a}\times\vec{b}=2\hat{i}-\hat{k}$

$\vec{a}.\vec{b}=3$

$\left|\vec{a}\times\vec{b}\right|^2+\left|\vec{a}.\vec{b}\right|^2=\left|\vec{a}\right|^2.\left|\vec{b}\right|^2$

$\Rightarrow 5+9=6\left|\vec{b}\right|^2$

$\Rightarrow |b|^2=\frac{7}{3}$

$$\left|\vec{a}-\vec{b}\right|=\sqrt{\left|\vec{a}\right|^2+\left|\vec{b}\right|^2-2\vec{a}.\vec{b}}=\sqrt{\frac{7}{3}}$$

Projection of $\vec{b}$ on $\vec{a}-\vec{b}=\dfrac{\vec{b}.\left(\vec{a}-\vec{b}\right)}{\left|\vec{a}-\vec{b}\right|}$

$$=\frac{\vec{b}.\vec{a}-|\vec{b}|^2}{\left|\vec{a}-\vec{b}\right|}=\frac{3-\frac{7}{3}}{\sqrt{\frac{7}{3}}}$$

$$=\frac{2}{\sqrt{21}}$$

93. Let $\vec{a}=\alpha\hat{i}+\hat{j}-\hat{k}$ and $\vec{b}=2\hat{i}+\hat{j}-\alpha\hat{k}$, $\alpha>0$. If the projection of $\vec{a}\times\vec{b}$ on the vector $-\hat{i}+2\hat{j}-2\hat{k}$ is 30, then α is equal to : **[JEE(Main)July2022]**

माना $\vec{a}=\alpha\hat{i}+\hat{j}-\hat{k}$ तथा $\vec{b}=2\hat{i}+\hat{j}-\alpha\hat{k}$, $\alpha>0$ यदि $\vec{a}\times\vec{b}$ का सदिश $-\hat{i}+2\hat{j}-2\hat{k}$ पर प्रक्षेप 30 है, तो α बराबर है: **[JEE(Main)July2022]**

(A) $\frac{15}{2}$ (B) 8 (C) $\frac{13}{2}$ (D) 7

Ans. D

Sol. $\vec{a}\times\vec{b}=(1-\alpha)\hat{i}+\left(\alpha^2-2\right)\hat{j}+(\alpha-2)\hat{k}$

Projection of $\vec{a}\times\vec{b}$ on $-\hat{i}+2\hat{j}-2\hat{k}$

$$=\frac{(\vec{a}\times\vec{b})\cdot(-\hat{i}+2\hat{j}-2\hat{k})}{3}=30$$

$$\Rightarrow 2\alpha^2-\alpha-91=0$$

$$\Rightarrow \alpha=7,-\frac{13}{2}$$

94. A vector $\vec{a}$ is parallel to the line of intersection of the plane determined by the vectors $\hat{i},\hat{i}+\hat{j}$ and the plane determined by the vectors $\hat{i}-\hat{j},\hat{i}+\hat{k}$. The obtuse angle between $\vec{a}$ and the vector $\vec{b}=\hat{i}-2\hat{j}+2\hat{k}$ is : **[JEE(Main)July2022]**

सदिशों $\hat{i},\hat{i}+\hat{j}$ द्वारा प्राप्त समतल तथा सदिशों $\hat{i}-\hat{j},\hat{i}+\hat{k}$ द्वारा प्राप्त समतल की प्रतिच्छेदन रेखा के समांतर एक सदिश $\vec{a}$ है। $\vec{a}$ तथा सदिश $\vec{b}=\hat{i}-2\hat{j}+2\hat{k}$ के बीच अधिक कोण है : **[JEE(Main)July2022]**

(A) $\frac{3\pi}{4}$ (B) $\frac{2\pi}{3}$ (C) $\frac{4\pi}{5}$ (D) $\frac{5\pi}{6}$

Ans. A

Sol. $\vec{n}_1 = \hat{i} \times \left(\hat{i} + \hat{j}\right) = \hat{i}$

$\vec{n}_2 = \left(\hat{i} + \hat{j}\right) \times \left(\hat{i} - \hat{j}\right)$

$= \hat{i} + \hat{j} - \hat{k}$

Line of intersection along $\vec{n}_1 \times \vec{n}_2$

$= \hat{k} \times \left(\hat{i} + \hat{j} - \hat{k}\right) = -\hat{i} + \hat{j}$

D.R of $\vec{a} = -\hat{i} + \hat{j}$

D.E of $\vec{b} = \hat{i} - 2\hat{j} + 2\hat{k}$

$\vec{a}.\vec{b} = -3$ and $\left(\vec{a} \wedge \vec{b}\right) = \theta$

$\cos\theta = \frac{-3}{\sqrt{2} \times 3}$

$\theta = \frac{3\pi}{4}$

95. Let $\vec{a} = \alpha\hat{i} + \hat{j} + \beta\hat{k}$ and $\vec{b} = 3\hat{i} - 5\hat{j} + 4\hat{k}$ be two vectors, such that $\vec{a} \times \vec{b} = -\hat{i} + 9\hat{i} + 12\hat{k}$. Then the projection of $\vec{b} - 2\vec{a}$ on $\vec{b} + \vec{a}$ is equal to : **[JEE(Main)July2022]**

माना दो सदिशों $\vec{a} = \alpha\hat{i} + \hat{j} + \beta\hat{k}$ तथा $\vec{b} = 3\hat{i} - 5\hat{j} + 4\hat{k}$ के लिए $\vec{a} \times \vec{b} = -\hat{i} + 9\hat{i} + 12\hat{k}$ है, तो $\vec{b} - 2\vec{a}$ का $\vec{b} + \vec{a}$ पर प्रक्षेप बराबर है : **[JEE(Main)July2022]**

(A) 2 (B) $\frac{39}{5}$ (C) 9 (D) $\frac{46}{5}$

Ans. D

Sol. Let $\vec{a} = a\hat{i} + \hat{j} + \beta\hat{k}, \vec{b} = 3\hat{i} - 5\hat{j} + 4\hat{k}$

$\vec{a} \times \vec{b} = -\hat{i} + 9\hat{j} + 12\hat{k}$

$$\Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \alpha & 1 & \beta \\ 3 & -5 & 4 \end{vmatrix}$$

$\Rightarrow (4+5\beta)\hat{i}+(3\beta-4\alpha)\hat{j}+(-5\alpha-3)\hat{k}$

$=-\hat{i}+9\hat{j}+12\hat{k}$

$\therefore 4+5\beta=-1, 3\beta-4\alpha=9, -5\alpha-3=12$

$\beta=-1,\ \alpha=-3$

$\therefore \vec{\alpha}=-3\hat{i}+\hat{j}-\hat{k}, \vec{b}=3\hat{i}-5\hat{j}+4\hat{k}$

$\therefore \vec{\alpha}+\vec{b}=-4\hat{j}+3\hat{k}$

$\left|\vec{\alpha}\right|^2=11, \left|\vec{b}\right|^2=50$

$\vec{\alpha}.\vec{b}=-9+(-5)-4=-18$

$\therefore$ Pojectile of $\left(2\vec{b}-2\vec{a}\right)$ on $\vec{a}+\vec{b}$ is

$$\frac{\left(\vec{b}-2\vec{a}\right).\left(\vec{a}+\vec{b}\right)}{\left|\vec{a}+\vec{b}\right|}$$

$$=\frac{\left|\vec{b}\right|^2-2\left|\vec{a}\right|^2-\left(\vec{a}.\vec{b}\right)}{\left|\vec{a}+\vec{b}\right|}=\frac{50-22-(-18)}{5}=\frac{46}{5}$$

Ans. $\left(\frac{46}{5}\right)$

96. Let $\vec{a}=2\hat{i}-\hat{j}+5\hat{k}$ and $\vec{b}=\alpha\hat{i}+\beta\hat{j}+2\hat{k}$. If $\left(\left(\vec{a}\times\vec{b}\right)\times\hat{i}\right).\hat{k}=\frac{23}{2}$, then $\left|\vec{b}\times 2\hat{j}\right|$ is equal to :

[JEE(Main)July2022]

माना $\vec{a}=2\hat{i}-\hat{j}+5\hat{k}$ तथा $\vec{b}=\alpha\hat{i}+\beta\hat{j}+2\hat{k}$ हैं। यदि $\left(\left(\vec{a}\times\vec{b}\right)\times\hat{i}\right).\hat{k}=\frac{23}{2}$ है, तो $\left|\vec{b}\times 2\hat{j}\right|$ बराबर है :

[JEE(Main)July2022]

(A) 4 (B) 5 (C) $\sqrt{21}$ (D) $\sqrt{17}$

Ans. B

Sol. $\vec{a}=2\hat{i}-\hat{j}+5\hat{k}, \vec{b}=\alpha\hat{i}+\beta\hat{j}+2\hat{k}$

$\left(\left(\vec{a}\times\vec{b}\right)\times\hat{i}\right).\hat{k}=\frac{23}{2}$, then $\left|\vec{b}\times 2\hat{j}\right|$ is

$\left(\left(\vec{a}.\hat{i}\right)\vec{b}-\left(\vec{b}.\hat{i}\right)\vec{a}\right).\hat{k}=\frac{23}{2}$

$$\left(\vec{a}.\hat{i}\right)\left(\vec{b}.\hat{i}\right)-\left(\vec{b}.\hat{i}\right)\left(\vec{a}.\hat{k}\right)=\frac{23}{2}$$

$$2\times 2-\alpha\times 5=\frac{23}{2}\Rightarrow 5\alpha=4-\frac{23}{2}\Rightarrow \alpha=\frac{-3}{2}$$

$$\vec{b}\times 2\hat{j}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ \alpha & \beta & 2\\ 0 & 2 & 0\end{vmatrix}=-4\hat{i}+2\alpha\hat{k}$$

$$\therefore\left|\vec{b}\times 2\hat{j}\right|=\sqrt{16+4\alpha^2}=\sqrt{16+4\times\frac{9}{4}}=5$$

97. Let the vectors $\vec{a}=(1+t)\hat{i}+(1-t)\hat{j}+\hat{k}$, $\vec{b}=(1-t)\hat{i}+(1+t)\hat{j}+2\hat{k}$ and $\vec{c}=t\hat{i}-t\hat{j}+\hat{k}$, $t\in R$ be such that for $\alpha, \beta, \gamma \in R$, $\alpha\vec{a}+\beta\vec{b}+\gamma\vec{c}=\vec{0}\Rightarrow\alpha=\beta=\gamma=0$. Then, the set of all values of t is :

[JEE(Main)July2022]

(A) a non-empty finite set (B) equal to N

(C) equal to R – {0} (D) equal to R

माना सदिश $\vec{a}=(1+t)\hat{i}+(1-t)\hat{j}+\hat{k}$, $\vec{b}=(1-t)\hat{i}+(1+t)\hat{j}+2\hat{k}$ तथा $\vec{c}=t\hat{i}-t\hat{j}+\hat{k}$, $t\in R$ इस प्रकार है कि $\alpha, \beta, \gamma \in R$, के लिए $\alpha\vec{a}+\beta\vec{b}+\gamma\vec{c}=\vec{0}\Rightarrow\alpha=\beta=\gamma=0$ तो t के सभी मानों का समुच्चय है :

[JEE(Main)July2022]

(A) एक अरिक्त परिमित समुच्चय है (B) N के बराबर है

(C) R – {0} के बराबर है (D) R के बराबर है

Ans. C

Sol. By its given condition

$:\vec{a},\vec{b},\vec{c}$ are linearly independent vectors

$$\Rightarrow\left[\vec{a}\,\vec{b}\,\vec{c}\right]\neq 0 \qquad \text{....(i)}$$

Now, $\left[\vec{a}\,\vec{b}\,\vec{c}\right]$

$$=\begin{vmatrix}1+t & 1-t & 1\\ 1-t & 1+t & 2\\ t & -t & 1\end{vmatrix}$$

$C_2\to C_1+C_2$

$$\begin{vmatrix} 1+t & 2 & 1 \\ 1-t & 2 & 2 \\ t & 0 & 1 \end{vmatrix}$$

$$= 2\begin{vmatrix} 1+t & 2 & 1 \\ 1-t & 2 & 2 \\ t & 0 & 1 \end{vmatrix}$$

$$= 2\left[(1+t)-(1-t)+t\right]$$

$$= 2[3t] = 6t$$

$$\left[\bar{a}\ \bar{b}\ \bar{c}\right] \neq 0 \Rightarrow t \neq 0$$

98. Let a vector $\vec{a}$ has magnitude 9. Let a vector $\vec{b}$ be such that for every $(x, y) \in R \times R - \{(0, 0)\}$, the vector $\left(x\vec{a} + y\vec{b}\right)$ is perpendicular to the vector $\left(6y\vec{a} - 18x\vec{b}\right)$. Then the value of $\left|\vec{a} \times \vec{b}\right|$ is equal to :

[JEE(Main)July2022]

माना एक सदिश $\vec{a}$ का परिमाण 9 है। माना एक सदिश $\vec{b}$ इस प्रकार है कि प्रत्येक $(x, y) \in R \times R - \{(0, 0)\}$ के लिए, सदिश $\left(x\vec{a} + y\vec{b}\right)$, सदिश $\left(6y\vec{a} - 18x\vec{b}\right)$ के लंबवत है। तो $\left|\vec{a} \times \vec{b}\right|$ का मान बराबर है : **[JEE(Main)July2022]**

(A) $9\sqrt{3}$ (B) $27\sqrt{3}$ (C) 9 (D) 81

Ans. B

Sol. $|\vec{a}| = 9 \ \& \ \left(x\vec{a} + y\vec{b}\right).\left(6y\vec{a} - 18x\vec{b}\right) = 0$

$$\Rightarrow 6xy|\bar{a}|^2 - 18x^2\left(\bar{a}.\bar{b}\right) + 6y^2\left(\bar{a}.\bar{b}\right) - 18xy|\bar{b}|^2 = 0$$

$$\Rightarrow 6xy\left(|\bar{a}|^2 - 3|\bar{b}|^2\right) + \left(\bar{a}.\bar{b}\right)\left(y^2 - 3x^2\right) = 0$$

This should hold $\forall x, y \in R \times R$

$$\therefore |\bar{a}|^2 = 3|\bar{b}|^2 \ \& \ \left(\bar{a}.\bar{b}\right) = 0$$

Now $\left|\bar{a} \times \bar{b}\right|^2 = |\bar{a}|^2|\bar{b}|^2 - \left(\bar{a}.\bar{b}\right)^2$

$$= \bar{a}.\frac{|\bar{a}|^2}{3}$$

$$\therefore |\bar{a} \times \bar{b}| = \frac{|\bar{a}|^2}{\sqrt{3}} = \frac{81}{\sqrt{3}} = 27\sqrt{3}$$

99. Let S be the set of all $a \in R$ for which the angle between the vectors $\vec{u} = a(\log_e b)\hat{i} - 6\hat{j} + 3\hat{k}$ and $\vec{v} = (\log_e b)\hat{i} + 2\hat{j} + 2a(\log_e b)\hat{k}, (b > 1)$ is acute. Then S is equal to : **[JEE(Main)July2022]**

माना सभी $a \in R$, जिनके लिए सदिशों $\vec{u} = a(\log_e b)\hat{i} - 6\hat{j} + 3\hat{k}$ तथा $\vec{v} = (\log_e b)\hat{i} + 2\hat{j} + 2a(\log_e b)\hat{k}, (b > 1)$ के बीच का कोण न्यून कोण है, का समुच्चय S है। तो S बराबर है : **[JEE(Main)July2022]**

(A) $\left(-\infty, -\frac{4}{3}\right)$ (B) Φ (C) $\left(-\frac{4}{3}, 0\right)$ (D) $\left(\frac{12}{7}, \infty\right)$

Ans. B

Sol. For angle to be acute

$\vec{u}.\vec{v} > 0$

$\Rightarrow a(\log_e b)^2 - 12 + 6a(\log_e b) > 0$

$\forall b > 1$

let $\log_e b = t \Rightarrow t > 0$ as $b > 1$

$y = at^2 + 6at - 12$ & $y > 0, \forall t > 0$

$\Rightarrow a \in \phi$

100. Let $\vec{a} = 3\hat{i} + \hat{j}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$. Let $\vec{c}$ be a vector satisfying $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b} + \lambda\vec{c}$. If $\vec{b}$ and $\vec{c}$ are non-parallel, then the value of λ is : **[JEE(Main)July2022]**

माना $\vec{a} = 3\hat{i} + \hat{j}$ तथा $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$ है। माना एक सदिश $\vec{c}$ के लिए $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b} + \lambda\vec{c}$ है। यदि $\vec{b}$ तथा $\vec{c}$ समांतर नहीं हैं, तो λ का मान है : **[JEE(Main)July2022]**

(A) –5 (B) 5 (C) 1 (D) –1

Ans. A

Sol. $\vec{a} = 3\hat{i} + \hat{j}, \vec{b} = \hat{i} + 2\hat{j} + \hat{k}$

As $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b} + \lambda\vec{c}$

$\Rightarrow \vec{a}.\vec{c}(\vec{b}) - (\vec{a}.\vec{b})\vec{c} = \vec{b} + \lambda\vec{c}$

$\Rightarrow \vec{a}.\vec{c} = 1, \vec{a}.\vec{b} = -\lambda$

$\Rightarrow (3\hat{i} + \hat{j}).(\hat{i} + 2\hat{j} + \hat{k}) = -\lambda$

$\Rightarrow \lambda = -5$

101. Let $\hat{a}$ and $\hat{b}$ be two unit vectors such that the angle between them is $\frac{\pi}{4}$. If θ is the angle between the vectors $\left(\hat{a}+\hat{b}\right)$ and $\left(\hat{a}+2\hat{b}+2\left(\hat{a}\times\hat{b}\right)\right)$, then the value of $164\cos^2\theta$ is equal to : **[JEE(Main)July2022]**

माना दो इकाई सदिशों $\hat{a}$ तथा $\hat{b}$ के बीच का कोण $\frac{\pi}{4}$ है। यदि सदिशों $\left(\hat{a}+\hat{b}\right)$ तथा $\left(\hat{a}+2\hat{b}+2\left(\hat{a}\times\hat{b}\right)\right)$ के बीच का कोण θ है, तो $164\cos^2\theta$ का मान बराबर है : **[JEE(Main)July2022]**

(A) $90+27\sqrt{2}$ (B) $45+18\sqrt{2}$ (C) $90+3\sqrt{2}$ (D) $54+90\sqrt{2}$

Ans. A

Sol. $\hat{a}\hat{}\hat{b}=\frac{\pi}{4}=\phi$

$\hat{a}\cdot\hat{b}=|\hat{a}||\hat{b}|\cos\phi$

$\hat{a}\cdot\hat{b}=\cos\phi=\frac{1}{\sqrt{2}}$

$\cos\theta=\frac{(\hat{a}+\hat{b})\cdot(\hat{a}+2\hat{b}+2(\hat{a}\times\hat{b}))}{|\hat{a}+\hat{b}||\hat{a}+2\hat{b}+2(\hat{a}\times\hat{b})|}$

$|\hat{a}+\hat{b}|^2=(\hat{a}+\hat{b})\cdot(\hat{a}+\hat{b})$

$|\hat{a}+\hat{b}|^2=2+2\hat{a}\cdot\hat{b}$

$=2+\sqrt{2}$

$\hat{a}\times\hat{b}=|\hat{a}||\hat{b}|\sin\phi\hat{n}$

$\hat{\mathbf{a}}\times\hat{\mathbf{b}}=\frac{\hat{n}}{\sqrt{2}}$ when $\hat{n}$ is vector $\perp\hat{a}$ and $\hat{b}$

let $\vec{c}=\hat{a}\times\hat{b}$

We know.

$\vec{c}\cdot\vec{a}=0$

$\vec{c}\cdot\vec{b}=0$

$|\hat{a}+2\hat{b}+2\vec{c}|^2$

$=1+4+\frac{(4)}{2}+4\hat{a}\cdot\hat{b}+8\hat{b}\cdot\vec{c}+4\vec{c}\cdot\hat{a}$

$=7+\frac{4}{\sqrt{2}}=7+2\sqrt{2}$

Now

$(\hat{a}+\hat{b})\cdot(\hat{a}+2\hat{b}+2\vec{c})$

$=|\hat{a}|^2+2\hat{a}\cdot\hat{b}+0+\hat{b}\cdot\hat{a}+2|\hat{b}|^2+0$

$=1+\frac{2}{\sqrt{2}}+\frac{1}{\sqrt{2}}+2$

$=3+\frac{3}{\sqrt{2}}$

$\cos\theta=\frac{3+\frac{3}{\sqrt{2}}}{\sqrt{2+\sqrt{2}}\sqrt{7+2\sqrt{2}}}$

$\cos^2\theta=\frac{9(\sqrt{2}+1)^2}{2(2+\sqrt{2})(7+2\sqrt{2})}$

$\cos^2\theta=\left(\frac{9}{2\sqrt{2}}\right)\frac{(\sqrt{2}+1)}{(7+2\sqrt{2})}$

$164\cos^2\theta=\frac{(82)(9)}{\sqrt{2}}\frac{(\sqrt{2}+1)}{(7+2\sqrt{2})}\frac{(7-2\sqrt{2})}{(7-2\sqrt{2})}$

$=\frac{(82)}{\sqrt{2}}\frac{(9)[7\sqrt{2}-4+7-2\sqrt{2}]}{(41)}$

$=(9\sqrt{2})[5\sqrt{2}+3]$

$=90+27\sqrt{2}$

102. If (2, 3, 9), (5, 2, 1), (1, λ, 8) and (λ, 2, 3) are coplanar, then the product of all possible values of λ is :

[JEE(Main)July2022]

यदि बिन्दु (2, 3, 9), (5, 2, 1), (1, λ, 8) तथा (λ, 2, 3) सह–तलीय है, तो λ के सभी संभव मानों का गुणनफल है :

[JEE(Main)July2022]

(A) $\frac{21}{2}$ (B) $\frac{59}{8}$ (C) $\frac{57}{8}$ (D) $\frac{95}{8}$

Ans. D

Sol. $A(2,3,9);B(5,2,1);C(1,\lambda,8);D(\lambda,2,3)$

$\left[\overrightarrow{AB}\quad\overrightarrow{AC}\quad\overrightarrow{AD}\right]=0$

$$\begin{vmatrix} 3 & -1 & -8 \\ -1 & \lambda-3 & -1 \\ \lambda-2 & -1 & -6 \end{vmatrix} = 0$$

$\Rightarrow [-6(\lambda-3)-1]-8(1-(\lambda-3)(\lambda-2))+(6+(\lambda-2)=0$

$3(-6\lambda+17)-8\left(-\lambda^2+5\lambda-5\right)+(\lambda+4)=8$

$8\lambda^2-57\lambda+95=0$

$\lambda_1\lambda_2=\frac{95}{8}$

103. Let $\vec{a},\vec{b},\vec{c}$ be three coplanar concurrent vectors such that angles between any two of them is same. If the product of their magnitudes is 14 and $\left(\vec{a}\times\vec{b}\right).\left(\vec{b}\times\vec{c}\right)+\left(\vec{b}\times\vec{c}\right).\left(\vec{c}\times\vec{a}\right)+\left(\vec{c}\times\vec{a}\right).\left(\vec{a}\times\vec{b}\right)=168$, then $|\vec{a}|+|\vec{b}|+|\vec{c}|$ is equal to: **[JEE(Main)July2022]**

माना तीन सहतलीय संगामी सदिश $\vec{a},\vec{b},\vec{c}$ इस प्रकार हैं कि किन्हीं भी दो भी बीच के कोण बराबर हैं। यदि इन सदिशों के परिमाणों का गुणनफल 14 है तथा $\left(\vec{a}\times\vec{b}\right).\left(\vec{b}\times\vec{c}\right)+\left(\vec{b}\times\vec{c}\right).\left(\vec{c}\times\vec{a}\right)+\left(\vec{c}\times\vec{a}\right).\left(\vec{a}\times\vec{b}\right)=168$ है, तो $|\vec{a}|+|\vec{b}|+|\vec{c}|$ बराबर है : **[JEE(Main)July2022]**

(A) 10 (B) 14 (C) 16 (D) 18

Ans. C

Sol. $\vec{a}\,\|\,\vec{b}\,\|\,\vec{c}\,|=14$

$\vec{a}\hat{}\,\vec{b}=\vec{b}\wedge\vec{c}=\vec{c}\hat{}\,\vec{a}=\theta=\frac{2\pi}{3}$

So, $\vec{a}\cdot\vec{b}=-\frac{1}{2}ab,\vec{b}\cdot\vec{c}=-\frac{1}{2}bc,\vec{a}\cdot\vec{c}.=-\frac{1}{2}ac$

(let)

$(\vec{a}\times\vec{b})\cdot(\vec{b}\times\vec{c})=(\vec{a}\cdot\vec{b})(\vec{b}\cdot\vec{c})-(\vec{a}\cdot\vec{c})(\vec{b}\cdot\vec{b})$

$=\frac{1}{4}ab^2c+\frac{1}{2}ab^2c=\frac{3}{4}ab^2c$

Similarly

$(\vec{b}\times\vec{c})\cdot(\vec{c}\times\vec{a})=\frac{3}{4}abc^2$

$(\vec{c}\times\vec{a})\cdot(\vec{a}\times\vec{b})=\frac{3}{4}a^2bc$

$168=\frac{3}{4}abc(a+b+c)$

So, (a + b + c) = 16

104. The largest value of a, for which the perpendicular distance of the plane containing the lines $\vec{r}=\left(\hat{i}+\hat{j}\right)+\lambda\left(\hat{i}+a\hat{j}-\hat{k}\right)$ and $\vec{r}=\left(\hat{i}+\hat{j}\right)+\mu\left(-\hat{i}+\hat{j}-a\hat{k}\right)$ from the point (2, 1, 4) is $\sqrt{3}$, is _____.

[JEE(Main)July2022]

a का अधिकतम मान, जिसके लिए रेखाओं $\vec{r}=\left(\hat{i}+\hat{j}\right)+\lambda\left(\hat{i}+a\hat{j}-\hat{k}\right)$ तथा $\vec{r}=\left(\hat{i}+\hat{j}\right)+\mu\left(-\hat{i}+\hat{j}-a\hat{k}\right)$ को अंतर्विष्ट करने वाले समतल की बिन्दु (2, 1, 4) से लंबवत् दूरी $\sqrt{3}$ है, है_____

[JEE(Main)July2022]

Ans. 2

Sol. $\vec{r}=\left(\hat{i}+\hat{j}\right)+\lambda\left(\hat{i}+a\hat{j}-\hat{k}\right)$

$\vec{r}=\left(\hat{i}+\hat{j}\right)+\mu\left(-\hat{i}+\hat{j}-a\hat{k}\right)$

D.R's of plane containing these lines is

$$\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & a & -1\\ -1 & 1 & -a\end{vmatrix}=\hat{i}\left(1-a^2\right)-\hat{j}(-a-1)+\hat{k}(1+a)$$

$\vec{n}=(1-a)\hat{i}+\hat{j}+\hat{k}$

One point in plane : (1, 1, 0)

$\therefore$ equation of plane is

$(1-a)(x-1)+(y-1)+(z-0)=0$

$(1-a)x+y+z+a-2=0$

$$\therefore D=\frac{|(1-a)2+1+4+a-2|}{\sqrt{(1-a)^2+1+1}}$$

$\Rightarrow|5-a|=\sqrt{3}.\sqrt{a^2-2a+3}$

$\Rightarrow a^2+2a-8=0$

$\Rightarrow a=2,-4$

$\therefore$ largest value of a = 2

105. Let $\vec{a},\vec{b},\vec{c}$ be three non-coplanar vectors such that $\vec{a}\times\vec{b}=4\vec{c},\ \vec{b}\times\vec{c}=9\vec{a}$ and $\vec{c}\times\vec{a}=\alpha\vec{b}$, $\alpha>0$. If $|\vec{a}|+|\vec{b}|+|\vec{c}|=36$, then α is equal to _____.

[JEE(Main)July2022]

माना तीन सदिश $\vec{a},\vec{b},\vec{c}$ सहतलीय नहीं है तथा $\vec{a}\times\vec{b}=4\vec{c},\ \vec{b}\times\vec{c}=9\vec{a}$, $\vec{c}\times\vec{a}=\alpha\vec{b}$, $\alpha>0$ हैं। यदि $|\vec{a}|+|\vec{b}|+|\vec{c}|=36$ है, तो α बराबर है _____

[JEE(Main)July2022]

Ans. 36

Sol. $\vec{a}+\vec{b}=4\vec{c}\Rightarrow \vec{a}.\vec{c}=0=\vec{b}.\vec{c}$

$\vec{b}\times\vec{c}=9\vec{a}\Rightarrow \vec{a}.\vec{b}=0=\vec{a}.\vec{c}$

$\therefore \vec{a},\vec{b},\vec{c}$ are mutually $\perp$ set of vectors.

$\Rightarrow |\vec{a}||\vec{b}|=4|\vec{c}|, |\vec{b}||\vec{c}|=9|\vec{a}| \ \& \ |\vec{c}||\vec{a}|=\alpha|\vec{b}|$

$$\Rightarrow \frac{|\vec{a}|}{|\vec{c}|}=\frac{4}{9}\frac{|\vec{c}|}{|\vec{a}|}$$

$$\Rightarrow \frac{|\vec{c}|}{|\vec{a}|}=\frac{3}{2}$$

$$\therefore \text{ If } |a|=\lambda, |c|=\frac{3\lambda}{2} \ \& \ |b|=6$$

Now $|a|+|b|+|c|=\frac{1}{36}$

$$\Rightarrow \frac{5}{2}\lambda+6=\frac{1}{36}, \lambda=\frac{-43}{18}=|a|$$

which gives negative value of λ or $|a|$ which is NOT possible & hence data seems to be wrong.

But if $|\vec{a}|+|\vec{b}|+|\vec{c}|=36$

$$\frac{5}{2}\lambda+6=36$$

$$\lambda=12$$

$$\alpha=\frac{|\vec{c}||\vec{a}|}{|\vec{b}|}=\frac{3\times12}{2}\times\frac{12}{6}$$

$$\alpha=36$$

106. Let $\vec{a}$ and $\vec{b}$ be two vectors such that $\left|\vec{a}+\vec{b}\right|^2=|\vec{a}|^2+2|\vec{b}|^2, \vec{a}.\vec{b}=3$ and $\left|\vec{a}\times\vec{b}\right|^2=75$. Then $|\vec{a}|^2$ is equal to _____. **[JEE(Main)July2022]**

माना सदिशों $\vec{a}$ तथा $\vec{b}$ के लिए $\left|\vec{a}+\vec{b}\right|^2=|\vec{a}|^2+2|\vec{b}|^2, \vec{a}.\vec{b}=3$ तथा $\left|\vec{a}\times\vec{b}\right|^2=75$ हैं, तो $|\vec{a}|^2$ बराबर है ____

[JEE(Main)July2022]

Ans. 14

Sol. $|\vec{a}+\vec{b}|^2=|\vec{a}|^2+2|\vec{b}|^2; \vec{a}\cdot\vec{b}=3$

As $|\vec{a}|^2+|\vec{b}|^2+2\vec{a}\cdot\vec{b}=|\vec{a}|^2+2|\vec{b}|^2$

$|\vec{b}|^2=2\vec{a}\cdot\vec{b}=6$

$|\vec{a}\times\vec{b}|^2=75$

$|\vec{a}|^2|\vec{b}|^2-(\vec{a}\cdot\vec{b})^2=75$

$6|\vec{a}|^2-9=75\Rightarrow|\vec{a}|^2=14$

3D Geometry

107. Let P be the plane containing the straight line $\frac{x-3}{9}=\frac{y+4}{-1}=\frac{z-7}{-5}$ and perpendicular to the plane containing the straight lines $\frac{x}{2}=\frac{y}{3}=\frac{z}{5}$ and $\frac{x}{3}=\frac{y}{7}=\frac{z}{8}$. If d is the distance of P from the point(2, –5, 11), then d^2 is equal to : **[JEE(Main)July2022]**

माना रेखा $\frac{x-3}{9}=\frac{y+4}{-1}=\frac{z-7}{-5}$, समतल P में स्थित है तथा समतल P, रेखाओं $\frac{x}{2}=\frac{y}{3}=\frac{z}{5}$ तथा $\frac{x}{3}=\frac{y}{7}=\frac{z}{8}$ को अंतर्विष्ट करने वाले समतल के लंबवत है। यदि P की बिन्दु (2, –5, 11) से दूरी d है, तो d^2 बराबर है : **[JEE(Main)July2022]**

(A) $\frac{147}{2}$ (B) 96 (C) $\frac{32}{3}$ (D) 54

Ans. C

Sol. Let <a, b, c> be direction ratios of plane containing

lines $\frac{x}{2}=\frac{y}{3}=\frac{z}{5}$ and $\frac{x}{3}=\frac{y}{7}=\frac{z}{8}$.

$\therefore$ 2a + 3b + 5c = 0 .....(i)

and 3a + 7b + 8c = 0 .....(ii)

from eq. (i) and (ii) : $\frac{a}{24-35}=\frac{b}{15-16}=\frac{c}{14-9}$

$\therefore$ D.R^S. of plane are < 11, 1, –5>

Let D.R^S of plane P be $<a_1, b_1, c_1>$ then.

$11a_1+b_1-5c_1=0$ .....(iii)

and $9a_1-b_1-5c_1=0$ .....(iv)

From eq. (iii) and (iv) :

$\frac{a_1}{-5-5}=\frac{b_1}{-45-55}=\frac{c_1}{-11-9}$

$\therefore$ D.A^5. of plane P are < 1, –1, 2>

Equation plane P is : $1(x-3)-1(y+4)+2(z-7)=0$

$\Rightarrow x-y+2z-21=0$

Distance from point (2, –5, 11) is $d=\frac{|2+5+22-2|}{\sqrt{6}}$

$\therefore \quad d^2=\frac{32}{3}$

108. A plane E is perpendicular to the two planes $2x-2y+z=0$ and $x-y+2z=4$, and passes through the point P(1, –1, 1). If the distance of the plane E from the point Q(a, a, 2) is $3\sqrt{2}$, then $(PQ)^2$ is equal to :

[JEE(Main)July2022]

एक समतल E दो समतलों $2x-2y+z=0$ तथा $x-y+2z=4$ के लंबवत है, तथा बिन्दु P(1, –1, 1) से होकर जाता है। समतल E की बिन्दु Q(a, a, 2) से दूरी $3\sqrt{2}$ है, तो $(PQ)^2$ बराबर है : **[JEE(Main)July2022]**

(A) 9 (B) 12 (C) 21 (D) 33

Ans. C

Sol. First plane, $P_1 = 2x-2y+z=0$, normal vector $\equiv \bar{n}_1 = \langle 2,-2,1\rangle$

Second plane, $P_2 \equiv x-y+2z=4$, normal vector $\equiv \bar{n}_2 = (1,-1,2)$

Plane perpendicular to P_1 and P_2 will have normal vector $\bar{n}_3$

Where $\bar{n}_3 = (\bar{n}_1 \times \bar{n}_2)$

Hence, $\bar{n}_3 = (-3,-3,0)$

Equation of plane E through P(1, –1, 1) and $\bar{n}_3$ as normal vector

$(x-1, y+1, z-1)\cdot(-3,-3,0)=0$

$\Rightarrow x+y=0 \equiv E$

Distance of PQ(a, a, 2) from $E=\left|\frac{2a}{\sqrt{2}}\right|$

as given, $\left|\frac{2a}{\sqrt{2}}\right| = 3\sqrt{2} \Rightarrow a=\pm 3$

Hence, $Q \equiv (\pm 3, \pm 3, 2)$

Distance $7Q = \sqrt{21} \Rightarrow (PQ)^2 = 21$

109. The shortest distance between the lines $\frac{x+7}{-6}=\frac{y-6}{7}=z$ and $\frac{7-x}{2}=y-2=z-6$ is :

[JEE(Main)July2022]

रेखाओं $\frac{x+7}{-6}=\frac{y-6}{7}=z$ तथा $\frac{7-x}{2}=y-2=z-6$ के बीच न्यूनतम दूरी है : **[JEE(Main)July2022]**

(A) $2\sqrt{29}$ (B) 1 (C) $\sqrt{\frac{37}{29}}$ (D) $\frac{\sqrt{29}}{2}$

Ans. A

Sol. $L_1 : \frac{x+7}{-6}=\frac{y-6}{7}=\frac{z-0}{1}$

Any point on it $\vec{a}_1$ (–7, 6, 0)

and L_1 is parallel to $\vec{b}_1$ (–6, 7, 1)

$L_2 : \frac{x-7}{-2}=\frac{y-2}{1}=\frac{z-6}{1}$

Any point on it, $\vec{a}_2$ (7, 2, 6)

and L_2 is parallel to $\vec{b}_2$ (–2, 1, 1)

Shortest distance between L_1 and L_2

$$=\left|\frac{(\vec{a}_1-\vec{a}_2)\cdot(\vec{b}_1\times\vec{b}_2)}{|\vec{b}_1\times\vec{b}_2|}\right|=\left|\frac{(-14,4,-6)\cdot(3,2,4)}{\sqrt{9+4+16}}\right|$$

$$=2\sqrt{29}$$

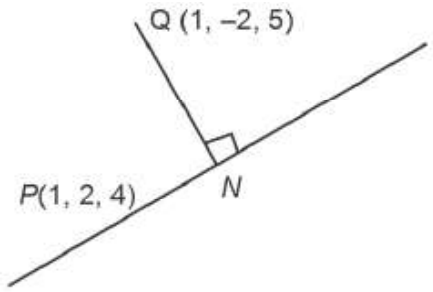
110. The length of the perpendicular from the point (1, –2, 5) on the line passing through (1, 2, 4) and parallel to the line x + y – z = 0 = x – 2y + 3z – 5 is : **[JEE(Main)July2022]**

बिन्दु (1, 2, 4) से होकर जाने वाली तथा रेखा x + y – z = 0 = x – 2y + 3z – 5 के समांतर रेखा की, बिन्दु (1, – 2, 5) से लंब की लंबाई है : **[JEE(Main)July2022]**

(A) $\sqrt{\frac{21}{2}}$ (B) $\sqrt{\frac{9}{2}}$ (C) $\sqrt{\frac{73}{2}}$ (D) 1

Ans. A

Sol.



The line $x + y - z = 0 = x - 2y + 3z - 5$ is parallel to the vector

$$\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ 1 & -2 & 3 \end{vmatrix} = (1, 4, -3)$$

Equation of line though P(1, 2, 3) and parallel to $\vec{b}$

$$\frac{x-1}{1} = \frac{y-2}{-4} = \frac{z-4}{-3}$$

Let $N \equiv (\lambda + 1, -4\lambda + 2, -3\lambda + 4)$

$\overrightarrow{QN} = (\lambda, -4\lambda + 4, -3\lambda - 1)$

$\overrightarrow{QN}$ is perpendicular to $\vec{b}$

$\Rightarrow (\lambda, -4\lambda + 4, -3\lambda - 1).(1, 4, -3) = 0$

$\Rightarrow \lambda = \frac{1}{2}$

Hence $\overrightarrow{QN} = \left(\frac{1}{2}, 2, \frac{-5}{2}\right)$ and $\left|\overrightarrow{QN}\right| = \sqrt{\frac{21}{2}}$

111. If the plane P passes through the intersection of two mutually perpendicular planes $2x + ky - 5z = 1$ and $3kx - ky + z = 5$, $k < 3$ and intercepts a unit length on positive x-axis, then the intercept made by the plane P on the y-axis is : **[JEE(Main)July2022]**

यदि दो परस्पर लंब समतलों $2x + ky - 5z = 1$ तथा $3kx - ky + z = 5$, $k < 3$ के प्रतिच्छेदन से होकर जाने वाले समतल P धनात्मक x-अक्ष पर अंतःखण्ड लंबाई एक है, तो P का y-अक्ष पर अंतः खण्ड है : **[JEE(Main)July2022]**

(A) $\frac{1}{11}$ (B) $\frac{5}{11}$ (C) 6 (D) 7

Ans. D

Sol. Two given planes mutually perpendicular

$2(3k) + k(-k) + (-5)\,1 = 0$

$k = 1, 5$

but $k < 3$ So $k = 1$

Plane passing through these planes is

$2x + y - 5z - 1 + \lambda(3x - y + z - 5) = 0$

$$\frac{x}{\frac{5\lambda+1}{2+3\lambda}}+\frac{y}{\frac{5\lambda+1}{1-\lambda}}+\frac{z}{\frac{5\lambda+1}{\lambda-5}}=1$$

Given $\frac{5\lambda+1}{2+3\lambda}=1\Rightarrow\lambda=\frac{1}{2}$

So intecept on y-axis is $=\frac{5\lambda+1}{1-\lambda}=7$

112. If the length of the perpendicular drawn from the point P(a, 4, 2), a > 0 on the line $\frac{x+1}{2}=\frac{y-3}{3}=\frac{z-1}{-1}$ is $2\sqrt{6}$ units and Q(α_1, α_2, α_3) is the image of the point P in this line, then $a+\sum_{i=1}^{3}\alpha_1$ is equal to :

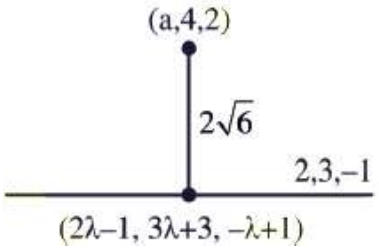
[JEE(Main)July2022]

यदि बिन्दु P(a, 4, 2), a > 0, से रेखा $\frac{x+1}{2}=\frac{y-3}{3}=\frac{z-1}{-1}$ पर डाले गए लंब की लंबाई $2\sqrt{6}$ इकाई है तथा इस रेखा में बिन्दु P का प्रतिबिम्ब Q(α_1, α_2, α_3) है, तो $a+\sum_{i=1}^{3}\alpha_1$ बराबर है : **[JEE(Main)July2022]**

(A) 7 (B) 8 (C) 12 (D) 14

Ans. B

Sol.



$(2\lambda-1-a)2+(3\lambda-1)3+(-\lambda-1)(-1)=0$

$\Rightarrow 4\lambda-2-2a+9\lambda-3+\lambda+1=0$

$\Rightarrow 14\lambda-4-2a=0$

$\Rightarrow 7\lambda-2-a=0$

and,

$(2\lambda-1-a)^2+(3\lambda-1)^2+(\lambda+1)^2=24$

$\Rightarrow (5\lambda-1)^2+(3\lambda-1)^2+(\lambda+1)^2=24$

$\Rightarrow 35\lambda^2-14\lambda-21=0$

$\Rightarrow (\lambda-1)(35\lambda+21)=0$

For, $\lambda = 1$ $\Rightarrow a = 5$

Let $(\alpha_1, \alpha_2, \alpha_3)$ be reflection of point P

$\alpha_1 + 5 = 2$ $\alpha_2 + 4 = 12$ $\alpha_3 + 2 = 0$

$\alpha_1 = -3$ $\alpha_2 = 8$ $\alpha_3 = -2$

$a + \alpha_1 + \alpha_2 + \alpha_3 = 8$

113. If the line of intersection of the planes $ax + by = 3$ and $ax + by + cz = 0$, $a > 0$ makes an angle 30° with the plane $y - z + 2 = 0$, then the direction cosines of the line are : **[JEE(Main)July2022]**

यदि समतलों $ax + by = 3$ तथा $ax + by + cz = 0$, $a > 0$ की प्रतिच्छेदन रेखा L, समतल $y - z + 2 = 0$ से 30º का कोण बनाती है, तो रेखा L की दिक् कोज्या हैं : **[JEE(Main)July2022]**

(A) $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0$ (B) $\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0$ (C) $\frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}}, 0$ (D) $\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0$

Ans. B

Sol. $\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a & b & 0 \\ a & b & c \end{vmatrix}$

$= bc\hat{i} - ac\hat{j}$

Direction ratios of line are **b, –a, 0**

Direction ratios of normal of the plane are **0, 1, – 1**

$$\cos 60^\circ = \left| \frac{-a}{\left|\sqrt{2}\right| \sqrt{b^2 + a^2}} \right| = \frac{1}{2}$$

$$\Rightarrow \left| \frac{a}{\sqrt{a^2 + b^2}} \right| = \frac{1}{\sqrt{2}}$$

$\Rightarrow b = \pm a$

So, D.R.'s can be $(\pm a, -a, 0)$

$\therefore$ D.C.'s can be $\pm\left(\frac{\pm 1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0\right)$

114. The foot of the perpendicular from a point on the circle $x^2 + y^2 = 1$, $z = 0$ to the plane $2x + 3y + z = 6$ lies on which one of the following curves? **[JEE(Main)July2022]**

वृत्त $x^2 + y^2 = 1$, $z = 0$ के एक बिन्दु से समतल $2x + 3y + z = 6$ पर डाले गए लंब का पाद निम्न में से किस वक्र पर है ? **[JEE(Main)July2022]**

(A) $(6x + 5y - 12)^2 + 4(3x + 7y - 8)^2 = 1$, $z = 6 - 2x - 3y$

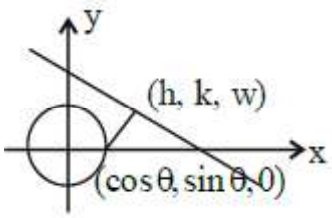
(B) $(5x + 6y - 12)^2 + 4(3x + 5y - 9)^2 = 1$, $z = 6 - 2x - 3y$

(C) $(6x + 5y - 14)^2 + 9(3x + 5y - 7)^2 = 1$, $z = 6 - 2x - 3y$

(D) $(5x + 6y - 14)^2 + 9(3x + 7y - 8)^2 = 1$, $z = 6 - 2x - 3y$

Ans. B

Sol.



$$\frac{h-\cos\theta}{2}=\frac{k-\sin\theta}{3}=\frac{w-0}{1}$$

$$=\frac{-1\left(2\cos\theta+3\sin\theta-6\right)}{14}$$

$$h=\cos\frac{-2\left(2\cos\theta+3\sin\theta-6\right)}{14}$$

$$=\frac{10\cos\theta-6\sin\theta+12}{14}$$

$$k=\sin\theta-\frac{3}{14}\left(2\cos\theta+3\sin\theta-6\right)$$

$$k=\frac{5\sin\theta-6\cos\theta+18}{14}$$

Elementaary $\sin\theta$ and $\cos\theta$

$$\left(5h+6k-12\right)^2+4\left(3h+5k-9\right)^2=1$$

115. Let the line $\frac{x-1}{\lambda}=\frac{y-2}{1}=\frac{z-3}{2}$ and $\frac{x+26}{-2}=\frac{y+18}{3}=\frac{z+28}{\lambda}$ be coplanar and P be the plane containing these two lines. Then which of the following points does NOT lie no P? **[JEE(Main)July2022]**

माना रेखाएँ $\frac{x-1}{\lambda}=\frac{y-2}{1}=\frac{z-3}{2}$ तथा $\frac{x+26}{-2}=\frac{y+18}{3}=\frac{z+28}{\lambda}$ सह–तलीय हैं तथा P वह समतल है जिस में यह दोनों रेखाएँ स्थित हैं। तो निम्न में से कौनसा बिन्दु, समतल P पर नहीं है ? **[JEE(Main)July2022]**

(A) (0, –2, –2) (B) (–5, 0, –1) (C) (3, –1, 0) (D) (0, 4, 5)

Ans. D

Sol. Given, $L_1 : \frac{x-1}{\lambda}=\frac{y-2}{1}=\frac{z-3}{2}$

and $L_2 : \frac{x+26}{-2}=\frac{y+18}{3}=\frac{z+28}{\lambda}$

are coplanar

$$\Rightarrow \begin{vmatrix} 27 & 20 & 31 \\ \lambda & 1 & 2 \\ -2 & 3 & \lambda \end{vmatrix} = 0$$

$\Rightarrow \lambda = 3$

Now, normal of plane P, which contains L_1 and L_2

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & 2 \\ -2 & 3 & 3 \end{vmatrix}$$

$= -3\hat{i} - 13\hat{j} + 11\hat{k}$

$\Rightarrow$ Equation of required plane P :

$3x + 13y - 11z + 4 = 0$

(0, 4, 5) does not lie on plane P.

116. A plane P is parallel to two lines whose direction ratios are –2, 1, –3 and –1, 2, –2 and it contains the point(2, 2, –2). Let P intersect the co-ordinate axes at the points A, B, C making the intercepts α, β, γ. If V is the volume of the tetrahedron OABC, where O is the origin, and $p = \alpha + \beta + \gamma$, then the ordered pair (V, p) is equal to : **[JEE(Main)July2022]**

एक समतल P, दो रेखाओं, जिनके दिक् अनुपात –2, 1, –3 तथा –1, 2, –2 हैं, के समान्तर है तथा बिन्दु (2, 2, –2) समतल P पर है। माना P निर्देशांक अक्षों को बिन्दुओं A, B, C पर काटता है तथा अंतःखण्ड α, β, γ बनाता है। यदि चतुष्फलक OABC का आयतन V है, जहाँ O मूल बिन्दु है, तथा $p = \alpha + \beta + \gamma$ है, तो क्रमित युग्म (V, p) बराबर है : **[JEE(Main)July2022]**

(A) (48, –13) (B) (24, –13) (C) (48, 11) (D) (24, –5)

Ans. B

Sol. Normal of plane P :

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 1 & -3 \\ -1 & 2 & -2 \end{vmatrix} = 4\hat{i} - \hat{j} - 3\hat{k}$$

Equation of plane P which passes through (2, 2, –2) is 4x – 7 – 3z – 12 = 0

Now, A (3, 0, 0), B (0, – 12, 0), C (0, 0, – 4)

$\Rightarrow \alpha = 3, \beta = -12, \gamma = -4$

$\Rightarrow p = \alpha + \beta + \gamma = -13$

Now, volume of tetrahedron OABC

$$V = \left| \frac{1}{6} \overrightarrow{OA}.\left(\overrightarrow{OB} \times \overrightarrow{OC} \right) \right| = 24$$

(V, p) = (24, –13)

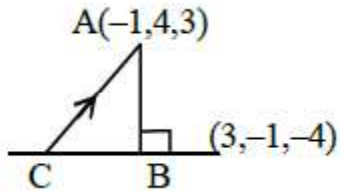
117. If the foot of the perpendicular from the point A(–1, 4, 3) on the plane P : 2x + my + nz = 4, is $\left(-2, \frac{7}{2}, \frac{3}{2}\right)$, then the distance of the point A from the plane P, measured parallel to a line with direction ratios 3, –1, –4, is equal to : **[JEE(Main)July2022]**

यदि बिन्दु A(–1, 4, 3) से समतल P : 2x + my + nz = 4, पर लंब का पाद $\left(-2, \frac{7}{2}, \frac{3}{2}\right)$ है, तो दिक् अनुपात 3, –1, –4 की एक रेखा के समांतर नापी गई बिन्दु A की समतल P से दूरी बराबर है : **[JEE(Main)July2022]**

(A) 1 (B) $\sqrt{26}$ (C) $2\sqrt{2}$ (D) $\sqrt{14}$

Ans. B

Sol.



Let B be foot of ⊥ coordinates of $B = \left(-2, \frac{7}{2}, \frac{3}{2}\right)$

Direction ratio of line AB is <2,1,3> so

m = 1, n = 3

So equation of AC is $\frac{x+1}{3}=\frac{y-4}{-1}=\frac{z-3}{-4}=\lambda$

So point C is $(3\lambda-1,-\lambda+4,-4\lambda+3)$. But C lies on the plane, so

$6\lambda-2-\lambda+4-12\lambda+9=4$

$\Rightarrow \lambda=1 \Rightarrow C(2,3,-1)$

$\Rightarrow AC=\sqrt{26}$

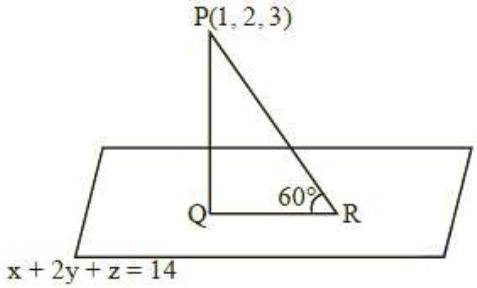
118. Let Q be the foot of perpendicular drawn from the point P(1, 2, 3) to the plane $x+2y+z=14$. If R is a point on the plane such that $\angle OPRQ = 60°$, then the area of ΔPQR is equal to : **[JEE(Main)July2022]**

माना बिन्दु P(1, 2, 3) से समतल $x+2y+z=14$ पर डाले गए लंब का पाद Q है। माना समतल पर बिन्दु R के लिए $\angle OPRQ = 60°$ है, तो ΔPQR का क्षेत्रफल बराबर : **[JEE(Main)July2022]**

(A) $\frac{\sqrt{3}}{2}$ (B) $\sqrt{3}$ (C) $2\sqrt{3}$ (D) 3

Ans. B

Sol.



Length of perpendicular

$PQ=\left|\frac{1+4+3-14}{\sqrt{6}}\right|=\sqrt{6}$

$QR=(PQ)\cot 60°=\sqrt{2}$

$\therefore$ Area of $\Delta PQR=\frac{1}{2}(PQ)(QR)=\sqrt{3}$

119. The line of shortest distance between the lines $\frac{x-2}{0}=\frac{y-1}{1}=\frac{z}{1}$ and $\frac{x-3}{2}=\frac{y-5}{2}=\frac{z-1}{1}$ makes an angle of $\cos^{-1}\left(\sqrt{\frac{2}{27}}\right)$ with the plane P : $ax-y-z=0$, $(a>0)$. If the image of the point (1, 1, –5) in the plane P is (α, β, γ), then $\alpha+\beta-\gamma$ is equal to _____. **[JEE(Main)July2022]**

रेखाओं $\frac{x-2}{0}=\frac{y-1}{1}=\frac{z}{1}$ तथा $\frac{x-3}{2}=\frac{y-5}{2}=\frac{z-1}{1}$ के बीच न्यूनतम दूरी की रेखा, समतल P : ax – y – z = 0, (a > 0) से $\cos^{-1}\left(\sqrt{\frac{2}{27}}\right)$ का कोण बनाती है। यदि बिन्दु (1, 1, –5) P में प्रतिबिंब (α, β, γ) है, तो $\alpha + \beta - \gamma$ बराबर है _____

[JEE(Main)July2022]

Ans. 3

Sol. DR's of line of shortest distance

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & 1 \\ 2 & 2 & 1 \end{vmatrix} = -\hat{i}+2\hat{j}-2\hat{k}$$

angle between line and plane is $\cos^{-1}\sqrt{\frac{2}{27}}=\alpha$

$$\cos\alpha=\left|\frac{-a-2+2}{\sqrt{4+4+1}\sqrt{a^2+1+1}}\right|=\frac{5}{3\sqrt{3}}$$

$\sqrt{3}\,|a|=5\sqrt{a^2+2}$

$3a^2=25a^2+50$

No value of (a)

120. Let Q and R be two points on the line $\frac{x+1}{2}=\frac{y+2}{3}=\frac{z-1}{2}$ at a distance $\sqrt{26}$ from the point P(4, 2, 7). Then the square of the area of the triangle PQR is _____. **[JEE(Main)July2022]**

माना रेखा $\frac{x+1}{2}=\frac{y+2}{3}=\frac{z-1}{2}$ पर बिन्दु P (4, 2, 7) से $\sqrt{26}$ दूरी पर दो बिन्दु Q तथा R है। तो त्रिभुज PQR के क्षेत्रफल का वर्ग बराबर है _____ **[JEE(Main)July2022]**

Ans. 153

Sol. Let $(2\lambda - 1, 3\lambda - 2, 2\lambda + 1)$ be any point on the line

$(2\lambda - 5)^2 + (3\lambda - 4)^2 + (2\lambda - 6)^2 = 26$

$\lambda = 1, 3$

Q (1, 1, 3) ; R (5, 7, 7) ; P (4, 2, 7)

Area of triangle $PQR=\frac{1}{2}\left|\overrightarrow{PQ}\times\overrightarrow{PR}\right|$

$= \sqrt{153}$

121. The plane passing through the line L : lx – y + 3(1 – l) z = 1, x + 2y – z = 2 and perpendicular to the plane 3x + 2y + z = 6 is 3x – 8y + 7z = 4. If θ is the acute angle between the line L and the y-axis, then 415 $\cos^2\theta$ is equal to _____. **[JEE(Main)July2022]**

रेखा L : lx – y + 3(1 – l) z = 1, x + 2y – z = 2 से होकर जाने वाले तथा समतल 3x + 2y + z = 6 के लम्बवत् समतल 3x – 8y + 7z = 4 है। यदि रेखा L तथा y-अक्ष के बीच न्यून कोण θ है, तो 415 $\cos^2\theta$ बराबर है _____

[JEE(Main)July2022]

Ans. 125

Sol. $\vec{n}_1 = \ell\hat{i} - \hat{j} + 3(1-\ell)\hat{k}$

$\vec{n}_2 = \hat{i} + 2\hat{j} - \hat{k}$

$$\text{Direction ratio of line} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \ell & -1 & 3(1-\ell) \\ 1 & 2 & -1 \end{vmatrix}$$

$= (6\ell - 5)\hat{i} + (3 - 2\ell)\hat{j} + (2\ell + 1)\hat{k}$

3x – 8y + 7z = 4 will contain the line

$(6\ell - 5)\hat{i} + (3 - 2\ell)\hat{j} + (2\ell + 1)\hat{k}$

Normal of 3x – 8y + 7z = 4 will be perpendicular to the line

$= 3(6\ell - 5) + (3 - 2\ell)(-8) + 7(2\ell + 1) = 0$

$\Rightarrow \ell = \frac{2}{3}$

$\therefore$ direction ratio of line $\left(-1, \frac{5}{3}, \frac{7}{3}\right)$

Angle with y axis

$$\cos\theta = \frac{5/3}{\sqrt{1 + \frac{25}{9} + \frac{49}{9}}}$$

$$\cos\theta = \frac{5}{\sqrt{83}}$$

$$\therefore 415\cos^2\theta = \frac{25}{83} \times 415 = 125$$

122. Let the line $\frac{x-3}{7}=\frac{y-2}{-1}=\frac{z-3}{-4}$ intersect the plane containing the lines $\frac{x-4}{1}=\frac{y+1}{-2}=\frac{z}{1}$ and $4ax - y + 5z - 7a = 0 = 2x - 5y - z - 3$, $a \in R$ at the point $P(\alpha, \beta, \gamma)$. Then the value of $\alpha + \beta + \gamma$ equals _____. **[JEE(Main)July2022]**

माना रेखा $\frac{x-3}{7}=\frac{y-2}{-1}=\frac{z-3}{-4}$ एक समतल, जिसमें रेखाऐं $\frac{x-4}{1}=\frac{y+1}{-2}=\frac{z}{1}$ तथा $4ax - y + 5z - 7a = 0 = 2x - 5y - z - 3$, $a \in R$ स्थित हैं, तो बिन्दु $P(\alpha, \beta, \gamma)$ पर काटती है। तो $\alpha + \beta + \gamma$ का मान है _____. **[JEE(Main)July2022]**

Ans. 12

Sol. Equation of plane

$4ax - y + 5z - 7a + \lambda (2x - 5y - z - 3) = 0$

this satisfy $(4, -1, 0)$

$16a + 1 - 7a + \lambda (8 + 5 - 3) = 0$

$9a + 1 + 10\lambda = 0$ ...(1)

Normal vector of the plane A is $(4a + 2\lambda, -1 - 5\lambda, 5 - \lambda)$ vector along the line which contained the plane A is $i - 2j + k$

$\therefore 4a + 2\lambda + 2 + 10\lambda + 5 - \lambda = 0$

$11\lambda + 4a + 7 = 0$ ..(2)

Solve (1) and (2) to get $a = 1$, $\lambda = -1$

Now equation of plane

$x + 2y + 3z - 2 = 0$

Let the point in the line $\frac{x-3}{7}=\frac{y-2}{-1}=\frac{z-3}{-4}=t$

is $(7t + 3, -t + 2, -4t + 3)$ satisfy the equation of plane A

$7t + 3 - 2t + 4 + 9 - 12t - 2 = 0$

$t = 2$

So $\alpha + \beta + \gamma = 2t + 8 = 12$

123. Let $P(-2, -1, 1)$ and $Q\left(\frac{56}{17}, \frac{43}{17}, \frac{111}{17}\right)$ be the vertices of the rhombus PRQS. If the direction ratios of the diagnonal RS are $\alpha, -1, \beta$, where both α and β are integers of minimum absolute values, then $\alpha^2 + \beta^2$ is equal to _____. **[JEE(Main)July2022]**

माना P(–2, –1, 1) तथा $Q\left(\frac{56}{17},\frac{43}{17},\frac{111}{17}\right)$ एक समचतुर्भुज PRQS के शीर्ष हैं। यदि विकर्ण RS के दिक्–अनुपात $\alpha, -1, \beta$ हैं, जहाँ α तथा β दोनों निम्नतम निरपेक्ष मान के पूर्णांक हैं, तो $\alpha^2 + \beta^2$ बराबर है _____

[JEE(Main)July2022]

Ans. 450

Sol. $RS \equiv (\alpha, -1, \beta)$

$$\text{DR of } PQ \equiv \left(\frac{56}{17}+2, \frac{43}{17}+1, \frac{111}{17}-1\right)$$

$$\equiv \left(\frac{90}{17}, \frac{60}{17}, \frac{94}{17}\right)$$

$$\frac{90}{17}\alpha + \frac{60}{17}(-1) + \frac{94}{17}\beta = 0$$

$$90\alpha + 94\beta = 60$$

$$\beta = \frac{60 - 90\alpha}{94}$$

$$\beta = \frac{30(2 - 3\alpha)}{94}$$

$$\beta = -30\frac{(3\alpha - 2)}{94}$$

$$\beta = \frac{\beta}{-15} = \frac{3\alpha - 2}{47}$$

$$\Rightarrow \beta = -15, \alpha = -15$$

$$\alpha^2 + \beta^2 = 225 + 225$$

$$= 450$$

124. Let a line with direction ratio a, –4a, –7 be perpendicular to the lines with direction ratios 3, –1, 2b and b, a, –2. If the point of intersection of the line $\frac{x+1}{a^2+b^2} = \frac{y-2}{a^2-b^2} = \frac{z}{1}$ and the plane x – y + z = 0 is (α, β, γ), then $\alpha + \beta + \gamma$ is equal to _____. **[JEE(Main)July2022]**

माना दिक् अनुपास a, –4a, –7 की एक रेखा, अनुपात 3, –1, 2b तथा b, a, –2 की रेखाओं के लंबवत है। यदि रेखा $\frac{x+1}{a^2+b^2} = \frac{y-2}{a^2-b^2} = \frac{z}{1}$ तथा समतल x – y + z = 0 का प्रतिच्छेदन बिन्दु (α, β, γ) है तो $\alpha + \beta + \gamma$ बराबर है _____

[JEE(Main)July2022]

Ans. 10

Sol. $(a, -4a, -7) \perp$ to $(3, -1, 2b)$

$a = 2b$ .....(i)

$(a, -4a, -7) \perp$ to $(b, a, -2)$

$3a + 4a - 14b = 0$

$ab - 4a^2 + 14 = 0$ .....(ii)

From Equations (i) and (ii)

$2b^2 - 16b^2 + 14 = 0$

$b^2 = 1$

$a^2 = 4b^2 = 4$

$$\frac{x+1}{5} = \frac{y-2}{3} = \frac{z}{1} = k$$

$\alpha = 5k - 1$, $\beta = 3k + 2$, $\gamma = k$

As (α, β, γ) satisfies $x - y + z = 0$

$5k - 1 - (3k + 2) + k = 0$

$k = 1$

$\therefore \alpha + \beta + \gamma = 9k + 1 = 10$

Exercise J - A

1. If $\vec{a},\vec{b}$ and $\vec{c}$ are unit vectors satisfying $|\vec{a}-\vec{b}|^2+|\vec{b}-\vec{c}|^2+|\vec{c}-\vec{a}|^2=9$, then $|2\vec{a}+5\vec{b}+5\vec{c}|$ is.

[JEE-2012]

यदि $\vec{a},\vec{b}$ तथा $\vec{c}$ इकाई सदिश है जो $|\vec{a}-\vec{b}|^2+|\vec{b}-\vec{c}|^2+|\vec{c}-\vec{a}|^2=9$ को संतुष्ट करते है, तो $|2\vec{a}+5\vec{b}+5\vec{c}|$ होगा।

[JEE-2012]

Ans. 3

Sol. $\left|\vec{a}-\vec{b}\right|^2+\left|\vec{b}-\vec{c}\right|^2+\left|\vec{c}-\vec{a}\right|^2=9$

$\Rightarrow \quad 6-2\Sigma\vec{a}.\vec{b}=9$

$\Rightarrow \quad \Sigma\vec{a}.\vec{b}=-\frac{3}{2}$ ...(1)

$\left|\vec{a}+\vec{b}+\vec{c}\right|^2\geq 0$

$\Sigma\vec{a}^2+2\Sigma\vec{a}.\vec{b}\geq 0$

$\Sigma\vec{a}.\vec{b}\geq -\frac{3}{2}$

for equality $|\vec{a}+\vec{b}+\vec{c}|=0$

$\Rightarrow \vec{a}+\vec{b}+\vec{c}=0$

$5\vec{b}+5\vec{c}=-5\vec{a}$

$2\vec{a}+5\vec{b}+5\vec{c}=-3\vec{a}$

$\left|2\vec{a}+5\vec{b}+5\vec{c}\right|=3|\vec{a}|=3$

2. If $\vec{a}$ and $\vec{b}$ are vectors such that $\left|\vec{a}+\vec{b}\right|=\sqrt{29}$ and $\vec{a}\times(2\hat{i}+3\hat{j}+4\hat{k})=(2\hat{i}+3\hat{j}+4\hat{k})\times\vec{b}$ then a possible value of $(\vec{a}+\vec{b}).(-7\hat{i}+2\hat{j}+3\hat{k})$ is :

[JEE-2012]

यदि $\vec{a}$ तथा $\vec{b}$ दो सदिश हैं इस प्रकार कि $\left|\vec{a}+\vec{b}\right|=\sqrt{29}$ तथा $\vec{a}\times(2\hat{i}+3\hat{j}+4\hat{k})=(2\hat{i}+3\hat{j}+4\hat{k})\times\vec{b}$ है तो $(\vec{a}+\vec{b}).(-7\hat{i}+2\hat{j}+3\hat{k})$ का संभव मान होगा :

[JEE-2012]

(A) 0 (B) 3 (C) 4 (D) 8

Ans. C

Sol. $(\vec{a}+\vec{b})\times(2\hat{i}+3\hat{j}+4\hat{k})=0$

$\Rightarrow \quad \vec{a}+\vec{b}=\lambda(2\hat{i}+3\hat{j}+4\hat{k})$

$|\vec{a}+\vec{b}|=\sqrt{29} \Rightarrow |\lambda|=1$

$\vec{a}+\vec{b}=(2\hat{i}+3\hat{j}+4\hat{k})$

$(\vec{a}+\vec{b}).(-7\hat{i}+2\hat{j}+3\hat{k})$

$=-14+6+12=4$

3. The point P is the intersection of the straight line joining the points Q(2, 3, 5) and R(1, –1, 4) with the plane $5x-4y-z=1$. If S is the foot of the perpendicular drawn from the point T(2, 1, 4) to QR, then the length of the line segment PS is : **[JEE-2012]**

यदि बिन्दु P, बिन्दु Q(2, 3, 5) तथा R(1, –1, 4) को जोड़ने वाली रेखा का तल $5x-4y-z=1$ के साथ प्रतिछेद बिन्दु है तथा बिन्दु S, बिन्दु T(2, 1, 4) से QR पर डाला गया लम्ब पाद है तो रेखाखण्ड PS की लम्बाई होगी : **[JEE-2012]**

(A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$ (C) 2 (D) $2\sqrt{2}$

Ans. A

Sol. Line QR : $\frac{x-2}{1}=\frac{y-3}{4}=\frac{z-5}{1}=\lambda$

Any point on line QR :

$(\lambda+2, 4\lambda+3, \lambda+5)$

$\therefore$ Point of intersection with plane :

$5\lambda+10-16\lambda-12-\lambda-5=1$

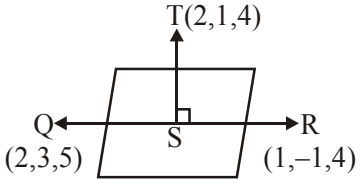
$\Rightarrow \quad \lambda=-\frac{2}{3}$

$\therefore \quad P\left(\frac{4}{3},\frac{1}{3},\frac{13}{3}\right)$

Also

$\because$ TQ = TR = $\sqrt{5}$

$\Rightarrow$ S is the mid-point of QR



$\Rightarrow \quad S\left(\frac{3}{2}, 1, \frac{9}{2}\right) \Rightarrow \quad PS=\frac{1}{\sqrt{2}}$ units

4. The equation of a plane passing through the line of intersection of the planes $x+2y+3z=2$ and $x-y+z=3$ and at a distance $\frac{2}{\sqrt{3}}$ from the point (3, 1, –1) is : **[JEE-2012]**

तल $x + 2y + 3z = 2$ तथा $x - y + z = 3$ की प्रतिछेदी रेखा से गुजरने वाले तथा बिन्दु $(3, 1, -1)$ से $\frac{2}{\sqrt{3}}$ दूरी पर स्थित तल का समीकरण होगा : **[JEE-2012]**

(A) $5x - 11y + z = 17$ (B) $\sqrt{2}x + y = 3\sqrt{2} - 1$ (C) $x + y + z = \sqrt{3}$ (D) $x - \sqrt{2}y = 1 - \sqrt{2}$

Ans. A

Sol. Let required plane be $(x + 2y + 3z - 2) + \lambda(x - y + z - 3) = 0$

$\because$ plane is at a distance $\frac{2}{\sqrt{3}}$ from the point $(3,1,-1)$.

$$\Rightarrow \left|\frac{(3+2-3-2)+\lambda(3-1-1-3)}{\sqrt{(1+\lambda)^2+(2-\lambda)^2+(3+\lambda)^2}}\right| = \frac{2}{\sqrt{3}}$$

$$\Rightarrow \lambda^2 = \frac{(1+\lambda)^2+(2-\lambda)^2+(3+\lambda)^2}{3}$$

$\Rightarrow 3\lambda^2 = 3\lambda^2 + 2\lambda - 4\lambda + 6\lambda + 14$

$$\Rightarrow \lambda = -\frac{7}{2}$$

$\therefore$ required plane is $(x + 2y + 3z - 2) + \left(-\frac{7}{2}\right)(x - y + z - 3) = 0$

$\Rightarrow 5x - 11y + z = 17$

5. If the straight line $\frac{x-1}{2} = \frac{y+1}{k} = \frac{z}{2}$ and $\frac{x+1}{5} = \frac{y+1}{2} = \frac{z}{k}$ are coplanar, then the plane(s) containing these two lines is(are) : **[JEE-2012]**

यदि रेखाऐं $\frac{x-1}{2} = \frac{y+1}{k} = \frac{z}{2}$ तथा $\frac{x+1}{5} = \frac{y+1}{2} = \frac{z}{k}$ समतलीय हैं तो इन रेखाओं को अन्तर्विष्ट करने वाले तल का समीकरण होगा : **[JEE-2012]**

(A) $y + 2z = -1$ (B) $y + z = -1$ (C) $y - z = -1$ (D) $y - 2z = -1$

Ans. BC

Sol. $(1, -1, 0); (-1, -1, 0)$

For coplanarity of lines

$$\begin{vmatrix} 2 & 0 & 0 \\ 2 & k & 2 \\ 5 & 2 & k \end{vmatrix} = 0 \Rightarrow 2(k^2 - 4) = 0$$

$\Rightarrow k = \pm 2$

for $k = 2$

Normal vector $\vec{n} = \hat{j} - \hat{k}$

$\therefore$ Required plane : $y - z = \lambda$

$\because$ Passes through $(1, -1, 0)$

$\Rightarrow \lambda = -1$

$\therefore y - z = -1$

for $k = -2$

$\vec{n} = \hat{j} + \hat{k}$

$\therefore$ Required plane : $y + z = \lambda$

$\because$ Passes through $(1, -1, 0)$

$\Rightarrow \lambda = -1$

$\therefore y + z = -1$

6. Let $\overrightarrow{PR} = 3\hat{i} + \hat{j} - 2\hat{k}$ and $\overrightarrow{SQ} = \hat{i} - 3\hat{j} - 4\hat{k}$ determine diagonals of a parallelogram PQRS and $\overrightarrow{PT} = \hat{i} + 2\hat{j} + 3\hat{k}$ be another vector. Then the volume of the parallelepiped determined by the vectors $\overrightarrow{PT}, \overrightarrow{PQ}$ and $\overrightarrow{PS}$ is :

[JEE(Adv.)2013]

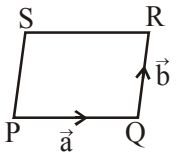
माना कि $\overrightarrow{PR} = 3\hat{i} + \hat{j} - 2\hat{k}$ तथा $\overrightarrow{SQ} = \hat{i} - 3\hat{j} - 4\hat{k}$ एक समांतर चतुर्भुज PQRS के विकर्ण निर्धारित करते है और $\overrightarrow{PT} = \hat{i} + 2\hat{j} + 3\hat{k}$ एक अन्य सदिश है। तब सदिशों $\overrightarrow{PT}, \overrightarrow{PQ}$ तथा $\overrightarrow{PS}$ द्वारा निर्धारित समान्तर षट्फलक का आयतन है :

[JEE(Adv.)2013]

(A) 5 (B) 20 (C) 10 (D) 30

Ans. C

Sol. $\vec{a} + \vec{b} = \overrightarrow{PR}$ and $\vec{a} - \vec{b} = \overrightarrow{QS}$



$\vec{a} = \dfrac{\overrightarrow{PR} + \overrightarrow{QS}}{2}$ and $\vec{b} = \dfrac{\overrightarrow{PR} - \overrightarrow{QS}}{2}$

$\vec{a} = 2\hat{i} - \hat{j} - 3\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$

$$\text{Volume} = \begin{vmatrix} 2 & -1 & -3 \\ 1 & 2 & 1 \\ 1 & 2 & 3 \end{vmatrix}$$

$2(4) + (3 - 1) - 3(2 - 2)$

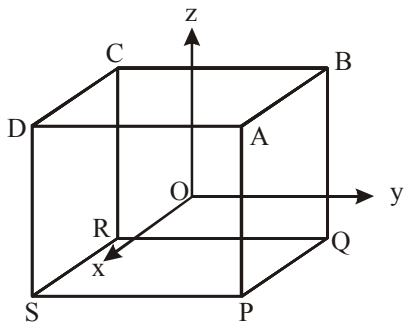
$8 + 2 = 10$

7. Consider the set of eight vectors $V = \left\{a\hat{i} + b\hat{j} + c\hat{k} : a, b, c \in \{-1, 1\}\right\}$. Three non-coplanar vectors can be chosen from be chosen from V in 2^p ways. Then p is : **[JEE(Adv.)2013]**

आठ सदिशों को समुच्चय $V = \left\{a\hat{i} + b\hat{j} + c\hat{k} : a, b, c \in \{-1, 1\}\right\}$ लीजिये। V से तीन असमतलीय सदिश 2^p प्रकार से का मान है : **[JEE(Adv.)2013]**

Ans. 5

Sol.



O is at the centre of cube

ABCDPQRS

The 8 vectors will represent

$\overrightarrow{OA}, \overrightarrow{OB}.....\overrightarrow{OD}, \overrightarrow{OP}, ...\overrightarrow{OS}$

any three out of these 8 will be coplanar

when two of them are collinear. There are 4 pairs of collinear vectors

$\overrightarrow{OA}$ and $\overrightarrow{OR}$, $\overrightarrow{OB}$ and $\overrightarrow{OS}$, $\overrightarrow{OC}$ and $\overrightarrow{OP}$, $\overrightarrow{OD}$ and $\overrightarrow{OQ}$ (it will generate $4 \times 6 = 24$ set of coplanar vectors) rest of the combination of 3 vectors will form three edges of a tetrahedron so they will be not coplanar.

So number of non-coplanar vectors

${}^8C_3 - 4.6 = 32$

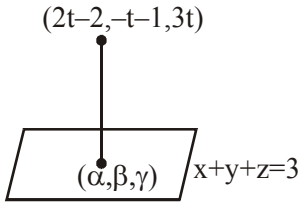
8. Perpendiculars are drawn from points on the line $\frac{x+2}{2} = \frac{y+1}{-1} = \frac{z}{3}$ to the plane x + y + z = 3. The feet of perpendiculars lie on the line : **[JEE(Adv.)2013]**

तल x + y + z = 3 पर रेखा $\frac{x+2}{2} = \frac{y+1}{-1} = \frac{z}{3}$ पर स्थित बिन्दुओं से लम्ब डाले जाते है। लम्ब पाद निम्न रेखा पर स्थित है : **[JEE(Adv.)2013]**

(A) $\frac{x}{5} = \frac{y-1}{8} = \frac{z-2}{-13}$ (B) $\frac{x}{2} = \frac{y-1}{3} = \frac{z-2}{-5}$ (C) $\frac{x}{4} = \frac{y-1}{3} = \frac{z-2}{-7}$ (D) $\frac{x}{2} = \frac{y-1}{-7} = \frac{z-2}{5}$

Ans. D

Sol. $\frac{\alpha-2t+2}{1}=\frac{\beta+t+1}{1}=\frac{\gamma-3t}{1}=k$



$\alpha = k + 2t - 2$

$\beta = k - t - 1$

$\gamma = k + 3t$

$\alpha + \beta + \gamma = 3$

$k = \frac{6-4t}{3}$

$\alpha = \frac{6-4t}{3} + 2t - 2 = \frac{2t}{3}$

$\beta = \frac{6-4t}{3} - t - 1 = \frac{3-7t}{3}$

$\gamma = \frac{6-4t}{3} + 3t = \frac{5t+6}{3}$

$\Rightarrow \frac{3\alpha}{2} = \frac{3\beta-3}{-7} = \frac{3\gamma-6}{5}$

$\Rightarrow \frac{x}{2} = \frac{y-3}{-7} = \frac{z-2}{5}$

9. A line l passing through the origin is perpendicular to the lines

$l_1 : (3+t)\hat{i} + (-1+2t)\hat{j} + (4+2t)\hat{k}, -\infty < t < \infty$ $\quad$ $l_2 : (3+2s)\hat{i} + (3+2s)\hat{j} + (2+s)\hat{k}, -\infty < s < \infty$

Then, the coordinate(s) of the point(s) on l_2 at a distance of $\sqrt{17}$ from the point of intersection of l and l_1 is (are):

[JEE(Adv.)2013]

एक रेखा l जो मूलबिन्दु से गुजरती है, रेखाओं

$l_1 : (3+t)\hat{i} + (-1+2t)\hat{j} + (4+2t)\hat{k}, -\infty < t < \infty$ $\quad$ $l_2 : (3+2s)\hat{i} + (3+2s)\hat{j} + (2+s)\hat{k}, -\infty < s < \infty$

पर लम्बवत है। तब l_2 पर स्थित बिन्दु के निर्देशांक, जो रेखाओं l तथा l_1 के प्रतिच्छेद बिन्दु से $\sqrt{17}$ की दूरी पर है, निम्न है :

[JEE(Adv.)2013]

(A) $\left(\frac{7}{3},\frac{7}{3},\frac{5}{3}\right)$ $\quad$ (B) $(-1,-1,0)$ $\quad$ (C) $(1,1,1)$ $\quad$ (D) $\left(\frac{7}{9},\frac{7}{9},\frac{8}{9}\right)$

Ans. BD

Sol. $\ell_1 : \vec{r} = (3,-1,4) + (1,2,2)t$

$\ell_2 : \vec{r} = (3,3,2) + (2,2,1)s$

vector perpendicular to ℓ_1 and ℓ_2 :

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 2 \\ 2 & 2 & 1 \end{vmatrix} = -2\hat{i} + 3\hat{j} - 2\hat{k}$$

$\therefore$ Equation of line ℓ : $\vec{r} = 0 + (-2, 3, -2)\lambda$

Point of intersection of ℓ_1 and ℓ :

$3 + t = -2\lambda$

$-1 + 2t = 3\lambda$.

$4 + 2t = -2\lambda$.

On solving we get $\lambda = -1$, $t = -1$

$\therefore$ Point of intersection of ℓ_1 & ℓ : P(2, –3, 2)

A point on ℓ_2 at distance of $\sqrt{17}$ from P :

$\Rightarrow$ $(1 + 2s)^2 + (6 + 2s)^2 + s^2 = 17$

$\Rightarrow$ $s = -\frac{10}{9}; s = -2$

for above s, point will be (B), (D)

10. Two lines L_1: $x = 5$, $\frac{y}{3-\alpha} = \frac{z}{-2}$ and L_2: $x = \alpha$, $\frac{y}{-1} = \frac{z}{2-\alpha}$ are coplanar. Then α can take value(s) :

[JEE(Adv.)2013]

दो रेखाएँ L_1: $x = 5$, $\frac{y}{3-\alpha} = \frac{z}{-2}$ तथा L_2: $x = \alpha$, $\frac{y}{-1} = \frac{z}{2-\alpha}$ समतलीय हैं। तब α का मान हो सकता है :

[JEE(Adv.)2013]

(A) 1 (B) 2 (C) 3 (D) 4

Ans. AD

Sol. $L_1 : \frac{x-5}{0} = \frac{y}{3-\alpha} = \frac{z}{-2}$

$L_2 : \frac{x-\alpha}{0} = \frac{y}{-1} = \frac{z}{2-\alpha}$

for lines to be coplanar

$$\begin{vmatrix} 5-\alpha & 0 & 0 \\ 0 & 3-\alpha & -2 \\ 0 & -1 & 2-\alpha \end{vmatrix} = 0$$

$\Rightarrow$ $(5 - \alpha)((3 - \alpha)(2 - \alpha) - 2) = 0$

$\Rightarrow$ $(5 - \alpha)(\alpha^2 - 5\alpha + 4) = 0$

$\Rightarrow$ $\alpha = 1, 4, 5$

11. Consider the line $L_1 : \frac{x-1}{2} = \frac{y}{-1} = \frac{z+3}{1}, L_2 : \frac{x-4}{1} = \frac{y+3}{1} = \frac{z+3}{2}$ and the planes P_1: 7x + y + 2z = 3, $P_2 : 3x + 5y - 6z = 4$. Let $ax + by + cz = d$ be the equation of the plane passing through the point of intersection of lines L_1 and L_2. and perpendicular to planes P_1 and P_2. **[JEE(Adv.)2013]**

Match Column – I with Column – II and select the correct answer using the code given below the lists :

रेखाएँ $L_1 : \frac{x-1}{2} = \frac{y}{-1} = \frac{z+3}{1}, L_2 : \frac{x-4}{1} = \frac{y+3}{1} = \frac{z+3}{2}$ का समतल P_1: 7x + y + 2z = 3, P_2 : 3x + 5y – 6z = 4 लीजिये। माना कि ax + by + cz = d रेखाओं L_1 व L_2 के प्रतिच्छेद बिन्दु से गुजरने वाला तथा समतल P_1 तथा P_2 के लम्बवत्, समतल का समीकरण है। **[JEE(Adv.)2013]**

Column – I को Column – II से सुमेलित कीजिए तथा सूचियों के नीचे दिए गए कोड का प्रयोग करके सही उत्तर चुनिए :

Column I	**Column II**
A. a =	p. 13
B. b =	q. –3
C. c =	r. 1
D. d =	s. –2

Codes :

(A) A-r; B-q; C-s; D-p (B) A-p; B-r; C-s; D-q

(C) A-r; B-q; C-p; D-s (D) A-q; B-s; C-p; D-r

Ans. A

Sol. $L_1 : \frac{x-1}{2} = \frac{y}{-1} = \frac{z+3}{1}$

Normal of plane P : $\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 7 & 1 & 2 \\ 3 & 5 & -6 \end{vmatrix}$

$= \hat{i}(-16) - \hat{j}(-42-6) + \hat{k}(32)$

$= -16\hat{i} + 48\hat{j} + 32\hat{k}$

$\Rightarrow \vec{n} = \hat{i} - 3\hat{j} - 2\hat{k}$

Point of intersection of L_1 and L_2

$2k_1 + 1 = k_2 + 4$

$-k_1 = k_2 - 3$

$1 = 3k_2 - 2$

$k_2 = 1$

Point of intersection (5, –2, –1)

Plane $(x - 5) - 3(y + 7) - 2(z + 1) = 0$

$x - 3y - 2z - 5 - 6 - 2 = 0$

$x - 3y - 2z = 13$

$\Rightarrow$ $a = 1, b = -3, c = -2, d = 13$

12. Match Column-I with Column-II and select the correct answer using the code given below the column :

[JEE(Adv.)2013]

	Column- I		**Column-II**
A.	Volume of parallelepiped determined by vectors $\vec{a}, \vec{b}$ and $\vec{c}$ is 2. Then the volume of the parallelepiped determined by vectors $2(\vec{a}\times\vec{b}), 3(\vec{b}\times\vec{c})$ and $(\vec{c}\times\vec{a})$ is	p.	100
B.	Volume of parallelepiped determined by vecors $\vec{a}, \vec{b}$ and $\vec{c}$ is 5. Then the volume of the parallelepiped determined by vectors $3(\vec{a}+\vec{b}), (\vec{b}+\vec{c})$ and $2(\vec{c}+\vec{a})$ is	q	. 30
C.	Area of a triangle with adjacent sides determined by vectors $\vec{a}$ and $\vec{b}$ is 20. Then the area of the triangle with adjacent sides determined by vectors $(2\vec{a}+3\vec{b})$ and $(\vec{a}-\vec{b})$ is	r.	24
D.	Area of a parallelogram with adjacent sides determined by vectors $\vec{a}$ and $\vec{b}$ is 30. Then the area of the parallelogram with adjacent sides determined by vectors $(\vec{a}+\vec{b})$ and $\vec{a}$ is	s.	60

Codes :

(A) A-s; B-q; C-r; D-p (B) A-q; B-r; C-p; D-s

(C) A-r; B-s; C-p; D-q (D) A-p; B-s; C-r; D-q

Column-I को Column-II से सुमेलित कीजिए और column के नीचे दिये गये कूट का प्रयोग कर सही उत्तर चुनिए :

[JEE(Adv.)2013]

	Column- I		**Column-II**
A.	सदिशों $\vec{a}, \vec{b}$ तथा $\vec{c}$ द्वारा निर्धारित संमान्तर षट्फलक का आयतन 2 है। तब सदिशों $2(\vec{a}\times\vec{b}), 3(\vec{b}\times\vec{c})$ तथा $(\vec{c}\times\vec{a})$ द्वारा निर्धारित समान्तर षट्फलक का आयतन है	p.	100
B.	सदिशों $\vec{a}, \vec{b}$ तथा $\vec{c}$ द्वारा निर्धारित समान्तर षट्फलक का आयतन 5 है। तब सदिशों $3(\vec{a}+\vec{b}), (\vec{b}+\vec{c})$ तथा $2(\vec{c}+\vec{a})$ द्वारा निर्धारित समांतर षट्फलक का आयतन है	q	. 30
C.	एक त्रिभुज का क्षेत्रफल जिसकी संलग्न भुजाएँ सदिशों $\vec{a}$ तथा $\vec{b}$ द्वारा निर्धारित है, 20 है। तब सदिशों $(2\vec{a}+3\vec{b})$ तथा $(\vec{a}-\vec{b})$ द्वारा निर्धारित संलग्न भुजाओं वाले त्रिभुज का क्षेत्रफल है।	r.	24

D. एक समांतर चतुर्भुज का क्षेत्रफल जिसकी संलग्न भुजाएँ सदिशों $\vec{a}$ तथा $\vec{b}$ द्वारा निर्धारित हैं 30 है। तब सदिशों $(\vec{a}+\vec{b})$ तथा $\vec{a}$ द्वारा निर्धारित संलग्न भुजाओं वाले समांतर चतुर्भुज का क्षेत्रफल है। s. 60

Codes :

(A) A-s; B-q; C-r; D-p (B) A-q; B-r; C-p; D-s

(C) A-r; B-s; C-p; D-q (D) A-p; B-s; C-r; D-q

Ans. C

Sol. A. Given $[\vec{a}\ \vec{b}\ \vec{c}]=2$

$$[2(\vec{a}\times\vec{b})\ \ 3(\vec{b}\times\vec{c})\ \ (\vec{c}\times\vec{a})]$$

$$=6[\vec{a}\times\vec{b}\ \ \vec{b}\times\vec{c}\ \ \vec{c}\times\vec{a}] = 6[\vec{a}\ \vec{b}\ \vec{c}]^2 = 24$$

B. Given $[\vec{a}\ \vec{b}\ \vec{c}]=5$

$$[3(\vec{a}+\vec{b})\ \ (\vec{b}+\vec{c})\ \ 2(\vec{c}+\vec{a})]$$

$$=12[\vec{a}\ \ \vec{b}\ \ \vec{c}]=60$$

C. Given $\frac{1}{2}|\vec{a}\times\vec{b}|=20 \Rightarrow \quad |\vec{a}\times\vec{b}|=40$

$$\left|\frac{1}{2}(2\vec{a}+3\vec{b})\times(\vec{a}-\vec{b})\right|=\frac{1}{2}\left|0+3\vec{b}\times\vec{a}-2\vec{a}\times\vec{b}\right|$$

$$=\frac{1}{2}|-5\vec{a}\times\vec{b}|=\frac{5}{2}|\vec{a}\times\vec{b}|=\frac{5}{2}.40=100$$

D. Given $|\vec{a}\times\vec{b}|=30$

$$(\vec{a}+\vec{b})\times\vec{a}|=|0+\vec{b}\times\vec{a}|=30$$

13. Let $\vec{x},\vec{y}$ and $\vec{z}$ be three vectors each of magnitude $\sqrt{2}$ and the angle between each pair of them is $\frac{\pi}{3}$. If $\vec{a}$ is a nonzero vector perpendicular to $\vec{x}$ and $\vec{y}\times\vec{z}$ and $\vec{b}$ is a nonzero vector perpendicular to $\vec{y}$ and $\vec{z}\times\vec{x}$, then :

[JEE(Adv.)2014]

माना कि सदिशों $\vec{x},\vec{y}$ तथा $\vec{z}$ में प्रत्येक का परिमाण $\sqrt{2}$ हैं तथा प्रत्येक युग्म (pair) के मध्य का कोण $\frac{\pi}{3}$ है। यदि शून्येतर (non-zero) सदिश $\vec{a}$ सदिशों $\vec{x}$ तथा $\vec{y}\times\vec{z}$ के लम्बवत् है एवं शून्येतर सदिश $\vec{b}$ सदिशों $\vec{y}$ तथा $\vec{z}\times\vec{x}$ के लम्बवत् है, तब :

[JEE(Adv.)2014]

(A) $\vec{b}=(\vec{b}\cdot\vec{z})(\vec{z}-\vec{x})$ (B) $\vec{a}=(\vec{a}\cdot\vec{y})(\vec{y}-\vec{z})$ (C) $\vec{a}\cdot\vec{b}=-(\vec{a}\cdot\vec{y})(\vec{b}\cdot\vec{z})$ (D) $\vec{a}=(\vec{a}\cdot\vec{y})(\vec{z}-\vec{y})$

Ans. ABC

Sol. $\vec{a}$ is in direction of $\vec{x}\times(\vec{y}\times\vec{z})$

i.e. $(\vec{x}.\vec{z})\vec{y}-(\vec{x}.\vec{y})\vec{z}$

$\Rightarrow \vec{a}=\lambda_1\left[2\times\frac{1}{2}(\vec{y}-\vec{z})\right]$

$\vec{a}=\lambda_1(\vec{y}-\vec{z})$ _____(1)

Now $\vec{a}.\vec{y}=\lambda_1(\vec{y}.\vec{y}-\vec{y}.\vec{z})$

$=l_1(2-1)$ Þ $l_1=\vec{a}.\vec{y}$ _____(2)

From (1) and (2), $\vec{a}=\vec{a}.\vec{y}(\vec{y}-\vec{z})$

Similarly, $\vec{b}=(\vec{b}.\vec{z})(\vec{z}-\vec{x})$

Now, $\vec{a}.\vec{b}=(\vec{a}.\vec{y})(\vec{b}.\vec{z})[(\vec{y}-\vec{z}).(\vec{z}.\vec{x})]$

$(\vec{a}.\vec{y})(\vec{b}.\vec{z})[1-1-2+1]$

$=-(\vec{a}.\vec{y})(\vec{b}.\vec{z})$

14. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three non-coplanar unit vectors such that the angle between every pair of them is $\frac{\pi}{3}$ If $\vec{a}\times\vec{b}+\vec{b}\times\vec{c}=p\vec{a}+q\vec{b}+r\vec{c}$, where p, q, and r are scalars, then the value of $\frac{p^2+2q^2+r^2}{q^2}$ is.

[JEE(Adv.)2014]

माना $\vec{a}$, $\vec{b}$ तथा $\vec{c}$ तीन विषमतलीय ईकाई सदिश है। इस प्रकार कि प्रत्येक युग्म के मध्य कोण $\frac{\pi}{3}$ है। यदि $\vec{a}\times\vec{b}+\vec{b}\times\vec{c}=p\vec{a}+q\vec{b}+r\vec{c}$, जहाँ p, q तथा r अदिश है, तो $\frac{p^2+2q^2+r^2}{q^2}$ का मान होगा। **[JEE(Adv.)2014]**

Ans. 4

Sol. $|\vec{a}|=|\vec{b}|=|\vec{c}|=1$

$\vec{a}\times\vec{b}+\vec{b}\times\vec{c}=p\vec{a}+q\vec{b}+r\vec{c}$

$\vec{a}.\left(\vec{b}\times\vec{c}\right)=p+q\left(\vec{a}.\vec{b}\right)+r\left(\vec{a}.\vec{c}\right)$

And $\left[\vec{a}\ \vec{b}\ \vec{c}\right]=\frac{1}{\sqrt{2}}$

$p+\frac{q}{2}+\frac{r}{2}=\left[\vec{a}\ \vec{b}\ \vec{c}\right]$ _____(1)

$\frac{p}{2}+q+\frac{r}{2}=0$ _____(2)

$\frac{p}{2}+\frac{q}{2}+r=\left[\vec{a}\,\vec{b}\,\vec{c}\right]$ _____(3)

Þ $p = r = -q$

$\frac{p^2+2q^2+r^2}{q^2}=4$

15. Match Column-I with Column-II and select the correct answer using the code given below the column :

[JEE(Adv.)2014]

Column-I	**Column-II**
A. Let $y(x)=\cos(3\cos^{-1}x)$, $x\in[-1,1]$, $x\neq\pm\frac{\sqrt{3}}{2}$. Then $\frac{1}{y(x)}\left\{(x^2-1)\frac{d^2y(x)}{dx^2}+x\frac{dy(x)}{dx}\right\}$ equals	p. 1
B. Let $A_1, A_2, \ldots\ldots, A_n (n>2)$ be the vertices of a regular polygon of n sides with its centre at the origin. Let $\vec{a}_k$ be the position vector of the point A_k, $k = 1, 2, \ldots\ldots n$. If $\left\lvert\sum_{k=1}^{n-1}(\vec{a}_k\times\vec{a}_{k+1})\right\rvert=\left\lvert\sum_{k=1}^{n-1}(\vec{a}_k\cdot\vec{a}_{k+1})\right\rvert$, then the minimum value of n is	q. 2
C. If the normal from the point P(h, 1) on the ellipse $\frac{x^2}{6}+\frac{y^2}{3}=1$ is perpendicular to the line $x + y = 8$, then the value of h is	r. 8
D. Number of positive solutions satisfying the equation $\tan^{-1}\left(\frac{1}{2x+1}\right)+\tan^{-1}\left(\frac{1}{4x+1}\right)=\tan^{-1}\left(\frac{2}{x^2}\right)$ is	s. 9

Codes:

(A) A-s; B-r; C-q; D-p

(B) A-q; B-s; C-r; D-p

(C) A-s; B-r; C-p; D-q

(D) A-q; B-s; C-p; D-r

Column-I को Column-II से सुमेलित कीजिए और column के नीचे दिये गये कूट का प्रयोग कर सही उत्तर चुनिए :

[JEE(Adv.)2014]

Column-I	**Column-II**
A. माना कि $y(x) = \cos(3\cos^{-1}x)$, $x \in [-1, 1]$, $x \neq \pm\frac{\sqrt{3}}{2}$, तो $\frac{1}{y(x)}\left\{(x^2-1)\frac{d^2y(x)}{dx^2}+x\frac{dy(x)}{dx}\right\}$ का मान है	p. 1
B. मानाकि $A_1, A_2, \ldots\ldots, A_n (n > 2)$ एक n भुजीय समबहुभुज के शीर्ष है जिसका केन्द्र मूलबिन्दु में है। माना कि $\vec{a}_k$ A_k, $k = 1, 2, \ldots\ldots n$. का स्थिति सदिश है। यदि $\left\lvert\sum_{k=1}^{n-1}(\vec{a}_k \times \vec{a}_{k+1})\right\rvert = \left\lvert\sum_{k=1}^{n-1}(\vec{a}_k \cdot \vec{a}_{k+1})\right\rvert$ है, तब n का न्यूनतम मान है।	q. 2
C. यदि दीर्घवृत्त $\frac{x^2}{6}+\frac{y^2}{3}=1$ पर बिन्दु P(h, 1) से खींचा गया अभिलंब, रेखा $x + y = 8$ पर लम्बवत् है, तो h का मान है।	r. 8
D. समीकरण $\tan^{-1}\left(\frac{1}{2x+1}\right)+\tan^{-1}\left(\frac{1}{4x+1}\right)=\tan^{-1}\left(\frac{2}{x^2}\right)$ को संतुष्ट करने वाले धनात्मक हलों की संख्या है।	s. 9

कूट :

(A) A-s; B-r; C-q; D-p
(B) A-q; B-s; C-r; D-p
(C) A-s; B-r; C-p; D-q
(D) A-q; B-s; C-p; D-r

Ans. A

Sol. (A) $y = \cos(3\cos^{-1}x)$

$$y' = \frac{3\sin\left(3\cos^{-1}x\right)}{\sqrt{1-x^2}}$$

$$\sqrt{1-x^2}\,y' = 3\sin\left(3\cos^{-1}x\right)$$

$$\Rightarrow \frac{-x}{\sqrt{1-x^2}}y' + \sqrt{1-x^2}\,y'' = 3\cos\left(3\cos^{-1}x\right)\cdot\frac{-3}{\sqrt{1-x^2}}$$

$$\Rightarrow -xy' + \left(1-x^2\right)y'' = -9y$$

$$\Rightarrow \frac{1}{y}\left[\left(x^2-1\right)y'' + xy'\right] = 9$$

(B) $\left(a_k \times a_{k+1}\right) = r^2\sin\frac{2\pi}{n}$

$$a_k \cdot a_{k+1} = r^2\cos\frac{2\pi}{n}$$

$$\Rightarrow \left|\sum_{k=1}^{n-1} \vec{a}_k \times \vec{a}_{k+1}\right| = \left|\sum_{k=1}^{n-1} a_k \cdot a_{k+1}\right|$$

$$\Rightarrow r^2(n-1)\sin\frac{2\pi}{n} = r^2(n-1)\cos\frac{2\pi}{n}$$

$$\tan\frac{2\pi}{n} = 1 \Rightarrow n = \frac{8}{4k+1}$$

$$\Rightarrow n = 8 .$$

(C) $\quad \frac{h^2}{6} + \frac{1^2}{3} = 1, \; h = \pm 2$

Tangent at (2, 1) is $\frac{2x}{6} + \frac{y}{3} = 1$ $x + y = 3$.

(D) $\quad \tan^{-1}\left(\frac{1}{2x+1}\right) + \tan^{-1}\frac{1}{4x+1} = \tan^{-1}\frac{2}{x^2}$

$$\tan^{-1}\left(\frac{3x+1}{4x^2+3x}\right) = \tan^{-1}\frac{2}{x^2}$$

$$\Rightarrow 3x^2 - 7x - 6 = 0$$

$$x = -\frac{2}{3}, 3.$$

16. From a point p(λ, λ, λ), perpendiculars PQ and PR are drawn respectively on the lines y = x, z = 1 and y = –x, z = –1. If P is such that ∠QPR is a right angle, then the possible value(s) of λ is(are) : **[JEE(Adv.)2014]**

बिन्दु p(λ, λ, λ) से रेखाओं y = x, z = 1 तथा y = –x, z = –1 पर डाले गये लम्ब (Perpendicular) क्रमशः PQ तथा PR है। यदि ∠QPR समकोण (right angle) है तो λ का (के) सम्भावित मान है (हैं) : **[JEE(Adv.)2014]**

(A) $\sqrt{2}$ (B) 1 (C) –1 (D) $-\sqrt{2}$

Ans. C

Sol. Line is

$$\frac{x-0}{1} = \frac{y-0}{1} = \frac{z-1}{0} = \alpha \qquad \text{_____(1)}$$

Q(α, α, 1)

Direction ratio of PQ are

$\lambda - \alpha, \lambda - \alpha, \lambda - 1$

Since PQ is perpendicular to (1)

$\therefore \quad \lambda - \alpha + \lambda - \alpha + 0 = 0$

$\lambda = \alpha$

$\therefore$ Direction ratio of PQ are

$0, 0, \lambda - 1$

Another line is

$$\frac{x-0}{-1} = \frac{y-0}{1} = \frac{z+1}{0} = \beta \qquad \text{_____(2)}$$

$\therefore$ R $(-\beta, \beta, -1)$

$\therefore$ Direction ratio of PR are

$\lambda + \beta, \lambda - \beta, \lambda + 1$

Since PQ is perpendicular to (2)

$\therefore$ $-\lambda - \beta + \lambda - \beta = 0$

$\beta = 0$

$\therefore$ R$(0, 0, -1)$

and Direction ratio of PQ are $\lambda, \lambda, \lambda + 1$

Since PQ $\perp$ PR

$\therefore$ $0 + 0 + \lambda^2 - 1 = 0 \Rightarrow \lambda = \pm 1 \Rightarrow$ B, C

For $\lambda = 1$ the point is on the line so it will be rejected.

$\Rightarrow$ $\lambda = -1$

17. Suppose that $\vec{p}, \vec{q}$ and $\vec{r}$ are three non-coplanar vectors in R^3. Let the components of a vector $\vec{s}$ along and $\vec{p}, \vec{q}$ and $\vec{r}$ be 4,3 and 5 respectively. If the components of this vector $\vec{s}$ along $(-\vec{p}+\vec{q}+\vec{r})$, $(\vec{p}-\vec{q}+\vec{r})$ and $(-\vec{p}-\vec{q}+\vec{r})$ are x, y and z respectively, then the value of $2x + y + z$ is. **[JEE(Adv.)2015]**

माना कि R^3 में $\vec{p}, \vec{q}$ और $\vec{r}$ तीन असमतलीय सदिश है। मानाकि सदिश $\vec{s}$ के घटक क्रमागत सदिशों $\vec{p}, \vec{q}$ एवं $\vec{r}$ के अनुदिश क्रमशः 4,3 और 5 हैं। यदि $\vec{s}$ के घटक क्रमागत सदिशों $(-\vec{p}+\vec{q}+\vec{r})$, $(\vec{p}-\vec{q}+\vec{r})$ एवं $(-\vec{p}-\vec{q}+\vec{r})$ के अनुदिश क्रमशः x, y और z हैं, तब $2x + y + z$ का मान है। **[JEE(Adv.)2015]**

Ans. 9

Sol. This question in seem to be wrong but examiner may think like this

$$\vec{S} = 4\vec{p} + 3\vec{q} + 5\vec{r}$$

$$\vec{S} = x(-\vec{p}+\vec{q}+\vec{r}) + y(\vec{p}-\vec{q}+\vec{r}) + z(-\vec{p}-\vec{q}+\vec{r})$$

$-x + y - z = 4$ .....(1)

$x - y - z = 3$ .....(2)

$x + y + z = 5$ .....(3)

add (1) and (2)

$$-2z = 7 \qquad \Rightarrow \qquad z = -\frac{7}{2}$$

$2x = 8 \quad \Rightarrow \quad x = 4$

$2x + y + z = 2(4) + 1 = 9$

18. Let ΔPQR be a triangle. Let $\vec{a} = \overrightarrow{QR}$, $\vec{b} = \overrightarrow{RP}$ and $\vec{c} = \overrightarrow{PQ}$. If $|\vec{a}| = 12$, $|\vec{b}| = 4\sqrt{3}$ and $\vec{b}.\vec{c} = 24$, then which of the following is (are) true? **[JEE(Adv.)2015]**

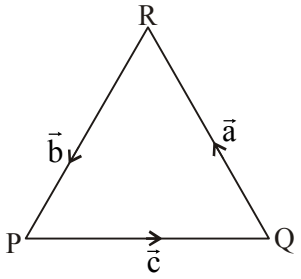
माना कि ΔPQR एक त्रिभुज है। माना कि $\vec{a} = \overrightarrow{QR}$, $\vec{b} = \overrightarrow{RP}$ और $\vec{c} = \overrightarrow{PQ}$ है। यदि $|\vec{a}| = 12$, $|\vec{b}| = 4\sqrt{3}$ तथा $\vec{b}.\vec{c} = 24$, तब निम्नलिखित में से कौन कौन से सही है? **[JEE(Adv.)2015]**

(A) $\frac{|\vec{c}|^2}{2} - |\vec{a}| = 12$ (B) $\frac{|\vec{c}|^2}{2} + |\vec{a}| = 30$

(C) $|\vec{a} \times \vec{b} + \vec{c} \times \vec{a}| = 48\sqrt{3}$ (D) $\vec{a}.\vec{b} = -72$

Ans. ACD

Sol. $\vec{a} + \vec{b} + \vec{c} = 0$



$\Rightarrow \quad \vec{b} + \vec{c} = -\vec{a}$

$\Rightarrow \quad 48 + \vec{c}^2 + 48 = 144$

$\Rightarrow \quad \vec{c}^2 = 48$

$\Rightarrow \quad |\vec{c}| = 4\sqrt{3}$

$\therefore \quad \frac{|\vec{c}^2|}{2} - |\vec{a}| = 24 - 12 = 12$ Ans. (A)

Further

$\vec{a} + \vec{b} = -\vec{c}$

$\Rightarrow \quad 144 + 48 + 2\vec{a}.\vec{b} = 48$

$\Rightarrow \quad \vec{a}.\vec{b} = -72$ Ans. (D)

$\because \quad \frac{|\vec{c}^2|}{2} - |\vec{a}| = 24 - 12 = 12$

$\Rightarrow \quad \vec{a} \times \vec{b} + \vec{a} \times \vec{c} = 0$

$\because \quad |\vec{a} \times \vec{b} + \vec{c} \times \vec{a}| = 2|\vec{a} \times \vec{b}| = 2.\sqrt{144.48 - (72)^2} = 48\sqrt{3}$ Ans. (C)

19. In R^3, consider the planes $P_1 : y = 0$ and $P_2 : x + z = 1$. Let P_3 be a plane, different from P_1 and P_2, which passes through the intersection of P_1 and P_2. If the distance of the point (0,1,0) from P_3 is 1 and the distance of a point (α, β, γ) from P_3 is 2, then which of the following relations is (are) true? **[JEE(Adv.)2015]**

(A) $2\alpha+\beta+2\gamma+2 = 0$ (B) $2\alpha-\beta+2\gamma+4=0$ (C) $2\alpha+\beta-2\gamma-10 = 0$ (D) $2\alpha - \beta + 2\gamma - 8 = 0$

मान लीजिए कि R^3 में $P_1 : y = 0$ और $P_2 : x + z = 1$ दो समतल है। माना कि P_3 एक समतल है जो समतल P_1 एवं P_2 से भिन्न है तथा P_1 एवं P_2 के प्रतिच्छेदन से जाता है। यदि बिन्दु (0,1,0) से P_3 की दूरी एक (1) है तथा बिन्दु (α, β, γ) से P_3 की दूरी दो (2) है, तब निम्नलिखित सम्बंध में कौन कौन से संतुष्टित होते है? **[JEE(Adv.)2015]**

(A) $2\alpha+\beta+2\gamma+2 = 0$ (B) $2\alpha-\beta+2\gamma+4=0$ (C) $2\alpha+\beta-2\gamma-10 = 0$ (D) $2\alpha - \beta + 2\gamma - 8 = 0$

Ans. BD

Sol. Let P_3 be $P_2 + \lambda P_1 = 0 \Rightarrow \quad x + \lambda y + z - 1 = 0$

Distance from (0, 1, 0) is 1

$$\therefore \quad \frac{0+\lambda+0-1}{\sqrt{1+\lambda^2+1}} = \pm 1$$

$$\lambda = -\frac{1}{2}$$

$\therefore$ Equation of P_3 is $2x - y + 2z - 2 = 0$

Dist. from (α, β, γ) is 3

$$\therefore \quad \left|\frac{2\alpha-\beta+2\gamma-2}{3}\right| = 2 \qquad \Rightarrow \quad 2\alpha-\beta+2\gamma-2 = \pm 6$$

$\therefore$ option (B, D) are correct.

20. In R^3, let L be a straight line passing through the origin. Suppose that all the points on L are at a constant distance from the two planes P_1: $x+2y-z+1=0$ and P_2: $2x-y+z-1 = 0$. Let M be the locus of the feet of the perpendiculars drawn from the points on L to the plane P_1. Which of the following points lie (s) on M? **[JEE(Adv.)2015]**

माना कि R^3 में L एक सरल रेखा है जो कि मूल बिन्दु से जाती है। माना कि L से सभी बिन्दु समतलों P_1:$x+2y-z + 1 = 0$ तथा $P_2 : 2x - y + z - 1 = 0$ से स्थिर दूरी पर है। माना कि L के बिन्दुओं से समतल P_1 पर डाले गए लम्बों के पादों का पथ M है। निम्नलिखित बिन्दुओं में से कौन सा बिन्दु पथ M पर स्थित है? **[JEE(Adv.)2015]**

(A) $\left(0, -\frac{5}{6}, -\frac{2}{3}\right)$ (B) $\left(-\frac{1}{6}, -\frac{1}{3}, \frac{1}{6}\right)$ (C) $\left(-\frac{5}{6}, 0, \frac{1}{6}\right)$ (D) $\left(-\frac{1}{3}, 0, \frac{2}{3}\right)$

Ans. AB

Sol. Let $\vec{v}$ be the vector along L

$$\text{then } \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -1 \\ 2 & -1 & 1 \end{vmatrix} = \hat{i} - 3\hat{j} - 5\hat{k}$$

So any point on line L is $A(\lambda, -3\lambda, -5\lambda)$

Foot of perpendicular from A to P, is

$$\frac{h-\lambda}{1}=\frac{k+3\lambda}{2}=\frac{l+5\lambda}{-1}=-\frac{(\lambda-6\lambda+5\lambda+1)}{1+4+1}=-\frac{1}{6}$$

$$h=\lambda-\frac{1}{6}, k=-3\lambda-\frac{1}{3}, l=-5\lambda+\frac{1}{6}$$

so foot is $\left(\lambda-\frac{1}{6},-3\lambda-\frac{1}{3},-5\lambda+\frac{1}{6}\right)$

So, (A,B)

21. Let $\hat{u}=u_1\hat{i}+u_2\hat{j}+u_3\hat{k}$ be a unit vector in R^3 and $\hat{\omega}=\frac{1}{\sqrt{6}}(\hat{i}+\hat{j}+2\hat{k})$. Given that there exists a vector $\vec{\upsilon}$ in R^3 such that $|\hat{u}\times\vec{\upsilon}|$ and $\hat{\omega}.(\hat{u}\times\vec{\upsilon})=1$. Which of the following statement(s) is(are) correct ?

[JEE(Adv.)2016]

(A) There is exactly one choice for such $\vec{\upsilon}$ (B) There are infinitely many choices for such $\vec{\upsilon}$

(C) If $\hat{u}$ lies in the xy-plane than $|u_1|=|u_2|$ (D) If $\hat{u}$ lies in the xz-plane than $2|u_1|=|u_3|$

माना कि R^3 में $\hat{u}=u_1\hat{i}+u_2\hat{j}+u_3\hat{k}$ एक मात्रक सदिश है और $\hat{\omega}=\frac{1}{\sqrt{6}}(\hat{i}+\hat{j}+2\hat{k})$ है। दिया हुआ है कि R^3 में सदिश $\vec{\upsilon}$ का अस्तित्व इस प्रकार है कि $|\hat{u}\times\vec{\upsilon}|$ और $\hat{\omega}.(\hat{u}\times\vec{\upsilon})=1$ है। निम्नलिखित में से कौन सा (से) कथन सही है (हैं) ?

[JEE(Adv.)2016]

(A) इस प्रकार के $\vec{\upsilon}$ के लिए ठीक चयन संभव है (B) इस प्रकार के $\vec{\upsilon}$ के लिए अनन्त चयन संभव है।

(C) यदि $\hat{u}$, xy-समतल पर है तब $|u_1|=|u_2|$ है (D) यदि $\hat{u}$, xz-समतल पर है तब $2|u_1|=|u_3|$ है

Ans. BC

Sol. $\hat{w}.(\hat{u}\times\vec{v})=1$

$\Rightarrow|\hat{w}||\hat{u}\times\vec{v}|\cos\alpha=1$

$\cos\alpha=1$

$\Rightarrow\hat{w}\perp\hat{u}$ and $\hat{w}\perp\vec{v}$

as it is given there exist a vector $\vec{v}$

$\Rightarrow\hat{w}$ must be $\perp$ to $\hat{u}$

hence infinite many such $\vec{v}$ exists.

if $\hat{u}=u_1\hat{i}+u_2\hat{j}$

$\vec{u}.\vec{w}=0\Rightarrow(u_1+u_2)=0$

$\Rightarrow|u_1|=|u_2|$

if $u=u_1\hat{i}+u_3\hat{k}$

$\vec{u}.\vec{w} = 0$

$u_1 + 2u_3 = 0$

$\Rightarrow |u_1| = 2|u_3|$.

22. Consider a pyramid OPQRS located in the first octant ($x \geq 0, y \geq 0, z \geq 0$) with O as origin, and OP and OR along the x-axis and the y-axis, respectively. The base OPQR of the pyramid is a square with OP = 3. The point S is directly above the mid-point T of diagonal OQ such that TS = 3. Then : **[JEE(Adv.)2016]**

(A) the acute angle between OQ and OS is $\frac{\pi}{3}$

(B) the equation of the plane containing the triangle OQS is $x - y = 0$

(C) the length of the perpendicular from P to the plane containing the triangle OQS = is $\frac{3}{\sqrt{2}}$

(D) the perpendicular distance from O to the straight line containing RS is $\sqrt{\frac{15}{2}}$

विचार कीजिये, एक सूच्याकार (pyramid) OPQRS जो प्रथम अष्टांशक ($x \geq 0, y \geq 0, z \geq 0$) में स्थित है, जिसमें O मूल बिन्दु तथा OP और OR क्रमशः x- अक्ष और y-अक्ष पर हैं। इस सूच्याकार का आधार OPQR एक वर्ग है जिसमें OP = 3 है। बिन्दु S कर्ण OQ के मध्यबिन्दु T के ठीक ऊपर इस प्रकार है कि TS = 3 है। तब : **[JEE(Adv.)2016]**

(A) OQ और OS के बीच का न्यूनकोण $\frac{\pi}{3}$ है

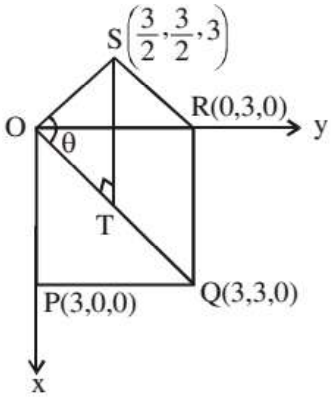
(B) त्रिभुज OQS को अंतर्विष्ट करने वाले समतल का समीकरण $x - y = 0$ है

(C) P से त्रिभुज OQS को अंतर्विष्ट करने वाले समतल पर लम्ब की लंबाई $\frac{3}{\sqrt{2}}$ है

(D) O से RS को अंतर्विष्ट करती हुई सरल रेखा की लम्बवत् दूरी $\sqrt{\frac{15}{2}}$ है

Ans. BCD

Sol. Points O, P, Q, R, S are (0, 0, 0), (3, 0, 0), (3, 3, 0) (0, 3, 0), $\left(\frac{3}{2}, \frac{3}{2}, 3\right)$ respectively



Angle between OQ and OS is $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$

Equation of plane containing the points O, Q and S is $x - y = 0$

Perpendicular distance from P(3, 0, 0) to the plane

$x - y = 0$ is $\left|\frac{3-0}{\sqrt{2}}\right| = \frac{3}{\sqrt{2}}$

Perpendicular distance from O(0, 0, 0) to the line RS:

$\frac{x}{1}=\frac{y-3}{-1}=\frac{z}{2}$ is $\sqrt{\frac{15}{2}}$

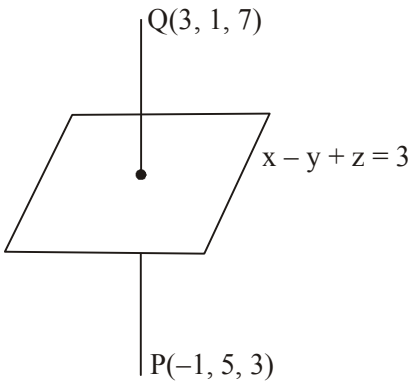
23. Let P be the image of the point (3, 1, 7) with respect to the plane $x-y+z=3$. Then the equation of the plane passing through P and containing the straight line $\frac{x}{1}=\frac{y}{2}=\frac{z}{1}$ is : **[JEE(Adv.)2016]**

माना कि बिन्दु (3, 1, 7) का, समतल $x-y+z=3$ के सापेक्ष प्रतिबिम्ब P है। तब बिन्दु P से गुजरने वाली और सरल रेखा $\frac{x}{1}=\frac{y}{2}=\frac{z}{1}$ को धारण करने वाले समतल का समीकरण है : **[JEE(Adv.)2016]**

(A) $x+y-3z=0$ (B) $3x+z=0$ (C) $x-4y+7z=0$ (D) $2x-y=0$

Ans. C

Sol.



Get image of Q(3,1,7) with respect to given plane i.e. P(–1,5,3)

Normal vector $= \overrightarrow{OP}\times\left(\hat{i}+2\hat{j}+\hat{k}\right)$

$= \hat{i}-4\hat{j}+7\hat{k}$

Equation of plane will be $x-4y+7z=\lambda$

Point (0, 0, 0) will lie in the plane

$\Rightarrow 0=\lambda$

So the required plane is $x-4y+7z=0$

24. Let O be the origin and let PQR be an arbitrary triangle. Then point S is such that $\overrightarrow{OP}.\overrightarrow{OQ}+\overrightarrow{OR}.\overrightarrow{OS}=\overrightarrow{OR}.\overrightarrow{OP}+\overrightarrow{OQ}.\overrightarrow{OS}=\overrightarrow{OQ}.\overrightarrow{OR}+\overrightarrow{OP}.\overrightarrow{OS}$. Then the triangle PQR has S as its : **[JEE(Adv.)2017]**

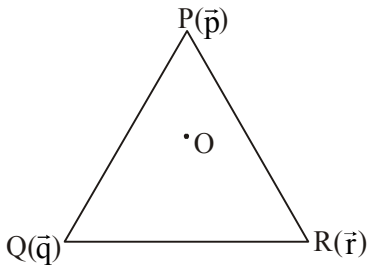
(A) incentre (B) circumcentre (C) orthocentre (D) centroid

माना कि O मूल बिन्दु है एवं PQR एक स्वेच्छिक त्रिभुज है। बिन्दु S इस प्रकार है, कि $\overrightarrow{OP}.\overrightarrow{OQ}+\overrightarrow{OR}.\overrightarrow{OS}=\overrightarrow{OR}.\overrightarrow{OP}+\overrightarrow{OQ}.\overrightarrow{OS}=\overrightarrow{OQ}.\overrightarrow{OR}+\overrightarrow{OP}.\overrightarrow{OS}$ तब बिन्दु S त्रिभुज PQR का है : **[JEE(Adv.)2017]**

(A) अन्तःकेन्द्र (B) परिवृत्तकेन्द्र (C) लम्बकेन्द्र (D) केन्द्रक

Ans. C

Sol.



$\vec{p}.\vec{q}+\vec{r}.\vec{s}=\vec{r}.\vec{p}+\vec{q}.\vec{s}=\vec{q}.\vec{r}+\vec{p}.\vec{s}$

$\Rightarrow \vec{p}(\vec{q}-\vec{r})-\vec{s}.(\vec{q}-\vec{r})=0 \Rightarrow \overrightarrow{PS}.\overrightarrow{QR}=0$

Similarly $\overrightarrow{PQ}.\overrightarrow{SR}=0$

$\Rightarrow$ S is orthocentre of the triangle

Paragraph for Question Nos. 25 to 26

Let O be the origin, and $\overrightarrow{OX},\overrightarrow{OY},\overrightarrow{OZ}$ be three unit vectors in the directions of the sides $\overrightarrow{QR},\overrightarrow{RP},\overrightarrow{PQ}$ respectievly, of a triangle PQR.

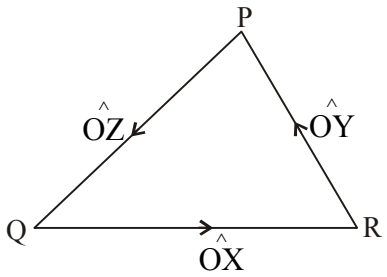
माना कि O मूल बिन्दु है एवं $\overrightarrow{OX},\overrightarrow{OY},\overrightarrow{OZ}$ क्रमशः त्रिभुज PQR की भुजायें $\overrightarrow{QR},\overrightarrow{RP},\overrightarrow{PQ}$ की दिशाओं में तीन इकाई सदिश (unit vectors) है।

25. $\left|\overrightarrow{OX}\times\overrightarrow{OY}\right|=$ **[JEE(Adv.)2017]**

(A) $\sin(P+R)$ (B) $\sin 2R$ (C) $\sin(P+Q)$ (D) $\sin(Q+R)$

Ans. C

Sol.



$\cos R=-\hat{OX}.\hat{OY}$

$\Rightarrow |\sin R|=\left|\hat{OX}\times\hat{OY}\right|$

$$\Rightarrow \left|\hat{OX}\times\hat{OY}\right| = |\sin R| = \left|\sin\left(\pi-(P+Q)\right)\right| = \left|\sin(P+Q)\right| = \sin(P+Q)$$

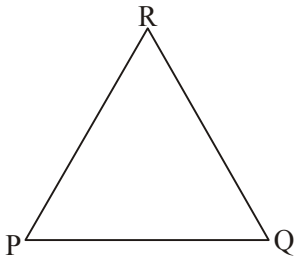
26. If the triangle PQR varies, then the minimum value of $\cos(P+Q)+\cos(Q+R)+\cos(R+P)$ is : **[JEE(Adv.)2017]**

यदि त्रिभुज PQR परिवर्ती है, तब $\cos(P+Q)+\cos(Q+R)+\cos(R+P)$ का न्यूनतम मान है : **[JEE(Adv.)2017]**

(A) $-\frac{3}{2}$ (B) $\frac{3}{2}$ (C) $\frac{5}{3}$ (D) $-\frac{5}{3}$

Ans. A

Sol.



$$\cos(P+Q)+\cos(Q+R)+\cos(R+P) = -\cos R-\cos P-\cos Q$$

In any triangle, max of $\cos P+\cos Q+\cos R = \frac{3}{2}$

So minimum value of the given expression is $-\frac{3}{2}$

27. The equation of the plane passing through the point (1, 1, 1) and perpendicular to the planes $2x+y-2z=5$ and $3x-6y-2z=7$, is : [JEE(Adv.)-2017]

समतलों $2x+y-2z=5$ तथा $3x-6y-2z=7$ के लम्बवत् और बिन्दु (1, 1, 1) से गुजरने वाले समतल का समीकरण है : [JEE(Adv.)-2017]

(A) $-14x+2y+15z=3$ (B) $14x-2y+15z=27$ (C) $14x+2y-15z=1$ (D) $14x+2y+15z=31$

Ans. D

Sol. Let plane be

$a(x-1)+b(y-1)+c(z-1)=0$

$$\text{Now, direction ratio of its normal} = \begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 1 & -2\\ 3 & -6 & -2\end{vmatrix} = \hat{i}(-14)-\hat{j}(2)+\hat{k}(-15)$$

So, $-14(x-1)-2(y-1)-15(z-1)=0$

$14x+2y+15z=31$

28. Let $\vec{a}$ and $\vec{b}$ be two unit vectors such that $\vec{a}\cdot\vec{b}=0$. For some x, y $\in$R, let $\vec{c}=x\vec{a}+y\vec{b}+(\vec{a}\times\vec{b})$. If $|\vec{c}|=2$ and the vector $\vec{c}$ is inclined at the same angle α to both $\vec{a}$ and $\vec{b}$, then the value of $8\cos^2\alpha$ is _____.

[JEE(Adv.)2018]

माना कि $\vec{a}$ और $\vec{b}$ दो ऐसे इकाई सदिश हैं कि $\vec{a}\cdot\vec{b}=0$ किन्ही x, y $\in$ R के लिये माना कि $\vec{c}=x\vec{a}+y\vec{b}+(\vec{a}\times\vec{b})$। यदि $|\vec{c}|=2$ और सदिश $\vec{c}$ सदिशों $\vec{a}$ तथा $\vec{b}$ दोनों के साथ समान कोण α बनाता है, तब $8\cos^2\alpha$ का मान है _____

[JEE(Adv.)2018]

Ans. 3

Sol. $\vec{c}=x\vec{a}+y\vec{b}+\vec{a}\times\vec{b}$ and $\vec{a}\cdot\vec{b}=0$

$\vec{a}\hat{}\vec{c}=\vec{b}\hat{}\vec{c}=\alpha$

$\vec{c}\cdot\vec{a}=\vec{c}\cdot\vec{b}=2\cos\alpha \quad\Rightarrow\quad x=y=2\cos\alpha$

$|\vec{c}|^2=x^2+y^2+|\vec{a}\times\vec{b}|^2=2\left(4\cos^2\alpha\right)+1-0$

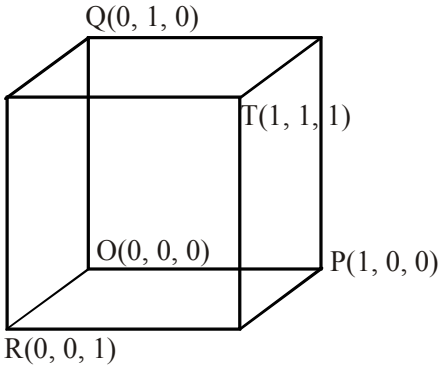
$4=8\cos^2\alpha+1 \quad\Rightarrow\quad 8\cos^2\alpha=3$

29. Consider the cube in the first octant with sides OP, OQ and OR of length 1, along the x-axis, y-axis and z-axis, respectively, where O(0,0,0) is the origin. Let $S\left(\frac{1}{2},\frac{1}{2},\frac{1}{2}\right)$ be the centre of the cube and T be the vertex of the cube opposite to the origin O such that S lies on the diagonal OT. If $\vec{p}=\overrightarrow{SP},\vec{q}=\overrightarrow{SQ},\vec{r}=\overrightarrow{SR}$ and $\vec{t}=\overrightarrow{ST}$, then the value of $\left|(\vec{p}\times\vec{q})\times(\vec{r}\times\vec{t})\right|$ is _____. **[JEE(Adv.)2018]**

प्रथम अष्ठांश (first octant) में एक ऐसे घन पर विचार कीजिये, जिसकी भुजाओं OP, OQ और OR की लम्बाई 1 है और जो क्रमशः x-अक्ष, y-अक्ष और z-अक्ष के अनुदिश हैं, जहाँ O(0,0,0) मूलबिन्दु है। माना कि घन का केन्द्र $S\left(\frac{1}{2},\frac{1}{2},\frac{1}{2}\right)$ है, और शीर्ष T मूलबिन्दु O के सम्मुख वाला वह शीर्ष है कि बिन्दु S विकर्ण OT पर स्थित है। यदि $\vec{p}=\overrightarrow{SP},\vec{q}=\overrightarrow{SQ},\vec{r}=\overrightarrow{SR}$ और $\vec{t}=\overrightarrow{ST}$, तब $\left|(\vec{p}\times\vec{q})\times(\vec{r}\times\vec{t})\right|$ का मान है _____ **[JEE(Adv.)2018]**

Ans. 0.5

Sol.



Point $S\left(\frac{1}{2},\frac{1}{2},\frac{1}{2}\right)$

point T(1, 1, 1)

$\vec{p}=\overrightarrow{SP}=\frac{\hat{i}-\hat{j}-\hat{k}}{2}$

$\vec{q}=\overrightarrow{SQ}=\frac{-\hat{i}+\hat{j}-\hat{k}}{2}$

$\vec{r}=\overrightarrow{SR}=\frac{-\hat{i}-\hat{j}+\hat{k}}{2}$

$\vec{t}=\overrightarrow{ST}=\frac{\hat{i}+\hat{j}+\hat{k}}{2}$

Now $\vec{p}\times\vec{q}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&-1&-1\\-1&1&-1\end{vmatrix}\times\frac{1}{4}=\frac{1}{4}(2\hat{i}+2\hat{j})=\frac{i+\hat{j}}{2}$

$\vec{r}\times\vec{t}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\-1&-1&1\\1&1&1\end{vmatrix}\times\frac{1}{4}=\frac{-2\hat{i}+2\hat{j}}{4}=\frac{-\hat{i}+\hat{j}}{2}$

Now $(\vec{p}\times\vec{q})\times(\vec{r}\times\vec{t})=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&1&0\\-1&1&0\end{vmatrix}\times\frac{1}{4}=\frac{\hat{k}}{2}\Rightarrow|(\vec{p}\times\vec{q})\times(\vec{r}\times\vec{t})|=\frac{1}{2}=0.5$

30. Let P_1: $2x + y - z = 3$ and P_2: $x + 2y + z = 2$ be two planes. Then, which of the following statement(s) is (are) TRUE? **[JEE(Adv.)2018]**

(A) The line of intersection of P_1 and P_2 has direction ratios 1,2, –1

(B) The line $\frac{3x-4}{9}=\frac{1-3y}{9}=\frac{z}{3}$ is perpendicular to the line of intersection of P_1 and P_2

(C) The acute angle between P_1 and P_2 is 60°

(D) If P_3 is the plane passing through the point (4, 2, –2) and perpendicular to the line of intersection of P_1 and P_2, then the distance of the point (2,1,1) from the plane P_3 is $\frac{2}{\sqrt{3}}$

माना कि P_1: $2x + y - z = 3$ और P_2: $x + 2y + z = 2$ दो समतल हैं। तब निम्नलिखित में से कौन सा (से) कथन सत्य है (हैं)?

[JEE(Adv.)2018]

(A) P_1 तथा P_2 की प्रतिच्छेदन रेखा के दिक्-अनुपात 1,2, –1 हैं

(B) रेखा $\frac{3x-4}{9}=\frac{1-3y}{9}=\frac{z}{3}$; P_1 और P_2 की प्रतिच्छेदन रेखा पर लम्बवत् है

(C) P_1 और P_2 के बीच का न्यून कोण 60° है

(D) यदि समतल P_3, बिन्दु (4, 2, –2) से गुजरता है तथा P_1 और P_2 की प्रतिच्छेदन रेखा के लम्बवत् है, तब बिन्दु (2,1,1) की समतल P_3 से दूरी $\frac{2}{\sqrt{3}}$ है

Ans. CD

Sol. Direction ratio of common line is $\vec{n}_1 \times \vec{n}_2$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 1 & 2 & 1 \end{vmatrix} = \hat{i}(3) - \hat{j}(3) + \hat{k}(3) = 3(\hat{i} - \hat{j} + \hat{k})$$

(B) $\frac{x-4/3}{3}=\frac{y-1/3}{-3}=\frac{z}{3}$

This is || to line of intersection

(C) $\cos\theta = \frac{\vec{x}_1 \cdot \vec{x}_2}{|\vec{x}_1||\vec{x}_2|} = \frac{2+2-1}{\sqrt{6}\sqrt{6}} = \frac{3}{6} = \frac{1}{2} \quad \Rightarrow \quad \theta = \frac{\pi}{3}$

(D) $P_3 : x - y + z = 1$ satisfy (4, 2, – 2)

$4 - 2 - 2 = 1 \quad \Rightarrow \quad x - y + z = 4$

distance of the point (2,1,1) from the plane $P_3 = \left|\frac{2-1+1-4}{\sqrt{3}}\right| \Rightarrow \frac{2}{\sqrt{3}}$

31. Let P be a point in the first octant, whose image Q in the plane $x + y = 3$ (that is, the line segment PQ is perpendicular to the plane $x + y = 3$ and the mid-point of PQ lies in the plane $x + y = 3$) lies on the z-axis. Let the distance of P from the x-axis be 5. If R is the image of P in the xy-plane, then the length of PR is _____ .

माना कि P प्रथम अष्ठांश (first octant) में एक बिन्दु है, जिसका समतल $x + y = 3$ में प्रतिबिम्ब Q (अर्थात रेखाखण्ड PQ समतल $x + y = 3$ के लम्बवत् है और PQ का मध्य बिन्दु समतल $x + y = 3$ में स्थित है) z-अक्ष पर स्थित है। माना कि P की x-अक्ष से दूरी 5 हैं यदि P का xy-समतल में प्रतिबिम्ब R है, तब PR की लम्बाई _____ है। **[JEE(Adv.)2018]**

Ans. 8

Sol. $P(\alpha, \beta, \gamma)$

$R(\alpha, \beta, -\gamma)$

$$\frac{x-\alpha}{1} = \frac{y-\beta}{1} = \frac{z-\gamma}{0} = \frac{-2(\alpha+\beta-3)}{2}$$

$x = 3 - \beta,\ y = 3 - \alpha,\ z = \gamma$

$Q(3 - \beta, 3 - \alpha, \gamma)$ lies on z-axis

$\therefore \quad \beta = 3,\ \alpha = 3$

$P(3, 3, \gamma)$ distance from x-axis is 5

$9 + \gamma^2 = 25$

$\gamma^2 = 16 \quad \Rightarrow \quad \gamma = 4$

$P(3, 3, 4) \quad \therefore \quad PR = 8$

$R(3, 3, -4)$

32. Let L_1 and L_2 denote the lines $\vec{r} = \hat{i} + \lambda\left(-\hat{i} + 2\hat{j} + 2\hat{k}\right)\lambda \in R$, $\vec{r} = \mu\left(2\hat{i} - \hat{j} + 2\hat{k}\right), \mu \in R$ respectively. If L_3 is line which is perpendicular to both L_1 and L_2 and cuts both of them then which of the following options describe (s) L_3? **[JEE(Adv.)2019**

माना कि L_1 और L_2 क्रमशः निम्न रेखाएं है $\vec{r} = \hat{i} + \lambda\left(-\hat{i} + 2\hat{j} + 2\hat{k}\right)\lambda \in R$ और $\vec{r} = \mu\left(2\hat{i} - \hat{j} + 2\hat{k}\right), \mu \in R$ यदि L_3 एक रेखा है जो L_1 और L_2 दोनों के लम्बवत् है और दोनों को काटती है, तब निम्नलिखित विकल्पों में से कौन सा (से) L_3 को निरूपित करता (करतें) है (है) ? **[JEE(Adv.)2019]**

(A) $\vec{r} = \frac{1}{3}\left(2\hat{i} + \hat{k}\right) + t\left(2\hat{i} + 2\hat{j} - \hat{k}\right), t \in R$

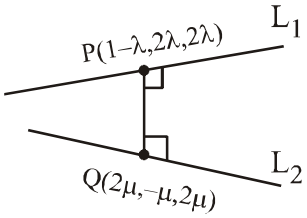
(B) $\vec{r} = \frac{2}{9}\left(2\hat{i} - \hat{j} + 2\hat{k}\right) + t\left(2\hat{i} + 2\hat{j} - \hat{k}\right), t \in R$

(C) $\vec{r} = \frac{2}{9}\left(4\hat{i} + \hat{j} + \hat{k}\right) + t\left(2\hat{i} + 2\hat{j} - \hat{k}\right), t \in R$

(D) $\vec{r} = t\left(2\hat{i} + 2\hat{j} - \hat{k}\right), t \in R$

Ans. ABC

Sol.



$\overrightarrow{PQ} = (1-\lambda-2\mu,\ 2\lambda+\mu,\ 2\lambda-2\mu)$

$PQ \perp L_1$ and $PQ \perp L_2$

$-(1-l-2m)+2(2l+m)+2(2l-m)=0$ $\quad 2(1-l-2m)-(2l+m)+2(2l-2m)=0$

$-1+l+2m+4l+2m+4l-4m=0$ $\quad 2-2l-4m-2l-m+4l-4m=0$

$9l=1, \quad l=\frac{1}{9}$ $\quad m=\frac{2}{9}$

$P\left(\frac{8}{9},\frac{2}{9},\frac{2}{9}\right)$; $\quad Q\left(\frac{4}{9},\frac{-2}{9},\frac{4}{9}\right)$

mid point of $PQ\left(\frac{2}{3},0,\frac{1}{3}\right)$

33. Three lines $L_1: \vec{r}=\lambda\hat{i},\lambda\in R$, $L_2: \vec{r}=\hat{k}+\mu\hat{j},\mu\in R$ and $L_3: \vec{r}=\hat{i}+\hat{j}+v\hat{k},v\in R$ are given. For which point (s) Q on L_2 can we find a point P on L_1 and a point R on L_3 so that P, Q and R are collinear :

[JEE(Adv.)2019]

तीन रेखाऐं $L_1: \vec{r}=\lambda\hat{i},\lambda\in R$, $L_2: \vec{r}=\hat{k}+\mu\hat{j},\mu\in R$ तथा $L_3: \vec{r}=\hat{i}+\hat{j}+v\hat{k},v\in R$ दी गयी है। L_2 के किस बिन्दु (किन बिन्दुओं) Q के लिए हम L_1 पर एक बिन्दु P और L_3 पर एक बिन्दु R प्राप्त कर सकते हैं ताकि P, Q और R सरेख (collinear) हों जाएँ:

[JEE(Adv.)2019]

(A) $\hat{k}+\hat{j}$ (B) $\hat{k}-\frac{1}{2}\hat{j}$ (C) $\hat{k}+\frac{1}{2}\hat{j}$ (D) $\hat{k}$

Ans. BC

Sol. P(l,0,0) Q(0,m,1) R(1,1,n)

P,Q,R are collinear

$\overrightarrow{PQ}=t\overrightarrow{QR}$ $t\in R$

$\overrightarrow{PQ}=(-\lambda,\mu,1)$

$\overrightarrow{QR}=(-1,\mu-1,1-\nu)$

$\frac{-\lambda}{-1}=\frac{\mu}{\mu-1}=\frac{1}{1-\nu}$

$\mu \neq 0,1$

So Q cannot be (0, 0, 1) and (0, 1, 1)

34. Let $\vec{a} = 2\hat{i} + \hat{j} - \hat{k}$ and $\hat{b} = \hat{i} + 2\hat{j} + \hat{k}$ be two vectors. Consider a vector $\vec{c} = \alpha\vec{a} + \beta\vec{b}$, a,b ∈R. If the projection of $\vec{c}$ on the vector $\left(\vec{a} + \vec{b}\right)$ is $3\sqrt{2}$ then the minimum value of $\left(\vec{c} - \left(\vec{a} \times \vec{b}\right)\right).\vec{c}$ equals_____

[JEE(Adv.)2019]

माना कि $\vec{a} = 2\hat{i} + \hat{j} - \hat{k}$ तथा $\hat{b} = \hat{i} + 2\hat{j} + \hat{k}$ दो सदिश (vector) हैं। माना कि एक सदिश $\vec{c} = \alpha\vec{a} + \beta\vec{b}$, a,b ∈ R है। यदि सदिश $\left(\vec{a} + \vec{b}\right)$ पर $\vec{c}$ का प्राक्षेप $3\sqrt{2}$ है, तब $\left(\vec{c} - \left(\vec{a} \times \vec{b}\right)\right).\vec{c}$ का निम्नतम (minimum) मान बराबर _____

[JEE(Adv.)2019]

Ans. 18

Sol. $\left|\vec{a}\right|^2 = \left|\vec{b}\right|^2 = 6 \quad \vec{a} + \vec{b} = 3\hat{i} + 3\hat{j}$

$\vec{a}.\vec{b} = 2 + 2 - 1 = 3$

Projection of $\vec{c}$ on $\vec{a} + \vec{b}$ is

$$\frac{\vec{c}.\left(\vec{a} + \vec{b}\right)}{\left|\vec{a} + \vec{b}\right|} = 3\sqrt{2}$$

$$\frac{\left(\alpha\vec{a} + \beta\vec{b}\right).\left(\vec{a} + \vec{b}\right)}{3\sqrt{2}} = 3\sqrt{2}$$

$9(\alpha + \beta) = 18$

$\alpha + \beta = 2$

Let $y = \left(\vec{c} - \left(\vec{a} \times \vec{b}\right)\right).\vec{c}$

$= \vec{c}.\vec{c} - \left(\alpha\vec{a} + \beta\vec{b}\right).\left(\vec{a} \times \vec{b}\right)$

$= \left|\vec{c}\right|^2 - 0$

$= \alpha^2\left|\vec{a}\right|^2 + \beta^2\left|\vec{b}\right|^2 + 2\alpha\beta\vec{a}.\vec{b}$

$= 6\alpha^2 + 6\beta^2 + 6\alpha\beta$

$= 6\left(\alpha^2 + \beta^2 + \alpha\beta\right) \qquad (\alpha = 2 - \beta)$

$= 6\left((2-\beta)^2 + \beta^2 + (2-\beta)\beta\right)$

$= 6\left(4-4\beta+\beta^2+\beta^2+2\beta-\beta^2\right)$

$= 6\left(\beta^2-2\beta+1+3\right)$

$y = 6\left((\beta-1)^2+3\right)$

$y_{min} = 6(0+3) = 18$

35. Three lines are given by $\vec{r} = \lambda\hat{i}, \lambda \in R$, $\vec{r} = \mu\left(\hat{i}+\hat{j}\right), \mu \in R$ and $\vec{r} = \upsilon\left(\hat{i}+\hat{j}+\hat{k}\right), \upsilon \in R$ Let the lines cut the plane x + y + z = 1 at the points A,B and C respectively. If the area of the triagle ABC is Δ then the value of $(6\Delta)^2$equas _____. **[JEE(Adv.)2019]**

तीन रेखाएं क्रमशः $\vec{r} = \lambda\hat{i}, \lambda \in R$, $\vec{r} = \mu\left(\hat{i}+\hat{j}\right), \mu \in R$ तथा $\vec{r} = \upsilon\left(\hat{i}+\hat{j}+\hat{k}\right), \upsilon \in R$ द्वारा दी गयी है। माना कि रेखाएं समतल (plane) x + y + z = 1 को क्रमशः बिन्दुओं A,B तथा C वा काटती है। यदि त्रिभुज ABC का क्षेत्रफल Δ है तब $(6\Delta)^2$ का मान बराबर_____ **[JEE(Adv.)2019]**

Ans. 0.75

Sol. Let A(l,0,0) b(m,m,0) C(u,u,u)

A,B,C lie on the plane x + y + z = 1 so

$l = 1,\ m = \frac{1}{2},\ u = \frac{1}{3}$

$A(1,0,0)\ B\left(\frac{1}{2},\frac{1}{2},0\right)\ \left(\frac{1}{3},\frac{1}{3},\frac{1}{3}\right)$

$\overrightarrow{AB} = \left(-\frac{1}{2},\frac{1}{2},0\right)$

$\overrightarrow{AC} = \left(-\frac{2}{3},\frac{1}{3},\frac{1}{3}\right)$

$$AB \times AC = \begin{vmatrix} i & j & k \\ -\frac{1}{2} & \frac{1}{2} & 0 \\ -\frac{2}{3} & \frac{1}{3} & \frac{1}{3} \end{vmatrix} = \frac{1}{6}\begin{vmatrix} i & j & k \\ -1 & 1 & 0 \\ -2 & 1 & 1 \end{vmatrix}$$

$= \frac{1}{6}\left(i(1-0)-\hat{j}\right)(-1-0)+k(-1+2)$

$$\overrightarrow{AB}\times\overrightarrow{AC}=\frac{1}{6}\left(\hat{i}+\hat{j}+\hat{k}\right)$$

$$\Delta AB=\frac{1}{2}\left|AB\times AC\right|$$

$$\Delta=\frac{1}{2}\cdot\frac{1}{6}\cdot\sqrt{3}=\frac{\sqrt{3}}{12}$$

$$(6\Delta)^2=\left(\frac{6}{12}\sqrt{3}\right)^2=\left(\frac{\sqrt{3}}{2}\right)^2=\frac{3}{4}=0.75$$

36. In a triangle PQR, let $\vec{a}=\overrightarrow{QR},\vec{b}=\overrightarrow{RP}$ and $\vec{c}=\overrightarrow{PQ}$. If $|\vec{a}|=3,\quad |\vec{b}|=4$ and $\frac{\vec{a}\cdot(\vec{c}-\vec{b})}{\vec{c}\cdot(\vec{a}-\vec{b})}=\frac{|\vec{a}|}{|\vec{a}|+|\vec{b}|}$, then the value of $|\vec{a}\times\vec{b}|^2$ is _____. **[JEE(Adv.)2020]**

एक त्रिभुज PQR में माना कि $\vec{a}=\overrightarrow{QR},\vec{b}=\overrightarrow{RP}$ एवं $\vec{c}=\overrightarrow{PQ}$ है। यदि $|\vec{a}|=3,\quad |\vec{b}|=4$ एवं $\frac{\vec{a}\cdot(\vec{c}-\vec{b})}{\vec{c}\cdot(\vec{a}-\vec{b})}=\frac{|\vec{a}|}{|\vec{a}|+|\vec{b}|}$ हैं, तब $|\vec{a}\times\vec{b}|^2$ का मान है _____ **[JEE(Adv.)2020]**

Ans. 108

Sol. $\vec{a}=\overrightarrow{QR},\ \vec{b}=\overrightarrow{RP},\ \vec{c}=\overrightarrow{PQ}$

$\Rightarrow\vec{a}+\vec{b}+\vec{c}=\vec{0}$

$\Rightarrow\vec{c}=-\vec{a}-\vec{b}$

Replacing in the given equation,

$$\frac{\vec{a}\cdot\left(-\vec{a}-\vec{b}-\vec{b}\right)}{\left(-\vec{a}-\vec{b}\right)\cdot\left(\vec{a}-\vec{b}\right)}=\frac{\left|\vec{a}\right|}{\left|\vec{a}\right|+\left|\vec{b}\right|}=\frac{3}{3+4}$$

$$\Rightarrow\ \frac{\left|\vec{a}\right|^2+2\vec{a}\cdot\vec{b}}{\left|\vec{a}\right|^2-\left|\vec{b}\right|^2}=\frac{3}{7}$$

$$\Rightarrow\ \frac{9+2\,\vec{a}\cdot\vec{b}}{9-16}=\frac{3}{7}$$

Now from Lagrange's Identity

$$\left|\vec{a}\times\vec{b}\right|^2=\left|\vec{a}\right|^2\left|\vec{b}\right|^2-\left(\vec{a}\cdot\vec{b}\right)^2$$

$= 9 \times 16 - (-6)^2$

$= 144 - 36 = 108$

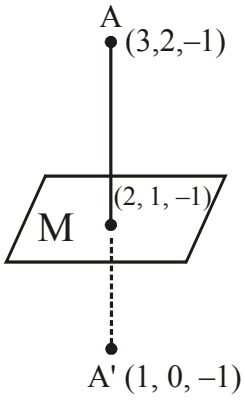
37. Let α , β , γ , δ be real numbers such that $\alpha^2+\beta^2+\gamma^2 \neq 0$ and $\alpha + \gamma = 1$. Suppose the point (3,2, –1) is the mirror image of the point (1,0, –1) with respect to the plane $\alpha x + \beta y + \gamma z = \delta$. Then which of the following statements is/are TRUE? **[JEE(Adv.)2020]**

माना कि वास्तविक संख्याएँ α , β , γ , δ इस प्रकार से हैं कि $\alpha^2+\beta^2+\gamma^2 \neq 0$ एवं $\alpha + \gamma = 1$ हैं। माना लीजिये कि समतल $\alpha x + \beta y + \gamma z = \delta$ के सापेक्ष बिंदु (1,0, –1) का दर्पण प्रतिबिम्ब बिन्दु (3,2, –1) है। तब निम्न में से कौनसा/से कथन सत्य है/हैं? **[JEE(Adv.)2020]**

(A) $\alpha + \beta = 2$ (B) $\delta - \gamma = 3$ (C) $\delta + \beta = 4$ (D) $\alpha + \beta + \gamma = \delta$

Ans. ABC

Sol.



Normal vector

$\vec{n} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$

$\vec{n} = \overrightarrow{A'A}$

$\vec{n} = 2\hat{i} + 2\hat{j} + 0\hat{k}$

$\frac{\alpha}{2} = \frac{\beta}{2} = \frac{\gamma}{0} = \lambda$

$\alpha + \gamma = 1$

$\alpha = 1 = \beta$ $\alpha = \beta$ and $\gamma = 0$

$\alpha x + \beta y + \gamma z = \delta$

m(2, 1, –1)

$2\beta + \beta + 0 = \delta$

$\delta = 3$

38. Let a and b be positive real numbers. Suppose $\overrightarrow{PQ} = a\hat{i} + b\hat{j}$ and $\overrightarrow{PS} = a\hat{i} - b\hat{j}$ are adjacent sides of a parallelogram PQRS. Let $\vec{u}$ and $\vec{v}$ be the projection vectors of $\vec{w} = \hat{i} + \hat{j}$ along $\overrightarrow{PQ}$ and $\overrightarrow{PS}$, respectively. If $|\vec{u}| + |\vec{v}| = |\vec{w}|$ and if the area of the parallelogram PQRS is 8, then which of the following statements is/are TRUE?

[JEE(Adv.)2020]

(A) a + b = 4

(B) a – b = 2

(C) The length of the diagonal PR of the parallelogram PQRS is 4

(D) $\vec{w}$ is an angle bisector of the vectors $\overrightarrow{PQ}$ and $\overrightarrow{PS}$

माना कि a और b धनात्मक वास्तविक संख्याएं हैं। मान लीजिये कि $\overrightarrow{PQ} = a\hat{i} + b\hat{j}$ एवं $\overrightarrow{PS} = a\hat{i} - b\hat{j}$ समांतर चतुर्भुज PQRS की संलग्न भुजाएँ हैं। माना कि $\vec{w} = \hat{i} + \hat{j}$ के $\overrightarrow{PQ}$ और $\overrightarrow{PS}$ पर प्रक्षेप सदिश क्रमशः $\vec{u}$ और $\vec{v}$ हैं। यदि $|\vec{u}| + |\vec{v}| = |\vec{w}|$ है एवं यदि समान्तर चतुर्भुज PQRS का क्षेत्रफल 8 है, तब निम्न में से कौनसा/से कथन सही है/हैं?

[JEE(Adv.)2020]

(A) a + b = 4

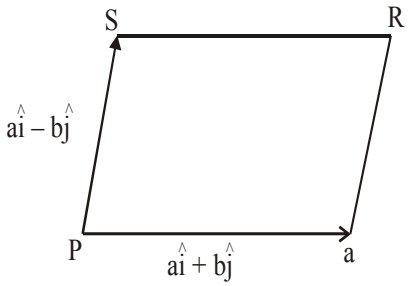
(B) a – b = 2

(C) समान्तर चर्तुभुज PQRS के विकर्ण PR की लम्बाई 4 है

(D) सदिशों $\overrightarrow{PQ}$ एवं $\overrightarrow{PS}$ का एक कोण समद्विभाजक $\vec{w}$ है

Ans. AC

Sol.



$$\vec{u} = \left(\frac{\vec{w}.\overrightarrow{PQ}}{|\overrightarrow{PQ}|}\right)\hat{PS} = \left(\frac{a-b}{\sqrt{a^2+b^2}}\right)\left(\frac{a\hat{j}-b\hat{j}}{\sqrt{a^2+b^2}}\right)$$

$$\vec{v} = \left(\frac{\vec{w}.\overrightarrow{PS}}{|\overrightarrow{PS}|}\right)\hat{PS} = \left(\frac{a-b}{\sqrt{a^2+b^2}}\right)\left(\frac{a\hat{j}-b\hat{j}}{\sqrt{a^2+b^2}}\right)$$

$$|\vec{u}| + |\vec{v}| = |\vec{w}|$$

$$\frac{a+b}{\sqrt{a^2+b^2}} + \frac{|a-b|}{\sqrt{a^2+b^2}} = \sqrt{2}$$

Case - I a > b

$2a = \sqrt{2}\sqrt{a^2+b^2}$

$2a^2 = a^2 + b^2$

$a^2 = b^2$

$a = b$

Case - II $a < b$

$2a = \sqrt{2}\sqrt{a^2+b^2}$

$a = b$

$$\text{Area of the parallelogram} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a & -b & 0 \\ a & b & 0 \end{vmatrix} = 8$$

$\left|\hat{k}.2ab\right| = 8$

$ab = 4$

and $a = b$

$a = b = 2$

(A) $a+b = 4$

(C) $\overrightarrow{PR} = \overrightarrow{PQ} + \overrightarrow{QR} = 2a\hat{i}$

$\left|\overrightarrow{PR}\right| = 4$

(D) $\overrightarrow{PQ} + \overrightarrow{PS} = 2a\hat{i} + \lambda\vec{\omega}$

$\overrightarrow{PQ} - \overrightarrow{PS} = 2b\hat{j} \neq \lambda\vec{\omega}$

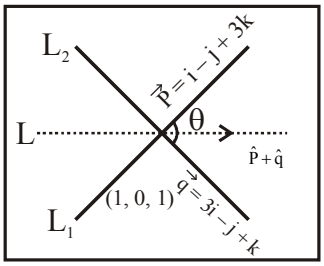
39. Let L_1 and L_2 be the following straight lines $L_1 : \frac{x-1}{1} = \frac{y}{-1} = \frac{z-1}{3}$ and $L_2 : \frac{x-1}{-3} = \frac{y}{-1} = \frac{z-1}{1}$. Suppose the straight line $L : \frac{x-\alpha}{l} = \frac{y-1}{m} = \frac{z-\gamma}{-2}$ lies in the plane containing L_1 and L_2, and passes through the point of intersection of L_1 and L_2. If the line L bisects the acute angle between the lines L_1 and L_2, then which of the following statements is/are TRUE? **[JEE(Adv.)2020]**

माना कि L_1 एवं L_2 निम्न सरल रेखाएँ हैं $L_1 : \frac{x-1}{1} = \frac{y}{-1} = \frac{z-1}{3}$ तथा $L_2 : \frac{x-1}{-3} = \frac{y}{-1} = \frac{z-1}{1}$ मान लीजिए कि सरल रेखा $L : \frac{x-\alpha}{l} = \frac{y-1}{m} = \frac{z-\gamma}{-2}$, L_1 एवं L_2 के समतल में स्थित है, और L_1 एवं L_2 के प्रतिच्छेद बिन्दु से जाती है। यदि रेखा L, रेखाओं L_1 एवं L_2 के बीच के न्यूनकोण को समद्विभाजित करती है, तब निम्न में से कौनसा (से) कथन सही है (हैं)? **[JEE(Adv.)2020]**

(A) $\alpha - \gamma = 3$ (B) $l + m = 2$ (C) $\alpha - \gamma = 1$ (D) $l + m = 0$

Ans. AB

Sol.



$\vec{p}\cdot\vec{q}=1>0\Rightarrow\theta\in\left(0,\frac{\pi}{2}\right)$

$\hat{p}+\hat{q}=\frac{1}{\sqrt{11}}(-2i-2j+4k)$

DR's of L = –2, –2, 4

OR

= 1, 1, –2

$L:\frac{x-\alpha}{\ell}=\frac{y-1}{m}=\frac{z-\gamma}{-2}$

$\ell = 1$

m = 1

Passes through (1, 0, 1)

$\frac{1-\alpha}{1}=\frac{0-1}{1}=\frac{1-\gamma}{-2}$

$\alpha = 2$

$\gamma = -1$

$\alpha - \gamma = 3$ (A)

$l + m = 2$ (B)

40. Let $\vec{u},\vec{v}$ and $\vec{w}$ be vectors in three-dimensional space, where $\vec{u}$ and $\vec{v}$ are unit vectors which are not perpendicular to each other and $\vec{u}.\vec{w}=1,\vec{v}.\vec{w}=1\vec{w}.\vec{w}=4$. If the volume of the parallelopiped, whose adjacent sides are represented by the vectors $\vec{u},\vec{v}$ and $\vec{w}$, is $\sqrt{2}$, then the value of $|3\vec{u}+5\vec{v}|$ is ___. **[JEE(Adv.)2021]**

माना $\vec{u},\vec{v}$ तथा $\vec{w}$ त्रिविम में सदिश है जहाँ $\vec{u}$ तथा $\vec{v}$ इकाई सदिश हैं जो एक दूसरे के लम्बवत् नहीं है तथा $\vec{u}.\vec{w}=1,\vec{v}.\vec{w}=1\vec{w}.\vec{w}=4$ यदि समान्तर षट्फलक जिसकी आसन्न भुजाएँ सदिश $\vec{u},\vec{v}$ व $\vec{w}$, द्वारा निरूपित की गयी है, का आयतन $\sqrt{2}$ है, तब $|3\vec{u}+5\vec{v}|$ का मान ___ है।

Ans. 7

Sol. $\vec{u}\cdot\vec{v}\neq 0$

$\vec{u}\cdot\vec{w}=1,\ \vec{v}\cdot\vec{w}=1,\ \vec{w}\cdot\vec{w}=4,\ [\vec{u}\vec{v}\vec{w}]=\sqrt{2}$

$|3\vec{u}+5\vec{v}|=?,\ \vec{u}\ \&\ \vec{v}$ are unit vectors

$[\vec{u}\vec{v}\vec{w}]=\sqrt{2}$

$\Rightarrow [\vec{u}\vec{v}\vec{w}]^2=2$

$$\Rightarrow \begin{vmatrix} \vec{u}\cdot\vec{u} & \vec{u}\cdot\vec{v} & \vec{u}\cdot\vec{w} \\ \vec{v}\cdot\vec{u} & \vec{v}\cdot\vec{v} & \vec{v}\cdot\vec{w} \\ \vec{w}\cdot\vec{u} & \vec{w}\cdot\vec{v} & \vec{w}\cdot\vec{w} \end{vmatrix}=2$$

$$\Rightarrow \begin{vmatrix} 1 & x & 1 \\ x & 1 & 1 \\ 1 & 1 & 4 \end{vmatrix}=2 \quad (\text{Let } \vec{u}\cdot\vec{v}=x)$$

$\Rightarrow 1(4-1)-x(4x-1)+1(x-1)=2$

$\Rightarrow 3-4x^2+x+x-1=2$

$\Rightarrow 4x^2-2x=0$

$\Rightarrow x=\left\{0,\frac{1}{2}\right\}\Rightarrow \vec{u}\cdot\vec{v}=0(\textit{rejected})\ \ or\ \ \vec{u}\cdot\vec{v}=\frac{1}{2}$

Now, $|3\vec{u}+5\vec{v}|^2=9\vec{u}\cdot\vec{u}+25\vec{v}\cdot\vec{v}+30\vec{u}\cdot\vec{v}$

41. Let O be the origin and $\overrightarrow{OA}=2\hat{i}+2\hat{j}+\hat{k}$, $\overrightarrow{OB}=\hat{i}-2\hat{j}+2\hat{k}$ and $\overrightarrow{OC}=\frac{1}{2}\left(\overrightarrow{OB}-\lambda\overrightarrow{OA}\right)$ for some $\lambda > 0$. If $\left|\overrightarrow{OB}\times\overrightarrow{OC}\right|=\frac{9}{2}$, then which of the following statements is (are) TRUE? **[JEE(Adv.)2021]**

(A) Projection of $\overrightarrow{OC}$ on $\overrightarrow{OA}$ is $-\frac{3}{2}$

(B) Area of the triangle OAB is $\frac{9}{2}$

(C) Area of the triangle ABC is $\frac{9}{2}$

(D) The acute angle between the diagonals of the parallelogram with adjacent sides $\overrightarrow{OA}$ and $\overrightarrow{OC}$ is $\frac{\pi}{3}$

माना O मूलबिन्दु है तथा किसी $\lambda > 0$ के लिए $\overrightarrow{OA} = 2\hat{i} + 2\hat{j} + \hat{k}$, $\overrightarrow{OB} = \hat{i} - 2\hat{j} + 2\hat{k}$ तथा $\overrightarrow{OC} = \frac{1}{2}\left(\overrightarrow{OB} - \lambda\,\overrightarrow{OA}\right)$। यदि $\left|\overrightarrow{OB} \times \overrightarrow{OC}\right| = \frac{9}{2}$, तब निम्न में से कौनसा/कौनसे कथन सही है/हैं? **[JEE(Adv.)2021]**

(A) $\overrightarrow{OC}$ का $\overrightarrow{OA}$ पर प्रक्षेप $-\frac{3}{2}$ है

(B) त्रिभुज OAB का क्षेत्रफल $\frac{9}{2}$ है

(C) त्रिभुज ABC का क्षेत्रफल $\frac{9}{2}$ है

(D) समान्तर चतुर्भुज जिसकी आसन्न भुजाऐं $\overrightarrow{OA}$ तथा $\overrightarrow{OC}$ है, के विकर्णों के मध्य न्यून कोण $\frac{\pi}{3}$ है

Ans. ABC

Sol. Let $\overrightarrow{OA} = \vec{a} = 2i + 2j + k$

$\overrightarrow{OB} = \vec{b} = i - 2j + 2k$

$\overrightarrow{OC} = \vec{c} = \frac{1}{2}\left(\vec{b} - \lambda\vec{a}\right)$

$\left|\overrightarrow{OB} \times \overrightarrow{OC}\right| = \frac{9}{2}$

$\left|\vec{b} \times \frac{\left(\vec{b} - \lambda\vec{a}\right)}{2}\right| = \frac{9}{2}$

$\vec{a} \times \vec{b} = \begin{vmatrix} i & j & k \\ 2 & 2 & 1 \\ 1 & -2 & 2 \end{vmatrix}$

$\vec{a} \times \vec{b} = 6i - 3j - 6k$

$\frac{|\lambda|}{2}.9 = \frac{9}{2} \Rightarrow \lambda \pm 1 (\lambda > 0)$

$\vec{a} \times \vec{b} = 3\left(2i - j - 2k\right)$

$\lambda = 1$

$\vec{c} = \frac{\vec{b} - \vec{a}}{2}$

(A) Projection of $\overrightarrow{OC}$ on $\overrightarrow{OA}$

$$=\frac{\vec{c}.\vec{a}}{|a|}=\frac{\left(\vec{b}-\vec{a}\right).\vec{a}}{2|\vec{a}|}=\frac{0-|\vec{a}|^2}{2|\vec{a}|}=\frac{-3}{2}$$

(B) Area of $\Delta OAB=\frac{1}{2}|\vec{a}\times\vec{b}|=\frac{9}{2}$

(C) Area of $\Delta ABC=\frac{1}{2}\left|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\right|$

$$=\frac{1}{2}\left|\vec{a}\times\vec{b}+\vec{b}\times\left(\frac{\vec{b}-\vec{a}}{2}\right)+\left(\frac{\vec{b}-\vec{a}}{2}\right)\times\vec{a}\right|$$

$$=\frac{1}{2}\left|\vec{a}\times\vec{b}\right|=\frac{9}{2}$$

(D) Let $\vec{d}_1=\frac{\vec{a}+\vec{c}}{2}=\frac{\vec{a}+\left(\frac{\vec{b}-\vec{a}}{2}\right)}{2}=\frac{\vec{a}+\vec{b}}{4}=\frac{3i+3k}{4}$

$$\vec{d}_2=\frac{\vec{a}-\vec{c}}{2}=\frac{\vec{a}-\left(\frac{\vec{b}-\vec{a}}{2}\right)}{2}=\frac{3\vec{a}-\vec{b}}{4}=\frac{5i+8j+k}{4}$$

$$\cos\theta=\left|\frac{\vec{d}_1.\vec{d}_2}{|\vec{d}_1||\vec{d}_2|}\right|$$

$$\cos\theta=\frac{1}{\sqrt{5}}$$

$$\theta=\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$$

As (ABC)

Paragraph Question Nos. 42 and 43

Let α, β and γ be real numbers such that the system of linear equations

$x + 2y + 3z = \alpha$

$4x + 5y + 6z = \beta$

$7x + 8y + 9z = \gamma - 1$

is consistent. Let |M| represent the determinant of the matrixs

$$M=\begin{bmatrix}\alpha & 2 & \gamma\\ \beta & 1 & 0\\ -1 & 0 & 1\end{bmatrix}$$

Let P be the plane containing all those (α, β, γ) for which the above system of linear equations is consistent, and D be the square of the distance of the point (0, 1, 0) from the plane P ?

माना α, β व γ वास्तविक संख्याऐं हैं जबकि रेखिक समीकरण निकाय

$x + 2y + 3z = \alpha$

$4x + 5y + 6z = \beta$

$7x + 8y + 9z = \gamma - 1$

सुसंगत है। माना |M| निम्न आव्यूह के सारणिक को निरूपित करता है

$$M=\begin{bmatrix}\alpha & 2 & \gamma\\ \beta & 1 & 0\\ -1 & 0 & 1\end{bmatrix}$$

माना P उन सभी (α, β, γ) का तल है जिनके लिए उपरोक्त रेखिक समीकरण निकाय सुंसगत है तथा D समतल P से बिन्दु (0,1,0) की दूरी का वर्ग है?

42. The value of |M| is _____. **[JEE(Adv.)2021]**

|M| का मान _____ है। **[JEE(Adv.)2021]**

Ans. 1

Sol. $$\Delta=\begin{vmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{vmatrix}=0$$

$\Rightarrow \Delta_x = 0$

$$\Rightarrow \begin{vmatrix}\alpha & 2 & 4\\ \beta & 1 & 0\\ \gamma-1 & 8 & 9\end{vmatrix}=0$$

$$|M|=\begin{vmatrix}\alpha & 2 & \gamma\\ \beta & 1 & 0\\ -1 & 0 & 1\end{vmatrix}$$

$|M| = \alpha - 2\beta + \gamma = 1$

43. The value of D is _____. **[JEE(Adv.)2021]**

D का मान _____ है। **[JEE(Adv.)2021]**

Ans. 1.5

Sol. Plane P $x - 2y + z = 1$

$$\sqrt{D} = \left|\frac{0-2+0-1}{\sqrt{1+4+1}}\right| = \frac{3}{\sqrt{6}}$$

$$D = \frac{9}{6} = 1.5$$

44. Let $\hat{i}, \hat{j}$ and $\hat{k}$ be the unit vectors along the three positive coordinate axes.

Let $\vec{a} = 3\hat{i} + \hat{j} - \hat{k}$

$\vec{b} = \hat{i} + b_2\hat{j} + b_3\hat{k}, \quad b_2, b_3 \in R$

$\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}, \quad c_1, c_2, c_3 \in R$

be three vectors such that $b_2b_3 > 0, \vec{a}.\vec{b} = 0$ and $\begin{pmatrix} 0 & -c_3 & c_2 \\ c_3 & 0 & -c_1 \\ -c_2 & c_1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} 3-c_1 \\ 1-c_2 \\ -1-c_3 \end{pmatrix}$. Then, which of the following is /are TRUE : **[JEE(Adv.)2022]**

माना $\hat{i}, \hat{j}$ तथा $\hat{k}$ इकाई सदिश है तो तीन धनात्मक निर्देशांक अक्षों के अनुदिश हैं। माना

Let $\vec{a} = 3\hat{i} + \hat{j} - \hat{k}$

$\vec{b} = \hat{i} + b_2\hat{j} + b_3\hat{k}, \quad b_2, b_3 \in R$

$\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}, \quad c_1, c_2, c_3 \in R$

तीन सदिश है जबकि $b_2b_3 > 0, \vec{a}.\vec{b} = 0$ तथा $\begin{pmatrix} 0 & -c_3 & c_2 \\ c_3 & 0 & -c_1 \\ -c_2 & c_1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} 3-c_1 \\ 1-c_2 \\ -1-c_3 \end{pmatrix}$. तब निम्नलिखित में से कौनसा/कौनसे विकल्प सत्य है/हैं : **[JEE(Adv.)2022]**

(A) $\vec{a}\,.\,\vec{c} = 0$ (B) $\vec{b}\,.\,\vec{c} = 0$ (C) $|\vec{b}| > \sqrt{10}$ (D) $|\vec{c}| \le \sqrt{11}$

Ans. BCD

Sol. $\vec{a} = 3\hat{i} + \hat{j} - \hat{k}$

$\vec{b} = \hat{i} + b_2\hat{j} + b_3\hat{k}$

$\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$

$$\begin{pmatrix} 0 & -c_3 & c_2 \\ c_3 & 0 & -c_1 \\ -c_2 & c_1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ b_2 \\ b_3 \end{pmatrix}=\begin{pmatrix} 3-c_1 \\ 1-c_2 \\ -1-c_3 \end{pmatrix}$$

multipily and comapre

$b_2c_3 - b_3c_2 = c_1 - 3$ .....(1)

$c_3 - b_3c_1 = 1 - c_2$ .....(2)

$c_2 - b_2c_1 = 1 + c_3$ .....(3)

$(1)\hat{i} - (2)\hat{j} + (3)\hat{k}$

$\hat{i}(b_2c_3 - b_3c_2) - \hat{j}(c_3 - b_3c_1) + \hat{k}(c_2 - b_2c_1)$

$= c_1\hat{i} + c_2\hat{j} + c_2\hat{k} - 3\hat{i} - \hat{j} + \hat{k}$

$\vec{b} \times \vec{c} = \vec{c} - \vec{a}$

Take dot product with $\vec{b}$

$0 = \vec{c}.\vec{b} - \vec{a}.\vec{b}$

$\vec{b}.\vec{c} = 0$

$\vec{b} \perp \vec{c}$

$\vec{b} \wedge \vec{c} = 90^\circ$

Take dot product with $\vec{c}$

$0 = |\vec{c}|^2 - \vec{a}.\vec{c}$

$\vec{a}.\vec{c} = |\vec{c}|^2$

$\vec{a}.\vec{c} \neq 0$

$\vec{b} \times \vec{c} = \vec{c} - \vec{a}$

Squaring

$|\vec{b}|^2 \ |\vec{c}|^2 = |\vec{c}|^2 + |\vec{a}|^2 - 2\vec{c}.\vec{a}$

$|\vec{b}|^2 \ |\vec{c}|^2 = |\vec{c}|^2 + 11 - 2|\vec{c}|^2$

$|\vec{b}|^2 \ |\vec{c}|^2 = 11 - |\vec{c}|^2$

$|\vec{c}|^2 = \dfrac{11}{|\vec{b}|^2 + 1}$

$|\vec{c}| \leq \sqrt{11}$

given $\vec{a}.\vec{b} = 0$

$b_2 - b_3 = -3$ also

$b_2^2 + b_3^2 - 2b_2b_3 = 9 \quad b_2b_3 > 0$

$b_2^2 + b_3^2 = 9 + 2b_2b_3$

$b_2^2 + b_3^2 = 9 + 2b_2b_3 > 9$

$b_2^2 + b_3^2 > 9$

$|\vec{b}| = \sqrt{1 + b_2^2 + b_3^2}$

$|\vec{b}| > \sqrt{10}$

45. Let P_1 and P_2 be two planes given by

$P_1 : 10x + 15y + 12z - 60 = 0,$

$P_2 : -2x + 5y + 4z - 20 = 0,$

Which of the following straight lines can be an edge of some tetrahedron whose two faces lie on P_1 and P_2?

[JEE(Adv.)2022]

माना दो समतल P_1 व P_2 है, जो

$P_1 : 10x + 15y + 12z - 60 = 0,$

$P_2 : -2x + 5y + 4z - 20 = 0,$

द्वारा दिये गये है। निम्नलिखित में से कौनसी सरल रेखाएं किसी चतुष्फलक का एक कोर हो सकती हैं जिसके दो फलक P_1 तथा P_2 पर स्थित हैं?

[JEE(Adv.)2022]

(A) $\frac{x-1}{0} = \frac{y-1}{0} = \frac{z-1}{5}$

(B) $\frac{x-6}{-5} = \frac{y}{2} = \frac{z}{3}$

(C) $\frac{x}{-2} = \frac{y-4}{5} = \frac{z}{4}$

(D) $\frac{x}{1} = \frac{y-4}{-2} = \frac{z}{3}$

Ans. ABD

Sol. Edge of Tetrahedron can be (i) common intersection line
(ii) Line totally in P_2/P_1 and meeting intersecting line.
(iii) Line intersecting both P_1 and P_2

(i) Common intersecting line is parallel to $\vec{n}_1 \times \vec{n}_2$

$10a + 15b + 12c = 0$

$-2a+5b+4c = 0$

i.e. it has direction ratio 0, –4, 5 and a point on it can be (0, 0, 5)

$\frac{x}{0} = \frac{y}{-4} = \frac{z-5}{5}$ Not possible

(ii) Line in P_2/P_1 must be perpendicular to their normals. option 'D' Line in plane P_2 and it should also cut common line

$\Rightarrow$ 0, -4λ, 5λ, + 5 should lie on D.

$\frac{0}{1} = \frac{-4\lambda - 4}{-2} = \frac{5\lambda + 5}{3}$ which is true for $\lambda = -1$.

Hence 'D' is correct.
(iii) In option 'C' is parallel to normal at 'P_2' but it is passing through (0, 4, 0) which is a point on intersecting lines. Hence, this can't be an edge of tetrahedron (A) and (B) cuts both P_1 and P_2 and do not passes through common intersecting lines hence are possible.
(A) and (B) are possible, Hence A, B, D are correct

46. Let S be the reflection of a point Q with respect to the plane given by $\vec{r} = -(t+p)\hat{i} + t\hat{j} + (1+p)\hat{k}$ where t, p are real parameters and $\hat{i}$, $\hat{j}$, $\hat{k}$ are the unit vectors along the three positive coordinate axes. If the position vectors of Q and S are $10\hat{i} + 15\hat{j} + 20\hat{k}$ and $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ respectively, then which of the following is/are TRUE ?
[JEE(Adv.)2022]

माना $\vec{r} = -(t+p)\hat{i} + t\hat{j} + (1+p)\hat{k}$ द्वारा दिये गये समतल के सापेक्ष बिन्दु Q का प्रतिबिम्ब S है, जहाँ t, p वास्तविक प्राचल हैं तथा $\hat{i}$, $\hat{j}$, $\hat{k}$ तीन धनात्मक निर्देशांक अक्षों के अनुदिश इकाई सदिश है। यदि Q तथा S के स्थिति सदिश क्रमशः $10\hat{i} + 15\hat{j} + 20\hat{k}$ तथा $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ है, तब निम्नलिखित में से कौनसा/कौनसे विकल्प सत्य है/हैं? **[JEE(Adv.)2022]**

(A) $3(\alpha + \beta) = -101$ (B) $3(\beta + \gamma) = -71$ (C) $3(\gamma + \alpha) = -86$ (D) $3(\alpha + \beta + \gamma) = -121$

Ans. ABC

Sol. $\vec{r} = \hat{k} + (-\hat{i} + \hat{k})p + t(-\hat{i} + \hat{j})$

$$\vec{n} = \begin{vmatrix} i & j & k \\ -1 & 0 & 1 \\ -1 & 1 & 0 \end{vmatrix} = -\hat{i} - \hat{j} - \hat{k}$$

Equation of Plane $x + y + z = 1$
Q = (10, 15, 20)
F = (10+d, 15+d, 20+d)
10+d + 10+d + 20+d =1
3d = –44

$$\frac{\alpha + 10}{2} = 10 + d$$

$3\alpha = 30 - 2(44) = -58$
$3\beta = 45 - 88 = -43$
$3\gamma = 60 - 88 = -28$

47. Let ℓ_1 and ℓ_2 be the lines $\vec{r}_1 = \lambda(\hat{i} + \hat{j} + \hat{k})$ and $\vec{r}_2 = (\hat{j} - \hat{k}) + \mu(\hat{i} + \hat{k})$, respectively. Let X be the set of all the planes H that contain the line ℓ_1. For a plane H, let $d(H)$ denote the smallest possible distance between the points of ℓ_2 and H. Let H_0 be a plane in X for which $d(H_0)$ is the maximum value of $d(H)$ as H varies over all planes in X. Match each entry in **List-I** to the correct entries in **List-II**. **[JEE(Adv.)2023]**

List-I	**List-II**
(P) The value of $d(H_0)$ is	(1) $\sqrt{3}$
(Q) The distance of the point (0,1,2) from H_0 is	(2) $\frac{1}{\sqrt{3}}$
(R) The distance of origin from H_0 is	(3) 0
(S) The distance of origin from the point of intersection of planes y = z, x = 1 and H_0 is	(4) $\sqrt{2}$
	(5) $\frac{1}{\sqrt{2}}$

माना कि ℓ_1 एवं ℓ_2 क्रमशः $\vec{r}_1 = \lambda(\hat{i}+\hat{j}+\hat{k})$ एवं $\vec{r}_2 = (\hat{j}-\hat{k})+\mu(\hat{i}+\hat{k})$ रेखाएं हैं। माना कि X उन सभी समतलों (planes) H का समुच्चय है जो रेखा ℓ_1 को अंतर्विष्ट (contain) करते हैं। समतल H के लिए माना कि $d(H)$, रेखा ℓ_2 के बिन्दुओं और H के बीच की न्यूनतम संभव (smallest possible) दूरी है। मान लीजिये कि d(H) का महत्तम संभव मान (maximum possible value), तब H समुच्चय X के सभी समतलों पर विचरण (vary) करता है, $d(H_0)$ है, जहाँ समतल H_0 समुच्चय X में है।

List-I की प्रत्येक प्रविष्टि (entry) का **List-II** की सही प्रविष्टियों (entries) से मिलान कीजिए। **[JEE(Adv.)2023]**

List-I	**List-II**
(P) $d(H_0)$ का मान है	(1) $\sqrt{3}$
(Q) बिन्दु (0,1,2) की H_0 से दूरी है	(2) $\frac{1}{\sqrt{3}}$
(R) मूल बिन्दु की H_0 से दूरी है	(3) 0
(S) मूल बिन्दु की समतल y = z, x = 1 एवं H_0 के प्रतिच्छेदन से दूरी है	(4) $\sqrt{2}$
	(5) $\frac{1}{\sqrt{2}}$

(A) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (5); (S) $\rightarrow$ (1)
(B) (P) $\rightarrow$ (5); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (3); (S) $\rightarrow$ (1)
(C) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (3); (S) $\rightarrow$ (2)
(D) (P) $\rightarrow$ (5); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (4); (S) $\rightarrow$ (2)

Sol. $\bar{r_1} = \lambda(i+j+k)$; $\bar{r_2} = (j-k)+\mu(i+k)$

H_0 will be plane containing l_1 & parallel to l_2.

$\therefore \perp$r vector of plane parallel to lines l_1 & l_2 is $\begin{vmatrix} i & j & k \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{vmatrix}$

$\Rightarrow i[1] - j[1-1] + k[-1]$

$\Rightarrow i - k$

$\therefore$ Plane will be $x + 0 \cdot y - z = \lambda$; passes through origin

$\therefore \lambda = 0$

$\therefore$ Plane $\Rightarrow x - z = 0 \rightarrow H_0$ Plane

(P) $d(H_0) = 1$ distance of point (0, 1, –1) from H

$$d = \frac{|0-(-1)|}{\sqrt{2}} = \frac{1}{\sqrt{2}} \quad P \rightarrow 5$$

(Q) $d = \left|\frac{0-2}{\sqrt{2}}\right| = \sqrt{2} \quad Q \rightarrow 4$

(R) $d = \left|\frac{0}{\sqrt{2}}\right| = 0 \quad R \rightarrow 3$

(S) Point of intersection will be (1,1,1)

$d = \sqrt{1+1+1} = \sqrt{3}$

(B)

48. Let the position vector of the points P, Q, R and S be $\vec{a} = \hat{i} + 2\hat{j} - 5\vec{k}$, $\vec{b} = 3\hat{i} + 6\hat{j} + 3\hat{k}$, $\vec{c} = \frac{17}{5}\hat{i} + \frac{16}{5}\hat{j} + 7\hat{k}$ and $\vec{d} = 2\hat{i} + \hat{j} + \hat{k}$, respectively. Then which of the following statements is true? **[JEE(Adv.)2023]**

माना कि बिन्दुओं P, Q, R एवं S के स्थिति सदिश (position vectors) क्रमशः $\vec{a} = \hat{i} + 2\hat{j} - 5\vec{k}$, $\vec{b} = 3\hat{i} + 6\hat{j} + 3\hat{k}$, $\vec{c} = \frac{17}{5}\hat{i} + \frac{16}{5}\hat{j} + 7\hat{k}$ और $\vec{d} = 2\hat{i} + \hat{j} + \hat{k}$, हैं। तब निम्न में से कौनसा कथन सत्य है? **[JEE(Adv.)2023]**

(A) The points P, Q, R and S are **NOT** coplanar.

बिन्दु P, Q, R एवं S सह्तलीय (Coplanar) नहीं हैं।

(B) $\frac{\vec{b}+2\vec{d}}{3}$ is the position vector of a point which divides PR internally in the ratio 5 : 4

सदिश $\frac{\vec{b}+2\vec{d}}{3}$ उस बिन्दु का स्थिति सदिश है जो PR को 5 : 4 के अनुपात में अंतः (Internally) विभाजित करता है।

(C) $\frac{\vec{b}+2\vec{d}}{3}$ is the position vector of a point which divides PR externally in the ratio 5 : 4

सदिश $\frac{\vec{b}+2\vec{d}}{3}$ उस बिन्दु का स्थिति सदिश है जो PR को 5 : 4 के अनुपात में बाह्य (Externally) विभाजित करता है।

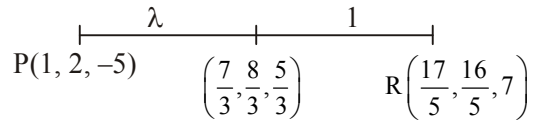
(D) The square of the magnitude of the vector $\vec{b} \times \vec{d}$ is 95

सदिश $\vec{b} \times \vec{d}$ के परिमाण (magnitude) का वर्ग (Square) 95 है।

Ans. B

Sol. $P(1, 2, -5)$, $Q(3, 6, 3)$, $R\left(\frac{17}{5}, \frac{16}{5}, 7\right)$, $S(2, 1, 1)$

$$\frac{\vec{b}+2\vec{d}}{3} = \frac{7i+8j+5k}{3}$$



$$\Rightarrow \frac{17\lambda}{5}+1=\frac{7}{3}(\lambda+1)$$

$$\Rightarrow 51\lambda+15=35\lambda+35$$

$$\Rightarrow 16\lambda=20 \qquad \Rightarrow \lambda=\frac{5}{4}$$

49. Let R^3 denote the three-dimensional space. Take two points P = (1,2,3) and Q = (4,2,7). Let dist(X, Y) denote the distance between two points X and Y in R^3. Let

$S = \{X \in R^3 : (\text{dist}(X, P))^2 - (\text{dist}(X, Q))^2 = 50\}$ and

$T = \{Y \in R^3 : (\text{dist}(Y, Q))^2 - (\text{dist}(Y, P))^2 = 50\}$.

Then which of the following statements is(are) TRUE ? **[JEE(Adv.)2024]**

(A) There is a triangle whose area is 1 and all of whose vertices are from S.

(B) There are two distinct points L and M in T such that each point on the line segment LM is also in T.

(C) There are infinitely many rectangles of perimeter 48, two of whose vertices are from S and the other two vertices are from T.

(D) There is a square of perimeter 48, two of whose vertices are from S and the other two vertices are from T.

माना कि R^3 त्रि–विमीय अंतरिक्ष (three-dimensional space) को दर्शाता है। दो बिन्दु P = (1,2,3) और Q = (4,2,7) लीजिये। माना कि dist(X, Y), R^3 के दो बिन्दुओं X और Y के बीच की दूरी को दर्शाता है। माना कि

$S = \{X \in R^3 : (\text{dist}(X, P))^2 - (\text{dist}(X, Q))^2 = 50\}$ और

$T = \{Y \in R^3 : (\text{dist}(Y, Q))^2 - (\text{dist}(Y, P))^2 = 50\}$.

हैं। तब निम्नलिखित कथनों में से कौनसा(से) सत्य है(हैं) ? **[JEE(Adv.)2024]**

(A) एक ऐसा त्रिभुज है जिसका क्षेत्रफल 1 है और जिसके सारे शीर्ष S से हैं।

(B) T में दो ऐसे भिन्न बिन्दु L और M हैं कि रेखाखण्ड LM में स्थित प्रत्येक बिन्दु भी T में है।

(C) परिमाप 48 के ऐसे अनंत आयत हैं जिनके दो शीर्ष S से हैं तथा अन्य दो शीर्ष T से हैं।

(D) परिमाप 48 का एक ऐसा वर्ग है जिसके दो शीर्ष S से हैं तथा अन्य दो शीर्ष T से हैं।

Ans. ABCD

Sol. P(1,2,3)

Q(4,2,7)

S : 6x + 8z = 105

T : 6x + 8z = 5

S & T represent two parallel plane

Now check the options

50. A straight line drawn from the point P(1, 3, 2), parallel to the line $\frac{x-2}{1}=\frac{y-4}{2}=\frac{z-6}{1}$, intersects the plane $L_1 : x-y+3z=6$ at the point Q. Another straight line which passes through Q and is perpendicular to the plane L_1 intersects the plane $L_2 : 2x-y+z=-4$ at the point R. Then which of the following statements is (are) TRUE?

[JEE(Adv.)2024]

(A) The length of the line segment PQ is $\sqrt{6}$

(B) The coordinates of R are (1, 6, 3)

(C) The centroid of the triangle PQR is $\left(\frac{4}{3},\frac{14}{3},\frac{5}{3}\right)$

(D) The perimeter of the triangle PQR is $\sqrt{2}+\sqrt{6}+\sqrt{11}$

बिंदु P(1, 3, 2) से, रेखा $\frac{x-2}{1}=\frac{y-4}{2}=\frac{z-6}{1}$ के समान्तर खींची गयी सरल रेखा (straight line), तल (plane) $L_1 : x-y+3z=6$ को बिंदु Q पर प्रतिच्छेदित (intersect) करती है। एक अन्य सरल रेखा जो बिन्दु Q से होकर जाती है और तल (plane) L_1 के लंबवत (perpendicular) है, तब (plane) $L_2 : 2x-y+z=-4$ को बिंदु R पर प्रतिच्छेदित करती है। तब निम्नलिखित कथनों में से कौनसा(से) सत्य है (हैं)? **[JEE(Adv.)2024]**

(A) रेखाखण्ड (line segment) PQ की लम्बाई $\sqrt{6}$ है

(B) R के निर्देशांक (coordinates) (1, 6, 3) हैं

(C) त्रिभुज (triangle) PQR का केन्द्रक (centroid) $\left(\frac{4}{3},\frac{14}{3},\frac{5}{3}\right)$ है

(D) त्रिभुज PQR का परिमाप (perimeter) $\sqrt{2}+\sqrt{6}+\sqrt{11}$ है

Ans. AC

Sol. P(1,3,2)

$\vec{r}=(1,3,2)+\lambda(1,2,1)$

Line intersects plane L_1 at Q

Let Q $(1+\lambda, 3+2\lambda, 2+\lambda)$

$(1+\lambda)-(3+2\lambda)+3(2+\lambda)=6$

$\Rightarrow \lambda = 1$

$\Rightarrow$ Q(2,5,3)

equations of line through Q

$\vec{r}=(2,5,3)+\mu(1,-1,3)$

Let $R(2+\mu, 5-\mu, 3+3\mu)$

R will lie on plane L_2

$2(2+\mu)-(5-\mu)+3+3\mu=-4$

$\Rightarrow \mu=-1$

$\Rightarrow R(1,6,0)$

(A) $PQ=\sqrt{6}$

(B) R(1,6,0)

(C) Centroid opf $\Delta PQR \equiv \left(\frac{4}{3}, \frac{14}{3}, \frac{5}{3}\right)$

(D) Perimeter of $\Delta PQR = \sqrt{6}+\sqrt{11}+\sqrt{13}$

51. Let $\overrightarrow{OP}=\frac{\alpha-1}{\alpha}\hat{i}+\hat{j}+\hat{k}$, $\overrightarrow{OQ}=\hat{i}+\frac{\beta-1}{\beta}\hat{j}+\hat{k}$ and $\overrightarrow{OR}=\hat{i}+\hat{j}+\frac{1}{2}\hat{k}$ be three vectors, where $\alpha, \beta \in R-\{0\}$ and O denotes the origin. If $\left(\overrightarrow{OP}\times\overrightarrow{OQ}\right)\cdot\overrightarrow{OR}=0$ and the point $(\alpha, \beta, 2)$ lies on the plane $3x+3y-z+l=0$, then the value of l is _____. **[JEE(Adv.)2024]**

माना कि $\overrightarrow{OP}=\frac{\alpha-1}{\alpha}\hat{i}+\hat{j}+\hat{k}$, $\overrightarrow{OQ}=\hat{i}+\frac{\beta-1}{\beta}\hat{j}+\hat{k}$ और $\overrightarrow{OR}=\hat{i}+\hat{j}+\frac{1}{2}\hat{k}$ तीन सदिश हैं, जहाँ $\alpha, \beta \in R-\{0\}$ और O मूल बिन्दु को दर्शाता है। यदि $\left(\overrightarrow{OP}\times\overrightarrow{OQ}\right)\cdot\overrightarrow{OR}=0$ और बिन्दु $(\alpha, \beta, 2)$ तल $3x+3y-z+l=0$ पर स्थित है, तब l का मान _____ है। **[JEE(Adv.)2024]**

Ans. 5

Sol.
$$\begin{vmatrix} \frac{\alpha-1}{\alpha} & 1 & 1 \\ 1 & \frac{\beta-1}{\beta} & 1 \\ 1 & 1 & \frac{1}{2} \end{vmatrix}=0$$

$$\left(\frac{\alpha-1}{\alpha}\right)\left[\frac{\beta-1}{2\beta}-1\right]+\frac{1}{2}+\left(1-\frac{(\beta-1)}{\beta}\right)=0$$

$$\frac{-(\alpha-1)(\beta+1)}{2\alpha\beta}+\frac{1}{2}\cdot\frac{\beta\alpha}{\alpha\beta}+\frac{1}{\beta}\cdot\frac{2\alpha}{2\alpha}=0$$

$-\alpha\beta-\alpha+\beta+1+\alpha\beta+2\alpha=0$

$\alpha+\beta+1=0$

and $3(\alpha+\beta)-2+\ell=0$

$\ell=5$

52. Let $\vec{p}=2\hat{i}+\hat{j}+3\hat{k}$ and $\vec{q}=\hat{i}-\hat{j}+\hat{k}$. If for some real numbers α, β, and γ, we have $15\hat{i}+10\hat{j}+6\hat{k}=\alpha(2\vec{p}+\vec{q})+\beta(\vec{p}-2\vec{q})+\gamma(\vec{p}\times\vec{q})$ then the value of γ is _____. **[JEE(Adv.)2024]**

माना कि $\vec{p}=2\hat{i}+\hat{j}+3\hat{k}$ और $\vec{q}=\hat{i}-\hat{j}+\hat{k}$ हैं। यदि कुछ वास्तविक संख्याओं (real numbers) α, β और γ के लिए, $15\hat{i}+10\hat{j}+6\hat{k}=\alpha(2\vec{p}+\vec{q})+\beta(\vec{p}-2\vec{q})+\gamma(\vec{p}\times\vec{q})$ है, तब γ का मान _____ है। **[JEE(Adv.)2024]**

Ans. 2

Sol. $\vec{p}=2\hat{i}+\hat{j}+3\hat{k}$

$\vec{q}=\hat{i}-\hat{j}+\hat{k}$ d

$15\hat{i}+10\hat{j}+6\hat{k}=\alpha(2\vec{p}+\vec{q})+\beta(\vec{p}-2\vec{q})+\gamma(\vec{p}\times\vec{q})$

Take dot with $\vec{p}\times\vec{q}$

$$\left(15\hat{i}+10\hat{j}+6\hat{k}\right).(\vec{p}\times\vec{q})=\alpha\left(2\vec{p}+\vec{q}\right).(\vec{p}\times\vec{q})+\beta\left(\vec{p}-2\vec{q}\right).(\vec{p}\times\vec{q})+\gamma(\vec{p}\times\vec{q}).(\vec{p}\times\vec{q})$$

$=0+0+\gamma|\vec{p}\times\vec{q}|^2$

Now, $\vec{p}\times\vec{q}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&1&3\\1&-1&1\end{vmatrix}$

$=\hat{i}(4)-\hat{j}(-1)+\hat{k}(-3)$

$=4\hat{i}+\hat{j}-3\hat{k}$

$|\vec{p}\times\vec{q}|=\sqrt{16+1+9}$

$\Rightarrow\sqrt{26}$

$\left(15\hat{i}+10\hat{j}+6\hat{k}\right)\left(4\hat{i}+\hat{j}-3\hat{k}\right)=\gamma 26$

$\frac{60+10-18}{26}=\gamma$

$\gamma=2$

53. Let $\gamma \in R$ be such that lines $L_1:\frac{x+11}{1}=\frac{y+21}{2}=\frac{z+29}{3}$ and $L_2:\frac{x+16}{3}=\frac{y+11}{2}=\frac{z+4}{\gamma}$ intersect. Let R_1 be the point of intersection of L_1 and L_2. Let $O=(0,0,0)$, and $\hat{n}$ denote a unit normal vector to the plane containing both the lines L_1 and L_2:

Match each entry in **List-I** to the correct entery in **List -II**. **[JEE(Adv.)2024]**

List -I	**List -II**
(P) γ equals	(1) $-\hat{i}-\hat{j}+\hat{k}$
(Q) A possible choice for $\hat{n}$ is	(2) $\sqrt{\frac{3}{2}}$
(R) $\overrightarrow{OR_1}$ equals	(3) 1
(S) A possible value of $\overrightarrow{OR_1}.\ \hat{n}$ is	(4) $\frac{1}{\sqrt{6}}\hat{i}-\frac{2}{\sqrt{6}}\hat{j}+\frac{1}{\sqrt{6}}\hat{k}$
	(5) $\sqrt{\frac{2}{3}}$

The correct option is :

माना कि $\gamma \in R$ इस प्रकार है कि रेखाएँ $L_1:\frac{x+11}{1}=\frac{y+21}{2}=\frac{z+29}{3}$ और $L_2:\frac{x+16}{3}=\frac{y+11}{2}=\frac{z+4}{\gamma}$ प्रतिच्छेदित करती हैं। माना कि L_1 और L_2 का प्रतिच्छेदन बिन्दु R_1 है। माना कि $O=(0,0,0)$ है, और $\hat{n}$, उस तल जिसमें L_1 और L_2 दोनों स्थित हैं, के एक मात्रक अभिलम्ब सदिश को दर्शाता है।

सूची-I की प्रत्येक प्रविष्टि का **सूची-II** की सही प्रविष्टि से मिलान कीजिए। **[JEE(Adv.)2024]**

सूची-I	**सूची -II**
(P) γ बराबर	(1) $-\hat{i}-\hat{j}+\hat{k}$
(Q) $\hat{n}$ का एक संभावित विकल्प है	(2) $\sqrt{\frac{3}{2}}$
(R) $\overrightarrow{OR_1}$ बराबर	(3) 1
(S) $\overrightarrow{OR_1}.\ \hat{n}$ का एक संभावित मान है	(4) $\frac{1}{\sqrt{6}}\hat{i}-\frac{2}{\sqrt{6}}\hat{j}+\frac{1}{\sqrt{6}}\hat{k}$
	(5) $\sqrt{\frac{2}{3}}$

सही विकल्प है :

(A) (P) $\rightarrow$ (3); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (1); (S) $\rightarrow$ (2)

(B) (P) $\rightarrow$ (5); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (1); (S) $\rightarrow$ (2)

(C) (P) $\rightarrow$ (3); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (1); (S) $\rightarrow$ (5)

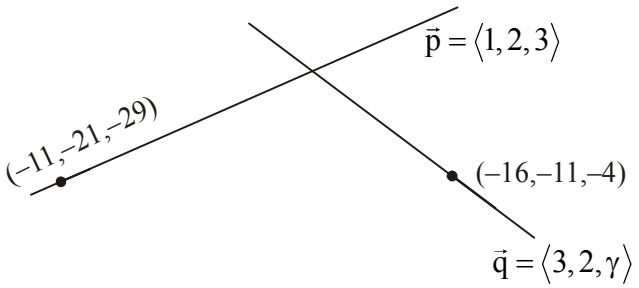
(D) (P) $\rightarrow$ (3); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (4); (S) $\rightarrow$ (5)

Ans. C

Sol. (P) Lines are intersecting.

Hence lines a coplanar.

$$\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$



$$\begin{vmatrix} 5 & -10 & -25 \\ 1 & 2 & 3 \\ 3 & 2 & \gamma \end{vmatrix} = 0$$

$\Rightarrow \gamma = 1$ Hence (P) $\rightarrow$ (3)

(Q) normal

$\vec{n} = \vec{p} \times \vec{q}$

$$= \begin{vmatrix} i & j & k \\ 1 & 2 & 3 \\ 3 & 2 & 1 \end{vmatrix} = -4\hat{i} + 8\hat{j} - 4\hat{k}$$

Hence a possible choice of $\hat{n}$ is $\frac{1}{\sqrt{6}}\hat{i} - \frac{2}{\sqrt{6}}\hat{j} + \frac{1}{\sqrt{6}}\hat{k}$, then (Q) $\rightarrow$ (4)

(R)

for POI of two lines

$< \lambda - 11, 2\lambda - 21, 3\lambda - 29 >$

$= < 3\mu - 16, 2\mu - 11, \mu - 4 >$

By comparing

$\mu = 5$

$\lambda = 10$

hence POI is (–1,–1,1).

then $\overrightarrow{OR_1} = -\hat{i} - \hat{j} + \hat{k}$

Hence (R) $\rightarrow$ (1)

(S) $\overrightarrow{OR_1} \cdot \hat{n}$

$$= \left(-\hat{i} - \hat{j} + \hat{k}\right) \cdot \left(\frac{\hat{i} - 2\hat{j} + \hat{k}}{\sqrt{6}}\right)$$

$$= \sqrt{\frac{2}{3}}$$

Hence, (S) $\rightarrow$ (5)

Overall correct option is (C).

54. Let L_1 be the line of intersection of the planes given by the equations

$2x + 3y + z = 4$ and $x + 2y + z = 5$.

Let L_2 be the line passing through the point P(2, –1, 3) and parallel to L_1. Let M denote the plane given by the equation

$2x + y - 2z = 6$.

Suppose that the line L_2 meets the plane M at the point Q. Let R be the foot of perpendicular drawn from P to the plane M.

Then which of the following statements is (are) TRUE ? **[JEE(Adv.)2025]**

(A) The length of the line segment PQ is $9\sqrt{3}$

(B) The length of the line segment QR is 15

(C) The area of ΔPQR is $\frac{3}{2}\sqrt{234}$

(D) The acute angle between the line segments PQ and PR is $\cos^{-1}\left(\frac{1}{2\sqrt{3}}\right)$

मान लीजिए कि L_1, समीकरणों

$2x + 3y + z = 4$ और $x + 2y + z = 5$.

द्वारा दिए गए समतलों की प्रतिच्छेदन रेखा है। मान लीजिए कि L_2 वह रेखा है जो बिन्दु P(2, –1, 3) से गुजरती है और L_1 के समांतर है। मान लीजिए कि M, समीकरण

$2x + y – 2z = 6$

द्वारा दिये गये समतल को दर्शाता है। मान लीजिए कि L_2, समतल M से बिन्दु Q पर मिलती है। मान लीजिए कि P से समतल M पर डाले गए लम्ब का पाद R है।

तब निम्नलिखित कथनों में से कौन सा (से) सत्य है (हैं) ? **[JEE(Adv.)2025]**

(A) रेखाखण्ड PQ की लम्बाई $9\sqrt{3}$ है

(B) रेखाखण्ड QR की लम्बाई 15 है

(C) त्रिभुज ΔPQR का क्षेत्रफल $\frac{3}{2}\sqrt{234}$ है

(D) रेखाखण्डों PQ और PR के बीच का न्यूनकोण $\cos^{-1}\left(\frac{1}{2\sqrt{3}}\right)$ है

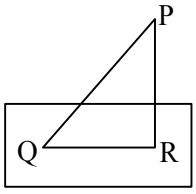
Ans. AC

Sol. $\vec{P}_1 = \text{direction vector of } L_2$

$$\vec{P}_1 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 1 \\ 1 & 2 & 1 \end{vmatrix} = \hat{i} - \hat{j} + \hat{k}$$

Equation of L_2

$\vec{r} = (2,-1,3) + \lambda(1,-1,1)$



$Q(2 + \lambda, –1 – \lambda, 3 + \lambda)$

Q will lie on $2x + y – 2z = 6$

$2(2 + \lambda) + (–1 – \lambda) – 2(3 + 2\lambda) = 6$

$\Rightarrow \lambda = –9$

$\Rightarrow Q(–7, 8, –6)$

Equation of PR

$\vec{r} = (2,-1,3) + \mu(2,1,-2)$

$R(2 + 2\mu, –1 + \mu, 3 – 2\mu)$

R will lie on $2x + y - 2z = 6$

$2(2 + 2\mu) + (-1 + \mu) - 2(3 - 2\mu) = 6$

$\Rightarrow \mu = 1$

$\Rightarrow R(4,0,1)$

$PQ = 9\sqrt{3}$ $\quad PR = 3$ $\quad QR = \sqrt{234}$

ΔPQR is right angled triangle

$$\text{Area} = \frac{1}{2}(QR)(PR) = \frac{3\sqrt{234}}{2}$$

θ = angle between line PQ & PR

$$\cos\theta = \frac{PR}{PQ} = \frac{3}{9\sqrt{3}} = \frac{1}{3\sqrt{3}}$$

55. For any two points M and N in the XY-plane, let $\overrightarrow{MN}$ denote the vector from M to N, and $\vec{0}$ denote the zero vector. Let P, Q and R be three distinct points in the XY-plane. Let S be a point inside the triangle ΔPQR such that

$\overrightarrow{SP} + 5\overrightarrow{SQ} + 6\overrightarrow{SR} = \vec{0}$.

Let E and F be the mid-points of the sides PR and QR, respectively. Then the value of

$$\frac{\text{length of the line segment EF}}{\text{length of the line segment ES}}$$

is _____. **[JEE(Adv.)2025]**

मान लीजिए कि XY-तल में किन्हीं दो बिन्दुओं M और N के लिए, M से N तक के सदिश को $\overrightarrow{MN}$ से दर्शाया जाता है, तथा शून्य सदिश को $\vec{0}$ से दर्शाया जाता है। मान लीजिए कि XY-तल में P, Q और R, तीन भिन्न बिन्दु हैं।

मान लीजिए कि S, त्रिभुज ΔPQR के आंतरिक भाग में इस प्रकार का एक बिन्दु है कि

$\overrightarrow{SP} + 5\overrightarrow{SQ} + 6\overrightarrow{SR} = \vec{0}$.

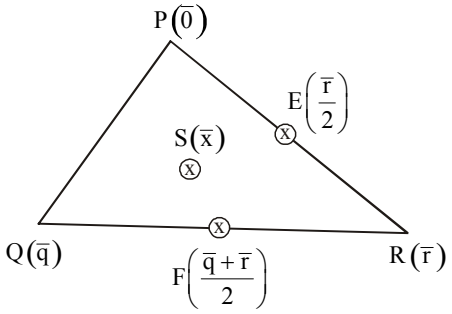
है। मान लीजिए कि E और F, क्रमशः PR और QR के मध्य बिन्दु हैं। तब

$$\frac{\text{रेखाखण्ड EF की लम्बाई}}{\text{रेखाखण्ड ES की लम्बाई}}$$

का मान _____ है। **[JEE(Adv.)2025]**

Ans. 1.2

Sol.



$$\overline{SP}+5\cdot\overline{SQ}+6\cdot\overline{SR}=0$$

$$0-\bar{x}+5\left(\bar{q}-\bar{x}\right)+6\left(\bar{r}-\bar{x}\right)=0$$

$$\Rightarrow \bar{x}=\frac{5\bar{q}+6\bar{r}}{12}$$

So $\quad \overline{EF}=\dfrac{\bar{q}}{2}$

and $\quad \overline{ES}=\dfrac{5\bar{q}}{12}$

$$\Rightarrow \frac{\left|\overline{EF}\right|}{\left|\overline{ES}\right|}=\frac{\left|\frac{\bar{q}}{2}\right|}{\left|\frac{5\bar{q}}{12}\right|}=\frac{6}{5}=1.2$$

56. Consider the vectors $\vec{x}=\hat{i}+2\hat{j}+3\hat{k},\vec{y}=2\hat{i}+3\hat{j}+\hat{k},$ and $\vec{z}=3\hat{i}+\hat{j}+2\hat{k}$. For two distinct positive real number α and β, define $\vec{X}=\alpha\vec{x}+\beta\vec{y}-\vec{z},\vec{Y}=\alpha\vec{y}+\beta\vec{z}-\vec{x},$ and $\vec{Z}=\alpha\vec{z}+\beta\vec{x}-\vec{y}$.

If the vectors $\vec{X},\vec{Y}$, and $\vec{Z}$ lie in a plane, then the value of $\alpha+\beta-3$ is _____. **[JEE(Adv.)2025]**

सदिशों (vectors) $\vec{x}=\hat{i}+2\hat{j}+3\hat{k},\vec{y}=2\hat{i}+3\hat{j}+\hat{k},$ और $\vec{z}=3\hat{i}+\hat{j}+2\hat{k}$ पर विचार कीजिए। दो भिन्न (distinct) धनात्मक (positive real number) α और β के लिए, $\vec{X}=\alpha\vec{x}+\beta\vec{y}-\vec{z},\vec{Y}=\alpha\vec{y}+\beta\vec{z}-\vec{x},$ और $\vec{Z}=\alpha\vec{z}+\beta\vec{x}-\vec{y}$ परिभाषित कीजिए। यदि सदिश $\vec{X},\vec{Y}$ और $\vec{Z}$ एक समतल (plane) पर स्थित हैं, तब $\alpha+\beta-3$ का मान _____ है। **[JEE(Adv.)2025]**

Ans. –2

Sol. $\vec{x}=(1,2,3),\vec{y}=(2,3,1),\vec{z}=(3,1,2)$

$$[\vec{X}\vec{Y}\vec{Z}] = \begin{vmatrix} \alpha & \beta & -1 \\ -1 & \alpha & \beta \\ \beta & -1 & \alpha \end{vmatrix} [\bar{x}\ \bar{y}\ \bar{z}] = 0$$

$$[\bar{x}\ \bar{y}\ \bar{z}] = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{vmatrix} = 5 - 2(1) + 3(-7) \neq 0$$

$$\Rightarrow \begin{vmatrix} \alpha & \beta & -1 \\ -1 & \alpha & \beta \\ \beta & -1 & \alpha \end{vmatrix} = 0 \qquad R_1 \to R_1 + R_2 + R_3$$

$$(\alpha+\beta-1)\begin{vmatrix} 1 & 1 & 1 \\ -1 & \alpha & \beta \\ \beta & -1 & \alpha \end{vmatrix} = 0$$

$\Rightarrow (\alpha + \beta - 1)(\alpha^2 + \beta + \alpha + \beta^2 + 1 - \alpha\beta) = 0$

$\Rightarrow \alpha + \beta = 1$ or $\alpha^2 + \alpha(1 - \beta) + \beta^2 + \beta + 1 = 0$

For $\alpha \in R^+$

$\Rightarrow D \geq 0$

$\Rightarrow (1 - \beta)^2 - 4(1)(\beta^2 + \beta + 1) \geq 0$

$\Rightarrow 1 + \beta^2 - 2\beta - 4\beta^2 - 4\beta - 4 \geq 0$

$\Rightarrow -3\beta^2 - 6\beta - 3 \geq 0$

$\Rightarrow \beta^2 + 2\beta + 1 \leq 0$

$\Rightarrow (\beta + 1)^2 \leq 0$

$\Rightarrow \beta = -1$ (not possible)

Hence $\alpha + \beta = 1$

57. Let $\vec{w} = \hat{i} + \hat{j} - 2\hat{k}$, and $\vec{u}$ and $\vec{v}$ be two vectors, such that $\vec{u} \times \vec{v} = \vec{w}$ and $\vec{v} \times \vec{w} = \vec{u}$. Let α, β, γ, and t be real numbers such that $\vec{u} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$, $-t\alpha + \beta + \gamma = 0$, $\alpha - t\beta + \gamma = 0$, and $\alpha + \beta - t\gamma = 0$.

Match each entry in **List-I** to the correct entry in **List-II** and choose the correct option.

[JEE(Adv.)2025]

List-I	**List-II**
(P) $\lvert\vec{v}\rvert^2$ is equal to	(1) 0
(Q) If $\alpha = \sqrt{3}$, then γ^2 is equal to	(2) 1
(R) If $\alpha = \sqrt{3}$, then $(\beta + \gamma)^2$ is equal to	(3) 2

(S) If $\alpha = \sqrt{2}$, then t + 3 is equal to (4) 3

(5) 5

The correct option is :

मान लीजिए कि $\vec{w} = \hat{i} + \hat{j} - 2\hat{k}$ है, तथा $\vec{u}$ और $\vec{v}$ दो इस प्रकार के सदिश हैं कि $\vec{u} \times \vec{v} = \vec{w}$ और $\vec{v} \times \vec{w} = \vec{u}$ हैं। मान लीजिए कि α, β, γ, और t इस प्रकार की वास्तविक संख्याएँ हैं कि $\vec{u} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$, $-t\alpha + \beta + \gamma = 0$, $\alpha - t\beta + \gamma = 0$, और $\alpha + \beta - t\gamma = 0$ हैं।

सूची-I की प्रत्येक प्रविष्टि का **सूची-II** की सही प्रविष्टि से मिलान कीजिए और सही विकल्प का चयन करें।

[JEE(Adv.)2025]

सूची-I	**सूची-II**
(P) $\|\vec{v}\|^2$ बराबर	(1) 0
(Q) यदि $\alpha = \sqrt{3}$ है, तब γ^2 बराबर	(2) 1
(R) यदि $\alpha = \sqrt{3}$ है, तब $(\beta + \gamma)^2$ बराबर	(3) 2
(S) यदि $\alpha = \sqrt{2}$ है, तब t + 3 बराबर	(4) 3
	(5) 5

The correct option is :

(A) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (4); (S) $\rightarrow$ (5)

(B) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (3); (S) $\rightarrow$ (5)

(C) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (4); (S) $\rightarrow$ (3)

(D) (P) $\rightarrow$ (5); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (1); (S) $\rightarrow$ (3)

Ans. A

Sol. $\vec{w} = i + j - 2k$

$\vec{u} \times \vec{v} = \vec{w}$ & $\vec{v} \times \vec{w} = \vec{u}$

$\Rightarrow \vec{u}, \vec{v}, \vec{w}$ are mutually $\perp$r.

Now, $\vec{v} \times \vec{w} = \vec{u}$

$(\vec{v} \times \vec{w}) \times \vec{v} = \vec{u} \times \vec{v}$

$(\vec{v} \cdot \vec{v})\vec{w} - (\vec{v} \cdot \vec{v})\vec{v} = \vec{u} \times \vec{v} = \vec{w}$

$\left(|\vec{v}|^2 - 1\right)\vec{w} - (\vec{w} \cdot \vec{v}) \cdot \vec{v} = 0$

$\Rightarrow |\vec{v}|^2 - 1 = 0$ & $\vec{w} \cdot \vec{v} = 0$

$\Rightarrow |\vec{v}|^2 = 1$ $\quad \therefore$ Angle bet $\vec{w}$&$\vec{v}$ is 90°

$P \rightarrow 2$

Again $\vec{v} \times \vec{w} = \vec{u}$

$|\vec{v} \times \vec{w}| = |\vec{u}|$

$|\vec{v}||\vec{w}| \sin 90^\circ = |\vec{u}|$

$1.\sqrt{1+1+4} = |\vec{u}|$

$|\vec{u}| = \sqrt{6}$

and $\vec{u} \cdot \vec{w} = 0$

$\alpha + \beta - 2\gamma = 0$

$$\left.\begin{array}{l} 8 - t\alpha + \beta + \gamma = 0 \\ \alpha - t\beta + \gamma = 0 \\ \alpha + \beta - t\gamma = 0 \end{array}\right\} \begin{array}{l} .....(1) \\ .....(2) \\ .....(3) \end{array} \text{ Given}$$

(ii) – (i) $\Rightarrow \alpha(1 + t) = (t + 1)\beta$

(iii) – (ii) $\Rightarrow \beta(1 + t) = (1 + t)\gamma$

$\therefore \alpha(1 + t) = (t + 1)\beta = (1 + t)\gamma$

$\Rightarrow t = -1$ or $\alpha = \beta = \gamma$

$\sqrt{\alpha^2 + \alpha^2 + \alpha^2} = \sqrt{6}$

$\alpha = \sqrt{2}$, $-t\alpha = -\alpha - \alpha$

$t = 2$

since $\alpha = \sqrt{3}$

$\Rightarrow t = -1$

$\Rightarrow \alpha + \beta + \gamma = 0$

$\alpha + \beta - 2\gamma = 0$

$\Rightarrow \alpha = 0$

$Q \rightarrow 1$

$\alpha + \beta = 0$

$\Rightarrow \beta + \gamma = -\alpha$

$(\beta + \gamma)^2 = \alpha^2$

$= 3$

$R \rightarrow 4$

(S) If $\alpha = \sqrt{2}; t = 2$

$t + 3 = 5$

$S \rightarrow 5$

Exercise C - I

1. Given $\vec{a} = x\hat{i} + y\hat{j} + 2\hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, $\vec{c} = \hat{i} + 2\hat{j}$; $(\vec{a} \wedge \vec{b}) = \frac{\pi}{2}$, $\vec{a} \cdot \vec{c} = 4$, then :

दिया गया है कि $\vec{a} = x\hat{i} + y\hat{j} + 2\hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, $\vec{c} = \hat{i} + 2\hat{j}$; $(\vec{a} \wedge \vec{b}) = \frac{\pi}{2}$, $\vec{a} \cdot \vec{c} = 4$, तो :

(A) $\left[\vec{a}\ \vec{b}\ \vec{c}\right]^2 = |\vec{a}|$ (B) $\left|\vec{a}\ \vec{b}\ \vec{c}\right| = |\vec{a}|$ (C) $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = 0$ (D) $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = |\vec{a}|^2$

Ans. D

Sol. $\vec{a} = (x, y, 2)$, $\vec{b} = (1, -1, 1)$, $\vec{c} = (1, 2, 0)$

$\vec{a} \cdot \vec{b} = 0 \Rightarrow x - y + 2 = 0$ ...(1) $(x, y) = (0, 2)$

$\vec{a} \cdot \vec{c} = 4 \Rightarrow x + 2y - 4 = 0$ ...(2)

$$\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right] = \begin{vmatrix} 0 & 2 & 2 \\ 1 & -1 & 1 \\ 1 & 2 & 0 \end{vmatrix} = 0 - 2(-1) + 2(3) = 8$$

$|\vec{a}|^2$

2. For a non zero vector $\vec{A}$ if the equations $\vec{A} \cdot \vec{B} = \vec{A} \cdot \vec{C}$ and $\vec{A} \times \vec{B} = \vec{A} \times \vec{C}$ hold simultaneously, then :

(A) $\vec{A}$ is perpendicular to $\vec{B} - \vec{C}$ (B) $\vec{A} - \vec{B}$

(C) $\vec{B} = \vec{C}$ (D) $\vec{C} = \vec{A}$

किसी अशून्य सदिश $\vec{A}$ के लिए यदि समीकरणें $\vec{A} \cdot \vec{B} = \vec{A} \cdot \vec{C}$ और $\vec{A} \times \vec{B} = \vec{A} \times \vec{C}$ सत्य है, तो :

(A) $\vec{A}$, $\vec{B} - \vec{C}$ के लम्बवत है (B) $\vec{A} - \vec{B}$

(C) $\vec{B} = \vec{C}$ (D) $\vec{C} = \vec{A}$

Ans. C

Sol. $\vec{A} \cdot \vec{B} = \vec{A} \cdot \vec{C} \Rightarrow \vec{A} \perp (\vec{B} - \vec{C})$

$\vec{A} \times \vec{B} = \vec{B} \times \vec{C} \Rightarrow \vec{A} \parallel (\vec{B} - \vec{C})$

$\vec{B} - \vec{C} = \vec{0} \Rightarrow \vec{B} = \vec{C}$

3. The volume of the parallelopiped constructed on the diagonals of the faces of the given rectangular parallelopiped is m times the volume of the given parallelopiped. Then m is equal to :

(A) 2 (B) 3 (C) 4 (D) None

दिये गये आयतीय समान्तर षट्फलक के पृष्ठों के विकर्णों से निर्मित समान्तर षट्फलक का आयतन दिये गये समान्तर षट्फलक के आयतन का m गुना है, तो m बराबर होगा :

(A) 2 (B) 3 (C) 4 (D) कोई नहीं

Ans. A

Sol. $V_{diagonal} = \left[\vec{a}+\vec{b} \quad \vec{b}+\vec{c} \quad \vec{c}+\vec{a}\right] = 2\left[\vec{a}\ \vec{b}\ \vec{c}\right]$

4. If $\vec{a} = \vec{b} + \vec{c}$, $\vec{b} \times \vec{d} = 0$ and $\vec{c}.\vec{d} = 0$, then $\dfrac{\vec{d} \times (\vec{a} \times \vec{d})}{\vec{d}^2}$ is equal to :

यदि $\vec{a} = \vec{b} + \vec{c}$, $\vec{b} \times \vec{d} = 0$ तथा $\vec{c}.\vec{d} = 0$ हो, तो $\dfrac{\vec{d} \times (\vec{a} \times \vec{d})}{\vec{d}^2}$ बराबर है :

(A) $\vec{a}$ (B) $\vec{b}$ (C) $\vec{c}$ (D) $\vec{d}$

Ans. C

Sol. $\vec{a} = \vec{b} + \vec{c}$ $\qquad \vec{b} \times \vec{d} = 0$

$\vec{a} \times \vec{d} = \vec{c} \times \vec{d} \Rightarrow \vec{d} \times \left(\vec{a} \times \vec{d}\right) = \vec{d} \times \left(\vec{c} \times \vec{d}\right)$

$\Rightarrow \vec{d} \times \left(\vec{a} \times \vec{d}\right) = \left|\vec{d}\right|^2 \vec{c} - \left(\vec{c} \cdot \vec{d}\right)\vec{d}$ $\qquad \left(\vec{c} \cdot \vec{d} = 0\right)$

$\Rightarrow \dfrac{\vec{d} \times \left(\vec{a} \times \vec{d}\right)}{\left|\vec{d}\right|^2} = \vec{c}$

5. The angles of a triangle, two of whose sides are represented by the vectors $\sqrt{3}\,(\hat{a} \times \vec{b})$ and $\vec{b} - (\hat{a}\ .\ \vec{b})\,\hat{a}$, where $\vec{b}$ is a non-zero vector and $\hat{a}$ is a unit vector in the direction of $\vec{a}$, are :

एक त्रिभुज की दो भुजाएँ सदिशों $\sqrt{3}\,(\hat{a} \times \vec{b})$ और $\vec{b} - (\hat{a}\ .\ \vec{b})\,\hat{a}$ से निरूपित होती है, जहाँ $\vec{b}$ अशून्य सदिश है और $\hat{a}$, $\vec{a}$ की दिशा में इकाई सदिश है, तो त्रिभुज के कोण हैं :

(A) $\tan^{-1}\left(\dfrac{1}{\sqrt{3}}\right);\ \tan^{-1}\left(\dfrac{1}{2}\right);\ \tan^{-1}\left(\dfrac{\sqrt{3}+2}{1-2\sqrt{3}}\right)$ (B) $\tan^{-1}\left(\sqrt{3}\right);\ \tan^{-1}\left(\dfrac{1}{\sqrt{3}}\right);\ \cot^{-1}(0)$

(C) $\tan^{-1}\left(\sqrt{3}\right);\ \tan^{-1}(2);\ \tan^{-1}\left(\dfrac{\sqrt{3}+2}{2\sqrt{3}-1}\right)$ (D) $\tan^{-1}\left(\sqrt{3}\right);\ \tan^{-1}\left(\sqrt{2}\right);\ \tan^{-1}\left(\dfrac{\sqrt{2}+3}{3\sqrt{2}-1}\right)$

Ans. B

Sol. $\overrightarrow{AB} = \sqrt{3}\left(\hat{a}\times\vec{b}\right)$

$\overrightarrow{AC} = \vec{b} - \left(\hat{a}\cdot\vec{b}\right)\hat{a}$

$\overrightarrow{AB}\cdot\overrightarrow{AC} = \sqrt{3}\left[\left(\vec{a}\times\vec{b}\right)\cdot\vec{b} - \left(\hat{a}\times\vec{b}\right)\cdot\hat{a}\left(\hat{a}\cdot\vec{b}\right)\right]$

$= 0$

$\angle A = 90°$

$\left|\overrightarrow{AB}\right|^2 = 3\left|\hat{a}\times\vec{b}\right|^2 = 3\left|\vec{b}\right|^2\sin^2\theta$

$\left|\overrightarrow{AC}\right|^2 = \left|\vec{b}\right|^2 + \left(\hat{a}\cdot\vec{b}\right)^2 - 2\left(\hat{a}\cdot\vec{b}\right)\left(\hat{a}\cdot\vec{b}\right)$

$|b|^2 - \left(\hat{a}\cdot\vec{b}\right)^2 = |b|^2\sin^2\theta$

$\frac{|AB|}{|AC|} = \sqrt{3} \Rightarrow \angle 60°, \angle B = 30°, \angle A = 90°$

6. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be non-coplanar unit vectors equally inclined to one another at an acute angle θ. Then $[\vec{a}\ \vec{b}\ \vec{c}]$ in terms of θ is equal to :

(A) $(1+\cos\theta)\sqrt{\cos 2\theta}$ (B) $(1+\cos\theta)\sqrt{1-2\cos 2\theta}$

(C) $(1-\cos\theta)\sqrt{1+2\cos\theta}$ (D) None of these

माना $\vec{a}$, $\vec{b}$ and $\vec{c}$ असमतलीय इकाई सदिश हैं और प्रत्येक एक दूसरे के साथ समान कोण θ बनाते है, तो $[\vec{a}\ \vec{b}\ \vec{c}]$ का मान θ के पदों में है :

(A) $(1+\cos\theta)\sqrt{\cos 2\theta}$ (B) $(1+\cos\theta)\sqrt{1-2\cos 2\theta}$

(C) $(1-\cos\theta)\sqrt{1+2\cos\theta}$ (D) इनमें से कोई नहीं

Ans. C

Sol. $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = \begin{vmatrix} \vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c} \\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c} \\ \vec{c}\cdot\vec{a} & \vec{c}\cdot\vec{b} & \vec{c}\cdot\vec{c} \end{vmatrix}$

$= \begin{vmatrix} 1 & \cos\theta & \cos\theta \\ \cos\theta & 1 & \cos\theta \\ \cos\theta & \cos\theta & 1 \end{vmatrix}$

$$= (1+2\cos\theta)\begin{vmatrix} 1 & 1 & 1 \\ \cos\theta & 1 & \cos\theta \\ \cos\theta & \cos\theta & 1 \end{vmatrix}$$

$= (1 + 2\cos\theta)\,(1\,(1 - \cos^2\theta) - 1\,(\cos\theta - \cos^2\theta) + 1\,(\cos^2\theta - \cos\theta))$

$= (1 + 2\cos\theta)\,(\cos^2\theta - 2\cos\theta + 1)$

$$\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right] = (1+2\cos\theta)(1-\cos\theta)^2$$

$$\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right] = (1-\cos\theta)\sqrt{1+2\cos\theta}$$

7. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are unit vectors, then $\left|\vec{a}-\vec{b}\right|^2 + \left|\vec{b}-\vec{c}\right|^2 + \left|\vec{c}-\vec{a}\right|^2$ does NOT exceed :

यदि $\vec{a}$, $\vec{b}$ तथा $\vec{c}$ इकाई सदिश हो, तो $\left|\vec{a}-\vec{b}\right|^2 + \left|\vec{b}-\vec{c}\right|^2 + \left|\vec{c}-\vec{a}\right|^2$ का अधिकतम मान है :

(A) 4 (B) 9 (C) 8 (D) 6

Ans. B

Sol. $\hat{a}, \hat{b}, \hat{c}$ are unit vectors.

$\therefore\ \hat{a}\cdot\hat{a} = \hat{b}\cdot\hat{b} = \hat{c}\cdot\hat{c} = 1$

Now, $x = \left|\hat{a}-\hat{b}\right|^2 + \left|\hat{b}-\hat{c}\right|^2 + \left|\hat{c}-\hat{a}\right|^2$

$= \hat{a}\cdot\hat{a} + \hat{b}\cdot\hat{b} - 2\hat{a}\cdot\hat{b} + \hat{b}\cdot\hat{b} + \hat{c}\cdot\hat{c} - 2\hat{b}\cdot\hat{c} + \hat{c}\cdot\hat{c} + \hat{a}\cdot\hat{a} - 2\hat{c}\cdot\hat{a}$

$= 6 - 2\left(\hat{a}.\hat{b} + \hat{b}.\hat{c} + \hat{c}.\hat{a}\right)$ _____(1)

Also

Also $\left|\hat{a}+\hat{b}+\hat{c}\right| \geq 0$

$\Rightarrow \hat{a}.\hat{a} + \hat{b}.\hat{b} + \hat{c}.\hat{c} + 2(\hat{a}.\hat{b} + \hat{b}.\hat{c} + \hat{c}.\hat{a}) \geq 0$

$\Rightarrow 3+2\,(\hat{a}.\hat{b} + \hat{b}.\hat{c} + \hat{c}.\hat{a}) \geq 0$

$\Rightarrow -2\,(\hat{a}.\hat{b} + \hat{b}.\hat{c} + \hat{c}.\hat{a}) \leq 3$

$\Rightarrow 6 - 2\,(\hat{a}.\hat{b} + \hat{b}.\hat{c} + \hat{c}.\hat{a}) \leq 9$ _____(2)

From (1) and (2), $x \leq 9$

$\therefore$ x does not exceed 9

8. Let $\vec{V} = 2\hat{i} + \hat{j} - \hat{k}$ and $\vec{W} = \hat{i} + 3\hat{k}$. If $\vec{U}$ is a unit vector, then the maximum value of the scalar triple product $\left[\vec{U}\ \vec{V}\ \vec{W}\right]$ is :

माना $\vec{V} = 2\hat{i} + \hat{j} - \hat{k}$ तथा $\vec{W} = \hat{i} + 3\hat{k}$ है। यदि $\vec{U}$ इकाई सदिश हो, तो अदिश त्रिक गुणन $\left[\vec{U}\ \vec{V}\ \vec{W}\right]$ का अधिकतम मान है :

(A) -1 (B) $\sqrt{10}+\sqrt{6}$ (C) $\sqrt{59}$ (D) $\sqrt{60}$

Ans. C

Sol. $\vec{v}\times\vec{w} = \begin{vmatrix} i & j & k \\ 2 & 1 & -1 \\ 1 & 0 & 3 \end{vmatrix} = (3,-7,-1)$

$\vec{u}\cdot(\vec{v}\times\vec{w}) = |\vec{u}||\vec{v}\times\vec{w}| = \sqrt{59}$

9. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{a}\cdot\vec{b} = 1$ and $\vec{a}\times\vec{b} = \hat{j} - \hat{k}$, then $\vec{b}$ =

यदि $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{a}\cdot\vec{b} = 1$ तथा $\vec{a}\times\vec{b} = \hat{j} - \hat{k}$, तो $\vec{b}$ =

(A) $\hat{i}$ (B) $\hat{i} - \hat{j} + \hat{k}$ (C) $2\hat{j} - \hat{k}$ (D) $2\hat{i}$

Ans. A

Sol. $\left(\vec{a}\times\vec{b}\right) = (0,1,-1), \quad \vec{a}\cdot\vec{b} = 1$

$\left(\vec{a}\times\vec{b}\right)\times\vec{a} = \vec{c}\times\vec{a} \qquad \vec{c} = (0,1,-1)$

$$\Rightarrow \vec{b} = \frac{\vec{c}\times\vec{a} + \left(\vec{a}\cdot\vec{b}\right)\vec{a}}{|\vec{a}|^2}$$

$$\vec{c}\times\vec{a} = \begin{vmatrix} i & j & k \\ 0 & 1 & -1 \\ 1 & 1 & 1 \end{vmatrix} = (2,-1,-1)$$

$$\vec{b} = \frac{(2,-1,-1)+(1,1,1)}{3} = \hat{i}$$

10. Let $\vec{A}$ be vector parallel to line of intersection of planes P_1 and P_2 through origin. P_1 is parallel to the vectors $2\hat{j} + 3\hat{k}$ and $4\hat{j} - 3\hat{k}$ and P_2 is parallel to $\hat{j} - \hat{k}$ and $3\hat{i} + 3\hat{j}$, then the angle between vector $\vec{A}$ and $2\hat{i} + \hat{j} - 2\hat{k}$ is:

माना सदिश $\vec{A}$ मूल बिन्दु से जाने वाले तल P_1 और P_2 की प्रतिच्छेद रेखा के समान्तर है। P_1, $2\hat{j} + 3\hat{k}$ और $4\hat{j} - 3\hat{k}$ के समान्तर है और P_2, $\hat{j} - \hat{k}$ तथा $3\hat{i} + 3\hat{j}$ के समान्तर है, तो सदिश $\vec{A}$ और $2\hat{i} + \hat{j} - 2\hat{k}$ के बीच का कोण है :

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{3}$

Ans. B

Sol. Normal to plane P_1 is

$\vec{n}_1 = (2\hat{j}+3\hat{k})\times(4\hat{j}-3\hat{k}) = -18\hat{i}$

Normal to plane P_2 is

$\vec{n}_2 = (\hat{j}-\hat{k})\times(3\hat{i}+3\hat{j}) = 3\hat{i}-3\hat{j}-3\hat{k}$

$\therefore$ $\vec{A}$ is parallel to $\pm(\vec{n}_1\times\vec{n}_2) = \pm(-54\hat{j}+54\hat{k})$

Now, angle between $\vec{A}$ and $2\hat{i}+\hat{j}-2\hat{k}$ is given by $\cos\theta = \pm\frac{(-54\hat{j}+54\hat{k}).(2\hat{i}+\hat{j}-2\hat{k})}{54\sqrt{2}.3} = \pm\frac{1}{\sqrt{2}}$

$\theta = \frac{\pi}{4}$ or $\frac{3\pi}{4}$.

11. A variable plane at a distance of 1 unit from the origin cuts the co-ordinate axes at A, B and C. If the centroid D(x, y, z) of triangle ABC satisfies the relation $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2} = k$, then the value of k is :

एक चर समतल मूल बिन्दु से इकाई दूरी पर है। यह निर्देशांक अक्षो को A, B व C पर प्रतिच्छेद करता है। केन्द्र (x, y, z) समीकरण $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2} = k$ को संतुष्ट करता है :

(A) 3 (B) 1 (C) 1/3 (D) 9

Ans. D

Sol. Let the equation of the plane ABCD be ax + by + cz + d = 0, the point A" be (α, β, γ) and the height of the parallelopiped ABCDA'B'C'D' be h.

$\Rightarrow$ $\frac{|a\alpha+b\beta+c\gamma+d|}{\sqrt{a^2+b^2+c^2}} = 90\%.h$

$\Rightarrow$ $a\alpha + b\beta + c\gamma + d = \pm 0.9h\sqrt{a^2+b^2+c^2}$

$\therefore$ locus is ax, + by + cz + d = $\pm 0.9h\sqrt{a^2+b^2+c^2}$

$\therefore$ locus of A" is a plane parallel to the plane ABCD.

12. If $\vec{a},\vec{b},\vec{c}$ and $\vec{d}$ are unit vectors such that $(\vec{a}\times\vec{b})\cdot(\vec{c}\times\vec{d}) = 1$ and $\vec{a}\cdot\vec{c} = \frac{1}{2}$, then :

(A) $\vec{a},\vec{b},\vec{c}$ are non-coplanar (B) $\vec{b},\vec{c},\vec{d}$ are non-coplanar

(C) $\vec{b},\vec{d}$ are non-parallel (D) $\vec{a},\vec{d}$ are parallel and $\vec{b},\vec{c}$ are parallel

यदि $\vec{a},\vec{b},\vec{c}$ तथा $\vec{d}$ ऐसे इकाई सदिश है जिनके लिये $\left(\vec{a}\times\vec{b}\right)\cdot\left(\vec{c}\times\vec{d}\right)=1$ तथा $\vec{a}\cdot\vec{c}=\frac{1}{2}$, तो :

(A) $\vec{a},\vec{b},\vec{c}$ असमतलीय है

(B) $\vec{b},\vec{c},\vec{d}$ असमतलीय है

(C) $\vec{b},\vec{d}$ असमान्तर है

(D) $\vec{a},\vec{d}$ समान्तर है एवं $\vec{b},\vec{c}$ समान्तर है

Ans. C

Sol. Given $a_1\vec{r}_1+a_2\vec{r}_2+.....+a_n\vec{r}_n=0$

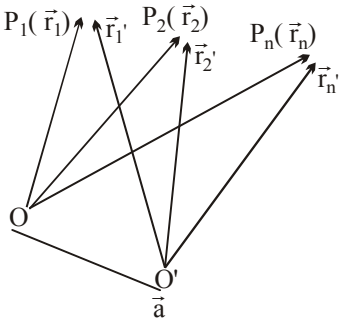
now $\vec{a}+\vec{r}_1'=\vec{r}_1$ and so on

hence $a_1(\vec{a}+\vec{r}_1')+a_2(\vec{a}+\vec{r}_2')+.....+a_n(\vec{a}+\vec{r}_n')=0$

$a_1\vec{r}_1'+a_2\vec{r}_2'+....+a_n\vec{r}_n'+\vec{a}(a_1+a_2+.....+a_n)=0$

hence $a_1\vec{r}_1'+a_2\vec{r}_2'+....+a_n\vec{r}_n'=0$

if $a_1+a_2+.....a_n=0$



13. Let $\vec{a}=\hat{i}+\hat{j}+\hat{k},\vec{b}=\hat{i}-\hat{j}+\hat{k}$ and $\vec{c}=\hat{i}-\hat{j}-\hat{k}$ be three vectors. A vector $\vec{v}$ in the plane of $\vec{a}$ and $\vec{b}$, whose projection on $\vec{c}$ is $\frac{1}{\sqrt{3}}$, is given by : [JEE-2011]

माना $\vec{a}=\hat{i}+\hat{j}+\hat{k},\vec{b}=\hat{i}-\hat{j}+\hat{k}$ तथा $\vec{c}=\hat{i}-\hat{j}-\hat{k}$ तीन सदिश हैं। $\vec{a}$ तथा $\vec{b}$ के तल में सदिश $\vec{v}$, जिसका $\vec{c}$ पर प्रक्षेप $\frac{1}{\sqrt{3}}$ है, होगा : [JEE-2011]

(A) $\hat{i}-3\hat{j}+3\hat{k}$ (B) $-3\hat{i}-3\hat{j}-\hat{k}$ (C) $3\hat{i}-\hat{j}+3\hat{k}$ (D) $\hat{i}+3\hat{j}-3\hat{k}$

Ans. C

Sol. $\vec{v}=x\vec{a}+y\vec{b}$

$=\hat{i}(x+y)+\hat{j}(x-y)+\hat{k}(x+y)$ _____(i)

Given, $\vec{v}.\hat{c}=\frac{1}{\sqrt{3}}$

$\Rightarrow \frac{x+y-x+y-x-y}{\sqrt{3}}=\frac{1}{\sqrt{3}}$

$y-x=1$

$\Rightarrow x-y=-1$ ....(ii)

using (ii) in (i) we get

$\vec{v}=(x+y)\hat{i}-\hat{j}+(x+y)\hat{k}$

ONE OR MORE THAN ONE CORRECT TYPE

14. If a, b, c are different real numbers and $a\hat{i} + b\hat{j} + c\hat{k}$, $b\hat{i} + c\hat{j} + a\hat{k}$ and $c\hat{i} + a\hat{j} + b\hat{k}$ are position vectors of three non-collinear points A, B and C, then :

(A) centroid of triangle ABC is $\frac{a+b+c}{3}\left(\hat{i} + \hat{j} + \hat{k}\right)$

(B) $\hat{i} + \hat{j} + \hat{k}$ is equally inclined to the three vectors

(C) perpendicular from the origin to the plane of triangle ABC meet at centroid

(D) triangle ABC is an equilateral triangle

यदि a, b, c भिन्न वास्तविक संख्याएँ हैं और $a\hat{i} + b\hat{j} + c\hat{k}$, $b\hat{i} + c\hat{j} + a\hat{k}$ और $c\hat{i} + a\hat{j} + b\hat{k}$ तीन असंरेखीय बिन्दुओं A, B और C के स्थिति सदिश हैं, तो :

(A) त्रिभुंज ABC का केन्द्रक $\frac{a+b+c}{3}\left(\hat{i} + \hat{j} + \hat{k}\right)$ हैं

(B) $\hat{i} + \hat{j} + \hat{k}$ तीनों सदिशों पर समान रूप से झुका हुआ है

(C) मूलबिन्दू से त्रिभुज ABC के समतल पर डाला गया लम्ब केन्द्रक पर मिलता है

(D) त्रिभुंज ABC समबाहु त्रिभुज है

Ans. ABCD

Sol. $\vec{P} = (a,b,c)$, $\vec{q} = (b,c,a)$, $\vec{r} = (c,a,b)$

(1) $\vec{g} = \left(\frac{a+b+c}{3}\right)(1,1,1)$

(2) angle with $(1,1,1) = \vec{S}$

$$\vec{S} \wedge \vec{p} = \vec{S} \wedge \vec{q} = \vec{S}\hat{\wedge} = \cos^{-1}\left(\frac{a+b+c}{\sqrt{3}\sqrt{a^2+b^2+c^2}}\right)$$

(3) Plane

$$\begin{vmatrix} x-a & y-b & z-c \\ b-a & c-b & a-c \\ c-a & a-b & b-c \end{vmatrix} = 0$$

$C_1 \to C_1 + C_2 + C_3$

$$\begin{vmatrix} x+y+z-(a+b+c) & y-b & z-c \\ 0 & c-b & a-c \\ 0 & a-b & b-c \end{vmatrix} = 0$$

$\Rightarrow x + y + z - (a + b + c) = 0$

foot of perpendicular $x = y = z = \dfrac{a+b+c}{3}$

(4) $|\vec{P}| = |\vec{q}| = |\vec{r}|$

15. If $\vec{z}_1 = a\hat{i} + b\hat{j}$ and $\vec{z}_2 = c\hat{i} + d\hat{j}$ are two vectors in $\hat{i}$ and $\hat{j}$ system, where $|\vec{z}_1| = |\vec{z}_2| = r$ and $\vec{z}_1 . \vec{z}_2 = 0$, then $\vec{w}_1 = a\hat{i} + c\hat{j}$ and $\vec{w}_2 = b\hat{i} + d\hat{j}$ satisfy:

(A) $|\vec{w}_1| = r$ (B) $|\vec{w}_2| = r$ (C) $\vec{w}_1 . \vec{w}_2 = 0$ (D) None of these

यदि $\hat{i}$ और $\hat{j}$ निकाय में दो सदिश $\vec{z}_1 = a\hat{i} + b\hat{j}$ एवं $\vec{z}_2 = c\hat{i} + d\hat{j}$ हैं, जहाँ $|\vec{z}_1| = |\vec{z}_2| = r$ और $\vec{z}_1 . \vec{z}_2 = 0$, तो $\vec{w}_1 = a\hat{i} + c\hat{j}$ और $\vec{w}_2 = b\hat{i} + d\hat{j}$ संतुष्ट करते है :

(A) $|\vec{w}_1| = r$ (B) $|\vec{w}_2| = r$ (C) $\vec{w}_1 . \vec{w}_2 = 0$ (D) इनमें से कोई नहीं

Ans. ABC

Sol. $a^2 + b^2 = c^2 + d^2 = r^2$ ...(1)

$ac + bd = 0 \Rightarrow a^2c^2 = b^2d^2 \Rightarrow \dfrac{a^2}{b^2} = \dfrac{d^2}{c^2}$

$\Rightarrow \dfrac{a^2+b^2}{a^2-b^2} = \dfrac{c^2+d^2}{d^2-c^2} \Rightarrow a^2 - b^2 = d^2 - c^2$ ...(2)

(1) + (2) $a^2 = d^2 \Rightarrow b^2 = c^2$

$|\vec{w}_1| = a^2 + b^2 = |w_2| = b^2 + d^2 = r^2$

$w_1 \cdot w_2 = ab + cd$

$a = -d \Rightarrow b = c$ ...(1)

$= 0$ $a = d \Rightarrow b = -c$ ...(2)

16. If $\vec{a}$, $\vec{b}$ are two non-collinear unit vectors and $\vec{a}$, $\vec{b}$, $x\vec{a} - y\vec{b}$ form a triangle, then :

(A) $x = -1; y = 1$ and $|\vec{a} + \vec{b}| = 2\cos\left(\dfrac{\vec{a} \wedge \vec{b}}{2}\right)$

(B) $x = -1; y = 1$ and $\cos(\vec{a} \wedge \vec{b}) + |\vec{a} + \vec{b}| \cos(\vec{a} \wedge -(\vec{a} + \vec{b})) = -1$

(C) $|\vec{a} + \vec{b}| = -2\cot\left(\dfrac{\vec{a} \wedge \vec{b}}{2}\right)\cos\left(\dfrac{\vec{a} \wedge \vec{b}}{2}\right)$ and $x = -1, y = 1$

(D) None of these

यदि $\vec{a}$ और $\vec{b}$ दो असंरेखीय इकाई सदिश हैं और $\vec{a}$, $\vec{b}$, $x\vec{a} - y\vec{b}$ एक त्रिभुज बनाते हैं, तो :

(A) $x = -1; y = 1$ तथा $\left|\vec{a}+\vec{b}\right| = 2\cos\left(\frac{\vec{a} \wedge \vec{b}}{2}\right)$

(B) $x = -1; y = 1$ तथा $\cos\left(\vec{a} \wedge \vec{b}\right) + \left|\vec{a}+\vec{b}\right|\cos\left(\vec{a} \wedge -\left(\vec{a} + \vec{b}\right)\right) = -1$

(C) $\left|\vec{a}+\vec{b}\right| = -2\cot\left(\frac{\vec{a} \wedge \vec{b}}{2}\right)\cos\left(\frac{\vec{a} \wedge \vec{b}}{2}\right)$ तथा $x = -1, y = 1$

(D) इनमें से कोई नहीं

Ans. AB

Sol. $\vec{P} + \vec{q} + \vec{r} = 0$ (triangle)

$(1+x)\vec{a} + (1-y)\vec{b} = 0 \Rightarrow x = -1, y = 1$

$\left|\vec{a}+\vec{b}\right|^2 = (2+2\cos\theta)^2 = 4\cos^2\left(\frac{\theta}{2}\right)$

$\left|\vec{a}+\vec{b}\right| = 2\cos\frac{\theta}{2}$ $\qquad\qquad$ $\theta = \vec{a} \wedge \vec{b}$

(2) $\cos\phi = -\frac{\vec{a}\cdot\left(\vec{a}+\vec{b}\right)}{|\vec{a}|\left|\vec{a}+\vec{b}\right|}$ $\qquad\qquad$ $|\vec{a}| = 1$

$\Rightarrow \vec{a}\cdot\vec{b} + \left|\vec{a}+\vec{b}\right|\cos\phi = -|\vec{a}|^2$

$\Rightarrow \cos\theta + \left|\vec{a}+\vec{b}\right|\cos\phi = -1$

17. The volume of a right triangular prism $ABCA_1B_1C_1$ is equal to 3. If the position vectors of the vertices of the base ABC are A(1, 0, 1), B(2,0, 0) and C(0, 1, 0), then position vectors of the vertex A_1 can be :

एक समकोण त्रिभुजकार प्रिज्म $ABCA_1B_1C_1$ का आयतन 3 है। यदि आधार ABC के शीर्षों के स्थिति सदिश A(1, 0, 1); B(2,0, 0) और C(0, 1, 0) हैं, तो शीर्ष A_1 का स्थिति सदिश हो सकता है :

(A) (2, 2, 2) (B) (0, 2, 0) (C) (0, −2, 2) (D) (0, −2, 0)

Ans. AD

Sol. Vol Prism $= \frac{1}{2}$ volume of parallelopiped

$\left[\vec{a} \quad \vec{b} \quad \vec{c}\right] = 2 \times \text{vol prism} = 6$

$\left|\vec{a}\times\vec{b}\right|\cdot|\vec{c}| = 6$

$$\vec{a}\times\vec{b}=\begin{vmatrix} i & j & k \\ 1 & 0 & -1 \\ 2 & -1 & 0 \end{vmatrix}=(-1,-2,-1)$$

$|\vec{c}|=\sqrt{6}$

equation $AA' = \vec{r}=(1,0,1)+\lambda(1,2,1)$

$\lambda = +1$ $A' = (2, 2, 2)$

18. Which of the following statement(s) is/are true?

(A) If $\vec{n}\cdot\vec{a}=0$, $\vec{n}\cdot\vec{b}=0$ and $\vec{n}\cdot\vec{c}=0$ for some non-zero vector $\vec{n}$, then $\left[\vec{a}\ \vec{b}\ \vec{c}\right]=0$.

(B) there exist a vector having direction angles $\alpha = 30^\circ$ and $\beta = 45^\circ$

(C) locus of point for which $x = 3$ and $y = 4$ is a line parallel to the Z-axis whose distance from the Z-axis is 5

(D) the vertices of a regular tetrahedron are OABC where 'O' is the origin. Then vector $\vec{OA}+\vec{OB}+\vec{OC}$ is perpendicular to the plane ABC.

निम्न में से कौनसा कथन सही है ?

(A) यदि किसी अशून्य सदिश $\vec{n}$ के लिए $\vec{n}\cdot\vec{a}=0$, $\vec{n}\cdot\vec{b}=0$ और $\vec{n}\cdot\vec{c}=0$ हो, तो $\left[\vec{a}\ \vec{b}\ \vec{c}\right]=0$.

(B) एक ऐसा सदिश विद्यमान है जिसके दिशात्मक कोण $\alpha = 30^\circ$ एवं $\beta = 45^\circ$ है

(C) बिन्दु का बिन्दुपथ जिसके लिए रेखा $x = 3$ और $y = 4$, z-अक्ष के समान्तर है और z- अक्ष से उसकी दूरी 5 है

(D) एक सम चतुष्फलक के शीर्ष OABC हैं जहाँ 'O' मूल बिन्दु है, तो सदिश $\vec{OA}+\vec{OB}+\vec{OC}$ समतल ABC के लम्बवत् है

Ans. ACD

Sol. $\vec{n}\cdot\vec{a}=\vec{n}\cdot\vec{b}=\vec{n}\cdot\vec{c}=0$

$\vec{n}\perp\vec{a},\vec{b},\vec{c}$ this is only possible if $\vec{a},\vec{b},\vec{c}$ are coplanar $\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]=0$

(B) $\alpha = 30^\circ, \beta = 45^\circ \Rightarrow \cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

$\Rightarrow \frac{3}{4}+\frac{1}{2}+\cos^2\gamma = 1$ (wong)

(C)

(D) $\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]=0$

DR's of plane ABC = $\overrightarrow{AB}\times\overrightarrow{AC}$

$\vec{n}=(\vec{b}-\vec{a})\times(\vec{c}-\vec{a})$

$=\vec{b}\times\vec{c}-\vec{a}\times\vec{c}-\vec{b}\times\vec{a}$

$=\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}$

Now, $\vec{n}\cdot\left(\vec{a}+\vec{b}+\vec{c}\right)=\left[\vec{b}\quad\vec{c}\quad\vec{a}\right]+\left[\vec{c}\quad\vec{a}\quad\vec{b}\right]+\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]$

$=3\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]=0$

$\vec{n}\perp\left(\vec{a}+\vec{b}+\vec{c}\right)$ (ACD)

19. $\hat{a}$ and $\hat{b}$ are two given unit vectors at right angle. The unit vector equally inclined with $\hat{a}$, $\hat{b}$ and $\hat{a}\times\hat{b}$ will be :

$\hat{a}$ एवं $\hat{b}$ दो इकाई सदिश हैं जिनके मध्य कोण समकोण है। $\hat{a}$, $\hat{b}$ और $\hat{a}\times\hat{b}$ के साथ समान कोण बनाने वाला इकाई सदिश हैं :

(A) $-\frac{1}{\sqrt{3}}\left(\hat{a}+\hat{b}+\hat{a}\times\hat{b}\right)$ (B) $\frac{1}{\sqrt{3}}\left(\hat{a}+\hat{b}+\hat{a}\times\hat{b}\right)$

(C) $\frac{1}{\sqrt{3}}\left(\hat{a}+\hat{b}-\hat{a}\times\hat{b}\right)$ (D) $-\frac{1}{\sqrt{3}}\left(\hat{a}+\hat{b}-\hat{a}\times\hat{b}\right)$

Ans. AB

Sol. $\vec{a}\perp\vec{b}$ & $\vec{a}\perp\vec{a}\times\vec{b}$ $(\cos\alpha=\cos\beta=\cos\gamma)$

$\vec{b}\perp\vec{a}\times\vec{b}$ $\vec{r}\,\hat{}\,\vec{a}=\alpha$

$\vec{r}=\lambda\vec{a}+\mu\vec{b}+\gamma\left(\vec{a}\times\vec{b}\right)$

angle is same $\Rightarrow\lambda=\mu=\gamma=\pm\frac{1}{\sqrt{3}}$

$\vec{r}=\pm\frac{1}{\sqrt{3}}\left(\vec{a}+\vec{b}+\vec{a}\times\vec{b}\right)$

20. Which of the following statement(s) is(are) incorrect?

(A) The relation $\left|(\vec{u}\times\vec{v})\right|=\left|\vec{u}\cdot\vec{v}\right|$ is only possible if atleast one of the vectors $\vec{u}$ and $\vec{v}$ is null vector

(B) Every vector contained in the line $\vec{r}(t)=\langle 1+2t,1+3t,1+4t\rangle$ is parallel to the vector $\langle 1,1,1\rangle$

(C) If scalar triple product of three vectors $\vec{u}$, $\vec{v}$, $\vec{w}$ is larger than $\left|\vec{u}\times\vec{v}\right|$ then $\left|\vec{w}\right|>1$

(D) The distance between the x-axis and the line $x = y = 1$ is $\sqrt{2}$

निम्न में से कौनसा कथन असत्य है?

(A) सम्बन्ध $\left|(\vec{u}\times\vec{v})\right|=\left|\vec{u}\cdot\vec{v}\right|$ तभी सम्भव है जब $\vec{u}$ तथा $\vec{v}$ में से कम से कम एक सदिश शून्य है

(B) रेखा $\vec{r}(t)=\langle 1+2t,1+3t,1+4t\rangle$ पर स्थित सभी सदिश, सदिश $\langle 1,1,1\rangle$ के समानान्तर है

(C) यदि $\vec{u}$, $\vec{v}$, $\vec{w}$ सदिशों का अदिश त्रिगुणन $\left|\vec{u}\times\vec{v}\right|$ से बड़ा है, तो $\left|\vec{w}\right|>1$

(D) x-अक्ष तथा रेखा $x = y = 1$ के मध्य दूरी $\sqrt{2}$ है

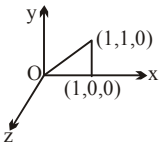
Ans. ABD

Sol. (A) Can be true if the angle between $\vec{u}$ and $\vec{v}$ is 45°.

(B) It is parallel to $\langle 2, 3, 4 \rangle$

(C) Think of the scalar triple product as the volume. If that is larger than the base area, the height has to be bigger than 1.

(D) $x = y = 1$ denotes a line passing through (1, 1, 0) and parallel to z-axis.



Hence the distance between this line and x-axis is 1 and its distance from the z-axis is $\sqrt{2}$.

21. Consider the family of planes $x + y + z = c$ where c is a parameter intersecting the coordinate axes at P, Q, R and α, β, γ are the angles made by each member of this family with positive x, y and z axis. Which of the following interpretations hold good for this family :

(A) each member of this family is equally inclined with the coordinate axes

(B) $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 1$

(C) $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 2$

(D) for $c = 3$ area of the triangle PQR is $3\sqrt{3}$ sq. units

समतलों के समूह $x + y + z = c$ के लिए जहाँ c एक प्राचल है निर्देशांक अक्षों को P, Q, R पर प्रतिच्छेद करता है तथा कोण α, β, γ तल द्वारा x, y तथा z-अक्षों के साथ क्रमशः बनाये गये है। निम्न में से कौनसे कथन एक इस समूह के लिए सही है :

(A) इसी समूह का प्रत्येक सदस्य निर्देशांक अक्षों से बराबर नत है

(B) $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 1$

(C) $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 2$

(D) $c = 3$ के लिए PQR का क्षेत्रफल $3\sqrt{3}$ वर्ग इकाई है

Ans. ABC

Sol. Vector normal to the plane

$$\vec{n} = \hat{i} + \hat{j} + \hat{k}$$

$$\vec{V}_x = \hat{i}\,; \quad \vec{V}_y = \hat{j}\,; \quad \vec{V}_z = \hat{k}$$

$$\cos(90 - \alpha) = \frac{\vec{V}_x \cdot \vec{n}}{|\vec{n}|} \quad \Rightarrow \quad \sin\alpha = \frac{1}{\sqrt{3}}$$

ly $\sin\beta = \frac{1}{\sqrt{3}}$ and $\sin\gamma = \frac{1}{\sqrt{3}}$

hence $\sum \sin^2\alpha = 1$ and $\sum \cos^2\alpha = 2$

Also plane is equally inclined with the coordinate axes.

Also $A = \frac{1}{2}\sqrt{9^2+9^2+9^2} = \frac{9\sqrt{3}}{2}$ $\Rightarrow$ (D) is not correct

22. Which of the following statement(s) is/are true in respect of the lines $\vec{r} = \vec{a} + \lambda\vec{b}$; $\vec{r} = \vec{c} + \mu\vec{d}$ where $\vec{b}\times\vec{d} \neq 0$:

(A) acute angle between the lines is $\cos^{-1}\left(\frac{|\vec{b}\cdot\vec{d}|}{|\vec{b}||\vec{d}|}\right)$

(B) The lines would intersect if $\left[\vec{c}\ \ \vec{b}\ \ \vec{d}\right] = \left[\vec{a}\ \ \vec{b}\ \ \vec{d}\right]$

(C) The lines will be skew if $\left|\vec{c}-\vec{a}\ \ \vec{b}\ \ \vec{d}\right| \neq 0$

(D) If the lines intersect at $\vec{r} = \vec{r}_0$ then the equation of the plane containing the lines is $\left[\vec{r}-\vec{r}_0\ \ \vec{b}\ \ \vec{d}\right] = 0$

निम्न में से कौनसे कथन रेखाओं $\vec{r} = \vec{a} + \lambda\vec{b}$; $\vec{r} = \vec{c} + \mu\vec{d}$ जहाँ $\vec{b}\times\vec{d} \neq 0$ के लिए सही है :

(A) रेखाओं के मध्य न्यूनकोण $\cos^{-1}\left(\frac{|\vec{b}\cdot\vec{d}|}{|\vec{b}||\vec{d}|}\right)$

(B) रेखाऐं प्रतिच्छेद करती है यदि $\left[\vec{c}\ \ \vec{b}\ \ \vec{d}\right] = \left[\vec{a}\ \ \vec{b}\ \ \vec{d}\right]$

(C) रेखाऐं विसमतलीय है यदि $\left|\vec{c}-\vec{a}\ \ \vec{b}\ \ \vec{d}\right| \neq 0$

(D) यदि रेखाऐं $\vec{r} = \vec{r}_0$ पर प्रतिच्छेद करती है, तो दोनों रेखाओं द्वारा बनाये तल का समीकरण $\left[\vec{r}-\vec{r}_0\ \ \vec{b}\ \ \vec{d}\right] = 0$ है

Ans. ABCD

Sol. $\vec{r} = \vec{a} + \lambda\vec{b}$ $\qquad \vec{b}\times\vec{d} \neq 0$

$\vec{r} = \vec{c} + \mu\vec{d}$

(A) $\cos\theta = \frac{|\vec{b}\cdot\vec{d}|}{|\vec{b}||\vec{d}|}$ $\qquad \theta \in (0, 90°)$

(B) Line intersects if $\qquad \left[\vec{a}-\vec{c}\ \ \vec{b}\ \ \vec{d}\right] = 0$

$$\left[\vec{a}\ \ \vec{b}\ \ \vec{d}\right] = \left[\vec{c}\ \ \vec{b}\ \ \vec{d}\right]$$

(C) for skewed lines $\qquad \left[\vec{a}-\vec{c}\ \ \vec{b}\ \ \vec{d}\right] \neq 0$

(D) $\vec{r} = \vec{r}_0$ (P.V. of intersection)

Plane $\Rightarrow \left[\vec{r}-\vec{r}_0\ \ \vec{b}\ \ \vec{d}\right] = 0$

23. Let $\vec{a}$ and $\vec{b}$ be two non-zero and non-collinear vectors then which of the following is/are always correct?

(A) $\vec{a}\times\vec{b}=\left[\vec{a}\ \vec{b}\ \hat{i}\right]\hat{i}+\left[\vec{a}\ \vec{b}\ \hat{j}\right]\hat{j}+\left[\vec{a}\ \vec{b}\ \hat{k}\right]\hat{k}$

(B) $\vec{a}\cdot\vec{b}=(\vec{a}\cdot\hat{i})(\vec{b}\cdot\hat{i})+(\vec{a}\cdot\hat{j})(\vec{b}\cdot\hat{j})+(\vec{a}\cdot\hat{k})(\vec{b}\cdot\hat{k})$

(C) if $\vec{u}=\hat{a}-(\hat{a}\cdot\hat{b})\hat{b}$ and $\vec{v}=\hat{a}\times\hat{b}$ then $|\vec{u}|=|\vec{v}|$

(D) if $\vec{c}=\vec{a}\times(\vec{a}\times\vec{b})$ and $\vec{d}=\vec{b}\times(\vec{a}\times\vec{b})$ then $\vec{c}+\vec{d}=\vec{0}$

माना $\vec{a}$ तथा $\vec{b}$ दो अशून्य एवं विसमरेखीय सदिश है, तो निम्न में से कौनसे सदैव सही होगें?

(A) $\vec{a}\times\vec{b}=\left[\vec{a}\ \vec{b}\ \hat{i}\right]\hat{i}+\left[\vec{a}\ \vec{b}\ \hat{j}\right]\hat{j}+\left[\vec{a}\ \vec{b}\ \hat{k}\right]\hat{k}$

(B) $\vec{a}\cdot\vec{b}=(\vec{a}\cdot\hat{i})(\vec{b}\cdot\hat{i})+(\vec{a}\cdot\hat{j})(\vec{b}\cdot\hat{j})+(\vec{a}\cdot\hat{k})(\vec{b}\cdot\hat{k})$

(C) यदि $\vec{u}=\hat{a}-(\hat{a}\cdot\hat{b})\hat{b}$ तथा $\vec{v}=\hat{a}\times\hat{b}$ तो $|\vec{u}|=|\vec{v}|$

(D) यदि $\vec{c}=\vec{a}\times(\vec{a}\times\vec{b})$ तथा $\vec{d}=\vec{b}\times(\vec{a}\times\vec{b})$ तो $\vec{c}+\vec{d}=\vec{0}$

Ans. ABC

Sol. (A) $\vec{r}=(\vec{r}\cdot\hat{i})\hat{i}+(\vec{r}\cdot\hat{j})\hat{j}+(\vec{r}\cdot\hat{k})\hat{k}$

put $\vec{r}=(\vec{a}\times\vec{b})$ $\Rightarrow$ (A) is correct

(B) put $\vec{r}=\vec{a}$ and $\vec{r}=\vec{b}$

take $\vec{a}\cdot\vec{b}$ $\Rightarrow$ (B) is correct

(C) $\vec{u}=\hat{b}\times(\hat{a}\times\hat{b})$

$|\vec{u}|=|\vec{a}\times\vec{b}|=|\vec{V}|$ $\Rightarrow$ (C) is correct

(D) is obviously wrong

24. The vector(s) which is/are coplanar with vectors $\hat{i}+\hat{j}+2\hat{k}$ and $\hat{i}+2\hat{j}+\hat{k}$, and perpendicular to the vector $\hat{i}+\hat{j}+\hat{k}$ is/are : [JEE-2011]

सदिश जो $\hat{i}+\hat{j}+2\hat{k}$ तथा $\hat{i}+2\hat{j}+\hat{k}$ के तल में है तथा $\hat{i}+\hat{j}+\hat{k}$ के लम्बवत् है होंगे/ होगा : [JEE-2011]

(A) $\hat{j}-\hat{k}$ (B) $-\hat{i}+\hat{j}$ (C) $\hat{i}-\hat{j}$ (D) $-\hat{j}+\hat{k}$

Ans. AD

Sol. $\vec{a}=\hat{i}+\hat{j}+2\hat{k}$ $\vec{b}=\hat{i}+2\hat{j}+\hat{k}$ $\vec{c}=\hat{i}+\hat{j}+\hat{k}$

$\vec{v}=\lambda((\vec{a}\times\vec{b})\times\vec{c})=\lambda((\vec{a}.\vec{c})\vec{b}-(\vec{b}.\vec{c})\vec{a}$

$\vec{v}=\lambda[4(\hat{i}+2\hat{j}+\hat{k})-4(\hat{i}+\hat{j}+2\hat{k})]$

$\vec{v}=4\lambda(\hat{j}-\hat{k})$

Exercise C - II

1. A variable plane is at a constant distance p from the origin and meets the coordinate axes in points A, B and C respectively. Through these points, planes are drawn parallel to the coordinates planes. Find the locus of their point of intersection.

एक चर तल जो कि मूल बिन्दु से नियत दूरी p पर स्थित है तथा निर्देशी अक्षों को बिन्दुओं A, B तथा C पर मिलती है। इन बिन्दुओं से निर्देश तलों के समानान्तर खीचे गये तलों के प्रतिच्छेद बिन्दु का बिन्दुपथ ज्ञात कीजिए।

Ans. $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=\frac{1}{p^2}$

Sol. Let plane is $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$ where a, b, c are intercepts on x–axis, y–axis and z–axis.

distance from origin

Given $d_{(0,0,0)} = P$

$$\frac{1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}}}=P$$

$$\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}=\frac{1}{P^2}$$

Planes parallel to co-ordinate plane

$x = a, y = b, c = z$

POI of these planes (a, b, c)

Locus of POI

$$\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}=\frac{1}{P^2}$$

$$\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=\frac{1}{P^2}$$

2. Let $\Pi : x+y-z-4=0$ be the equation of a plane and A be the point with position vector $\hat{i}+2\hat{j}-3\hat{k}$. L is a line which passes through the point (1, 2, 3) with direction ratios 3, – 1 and 4. If the distance of the point A from the line L measured parallel to the plane Π is d_1 and the distance of the point A from the plane Π measured parallel to the line L is d_2, then find the value of $\sqrt{d_1^{\,2}-d_2^{\,2}}$.

माना $\Pi : x + y – z – 4 = 0$ एक तल का समीकरण है तथा A एक बिन्दु है जिसका स्थिति सदिश $\hat{i}+2\hat{j}-3\hat{k}$ है। L एक रेखा है जो बिन्दु (1, 2, 3) से गुजरती है तथा जिसके दिक् अनुपात 3, – 1 तथा 4 है। यदि बिन्दु A की रेखा L से तल Π के समान्तर नापी गई दूरी d_1 है तथा बिन्दु A की तल Π से रेखा L से समान्तर नापी गई दूरी d_2 है तो $\sqrt{d_1^{\,2}-d_2^{\,2}}$ का मान ज्ञात कीजिये।

Ans. 10

Sol. $L: \frac{x-1}{3}=\frac{y-2}{-1}=\frac{z-3}{4}=r$

$A: (1, 2, -3)$; $x + y - z = 6$

Any point say B on line is

$3r + 1, 2 - r, 4r + 3$

$\overrightarrow{AB} = (3r)\hat{i} + (-r)\hat{j} + (4r+6)\hat{j}$

$\vec{n} = \hat{i} + \hat{j} - \hat{k}$

$\overrightarrow{AB} \cdot \vec{n} = 0 \Rightarrow 3r - r - 4r - 6 = 0$

$2r = -6 \Rightarrow r = -3.$

Hence B is $(-8, 5, -9)$

$AP = \sqrt{81+9+36} = \sqrt{126} \Rightarrow d_1^2 = 126.$

Again line through $(1, 2, -3)$ with dr's $3, -1, 4$

$\frac{x-1}{3}=\frac{y-2}{-1}=\frac{2+3}{4}=\lambda$

$\left.\begin{array}{l} x = 3\lambda + 1 \\ y = 2 - \lambda \\ z = 4\lambda - 3 \end{array}\right]$ must satisfy $x + y - z = 4.$

$3\lambda + 1 + 2 - \lambda - 4\lambda + 3 = 4$

$-2\lambda + 6 = 4 \Rightarrow \lambda = 1$

Hence the point on the plane is $(4, 1, 1)$

$\therefore$ Distance of A from the plane measured parallel to the line $= \sqrt{9+1+16} = \sqrt{26}$

$\Rightarrow$ $d_2^2 = 26$

Hence $\sqrt{d_1^2 - d_2^2} = \sqrt{126-26} = \sqrt{100} = 10.$ Ans.

3. The position vectors of the four angular points of a tetrahedron OABC are (0, 0, 0); (0, 0, 2); (0, 4, 0) and (6, 0, 0) respectively. A point P inside the tetrahedron is at the same distance 'r' from the four plane faces of the tetrahedron. Find the value of 'r'.

चतुष्फलक OABC के चारो कोणीय बिन्दुओं के निर्देशांक क्रमशः (0, 0, 0); (0, 0, 2); (0, 4, 0) तथा (6, 0, 0) चतुष्फलक के अन्दर चारों तलीय फलकों से समदूरस्थ (दूरी 'r') बिन्दु P है। r का मान बताइये।

Ans. $\frac{2}{3}$

Sol. Angular point OABC are (0, 0, 0), (0, 0, 2), (0, 4, 0) & (6, 0, 0)

Let centre of sphere be (r, r, r)

Equation of plane passing ABC is

$$\frac{x}{6}+\frac{y}{4}+\frac{z}{2}=1$$

$$r=\left|\frac{\frac{r}{6}+\frac{r}{4}+\frac{r}{2}-1}{\sqrt{\frac{1}{6^2}+\frac{1}{4^2}+\frac{1}{2^2}}}\right|$$

$7r = \pm (11r - 12)$

$r=\frac{2}{3}$, $r = 3$ (not satisfied)

4. The line $\frac{x+6}{5}=\frac{y+10}{3}=\frac{z+14}{8}$ is the hypotenuse of an isosceles right angled triangle whose opposite vertex is (7, 2, 4). Find the equation of the remaining sides.

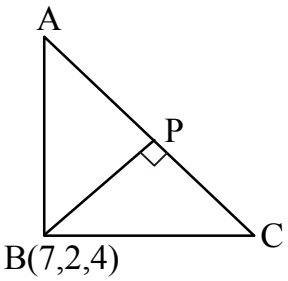
रेखा $\frac{x+6}{5}=\frac{y+10}{3}=\frac{z+14}{8}$ एक समद्विबाहु समकोण त्रिभुज का विकर्ण है। जिसके समाने का शीर्ष (7, 2, 4) है। बाकी भुजाओं का समीकरण ज्ञात कीजिए।

Ans. $\frac{x-7}{3}=\frac{y-2}{6}=\frac{z-4}{2}$; $\frac{x-7}{2}=\frac{y-2}{-3}=\frac{z-4}{6}$

Sol. $P = (5k - 6, 3k - 10, 8k - 14)$

Now $\overrightarrow{BP}\cdot\overrightarrow{AC}=0$

$5(5k - 13) + 3(3k - 12) + 8(8k - 18) = 0$



$$k=\frac{245}{98}=\frac{5}{2}$$

$$P=\left(\frac{13}{2},-\frac{5}{2},6\right),\ \text{Now length BP}=\frac{\sqrt{98}}{2}$$

Now $A,C=\left(\frac{13}{2}\pm\frac{5}{\sqrt{98}}.\frac{\sqrt{98}}{2},-\frac{5}{2}\pm\frac{3}{\sqrt{98}.}\frac{\sqrt{98}}{2},6\pm\frac{8}{\sqrt{98}},\frac{\sqrt{98}}{2}\right)$

$A = (9, -1, 10)$

$C = (4, -4, 2)$

Equation of BC $\frac{x-7}{3}=\frac{y-2}{6}=\frac{z-4}{2}$

Equation of AC $\frac{x-7}{2}=\frac{y-2}{-3}=\frac{z-4}{6}$

5. Consider a plane passing through three points A(a, 0, 0), B(0, b, 0), C(0, 0, c) with a > 0, b > 0, c > 0. Let d be the distance between the origin O and the plane and m be the distance between the origin O and the point M(a, b, c). If a, b, c vary in the range of any positive numbers, then find the minimum value of $\left(\frac{m}{d}\right)^2$.

माना एक तल बिन्दु A(a, 0, 0), B(0, b, 0), C(0, 0, c) जहाँ a > 0, b > 0, c > 0, से गुजरता है। बिन्दु O के तल से दूरी d है तथा बिन्दु O तथा M(a, b, c) के मध्य दूरी m है। यदि a, b, c धनात्मक संख्याऐं हैं तो $\left(\frac{m}{d}\right)^2$ का न्यूनतम मान ज्ञात कीजिये।

Ans. 9

Sol. Plane $P = \frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$

$$d_{(0,0,0)} = d = \frac{1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}}}$$

$$\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}=\frac{1}{d^2}$$

$m^2 = a^2 + b^2 + c^2$

Let $\varepsilon = \left(\frac{m}{d}\right)^2$

$$\varepsilon = \left(a^2+b^2+c^2\right)\left(\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}\right)$$

Find $\varepsilon_{min.}$

$$\varepsilon = 3+\frac{a^2}{b^2}+\frac{a^2}{c^2}+\frac{b^2}{a^2}+\frac{b^2}{c^2}+\frac{c^2}{a^2}+\frac{c^2}{b^2}$$

use AM & GM in last 6 terms

AM ≥ GM

$$\frac{\frac{a^2}{b^2}+\frac{a^2}{c^2}+\frac{b^2}{a^2}+\frac{b^2}{c^2}+\frac{c^2}{a^2}+\frac{c^2}{b^2}}{6} \geq (1)^{\frac{1}{6}}$$

$$\frac{a^2}{b^2}+\frac{a^2}{c^2}+\frac{b^2}{a^2}+\frac{b^2}{c^2}+\frac{c^2}{a^2}+\frac{c^2}{b^2} \geq 6$$

$\varepsilon_{min.} = 3 + 6 = 9$

6. Let A_1, A_2, A_3, A_4 be the areas of the triangular faces of a tetrahedron, and h_1, h_2, h_3, h_4 be the corresponding altitudes of the tetrahedron. If volume of tetrahedron is 5 cubic units, then find the minimum value of $(A_1 + A_2 + A_3 + A_4)(h_1 + h_2 + h_3 + h_4)$ (in cubic units).

माना A_1, A_2, A_3, A_4 एक चतुष्फलक के त्रिभुजाकार फलकों के क्षेत्रफल हैं एवं h_1, h_2, h_3, h_4 उनके संगत शीर्षलम्बों की लम्बाईयां हैं। यदि चतुष्फलक का आयतन 5 है तो $(A_1 + A_2 + A_3 + A_4)(h_1 + h_2 + h_3 + h_4)$ का न्यूनतम मान ज्ञात कीजिए।

Ans. 240

Sol. $\text{Volume}(V) = \frac{1}{3}A_1h_1$ or $h_1 = \frac{3V}{A_1}$

$h_2 = \frac{3V}{A_2}, h_3 = \frac{3V}{A_3}$ and $h_4 = \frac{3V}{A_4}$

$\left(A_1 + A_2 + A_3 + A_4\right)\left(h_1 + h_2 + h_3 + h_4\right)$

$= \left(A_1 + A_2 + A_3 + A_4\right)\left(\frac{3V}{A_1} + \frac{3V}{A_2} + \frac{3V}{A_3} + \frac{3V}{A_4}\right) = 3V\left(A_1 + A_2 + A_3 + A_4\right)\left(\frac{1}{A_1} + \frac{1}{A_2} + \frac{1}{A_3} + \frac{1}{A_4}\right)$

Now using A.M.-H.M inequality in A_1, A_2, A_3, A_4

$\frac{A_1 + A_2 + A_3 + A_4}{4} \geq \frac{4}{\left(\frac{1}{A_1} + \frac{1}{A_2} + \frac{1}{A_3} + \frac{1}{A_4}\right)}$

$\left(A_1 + A_2 + A_3 + A_4\right)\left(\frac{1}{A_1} + \frac{1}{A_2} + \frac{1}{A_3} + \frac{1}{A_4}\right) \geq 16$

Hence, the minimum value of $\left(A_1 + A_2 + A_3 + A_4\right)\left(h_1 + h_2 + h_3 + h_4\right) = 3V(16) = 48V = 48(5) = 240$

7. The vector $\vec{OP} = \hat{i} + 2\hat{j} + 2\hat{k}$ turns through a right angle, passing through the positive x-axis on the way. Find the vector in its new position.

यदि सदिश $\vec{OP} = \hat{i} + 2\hat{j} + 2\hat{k}$ को समकोण पर घुमाने के दौरान धनात्मक x-अक्ष को काटता है तो सदिश को नई स्थिति में ज्ञात कीजिये।

Ans. $\frac{4}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j} - \frac{1}{\sqrt{2}}\hat{k}$

Sol. $\overrightarrow{OP} = \hat{i} + 2\hat{j} + 2\hat{k}$

after rotation of $\overrightarrow{OP}$, let new vector is $\overrightarrow{OP}'$

Now $\overrightarrow{OP}$, $\hat{i}$, $\overrightarrow{OP}'$ will be coplanar

So $\overrightarrow{OP}' = \left|\overrightarrow{OP}\right| \dfrac{(\overrightarrow{OP} \times \hat{i}) \times \overrightarrow{OP}}{\left|(\overrightarrow{OP} \times \hat{i}) \times \overrightarrow{OP}\right|}$ $\left[\because \left|\overrightarrow{OP}\right| = \left|\overrightarrow{OP}'\right|\right]$

But $(\overrightarrow{OP} \times \hat{i}) \times \overrightarrow{OP} = 8\hat{i} - 2\hat{j} - 2\hat{k}$

$\Rightarrow \overrightarrow{OP}' = \dfrac{3(8\hat{i} - 2\hat{j} - 2\hat{k})}{2 \times 3\sqrt{2}}$

or $\overrightarrow{OP}' = \dfrac{4}{\sqrt{2}}\hat{i} - \dfrac{1}{\sqrt{2}}\hat{j} - \dfrac{1}{\sqrt{2}}\hat{k}$

8. If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a}| = 1$, $|\vec{b}| = 4$, $\vec{a} \cdot \vec{b} = 2$. If $|\vec{a}| = 1$, $|\vec{b}| = 4$, $\vec{a} \cdot \vec{b} = 2$, $\vec{c} = (2\vec{a} \times \vec{b}) - 3\vec{b}$ then find the angle between $\vec{b}$ and $\vec{c}$.

यदि दो सदिश $\vec{a}$ तथा $\vec{b}$ इस प्रकार है कि $|\vec{a}| = 1$, $|\vec{b}| = 4$, $\vec{a} \cdot \vec{b} = 2$ यदि $\vec{c} = (2\vec{a} \times \vec{b}) - 3\vec{b}$ तो सदिश $\vec{b}$ एवं $\vec{c}$ के बीच का कोण बताइये।

Ans. $\dfrac{5\pi}{6}$

Sol. $|\vec{a}| = 1$

$|\vec{b}| = 4$

$|\vec{c}| = \left(2\vec{a} \times \vec{b}\right) - 3\vec{b}$

$\vec{a} \cdot \vec{b} = 2$

$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta$

$2 = 1 \times 4 \cos\theta$

$\cos\theta = \dfrac{1}{2} \Rightarrow \theta = 60°$

$|\vec{c}|^2 = \left|2\vec{a} \times \vec{b}\right|^2 + 9|\vec{b}|^2 - 2 \times 3\vec{a} \cdot \left(2\vec{a} \times \vec{b}\right)$

$|\vec{c}|^2 = 4\left|\vec{a} \times \vec{b}\right|^2 + 9|\vec{b}|^2$

$$|\vec{c}|^2 = 4\left|1\times 4\times\frac{\sqrt{3}}{2}\right|^2 + 9\times 16$$

$$|\vec{c}|^2 = 4\times 4\times 3 + 9\times 16$$

$$|\vec{c}|^2 = 16\times 12$$

$$|\vec{c}| = 8\sqrt{3}$$

$$\vec{b}\cdot\vec{c} = \vec{b}\cdot\left(2\vec{a}\times\vec{b}\right) - 3\vec{b}\cdot\vec{b}$$

$$4\times 8\sqrt{3}\cos\phi = 0 - 3\times 16$$

$$\cos\phi = -\frac{\sqrt{3}}{2}$$

$$\phi = \frac{5\pi}{6}$$

9. The pv's of the four angular points of a tetrahedron are: A $\left(\hat{j}+2\hat{k}\right)$; B $\left(3\hat{i}+\hat{k}\right)$; C $\left(4\hat{i}+3\hat{j}+6\hat{k}\right)$ and D $\left(2\hat{i}+3\hat{j}+2\hat{k}\right)$. Find :

(i) the perpendicular distance from A to the line BC.
(ii) the volume of the tetrahedron ABCD.
(iii) the perpendicular distance from D to the plane ABC.
(iv) the shortest distance between the lines AB and CD.

यदि चार बिन्दुओं की स्थिति सदिश क्रमशः A $\left(\hat{j}+2\hat{k}\right)$; B $\left(3\hat{i}+\hat{k}\right)$; C $\left(4\hat{i}+3\hat{j}+6\hat{k}\right)$ तथा D $\left(2\hat{i}+3\hat{j}+2\hat{k}\right)$ है तो ज्ञात कीजिए :

(i) A से रखा BC की लम्बवत् दूरी
(ii) चतुष्फलक ABCD का आयतन
(iii) D की तल ABC से लम्बवत् दूरी
(iv) AB एवं CD के मध्य न्यूनतम दूरी

Ans. (i) $\frac{6}{7}\sqrt{14}$ (ii) 6 (iii) $\frac{3}{5}\sqrt{10}$ (iv) $\sqrt{6}$

Sol. A(0,1,2)
B(3,0,1)
C(4,3,6)
D(2,3,2)

$\overrightarrow{AB}(3,-1,-1)$ $\overrightarrow{BC}(1,3,5)$ $\overrightarrow{CD}(-2,0,-4)$

(i) Line BC $\frac{x-3}{1}=\frac{y-0}{3}=\frac{z-1}{5}=\lambda$

Point on line BC P(3 + λ, 3λ, 1 + 5λ)

Let P is FOP from A to line BC

$\overrightarrow{AP}(3+\lambda,3\lambda,1+5\lambda)$

$\overrightarrow{AP}\cdot\overrightarrow{BC}=0$

$3 + \lambda + 9\lambda - 3 + 25\lambda - 5 = 0$

$35\lambda = 5 \Rightarrow \lambda = \frac{1}{7}$

$\overrightarrow{AP}\left(\frac{22}{7},-\frac{4}{7},-\frac{2}{7}\right)$

$\left|\overrightarrow{AP}\right| = \sqrt{\frac{484+16+4}{49}} = \sqrt{\frac{504}{49}} = \sqrt{\frac{36\times14}{49}} = \frac{6}{7}\sqrt{14}$

(ii) Volume of tetrahedron

$V = \frac{1}{6}[AB \quad AC \quad AD]$

$$\begin{vmatrix} 3 & -1 & -1 \\ 4 & 2 & 4 \\ 2 & 2 & 0 \end{vmatrix}$$

$= \frac{1}{6}\left|3(-8)+1(-8)-1(8-4)\right|$

$= \frac{1}{6}\left|24+8+4\right|$

$= \frac{1}{6}\times 36 = 6$

(iii) Plane ABC

$$\text{Normal to ABC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & -1 \\ 1 & 3 & 5 \end{vmatrix}$$

$= \hat{i}(-5+3)-\hat{j}(15+1)+\hat{k}(10)$

$= -2\hat{i}-16\hat{j}+10\hat{k}$

<1, 8, –5>

Plane

$x + 8y - 5z = -2$

D(2,3,2)

$$d_{(2,3,2)} = \left|\frac{2+24-10}{\sqrt{1+64+25}}\right| = \frac{16}{\sqrt{90}} = \frac{16}{3\sqrt{10}} = \frac{8}{15}\sqrt{10}$$

(iv) Line AB $\frac{x}{3} = \frac{y-1}{-1} = \frac{z-2}{-1}$

Line CD $\frac{x-2}{-2} = \frac{y-3}{0} = \frac{z-2}{-4}$

$$\text{Shortest distance} = \overrightarrow{AC}\cdot\frac{\left(\overrightarrow{AB}\times\overrightarrow{CD}\right)}{\left|\overrightarrow{AB}\times\overrightarrow{CD}\right|}$$

$$\overrightarrow{AB}\times\overrightarrow{CD} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & -1 \\ -2 & 0 & -4 \end{vmatrix}$$

$$\overrightarrow{AB}\times\overrightarrow{CD} = \left(+4\hat{i}+4\hat{j}-2\hat{k}\right)$$

$$= \hat{i}(+4) - \hat{j}(-12-2) + \hat{k}(-2)$$

$$\text{S.D.} = \left|(4,2,4)\cdot\frac{(+4,+14,-2)}{\sqrt{16+196+4}}\right|$$

$$= \left|\frac{16+28-8}{\sqrt{120}}\right|$$

$$= \left|\frac{36}{\sqrt{216}}\right|$$

$$= \left|\frac{36}{6\sqrt{6}}\right| = \sqrt{6}$$

10. If $\vec{a} = a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$; $\vec{b} = b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$ and $\vec{c} = c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$ then show that the value of the scalar triple product $\left[n\vec{a}+\vec{b} \quad n\vec{b}+\vec{c} \quad n\vec{c}+\vec{a}\right]$ is $(n^3+1)\begin{vmatrix} \vec{a}\cdot\hat{i} & \vec{a}\cdot\hat{j} & \vec{a}\cdot\hat{k} \\ \vec{b}\cdot\hat{i} & \vec{b}\cdot\hat{j} & \vec{b}\cdot\hat{k} \\ \vec{c}\cdot\hat{i} & \vec{c}\cdot\hat{j} & \vec{c}\cdot\hat{k} \end{vmatrix}$.

यदि $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$; $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ तथा $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ तो सिद्ध कीजिए कि $\left[n\vec{a} + \vec{b} \quad n\vec{b} + \vec{c} \quad n\vec{c} + \vec{a}\right]$

का मान $(n^3 + 1)$ $\begin{vmatrix} \vec{a}\cdot\hat{i} & \vec{a}\cdot\hat{j} & \vec{a}\cdot\hat{k} \\ \vec{b}\cdot\hat{i} & \vec{b}\cdot\hat{j} & \vec{b}\cdot\hat{k} \\ \vec{c}\cdot\hat{i} & \vec{c}\cdot\hat{j} & \vec{c}\cdot\hat{k} \end{vmatrix}$ होगा।

Ans.

Sol. $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

$\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

$\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$

$[\vec{a}\,\vec{b}\,\vec{c}]$ is written as $\begin{vmatrix} \vec{a}.\hat{i} & \vec{a}.\hat{j} & \vec{a}.\hat{k} \\ \vec{b}.\hat{i} & \vec{b}.\hat{j} & \vec{b}.\hat{k} \\ \vec{c}.\hat{i} & \vec{c}.\hat{j} & \vec{c}.\hat{k} \end{vmatrix}$

Now $\{(n\vec{a} + \vec{b}) \times (n\vec{b} + \vec{c})\}.(n\vec{c} + \vec{a})$

$= \{n^2(\vec{a} \times \vec{b}) + n(\vec{a} \times \vec{c}) + \vec{b} \times \vec{c}\} . (n\vec{c} + \vec{a})$

$= n^3[\vec{a}\,\vec{b}\,\vec{c}] + [\vec{b}\,\vec{c}\,\vec{a}]$

$= (n^3 + 1)\ [\vec{a}\,\vec{b}\,\vec{c}]$

11. ABCD is a tetrahedron with pv's of its angular points as A(–5, 22, 5); B(1, 2, 3); C(4, 3, 2) and D(–1, 2, – 3). If the area of the triangle AEF where the quadrilaterals ABDE and ABCF are parallelograms is $\sqrt{S}$ then find the value of S.

ABCD एक चतुष्फलक है तथा इनके शीर्ष की स्थिति सदिश क्रमशः A(–5, 22, 5); B(1, 2, 3); C(4, 3, 2) एवं D(–1, 2, – 3) है। यदि ABDF एवं ABCF एक समान्तर चतुर्भुज है तथा त्रिभुज AEF का क्षेत्रफल $\sqrt{S}$ हो तो S का मान बताइये।

Ans. 110

Sol. A(–5,22,5)
B(1,2,3)
C(4,3,2)
D(–1,2,–3)
ABDE is parallelogram
Let E(α, β, γ)

Mid–point of AD = Mid–point of BE

$$(-3,12,1)=\left(\frac{\alpha+1}{2},\frac{\beta+2}{2},\frac{\gamma+3}{2}\right)$$

$\alpha = -7$

$\beta = 22$

$\gamma = -1$

E(–7, 22, –1)

$$\overrightarrow{AE}=(-2,0,-6)$$

ABCF is a parallelogram

Let F(a,b,c)

Mid–point of AC = mid–point of BF

$$\left(-\frac{1}{2},\frac{25}{2},\frac{7}{2}\right)=\left(\frac{1+a}{2},\frac{2+b}{2},\frac{3+c}{2}\right)$$

a = –2

b = 23

c = 4

F(–2,23,4)

$$\overrightarrow{AF}=(3,1,-1)$$

Area of ΔAEF

$$=\frac{1}{2}\left|\overrightarrow{AE}\times\overrightarrow{AF}\right|$$

$$=\frac{1}{2}\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ -2 & 0 & -6\\ 3 & 1 & -1\end{vmatrix}$$

$$=\frac{1}{2}\left|\hat{i}(6)-\hat{j}(2+18)+\hat{k}(-2)\right|$$

$$=\frac{1}{2}\sqrt{36+400+4}$$

$$=\frac{1}{2}\sqrt{440}$$

$$=\sqrt{110}$$

12. If volume of a regular tetrahedron of edge length k is V and shortest distance between any pair of opposite edges of same regular tetrahedron is d, then find the value of $\frac{d^3}{V}$.

एक समान्तर चतुष्फलक जिसके भुजा की लम्बाई k है का आयतन V है। यदि किन्हीं दो विपरीत भुजाओं के मध्य न्यूनतम दूरी d है तो $\frac{d^3}{V}$ का मान ज्ञात कीजिये।

Ans. 3

Sol. $V = \frac{1}{6}\left|\left[\vec{a}\ \vec{b}\ \vec{c}\right]\right|$ = volume of tetrahedron OABC

Now, $\left[\vec{a}\ \vec{b}\ \vec{c}\right]^2 = \begin{vmatrix} \vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c} \\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c} \\ \vec{c}\cdot\vec{a} & \vec{c}\cdot\vec{b} & \vec{c}\cdot\vec{c} \end{vmatrix} = k^6 \begin{vmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 \end{vmatrix} = \frac{k^6}{2} \Rightarrow \left|\left[\vec{a}\ \vec{b}\ \vec{c}\right]\right| = \frac{k^3}{\sqrt{2}}$

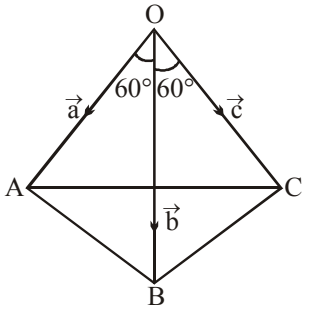
So, $V = \frac{k^3}{6\sqrt{2}}$

Shortest distance between OA and BC, is

$d = \left|\frac{\left(\vec{b}-\vec{a}\right)\cdot\left(\vec{a}\times\left(\vec{c}-\vec{b}\right)\right)}{\left|\vec{a}\times\left(\vec{c}-\vec{b}\right)\right|}\right| = \left|\frac{\left[\vec{a}\ \vec{c}\ \vec{b}\right]}{\left|\vec{a}\times\left(\vec{c}-\vec{b}\right)\right|}\right| = \frac{k^3/\sqrt{2}}{k^2} = \frac{k}{\sqrt{2}}$, as

$\left|\vec{a}\times\left(\vec{c}-\vec{b}\right)\right|^2 = \vec{a}^2(\vec{c}^2-\vec{b}^2) - \left(\vec{a}\cdot\left(\vec{c}-\vec{b}\right)\right)^2 = \vec{a}^2(\vec{c}^2+\vec{b}^2-2\vec{c}\cdot\vec{b}) - \left(\vec{a}\cdot\left(\vec{c}-\vec{b}\right)\right)^2 = k^4 + k^4 - \frac{2k^2k^2}{2} = k^4 \Rightarrow$

$\left|\vec{a}\times\left(\vec{c}-\vec{b}\right)\right| = k^2$.



Hence $\frac{d^3}{V} = \frac{\left(\frac{k}{\sqrt{2}}\right)^3}{\frac{1}{6\sqrt{2}}k^3} = \frac{1}{2\sqrt{2}}\left(6\sqrt{2}\right) = 3$. **Ans.**

13. Find the scalars α and β if $\vec{a}\times(\vec{b}\times\vec{c}) + (\vec{a}.\vec{b})\vec{b} = (4-2\beta-\sin\alpha)\vec{b} + (\beta^2-1)\vec{c}$ and $(\vec{c}.\vec{c})\vec{a} = \vec{c}$ while $\vec{b}$ and $\vec{c}$ are non zero non collinear vectors.

दिया है $\vec{a}\times(\vec{b}\times\vec{c}) + (\vec{a}.\vec{b})\vec{b} = (4-2\beta-\sin\alpha)\vec{b} + (\beta^2-1)\vec{c}$ एवं $(\vec{c}.\vec{c})\vec{a} = \vec{c}$ यदि $\vec{b}$ एवं $\vec{c}$ अशून्य तथा असंरेखीय है तो α एवं β के मान ज्ञात कीजिए।

Ans. $\alpha = n\pi + \frac{(-1)^n\pi}{2}, n \in I \& \beta = 1$

Sol. $\vec{a}\times\left(\vec{b}\times\vec{c}\right)+\left(\vec{a}\cdot\vec{b}\right)\vec{b}=\left(4-2\beta-\sin\alpha\right)\vec{b}+\left(\beta^2-1\right)\vec{c}$

$\left(\vec{c}\cdot\vec{c}\right)\vec{a}=\vec{c}$

$\vec{a}\times\left(\vec{b}\times\vec{c}\right)=\vec{b}\left(\vec{a}\cdot\vec{c}\right)-\vec{c}\left(\vec{a}\cdot\vec{b}\right)$

$\vec{a}\times\left(\vec{b}\times\vec{c}\right)+\left(\vec{a}\cdot\vec{b}\right)\vec{b}=\vec{b}\left(\vec{a}\cdot\vec{c}+\vec{a}\cdot\vec{b}\right)-\vec{c}\left(\vec{a}\cdot\vec{b}\right)$

By comparing with given relation

$\vec{a}\cdot\vec{c}+\vec{a}\cdot\vec{b}=4-2\beta-\sin\alpha$ ...(1)

$-\vec{a}\cdot\vec{b}=\beta^2-1$ ...(2)

Now $\left(\vec{c}\cdot\vec{c}\right)\vec{a}=\vec{c}$

$\left(\vec{c}\cdot\vec{c}\right)\vec{a}\cdot\vec{c}=\vec{c}\cdot\vec{c}$

$\vec{a}\cdot\vec{c}=1$ ...(3)

Using eq (1), (2) & (3)

$1=\beta^2-1+4-2\beta-\sin\alpha$

$\beta^2-2\alpha+2=\sin\alpha$

$(\beta-1)^2+1=\sin\alpha$

$\sin\alpha=1$ $\beta=1$

$\alpha=n\pi+(-1)^n\frac{\pi}{2}$ $\beta=1$

14. Let $\vec{a}=\alpha\hat{i}+2\hat{j}-3\hat{k}$, $\vec{b}=\hat{i}+2\alpha\hat{j}-2\hat{k}$ and $\vec{c}=2\hat{i}-\alpha\hat{j}+\hat{k}$. Find the value(s) of α, if any, such that $\left\{\left(\vec{a}\times\vec{b}\right)\times\left(\vec{b}\times\vec{c}\right)\right\}\times\left(\vec{c}\times\vec{a}\right)=0$. Find the vector product when $\alpha=0$.

यदि $\vec{a}=\alpha\hat{i}+2\hat{j}-3\hat{k}$, $\vec{b}=\hat{i}+2\alpha\hat{j}-2\hat{k}$ तथा $\vec{c}=2\hat{i}-\alpha\hat{j}+\hat{k}$ यदि $\left\{\left(\vec{a}\times\vec{b}\right)\times\left(\vec{b}\times\vec{c}\right)\right\}\times\left(\vec{c}\times\vec{a}\right)=0$ है तो α का मान ज्ञात करो। यदि $\alpha=0$ तो उनका सदिश गुणनफल बताओ।

Ans. $\alpha=2/3$; if $\alpha=0$ then vector product is $-60\left(2\hat{i}+\hat{k}\right)$

Sol. $\vec{a}=\langle\alpha, 2, -3\rangle$

$\vec{b}=\langle1, 2\alpha, -2\rangle$

$\vec{c}=\langle2, -\alpha, 1\rangle$

Case-I

$\left(\vec{a}\times\vec{b}\right)\times\left(\vec{b}\times\vec{c}\right)=\lambda\left(\vec{c}\times\vec{a}\right)$

dot with b

$\vec{b}\left(\left(\vec{a}\times\vec{b}\right)\times\left(\vec{b}\times\vec{c}\right)\right)=\lambda\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]$

$\left(\vec{b}\times\left(\vec{a}\times\vec{b}\right)\cdot\left(\vec{b}\times\vec{c}\right)\right)=\lambda\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]$

$\left(\vec{a}\left(\vec{b}\cdot\vec{b}\right)-\vec{b}\left(\vec{a}\cdot\vec{b}\right)\right)\cdot\left(\vec{b}\times\vec{c}\right)=\lambda\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]$

$\left(\vec{b}\cdot\vec{b}\right)\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]-\left[\vec{a}\quad\vec{b}\right]\times 0=\lambda\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]$

$\vec{b}\cdot\vec{b}\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]=\lambda\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]$

$\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]=0$

$$\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]=\begin{vmatrix}\alpha & 2 & -3\\ 1 & 2\alpha & -2\\ 2 & -\alpha & 1\end{vmatrix}=0$$

$= \alpha(2\alpha - 2\alpha) - 2(1 + 4) - 3\,(-\alpha - 4\alpha) = 0$

$0 - 10 + 15\alpha = 0$

$\alpha=\frac{2}{3}$

Case-II : $\vec{c}\times\vec{a}=0$ Not possible

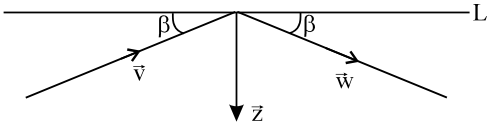
Case-III : $\left(\vec{a}\times\vec{b}\right)=\lambda\left(\vec{b}\times\vec{c}\right)$

dot with $\vec{c}$

$\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]=0$

15. The figure shows non zero vector $\vec{v}$, $\vec{w}$ and $\vec{z}$ with orthogonal to the line L, and $\vec{v}$ and $\vec{w}$ making equal angles β with the line L. Assuming $|\vec{v}|=|\vec{w}|$, if the vector $\vec{w}$ is expressed as a linear combination of $\vec{v}$ and $\vec{z}$ as $\vec{w}=x\vec{v}+y\vec{z}$. Find the value of x and y.

दिखाए गये चित्रनुसार तीन अशून्य सदिश $\vec{v}$, $\vec{w}$ एवं $\vec{z}$ है जहाँ $\vec{z}$ रेखा L के लम्बवत् (orthogonal) है तथा $\vec{v}$ एवं $\vec{w}$ रेखा L के साथ समान कोण β बनाती है। $|\vec{v}|=|\vec{w}|$, $\vec{w}$ को $\vec{v}$ तथा $\vec{z}$ के रेखिक संचय के रूप में व्यक्त किया जा सके अर्थात् $\vec{w}=x\vec{v}+y\vec{z}$ तो x एवं y का मान ज्ञात कीजिए।



Ans. $x = 1$ and $y=\frac{2|\vec{v}|\sin\beta}{|\vec{z}|}$

Sol. $\vec{w}=x\vec{v}+y\vec{z}$ $|\vec{v}|=|\vec{w}|$

dot $\vec{v}$

$\vec{w}\cdot\vec{v} = x|\vec{v}|^2 + yz\cdot\vec{v}$

$|\vec{v}||\vec{w}|\cos 2\beta = x|\vec{v}|^2 + y|\vec{z}||\vec{v}|\cos(90+\beta)$

$|\vec{w}|\cos 2\beta = x|\vec{w}| - y|\vec{z}|\sin\beta$ ...(1)

$\vec{w} = x\vec{v} + y\vec{z}$

dot with $\vec{w}$

$|\vec{w}|^2 = x|\vec{v}||\vec{w}|\cos 2\beta + y|\vec{z}||\vec{w}|\cos(90-\beta)$

$|\vec{w}| = x|\vec{w}|\cos 2\beta + y|\vec{z}|\sin\beta$ ...(2)

(1) + (2)

$|\vec{w}|(1+\cos 2\beta) = x(1+\cos 2\beta)|\vec{w}|$

$x = 1$

$y = \dfrac{|\vec{w}|(1-\cos 2\beta)}{|\vec{z}|\sin\beta}$

$= \dfrac{|\vec{w}|2\sin\beta}{|\vec{z}|}$

16. Consider the non zero vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ such that no three of which are coplanar then prove that $\vec{a}[\vec{b}\,\vec{c}\,\vec{d}] + \vec{c}[\vec{a}\,\vec{b}\,\vec{d}] = \vec{b}[\vec{a}\,\vec{c}\,\vec{d}] + \vec{d}[\vec{a}\,\vec{b}\,\vec{c}]$. Hence prove that if $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ represent the position vectors of the vertices of a plane quadrilateral then $\dfrac{[\vec{b}\,\vec{c}\,\vec{d}] + [\vec{a}\,\vec{b}\,\vec{d}]}{[\vec{a}\,\vec{c}\,\vec{d}] + [\vec{a}\,\vec{b}\,\vec{c}]} = 1$.

यदि चार अशून्य सदिश $\vec{a}$, $\vec{b}$, $\vec{c}$ तथा $\vec{d}$ इस प्रकार से है कि उनमें से कोई भी तीन समतलीय नही हो तो सिद्ध करो कि $\vec{a}[\vec{b}\,\vec{c}\,\vec{d}] + \vec{c}[\vec{a}\,\vec{b}\,\vec{d}] = \vec{b}[\vec{a}\,\vec{c}\,\vec{d}] + \vec{d}[\vec{a}\,\vec{b}\,\vec{c}]$. $\vec{a}$, $\vec{b}$, $\vec{c}$ तथा $\vec{d}$ यदि एक चतुर्भुज के शीर्षो की स्थिति सदिश है तो सिद्ध कीजिए कि $\dfrac{[\vec{b}\,\vec{c}\,\vec{d}] + [\vec{a}\,\vec{b}\,\vec{d}]}{[\vec{a}\,\vec{c}\,\vec{d}] + [\vec{a}\,\vec{b}\,\vec{c}]} = 1$

Ans.

Sol. (1) $\vec{a} = x\vec{b} + y\vec{c} + z\vec{d}$

$[\vec{a}\ \ \vec{b}\ \ \vec{c}] = z[\vec{d}\ \ \vec{b}\ \ \vec{c}] = z[\vec{b}\ \ \vec{c}\ \ \vec{d}]$

$\left[\vec{a}\ \ \vec{c}\ \ \vec{d}\right] = x\left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right]$

$\left[\vec{a}\ \ \vec{b}\ \ \vec{d}\right] = y\left[\vec{c}\ \ \vec{b}\ \ \vec{d}\right] = -y\left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right]$

LHS $\vec{a}\left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right] = \vec{c}\left[\vec{a}\ \ \vec{b}\ \ \vec{d}\right]$

$\vec{a}\left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right] = -y\vec{c}\left[\vec{b}\ \ +\vec{c}\ \ \vec{d}\right]$

RHS $\vec{b}\left[\vec{a}\ \ \vec{c}\ \ \vec{d}\right] = \vec{b}x\left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right]$

$+\vec{d}\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right] = +z\vec{d}\left[\vec{b}\ \ +\vec{c}\ \ +\vec{d}\right]$

putting in equation

$\left(\vec{b}\ \ \vec{c}\ \ \vec{d}\right)\left(\vec{a}\ \ -y\vec{c}\right) = \left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right]\left[x\vec{b}\ \ +z\vec{d}\right]$

$\vec{a} = x\vec{b} + y\vec{c} + z\vec{d}$

(2) $\vec{a}\ \ \vec{b}\ \ \vec{c}\ \ \vec{d}$ $\Rightarrow$ PV of vertices of a plane quadrilateral

$\left[\left(\vec{a}-\vec{b}\right)\times\left(\vec{b}-\vec{c}\right)\right].\left(\vec{c}-\vec{d}\right) = 0$

$\left(\vec{a}\times\vec{b} - \vec{a}\times\vec{c} + \vec{b}\times\vec{c}\right)\cdot\left(\vec{c}-\vec{d}\right) = 0$

$\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right] - \left[\vec{a}\ \ \vec{b}\ \ \vec{d}\right] + \left[\vec{a}\ \ \vec{c}\ \ \vec{d}\right] - \left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right] = 0$

$$\frac{\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right] + \left[\vec{a}\ \ \vec{c}\ \ \vec{d}\right]}{\left[\vec{a}\ \ \vec{b}\ \ \vec{d}\right] + \left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right]} = 1$$

17. Let $\vec{A} = \hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{B} = 2\hat{i} + \hat{j} - \hat{k}$, $\vec{C} = \hat{j} + \hat{k}$. If the vector $\vec{B}\times\vec{C}$ can be expressed as a linear combination $\vec{B}\times\vec{C} = x\vec{A} + y\vec{B} + z\vec{C}$ where x, y, z are scalars, then find the value of $(100x + 10y + 8z)$.

माना $\vec{A} = \hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{B} = 2\hat{i} + \hat{j} - \hat{k}$, $\vec{C} = \hat{j} + \hat{k}$. यदि सदिश $\vec{B}\times\vec{C}$ को रैखिक संचय के रूप में $\vec{B}\times\vec{C} = x\vec{A} + y\vec{B} + z\vec{C}$ तरह से व्यक्त किया जा सके जहाँ x, y, z अदिश है तो $(100x + 10y + 8z)$ का मान ज्ञात कीजिये।

Ans. 101

Sol. $\vec{A}(1,-2,3)$

$\vec{B}(2,1,-1)$

$\vec{C}(0,1,1)$

$\vec{B}\times\vec{C} = x\vec{A} + y\vec{B} + z\vec{C}$

$$\vec{B}\times\vec{C}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 1 & -1\\ 0 & 1 & 1\end{vmatrix}=2(1,-1,1)$$

$2(1,-1,1) = x(1,-2,3) + y(2,1,-1) + z(0,1,1)$

$x + 2y = 2$

$-2x + y + z = -2$

$3x - y + z = 2$

Solving

$x = 1,\ y = \frac{1}{2},\ z = -\frac{1}{2}$

$100x + 10y + 8z = 100 + 5 - 4 = 101$

18. Let $\vec{a}=\begin{bmatrix}1\\0\\-3\end{bmatrix}$; $\vec{b}=\begin{bmatrix}2\\1\\0\end{bmatrix}$; $\vec{c}=\begin{bmatrix}1\\-1\\1\end{bmatrix}$. Find the numbers α, β, γ such that $\alpha\vec{a}+\beta\vec{b}+\gamma\vec{c}=\begin{bmatrix}-2\\-5\\6\end{bmatrix}$.

दिया है $\vec{a}=\begin{bmatrix}1\\0\\-3\end{bmatrix}$; $\vec{b}=\begin{bmatrix}2\\1\\0\end{bmatrix}$; $\vec{c}=\begin{bmatrix}1\\-1\\1\end{bmatrix}$. यदि α, β, γ तीन अदिश ऐसे है कि $\alpha\vec{a}+\beta\vec{b}+\gamma\vec{c}=\begin{bmatrix}-2\\-5\\6\end{bmatrix}$ तो α, β, γ ज्ञात कीजिए।

Ans. $\alpha = -1, \beta = -2, \gamma = 3$

Sol. $\vec{a}=\begin{bmatrix}1\\0\\-3\end{bmatrix}$ $\vec{b}=\begin{bmatrix}2\\1\\0\end{bmatrix}$ $\vec{c}=\begin{bmatrix}1\\-1\\1\end{bmatrix}$

$$\alpha\vec{a}+\beta\vec{b}+\gamma\vec{c}=\begin{bmatrix}-2\\-5\\6\end{bmatrix}$$

$\alpha + 2\beta + \gamma = -2$

$\beta - \gamma = -5$

$-3\alpha + \gamma = 6$

Solving them

$\alpha = -1$

$\beta = -2$

$\gamma = 3$

19. Let a 3 dimensional vector $\vec{V}$ satisfies the condition, $2\vec{V} + \vec{V} \times (\hat{i} + 2\hat{j}) = 2\hat{i} + \hat{k}$. If $3|\vec{V}| = \sqrt{m}$ where $m \in N$, then find m.

माना एक त्रिविम सदिश $\vec{V}$ प्रतिबन्ध $2\vec{V} + \vec{V} \times (\hat{i} + 2\hat{j}) = 2\hat{i} + \hat{k}$ को सन्तुष्ट करता है। यदि $3|\vec{V}| = \sqrt{m}$ जहाँ $m \in N$ तो m ज्ञात कीजिये।

Ans. 6

Sol. $2\vec{V} + \vec{V} \times \left(\hat{i} + 2\hat{j}\right) = 2\hat{i} + \hat{k}$

$3|\vec{V}| = \sqrt{m}$

dot with $\left(\hat{i} + 2\hat{j}\right)$

$2\vec{V} \cdot \left(\hat{i} + 2\hat{j}\right) = 2$

$\vec{V} \cdot \left(\hat{i} + 2\hat{j}\right) = 1$

$\left|2\vec{V} + \vec{V} \times \left(\hat{i} + 2\hat{j}\right)\right|^2 = \left(2\hat{j} + \hat{k}\right)^2$

$4|\vec{V}|^2 + \left(\vec{V} \times \left(\hat{i} + 2\hat{j}\right)\right) \cdot \left(\vec{V} \times \left(\hat{i} + 2\hat{j}\right)\right) + 2\vec{V} \cdot \left(\vec{V} \times \left(\hat{i} + 2\hat{j}\right)\right) = \left(\sqrt{4+1}\right)^2$

$4|\vec{V}|^2 + \vec{V} \cdot \left(\left(\hat{i} + 2\hat{j}\right) \times \left(\vec{V}\left(\hat{i} + 2\hat{j}\right)\right)\right) = 5$

$4|\vec{V}|^2 + \vec{V} \cdot \left(\vec{V}(1+4) - \left(\hat{i} + 2\hat{j}\right)\left(\vec{V} \cdot \left(\hat{i} + 2\hat{j}\right)\right)\right)$

$4|\vec{V}|^2 + \vec{V} \cdot \left(5\vec{V} - \left(\hat{i} + 2\hat{j}\right)(1)\right) = 5$

$9|\vec{V}|^2 - \vec{V} \cdot \left(\hat{i} + 2\hat{j}\right) = 5$

$9|\vec{V}|^2 = 6$

$3|\vec{V}| = \sqrt{6}$

20. If $\vec{x}$, $\vec{y}$ are two non-zero and non-collinear vectors satisfying

$[(a-2)\alpha^2 + (b-3)\alpha + c]\vec{x} + [(a-2)\beta^2 + (b-3)\beta + c]\vec{y} + [(a-2)\gamma^2 + (b-3)\gamma + c](\vec{x} \times \vec{y}) = 0$
where α, β, γ are three distinct real numbers, then find the value of $(a^2 + b^2 + c^2)$.

यदि $\vec{x}$, $\vec{y}$ दो अशून्य विषम रेखिये सदिश हैं जो

$[(a-2)\alpha^2 + (b-3)\alpha + c]\vec{x} + [(a-2)\beta^2 + (b-3)\beta + c]\vec{y} + [(a-2)\gamma^2 + (b-3)\gamma + c](\vec{x} \times \vec{y}) = 0$
को सन्तुष्ट करते हैं जहाँ α, β, γ तीन भिन्न वास्तविक संख्याऐं हैं तो $(a^2 + b^2 + c^2)$ का मान ज्ञात कीजिये।

Ans. 13

Sol. Clearly, $(a-2)\alpha^2 + (b-3)\alpha + c = 0$

$(a-2)\beta^2 + (b-3)\beta + c = 0$ } (As $\vec{x}, \vec{y}$ and $(\vec{x} \times \vec{y})$ are non coplanar.) and

$(a-2)\gamma^2 + (b-3)\gamma + c = 0$

$\therefore$ The equation $(a-2)x^2 + (b-3)x + c = 0$ has 3 distinct roots.

$\therefore$ Above equation will be an identity

$\therefore$ $a - 2 = 0 \Rightarrow a = 2$

$b - 3 = 0 \Rightarrow b = 3$

$c = 0$

$\therefore$ $a^2 + b^2 + c^2 = 13$ Ans.

21. The diagonals of a parallelogram are given by vectors $2\hat{i} + 3\hat{j} - 6\hat{k}$ and $3\hat{i} - 4\hat{j} - \hat{k}$. Determine its sides and also the area.

किसी समान्तर चतुर्भुज के विकर्णो के सदिश क्रमशः $2\hat{i} + 3\hat{j} - 6\hat{k}$ तथा $3\hat{i} - 4\hat{j} - \hat{k}$ है। उसकी भुजाओं तथा क्षेत्रफल ज्ञात कीजिए।

Ans. $\frac{1}{2}\left(5\hat{i} - \hat{j} - 7\hat{k}\right), \frac{1}{2}(-\hat{i} + 7\hat{j} - 5\hat{k})$; $\frac{1}{2}\sqrt{1274}$ sq. units

Sol. $\vec{a} + \vec{b} = (2, 3, -6) = \vec{d}_1$

$\vec{a} - \vec{b} = (3, 4, -1) = \vec{d}_2$

$\vec{a} = \left(\frac{5}{2}, -\frac{1}{2}, -\frac{7}{2}\right)$

$\vec{b} = \left(-\frac{1}{2}, +\frac{7}{2}, -\frac{5}{2}\right)$

Area $= \vec{d}_1 \times \vec{d}_2$

$$= \frac{1}{2}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -6 \\ 3 & -4 & -1 \end{vmatrix}$$

$$= \frac{1}{2}\left|\hat{i}(-3-24) - \hat{j}(-2+18) + \hat{k}(-8-9)\right|$$

$$= \frac{1}{2}\left|-27\hat{i} - 16\hat{j} - 17\hat{k}\right|$$

$$= \frac{1}{2}\sqrt{1274}$$

22. Find the value of λ such that a, b, c are all non-zero and

$\left(-4\hat{i}+5\hat{j}\right)a+(3\hat{i}-3\hat{j}+\hat{k})b+(\hat{i}+\hat{j}+3\hat{k})c=\lambda(a\hat{i}+b\hat{j}+c\hat{k})$

a, b, c अशून्य संख्याओं के लिये λ का मान ज्ञात कीजिये। कि $\left(-4\hat{i}+5\hat{j}\right)a+(3\hat{i}-3\hat{j}+\hat{k})b+(\hat{i}+\hat{j}+3\hat{k})c=\lambda(a\hat{i}+b\hat{j}+c\hat{k})$ है।

Ans. $\lambda = 0,\ \lambda = -2 \pm \sqrt{29}$

Sol. $-4a + 3b + c = a\lambda$

$(-4 - \lambda)\, a + 3b + c = 0$ ...(i)

$5a - 3b + xc = \lambda b$

$5a - (3 + \lambda)b + c = 0$ ...(ii)

$b + 3c = \lambda c$

$b + (3 - \lambda)c = 0$ ...(iii)

Homogeneous system with non–zero solution

$$\begin{vmatrix} -4-\lambda & 3 & 1 \\ 5 & -3-\lambda & 1 \\ 0 & 1 & 3-\lambda \end{vmatrix} = 0$$

$(-4 - \lambda)\,((-3 - \lambda)\,(3 - \lambda) - 1) - 5\,(9 - \lambda - 1) = 0$

$(4 + \lambda)\,(9 - \lambda^2 + 1) - 5\,(8 - \lambda) = 0$

$40 - 4\lambda^2 - 10\lambda - \lambda^3 - 40 + 5\lambda = 0$

$\lambda^3 + 4\lambda^2 + 15\lambda = 0$

$d = 0 \qquad \lambda^2 + 4\lambda - 15 = 0$

$$\lambda = \frac{-4 \pm \sqrt{16+60}}{4}$$

$\lambda = -4 \pm \sqrt{29}$

23. Find the vector $\vec{r}$ which is perpendicular to $\vec{a}=\hat{i}-2\hat{j}+5\hat{k}$ and $\vec{b}=2\hat{i}+3\hat{j}-\hat{k}$ and $\vec{r}\cdot\left(2\hat{i}+\hat{j}+\hat{k}\right)+8=0$.

$\vec{a}=\hat{i}-2\hat{j}+5\hat{k}$ तथा $\vec{b}=2\hat{i}+3\hat{j}-\hat{k}$ के लम्बवत् सदिश $\vec{r}$ इस प्रकार है कि $\vec{r}\cdot\left(2\hat{i}+\hat{j}+\hat{k}\right)+8=0$

Ans. $\vec{r}=-13\hat{i}+11\hat{j}+7\hat{k}$

Sol. $\vec{r}=\lambda\left(\vec{a}\times\vec{b}\right)$

$$=\lambda\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 5 \\ 2 & 3 & -1 \end{vmatrix}$$

$=\lambda\left(-13\hat{i}+11\hat{j}+7\hat{k}\right)$

$r \cdot (2,1,1) = -8$
$\lambda(-26 + 11 + 7) = -8$
$\lambda(-8) = -8$
$\lambda = 1$

24. Two vertices of a triangle are at $-\hat{i} + 3\hat{j}$ and $2\hat{i} + 5\hat{j}$ and its orthocentre is at $\hat{i} + 2\hat{j}$. Find the position vector of third vertex.

किसी त्रिभुज के दो शीर्षो $-\hat{i} + 3\hat{j}$ तथा $2\hat{i} + 5\hat{j}$ और लम्बकेन्द्र $\hat{i} + 2\hat{j}$ है। तो तीसरे शीर्ष का स्थिति सदिश ज्ञात कीजिए।

Ans. $\frac{5}{7}\hat{i} + \frac{17}{7}\hat{j}$

Sol. Orthocenter O (1,2)
DR′s of BE = (1, 3)
DR′s of BF = (2, –1)
DR′s of AC = $(\alpha + 1, \beta - 3)$
DR′s of BC = $((\alpha - 2), (\beta - 5)$
BE $\perp$ AC
$(1, 3) \cdot (\alpha + 1, \beta - 3) = 0$
$\alpha + 1 + 3\beta - 9 = 0$
$\alpha + 3\beta = 8$ ...(i)
BF $\perp$ BC
$(2, -1) \cdot (\alpha - 2, \beta - 5) = 0$
$2\alpha - 4 - \beta + 5 = 0$
$2\alpha - \beta = -1$ ...(ii)
equation (i) + 3× (ii)
$7\alpha = 5$

$$\alpha = \frac{5}{7}$$

$$\beta = \frac{10}{7} + 1 = \frac{17}{7}$$

Third vertex $\left(\frac{5}{7}, \frac{17}{7}\right)$

25. Let $\vec{A}(t) = f_1(t)\hat{i} + f_2(t)\hat{j}$ and $\vec{B}(t) = g_1(t)\hat{i} + g_2(t)\hat{j}$, $t \in [0, 1]$, where f_1, f_2, g_1, g_2 are continuous functions. If $\vec{A}(t)$ and $\vec{B}(t)$ are nonzero vectors for all t and $\vec{A}(0) = 2\hat{i} + 3\hat{j}$, $\vec{A}(1) = 6\hat{i} + 2\hat{j}$, $\vec{B}(0) = 3\hat{i} + 2\hat{j}$ and $\vec{B}(1) = 2\hat{i} + 6\hat{j}$, then show that $\vec{A}(t)$ and $\vec{B}(t)$ are parallel for some t.

माना कि $\vec{A}(t) = f_1(t)\hat{i} + f_2(t)\hat{j}$ तथा $\vec{B}(t) = g_1(t)\hat{i} + g_2(t)\hat{j}$, $t \in [0, 1]$, है जहाँ f_1, f_2, g_1, g_2 सतत् फलन है यदि $\vec{A}(t)$ और $\vec{B}(t)$ सभी t के लिये अशून्य सदिश और $\vec{A}(0) = 2\hat{i} + 3\hat{j}$, $\vec{A}(1) = 6\hat{i} + 2\hat{j}$, $\vec{B}(0) = 3\hat{i} + 2\hat{j}$ एवं $\vec{B}(1) = 2\hat{i} + 6\hat{j}$ हो, तो प्रदर्शित कीजिए कि कुछ t के लिये $\vec{A}(t)$ और $\vec{B}(t)$ समान्तर है।

Ans.

Sol. $\vec{A}(t)$ is parallel to $\vec{B}(t)$ for some $t \in [0,1]$ if and only if $\frac{f_1(t)}{g_1(t)} = \frac{f_2(t)}{g_2(t)}$ for some $t \in [0,1]$

or $f_1(t).g_2(t) = f_2(t).g_1(t)$ for some $t \in [0,1]$

Let $h(t) = f_1(t).g_2(t) - f_2(t).g_1(t)$

$h(0) = f_1(0).g_2(0) - f_2(0).g_1(0)$

$= 2 \times 2 - 3 \times 3 = -5 < 0$

$h(1) = f_1(1).g_2(1) - f_2(1).g_1(1)$

$= 6 \times 6 - 2 \times 2 = 32 > 0$

Since h is a continuous function, and $h(0).h(1) < 0$

$\Rightarrow$ there is some $t \in [0,1]$ for which $h(t)=0$ i.e., $\vec{A}(t)$ and $\vec{B}(t)$ are parallel vectors for this t.

26. Let V be the volume of the parallelopiped formed by the vectors $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$, $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$. If a_r, b_r, c_r, where r = 1, 2, 3, are non-negative real numbers and $\sum_{r=1}^{3}(a_r + b_r + c_r) = 3L$, show that $V \le L^3$.

माना सदिशों $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$, $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ से निर्मित समान्तर षट्फलक का आयतन V है। यदि a_r, b_r, c_r, जहाँ r = 1, 2, 3, अऋणात्मक वास्तविक संख्याएँ है और $\sum_{r=1}^{3}(a_r + b_r + c_r) = 3L$, तो प्रदर्शित कीजिये कि $V \le L^3$

Ans.

Sol. Given that, $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

$\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

$\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$

where a_r, b_r, c_r, r = 1,2,3 are all non negative real numbers.

Also $\sum_{r=1}^{3}(a_r + b_r + c_r) = 3L$

To prove $V \le L^3$ Where V is vol. of parallelopiped formed by the vectors $\vec{a}, \vec{b}$ and $\vec{c}$

$\therefore$ We have $V = [\vec{a}\,\vec{b}\,\vec{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$

$\Rightarrow V = (a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2) - (a_1b_3c_2 + a_2b_1c_3 + a_3b_2c_1)$....(1)

Now we know that AM ≥ GM

$$\therefore \frac{(a_1 + b_1 + c_1) + (a_2 + b_2 + c_2) + (a_3 + b_3 + c_3)}{3}$$

$$\geq [(a_1 + b_1 + c_1)(a_2 + b_2 + c_2)(a_3 + b_3 + c_3)]^{1/3}$$

$$\Rightarrow \frac{3L}{3} \geq [(a_1+b_1+c_1)(a_2+b_2+c_2)(a_3+b_3+c_3)]^{1/3}$$

$\Rightarrow L^3 \geq (a_1+b_1+c_1)(a_2 + b_2 + c_2)(a_3 + b_3 + c_3)$

$= a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2 + 24$ more such terms

$\geq a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2$

$[\because a_r, b_r, c_r \geq 0$ or $r = 1,2,3]$

$\geq (a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2)$

$- (a_1b_3c_2 + a_2b_1c_3 + a_3b_2c_1)$ [same reason]

= V from (1)

Thus, $L^3 \geq V$ Hence Proved

27. If $\vec{u}, \vec{v}, \vec{w}$ are three non-coplanar unit vectors and α, β, γ are the angles between $\vec{u}$ and $\vec{v}$, $\vec{v}$ and $\vec{w}$, $\vec{w}$ and $\vec{u}$ respectively and $\vec{x}, \vec{y}, \vec{z}$ are unit vectors along the bisectors of the angles α, β, γ respectively. Prove that

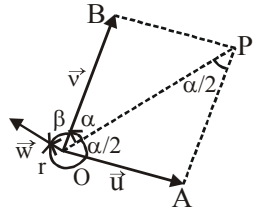
$$\left[\vec{x}\times\vec{y}\ \ \vec{y}\times\vec{z}\ \ \vec{z}\times\vec{x}\right] = \frac{1}{16}\left[\vec{u}\ \vec{v}\ \vec{w}\right]^2 \sec^2\frac{\alpha}{2}\ \sec^2\frac{\beta}{2}\ \sec^2\frac{\gamma}{2}$$

यदि $\vec{u}, \vec{v}, \vec{w}$ तीन विषमतलीय इकाई सदिश है और α, β, γ क्रमशः $\vec{u}$ और $\vec{v}$, $\vec{v}$ और $\vec{w}$, $\vec{w}$ और $\vec{u}$ के बीच के कोण है और $\vec{x}, \vec{y}, \vec{z}$ क्रमशः α, β, γ के लम्बअर्द्धको की दिशा में इकाई सदिश है। सिद्ध कीजिये कि

$$\left[\vec{x}\times\vec{y}\ \ \vec{y}\times\vec{z}\ \ \vec{z}\times\vec{x}\right] = \frac{1}{16}\left[\vec{u}\ \vec{v}\ \vec{w}\right]^2 \sec^2\frac{\alpha}{2}\ \sec^2\frac{\beta}{2}\ \sec^2\frac{\gamma}{2}$$

Ans.

Sol. Given that u, ν, ω are three non coplanar unit vectors. Angle between $\vec{u}$ and $\vec{v}$ is α, between $\vec{v}$ and $\vec{\omega}$ is β and between $\vec{\omega}$ and $\vec{u}$ it is γ. In fig. $\overrightarrow{OA}$ and $\overrightarrow{OB}$ represent $\vec{u}$ and $\vec{v}$. Let P be a pt. on angle bisector of $\angle AOB$ such that OAPB is a parallelogram.



Also $\angle POA = \angle BOP = \alpha/2$

$\therefore \angle APO = \angle BOP = \alpha/2$ (Alternate angles)

$\therefore$ In Δ OAP, OA = AP

$\therefore \quad \overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP} = \vec{u} + \vec{v}$

$\therefore \quad \overrightarrow{OP} = \dfrac{\vec{u}+\vec{v}}{|\vec{u}+\vec{v}|}$ i.e. $\vec{x} = \dfrac{\vec{u}+\vec{v}}{|\vec{u}+\vec{v}|}$

But $|\vec{u}+\vec{v}|^2 = (\vec{u}+\vec{v}).(\vec{u}+\vec{v})$

$= 1 + 1 + 2\,\vec{u}.\vec{v}$

$[\because \ |\vec{u}| = |\vec{v}| = 1]$

$= 2 + 2\cos\alpha = 4\cos^2\alpha/2.$

$\therefore \ |\vec{u}| = |\vec{v}| = 2\cos\alpha/2$

$\Rightarrow \quad \vec{x} = \dfrac{1}{2}\sec(\alpha/2)\ (\vec{u}+\vec{v})$

Similarly, $\vec{y} = \dfrac{1}{2}\sec(\beta/2)\ (\vec{v}+\vec{\omega})$

$$\vec{z} = \frac{1}{2}\sec(\gamma/2)(\vec{\omega}+\vec{u})$$

Now consider $[\vec{x}\times\vec{y}\ \ \vec{y}\times\vec{z}\ \ \vec{z}\times\vec{x}]$

$= (\vec{x}\times\vec{y}).[(\vec{y}\times\vec{z})\times(\vec{z}\times\vec{x})]$

$= (\vec{x}\times\vec{y}).[\{(\vec{y}\times\vec{z}).\vec{x}\}\vec{z} - \{(\vec{y}\times\vec{z}).\vec{z}\}\vec{x}]$

[Using defⁿ of vector triple product,]

$= (\vec{x}\times\vec{y}).[[\vec{x}\,\vec{y}\,\vec{z}]\vec{z} - 0]$

$= [\vec{x}\,\vec{y}\,\vec{z}][\vec{x}\,\vec{y}\,\vec{z}] \quad [\because \ [\vec{y}\,\vec{z}\,\vec{z}] = 0]$

$= [\vec{x}\,\vec{y}\,\vec{z}]^2 \qquad ...(i)$

Also $\quad [\vec{x}\,\vec{y}\,\vec{z}] = \left[\dfrac{1}{2}\sec\dfrac{\alpha}{2}(\vec{u}+\vec{v}) \ \ \dfrac{1}{2}\sec\dfrac{\beta}{2}\right.$

$\left.(\vec{v}+\vec{\omega}) \ \ \dfrac{1}{2}\sec(\gamma/2)(\vec{w}+\vec{u}))\right]$

$= \dfrac{1}{8}\sec(\alpha/2)\sec(\beta/2)\sec(\gamma/2)[\vec{u}+\vec{v}\ \ \vec{v}+\vec{\omega}\ \ \vec{\omega}+\vec{u}]$

$= \dfrac{1}{8}\sec(\alpha/2)\sec(\beta/2)\sec(\gamma/2)$

$[(\vec{u}+\vec{v}).(\vec{v}+\vec{\omega})\times(\vec{\omega}+\vec{u})]$

$= \dfrac{1}{8}\sec(\alpha/2)\sec(\beta/2)\sec(\gamma/2)$

$$[(\vec{u}+\vec{v}).(\vec{v}\times\vec{\omega}+\vec{v}\times\vec{u}+\vec{\omega}\times\vec{u})]$$

$$= \frac{1}{8}\sec(\alpha/2)\sec(\beta/2)\sec(\gamma/2)[\vec{u}.\vec{v}\times\vec{\omega}+\vec{v}.\vec{\omega}\times\vec{u}]$$

$(\because\ [\vec{a}\,\vec{b}\,\vec{c}]=0$ when ever any two vectors are same)

$$= \frac{1}{8}\sec(\alpha/2)\sec(\beta/2)\sec(\gamma/2)\,2[\vec{u}\,\vec{v}\,\vec{\omega}]$$

$$= \frac{1}{4}\sec(\alpha/2)\sec(\beta/2)\sec(\gamma/2)\,2[\vec{u}\,\vec{v}\,\vec{\omega}]$$

$$\therefore\ [\vec{x}\,\vec{y}\,\vec{z}]^2 = \frac{1}{16}[\vec{u}\,\vec{v}\,\vec{\omega}]^2\sec^2\alpha/2\ \sec^2\beta/2\ \sec^2\gamma/2$$

....(ii)

From (i) and (ii),

$$[\vec{x}\times\vec{y}\quad\vec{y}\times\vec{z}\quad\vec{z}\times\vec{x}]$$

$$= \frac{1}{16}[\vec{u}\,\vec{v}\,\vec{\omega}]^2\qquad\sec^2\alpha/2\ \sec^2\beta/2\ \sec^2\gamma/2.$$

28. $\vec{a},\vec{b},\vec{c}$ and $\vec{d}$ are the position vectors of the points A ≡ (x, y, z) ; B ≡ (y, – 2z, 3x) ; C ≡ (2z, 3x, – y) and D ≡ (1, –1, 2) respectively. If $|\vec{a}| = 2\sqrt{3}$; $(\vec{a}\,\hat{\wedge}\,\vec{b}) = (\vec{a}\,\hat{\wedge}\,\vec{c})$; $(\vec{a}\,\hat{\wedge}\,\vec{d}) = \frac{\pi}{2}$ and $(\vec{a}\,\hat{\wedge}\,\hat{j})$ is obtuse, then find x, y, z.

यदि A ≡ (x, y, z) ; B ≡ (y, – 2z, 3x) ; C ≡ (2z, 3x, – y) एवं D≡(1,–1, 2) की स्थिति सदिश क्रमशः $\vec{a},\vec{b},\vec{c}$ एवं $\vec{d}$ है यदि $|\vec{a}| = 2\sqrt{3}$; $(\vec{a}\,\hat{\wedge}\,\vec{b}) = (\vec{a}\,\hat{\wedge}\,\vec{c})$; $(\vec{a}\,\hat{\wedge}\,\vec{d}) = \frac{\pi}{2}$ तथा $(\vec{a}\,\hat{\wedge}\,\hat{j})$ अधिक कोणीय है तो x, y, z का मान ज्ञात कीजिए।

Ans. x = 2, y = – 2, z = – 2

Sol. A (x, y, z) $\qquad |\vec{a}| = 2\sqrt{3}$

B (y, –2z, 3x)

C (2z, 3x, –y)

D (1, –1, 2)

$(\vec{a}\,\hat{\wedge}\,\vec{b}) = (\vec{a}\,\hat{\wedge}\,\vec{c}) = \theta$

$(\vec{a}\,\hat{\wedge}\,\vec{d}) = \frac{\pi}{2}$

$(\hat{a}\,\hat{\wedge}\,\hat{j}) > 90^\circ$

$\vec{a}\cdot\vec{d} = 0$

$x - y + 2z = 0$ ...(i)

$|\vec{a}|^2 = x^2 + y^2 + z^2$

$|\vec{b}|^2 = y^2 + 4z^2 + 9x^2$

$|\vec{c}| = 4z^2 + 9x^2 + y^2$

$|\vec{b}| = |\vec{c}|$

$$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|} = \frac{\vec{a}\cdot\vec{c}}{|\vec{a}||\vec{c}|}$$

$xy - 2yz + 3xz = 2xz + 3xy - yz$
$2xy + yz - xz = 0$ ...(ii)
$x^2 + y^2 + z^2 = 12$ ...(iii)
$x - y + 2z = 0$
$y = x + 2z$
$2xy + yz - xz = 0$
$(2x + z)\, y = xz$
$(2x + z)\,(x + 2z) = xz$
$2x^2 + 5xz + 2z^2 = xz$
$2x^2 + 4xz + 2z^2 = 0$
$x^2 + 2xz + z^2 = 0$
$x = -z$
$y = z$
$x = -y = -z$
$x^2 + y^2 + z^2 = 12$
$3x^2 = 12$
$x = \pm 2$
$\vec{a} \wedge \vec{j}$ = obtuse $z < 0$
$x = 2, z = -2, y = -2$

29. If $\vec{r}$ and $\vec{s}$ are non zero constant vectors and the scalar b is chosen such that $|\vec{r} + b\vec{s}|$ is minimum, then show that the value of $|b\vec{s}|^2 + |\vec{r} + b\vec{s}|^2$ is equal to $|\vec{r}|^2$.

यदि $\vec{r}$ एवं $\vec{s}$ अशून्य तथा स्थिर सदिश है। एक अदिश b ऐसा चुना जाता है कि $|\vec{r} + b\vec{s}|$ न्यूनतम है तो सिद्ध करो कि $|b\vec{s}|^2 + |\vec{r} + b\vec{s}|^2$ का मान $|\vec{r}|^2$ होगा।

Ans.

Sol. $|\vec{r} + b\vec{s}|$ is minimum

Let f(b) = $\sqrt{\vec{r}^2 + \vec{b}^2\vec{s}^2 + 2\vec{r}.b\vec{s}}$

for maxima & minima

$$f'(b) = \frac{2b\vec{s}^2 + 2\vec{r}.\vec{s}}{\sqrt{\vec{r}^2 + b^2\vec{s}^2 + 2b\vec{r}.\vec{s}}} = 0$$

$$b = -\frac{\vec{r}.\vec{s}}{\vec{s}^2}$$

$$\left|b\vec{s}\right|^2 + \left|\vec{r} + b\vec{s}\right|^2 = b^2\vec{s}^2 + \vec{r}^2 + b^2\vec{s}^2 + 2b\vec{r}.\vec{s}$$

$$= 2b^2\vec{s}^2 + \vec{r}^2 - 2b^2\vec{s}^2 = |\vec{r}|^2$$

30. Given four non zero vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$. The vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are coplanar but not collinear pair by pair and vector $\vec{d}$ is not coplanar with vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ and $(\hat{\vec{a}\vec{b}}) = (\hat{\vec{b}\vec{c}}) = \frac{\pi}{3}, (\hat{\vec{d}\vec{a}}) = \alpha, (\hat{\vec{d}\vec{b}}) = \beta$ then prove that $(\hat{\vec{d}\,\vec{c}}) = \cos^{-1}(\cos\beta - \cos\alpha)$.

चार अशून्य सदिश $\vec{a}$, $\vec{b}$, $\vec{c}$ तथा $\vec{d}$ दिया हुआ है यदि सदिश $\vec{a}$, $\vec{b}$ तथा $\vec{c}$ समतलीय है लेकिन युग्म में संरेखीय नही है तथा $\vec{d}$ सदिश $\vec{a}$, $\vec{b}$ एवं $\vec{c}$ से समतलीय नही है। यदि $(\hat{\vec{a}\vec{b}}) = (\hat{\vec{b}\vec{c}}) = \frac{\pi}{3}, (\hat{\vec{d}\vec{a}}) = \alpha, (\hat{\vec{d}\vec{b}}) = \beta$ तो सिद्ध करो कि $(\hat{\vec{d}\,\vec{c}}) = \cos^{-1}(\cos\beta - \cos\alpha)$

Ans.

Sol. $\vec{a} \wedge \vec{b} = \vec{b} \wedge \vec{c} = \frac{\pi}{3}$

$\vec{d} \wedge \vec{a} = \alpha$

$\vec{d} \wedge \vec{b} = \beta$

$\vec{a}, \vec{b}, \vec{c}$ coplanar but not collinear.

$\vec{d}$ not coplanar with $\vec{a}, \vec{b}, \vec{c}$.

Let $\vec{d} \wedge \vec{c} = \theta$

$\vec{c} = x\vec{a} + y\vec{b}$

dot with $\vec{b}$

$\vec{b}\cdot\vec{c} = x\vec{a}\cdot\vec{b} + y\left|\vec{b}\right|^2$

dot with $\vec{d}$

$\vec{d}\cdot\vec{c} = x\vec{d}\cdot\vec{a} + y\vec{d}\cdot\vec{b}$

$\left|\vec{d}\right|\left|\vec{c}\right|\cos\gamma = x\left|\vec{d}\right|\left|\vec{a}\right|\cos\alpha + y\left|\vec{d}\right|\left|\vec{b}\right|\cos\beta$

$$\cos\gamma = x\frac{\left|\vec{a}\right|}{\left|\vec{c}\right|}\cos\alpha + y\frac{\left|\vec{b}\right|}{\left|\vec{c}\right|}\cos\beta \qquad ...(1)$$

dot with $\vec{b}$

$\vec{b}\cdot\vec{c} = x\vec{b}\cdot\vec{a} + y\vec{b}\cdot\vec{b}$

$$\left|\vec{b}\right|\left|\vec{c}\right|\cos\frac{\pi}{3} = x\left|\vec{b}\right|\left|\vec{a}\right|\cos\frac{\pi}{3} + y\left|\vec{b}\right|\left|\vec{b}\right|$$

$$\frac{1}{2} = \frac{x}{2}\frac{\left|\vec{a}\right|}{\left|\vec{c}\right|} + \frac{y\left|\vec{b}\right|}{\left|\vec{c}\right|} \qquad ...(2)$$

dot with $\vec{c}$

$$\left|\vec{c}\right|\left|\vec{c}\right| = x\left|\vec{a}\right|\left|\vec{c}\right|\cos\frac{2\pi}{3} + y\left|\vec{b}\right|\left|\vec{c}\right|\cos\frac{\pi}{3}$$

$$2 = -\frac{x\left|\vec{a}\right|}{\left|\vec{c}\right|} + \frac{y\left|\vec{b}\right|}{\left|\vec{c}\right|} \qquad ...(3)$$

from equation (2) & (3)

$$3 = \frac{3y\left|\vec{b}\right|}{\left|\vec{c}\right|}$$

$$\frac{y\left|\vec{b}\right|}{\left|\vec{c}\right|} = 1$$

$$\frac{x\left|\vec{a}\right|}{\left|\vec{c}\right|} = -1$$

Put these in equation (1)

$\cos\gamma = -\cos\alpha + \cos\beta$

$\cos\gamma = \cos\beta - \cos\alpha$

$\gamma = \cos^{-1}\left(\cos\beta - \cos\alpha\right)$

31. The position vectors of the vertices A, B and C of a tetrahedron are (1, 1, 1), (1, 0, 0) and (3, 0, 0) respectively. The altitude from the vertex D to the opposite face ABC meets the median line through A of the triangle ABC at a point E. If the length of side AD is 4 and volume of the tetrahedron is $\frac{2\sqrt{2}}{3}$ then find the all possible position vectors of the point E.

किसी चतुष्फलक के शीर्ष A, B एवं C की स्थिति सदिश क्रमशः (1, 1, 1), (1, 0, 0) तथा (3, 0, 0) है यदि शीर्ष D से विपरीत तल ABC पर लम्ब डाला जाता है तथा यह त्रिभुज ABC के शीर्ष A से जाने वाली माध्यिका पर बिन्दु E पर मिलता है यदि AD की लम्बाई 4 तथा चतुष्फलक का आयतन $\frac{2\sqrt{2}}{3}$ है तो E की स्थिति सदिश बताओं।

Ans. $(-1, 3, 3)$ & $(3, -1, -1)$

Sol. As, F is the mid point of BC. So, $\vec{F} = 2\hat{i}$

As, AE is perpendicular to DE. Let E divides AF in the ratio $\lambda : 1$

$$\vec{E} = \frac{2\lambda\hat{i} + 1(\hat{i} + \hat{j} + \hat{k})}{\lambda + 1} = \frac{2\lambda + 1}{\lambda + 1}\hat{i} + \frac{1}{\lambda + 1}\hat{j} + \frac{1}{\lambda + 1}\hat{k}$$

$$\text{Volume of tetrahedron} = 1/3 \times \text{Area of base} \times \text{Height} = \frac{2\sqrt{2}}{3}$$

$$\text{Area of base} = \frac{1}{2}|\overrightarrow{BC} X \overrightarrow{BA}| = \sqrt{2}$$

$$\frac{1}{3} \cdot \sqrt{2} \cdot DE = 2\frac{\sqrt{2}}{3}$$

Hence $DE = 2$

As $\angle ADE$ is right angle

So $AD^2 = AE^2 + DE^2$

$AE^2 = 12$

$$\overrightarrow{AE} = \vec{E} - \vec{A} = \frac{\lambda}{\lambda + 1}\hat{i} - \frac{\lambda}{\lambda + 1}\hat{j} - \frac{\lambda}{\lambda + 1}\hat{k}$$

$$|\overrightarrow{AE}|^2 = 12$$

$$3\frac{\lambda^2}{(\lambda + 1)^2} = 12$$

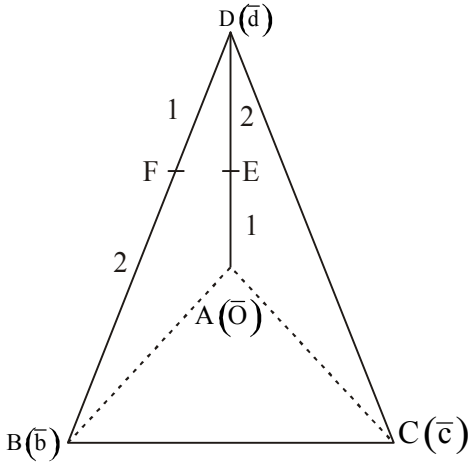
On solving $\lambda = -2, -\frac{2}{3}$

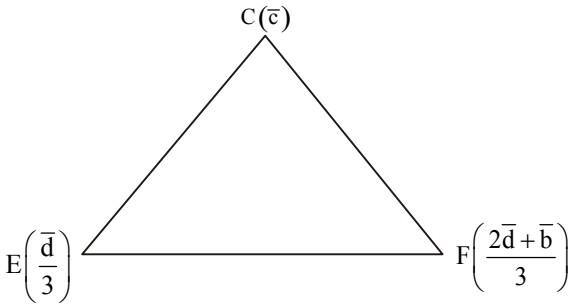
32. The length of the edge of the regular tetrahedron $D-ABC$ is 'a'. Point E and F are taken on the edges AD and BD respectively such that E divides $\vec{DA}$ and F divides $\vec{BD}$ in the ratio 2:1 each. Then find the area of triangle CEF.

एक समचतुष्फलक D – ABC की भुजा 'a' है यदि भुजा AD एवं BD पर बिन्दु E तथा F इस प्रकार से लिया जा रहा है कि fcUnq E $\vec{DA}$ को तथा बिन्दु F सदिश $\vec{BD}$ को 2:1 के अनुपात में बाट रही है तो त्रिभुज CEF का क्षेत्रफल बताइये।

Ans. $\frac{5a^2}{12\sqrt{3}}$ sq. units

Sol.





$c(\bar{c}), E\left(\frac{\bar{d}}{3}\right)$ and $F\left(\frac{2\bar{d}+\bar{b}}{3}\right)$

$\bar{c} \cap \bar{d} = \bar{b} \wedge \bar{d} = \bar{b} \wedge \bar{c} = 60$

$$\overline{CE} = \left|\frac{\bar{d}}{3} - \bar{c}\right| = \sqrt{\frac{a^2}{9} + a^2 - \frac{2}{3}a^2 \cdot \frac{1}{2}} = \frac{\sqrt{7}a}{3}$$

$$\overline{CF} = \left|\frac{2\bar{d}+\bar{b}}{3} - \bar{c}\right| = \sqrt{\frac{4a^2}{9} + \frac{a^2}{9} + a^2 + 2\left(\frac{2}{9}a^2 \cdot \frac{1}{2} - \frac{a^2}{3} \cdot \frac{1}{2} - \frac{2a^2}{3} \cdot \frac{1}{2}\right)}$$

$$= \frac{\sqrt{7}a}{3}$$

$$\overline{EF} = \left|\frac{\bar{b}+\bar{d}}{3}\right| = \frac{1}{3}\sqrt{a^2 + a^2 + 2 \cdot a^2 \cdot \frac{1}{2}} = \frac{\sqrt{3}a}{3} = \frac{a}{\sqrt{3}}$$

Now, use heron's formula for area.

33. Given three points on the xy plane on O(0, 0), A(1, 0) and B(–1, 0). Point P is moving on the plane satisfying the condition $\left(\overrightarrow{PA}\cdot\overrightarrow{PB}\right)+3\left(\overrightarrow{OA}\cdot\overrightarrow{OB}\right)=0$. If the maximum and minimum values of $\left|\overrightarrow{PA}\right|\left|\overrightarrow{PB}\right|$ are M and *m* respectively then find the value of $M^2 + m^2$.

xy तल पर O(0, 0), A(1, 0) तथा B(–1, 0) दिया हुआ है इसी तल पर एक बिन्दु P इस प्रकार से गति कर रहा है कि $\left(\overrightarrow{PA}\cdot\overrightarrow{PB}\right)+3\left(\overrightarrow{OA}\cdot\overrightarrow{OB}\right)=0$. यदि $\left|\overrightarrow{PA}\right|\left|\overrightarrow{PB}\right|$ का अधिकतम तथा न्यूनतम मान M एवं *m* है तो $M^2 + m^2$ का मान बताओं।

Ans. 34

Sol. O (0,0), A(1,0) & B (–1, 0)

Let P (x,y)

$\overrightarrow{PA} = (1-x)\hat{i} - y\hat{j}$

$\overrightarrow{PB} = -(1+x)\hat{i} - y\hat{j}$

$\overrightarrow{PA}.\overrightarrow{PB} + 3\ \overrightarrow{OA}.\overrightarrow{OB} = 0$

$\Rightarrow (x^2 - 1) + y^2 - 3 = 0$

$x^2 + y^2 = 4$ _____(1)

$\left|\overrightarrow{PA}\right|\left|\overrightarrow{PB}\right| = \sqrt{(x-1)^2+y^2}\sqrt{(x+1)^2+y^2}$

$= \sqrt{5-2x}\cdot\sqrt{5+2x}$

$= \sqrt{25-4x^2}$, $x\in(-2,2)$ (from (1))

so M = 5, m = 3

$\Rightarrow M^2 + m^2 = 25 + 9 = 34$

Paragraph for Question Nos. 34 to 36

Let $|\vec{a}| = |\vec{b}| = 2$ and $|\vec{c}| = 1$. Also $(\vec{a}-\vec{c})\cdot(\vec{b}-\vec{c}) = 0$ and $|\vec{a}-\vec{b}|^2 + |\vec{a}+\vec{b}|^2 = 16$

माना $|\vec{a}| = |\vec{b}| = 2$ तथा $|\vec{c}| = 1$ साथ ही $(\vec{a}-\vec{c})\cdot(\vec{b}-\vec{c}) = 0$ तथा $|\vec{a}-\vec{b}|^2 + |\vec{a}+\vec{b}|^2 = 16$

34. $|\vec{a}-\vec{b}|^2 + 2\vec{c}\cdot(\vec{a}+\vec{b})$ has the value equal to :

$|\vec{a}-\vec{b}|^2 + 2\vec{c}\cdot(\vec{a}+\vec{b})$ का मान बराबर होगा :

(A) 12 (B) 10 (C) 8 (D) 6

Ans. B

35. $\left|\vec{a}+\vec{b}-\vec{c}\right|^2$ equals :

$\left|\vec{a}+\vec{b}-\vec{c}\right|^2$ बराबर है :

(A) 5 (B) 6 (C) 7 (D) 8

Ans. C

36. Difference of the maximum and minimum value of $\left|\vec{a}+\vec{b}\right|$ is equal to :

$\left|\vec{a}+\vec{b}\right|$ के अधिकतम एवं न्यूनतम मान का अंतर है :

(A) 2 (B) 3 (C) 4 (D) 1

Ans. A

Sol. Given $\left|\vec{a}-\vec{b}\right|^2+\left|\vec{a}+\vec{b}\right|^2=16$ ...(1) ; $|\vec{a}|=|\vec{b}|=2$ and $|\vec{c}|=1$

Also, $\left|\vec{a}-\vec{b}\right|^2=\left|(\vec{a}-\vec{c})-(\vec{b}-\vec{c})\right|^2$

$\left|\vec{a}-\vec{b}\right|^2=(\vec{a}-\vec{c})^2+(\vec{b}-\vec{c})^2-\underbrace{2(\vec{a}-\vec{c})\cdot(\vec{b}-\vec{c})}_{\text{zero}}$

$=\vec{a}^2+\vec{c}^2-2\vec{a}\cdot\vec{c}+\vec{b}^2+\vec{c}^2-2\vec{b}\cdot\vec{c}$

$\left|\vec{a}-\vec{b}\right|^2=10-2\left(\vec{a}\cdot\vec{c}+\vec{b}\cdot\vec{c}\right)$ ...(2)

$\Rightarrow \left|\vec{a}-\vec{b}\right|^2+2\vec{c}\cdot\left(\vec{a}+\vec{b}\right)=10 \Rightarrow$ B Ans.

subsituting $\left|\vec{a}-\vec{b}\right|^2=16-\left|\vec{a}+\vec{b}\right|^2$ from (1) in (2)

we get, $16-\left|\vec{a}+\vec{b}\right|^2=10-2\left(\vec{a}\cdot\vec{c}+\vec{b}\cdot\vec{c}\right)$

$\left|\vec{a}+\vec{b}\right|^2-2\vec{c}\cdot(\vec{a}+\vec{b})=6$ .....(3)

$\left|\vec{a}+\vec{b}\right|^2-2\vec{c}\cdot(\vec{a}+\vec{b})+\vec{c}^2=6+\vec{c}^2=7$

$\left|\vec{a}+\vec{b}-\vec{c}\right|^2=7 \Rightarrow$ C

Again, using triangle inequality, we have

$\left|\vec{a}+\vec{b}\right|-|\vec{c}| \le \left|\vec{a}+\vec{b}-\vec{c}\right| \le \left|\vec{a}+\vec{b}\right|+|\vec{c}|$

Hence $\left|\vec{a}+\vec{b}-\vec{c}\right|-|\vec{c}| \le \left|\vec{a}+\vec{b}\right| \le \left|\vec{a}+\vec{b}-\vec{c}\right|+|\vec{c}|$

$\sqrt{7}-1 \le \left|\vec{a}+\vec{b}\right| \le \sqrt{7}+1$

$\therefore \left|\vec{a}+\vec{b}\right|_{max} = \sqrt{7}+1$

and $\left|\vec{a}+\vec{b}\right|_{min} = \sqrt{7}-1$

$\therefore$ Difference $= 2 \Rightarrow$ A

Note: $\sqrt{16-(8+2\sqrt{7})} \le \left|\vec{a}-\vec{b}\right| \le \sqrt{16-(8-2\sqrt{7})}$

$\Rightarrow \sqrt{7}-1 \le \left|\vec{a}-\vec{b}\right| \le 1+\sqrt{7}$

Paragraph for Question Nos. 37 to 39

Consider the three vectors $\vec{p}$, $\vec{q}$ and $\vec{r}$ such that

$\vec{p}=\hat{i}+\hat{j}+\hat{k}$; $\vec{q}=\hat{i}-\hat{j}+\hat{k}$

$\vec{p}\times\vec{r}=\vec{q}+c\vec{p}$ and $\vec{p}\cdot\vec{r}=2$

माना कि सदिश $\vec{p}$, $\vec{q}$ तथा $\vec{r}$

$\vec{p}=\hat{i}+\hat{j}+\hat{k}$; $\vec{q}=\hat{i}-\hat{j}+\hat{k}$

$\vec{p}\times\vec{r}=\vec{q}+c\vec{p}$ तथा $\vec{p}\cdot\vec{r}=2$

37. The value of $[\vec{p}\ \vec{q}\ \vec{r}]$ is :

(A) $-\frac{5\sqrt{2}c}{|\vec{r}|}$ (B) $-\frac{8}{3}$ (C) 0 (D) greater then zero

$[\vec{p}\ \vec{q}\ \vec{r}]$ का मान होगा :

(A) $-\frac{5\sqrt{2}c}{|\vec{r}|}$ (B) $-\frac{8}{3}$ (C) 0 (D) 0 से अधिक

Ans. B

Sol. Given $\vec{p}=\hat{i}+\hat{j}+\hat{k}$

$\vec{q}=\hat{i}-\hat{j}+\hat{k}$

$\vec{p}\times\vec{r}=\vec{q}+c\vec{p}$ and $\vec{p}\,.\,\vec{r}=2$

cross with $\vec{p}$ dot with $\vec{p}$

$\therefore \quad \vec{p} \times (\vec{p} \times \vec{r}) = \vec{p} \times (\vec{q} + c\vec{p})$ and $\vec{p} \cdot (\vec{p} \times \vec{r}) = \vec{p} \cdot (\vec{q} + c\vec{p})$

$\therefore \quad (\vec{p} \cdot \vec{r})\vec{p} - (\vec{p} \cdot \vec{p})\vec{r} = \vec{p} \times \vec{q} + c\vec{0}$ $0 = \vec{p} \cdot \vec{q} + c\,(\vec{p} \cdot \vec{p})$

$\therefore \quad (\vec{p} \cdot \vec{p})\vec{r} = (\vec{p} \cdot \vec{r})\,\vec{p} - \vec{p} \times \vec{q}$ ... (i) $c = -\dfrac{\vec{p} \cdot \vec{q}}{\vec{p} \cdot \vec{p}}$ ... (ii)

But $\vec{p} \cdot \vec{p} = |\vec{p}| = p^2 = 3$ ... (iii)

$\vec{p} \cdot \vec{q} = 1 - 1 + 1 = 1$ ... (iv)

$$\vec{p} \times \vec{q} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & -1 & 1 \end{vmatrix} = 2\hat{i} - 2\hat{k} \quad \text{... (v)}$$

using (iii), (iv), (v) in (i) and (ii), we get

$3\vec{r} = 2\vec{p} - 2i + 2k$ and $c = -\dfrac{1}{3}$ ....(vii)

$\therefore \quad \vec{r} = \dfrac{1}{3}\left[2\hat{i} + 2\hat{j} + 2\hat{k} - 2\hat{i} + 2\hat{k}\right]$

$\therefore \quad \vec{r} = \dfrac{2}{3}(\hat{j} + 2\hat{k})$ ....(vi)

Now $$[\vec{p}\ \vec{q}\ \vec{r}] = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 0 & \frac{1}{2} & \frac{4}{3} \end{vmatrix} = \left(\frac{-4}{3} - \frac{2}{3}\right) - 1\left(\frac{4}{3}\right) + 1\left(\frac{2}{3}\right)$$

$\therefore \quad [\vec{p}\ \vec{q}\ \vec{r}] = -2 - \dfrac{2}{5} = -\dfrac{8}{3}$ ... (viii)

38. If $\vec{x}$ is a vector such that $[\vec{p}\,\vec{q}\,\vec{r}]\vec{x} = (\vec{p} \times \vec{q}) \times \vec{r}$, then $\vec{x}$ is :

(A) $c\left(\hat{i} - 2\hat{j} + \hat{k}\right)$ (B) a unit vector

(C) indeterminate, as $[\vec{p}\,\vec{q}\,\vec{r}]$ (D) $-\dfrac{1}{2}(\hat{i} - 2\hat{j} + \hat{k})$

यदि $\vec{x}$ इस प्रकार है कि $[\vec{p}\,\vec{q}\,\vec{r}]\vec{x} = (\vec{p} \times \vec{q}) \times \vec{r}$, तो $\vec{x}$ है :

(A) $c\left(\hat{i} - 2\hat{j} + \hat{k}\right)$ (B) एक इकाई सदिश

(C) $[\vec{p}\,\vec{q}\,\vec{r}]$ के समान अनिरधार्य (D) $-\dfrac{1}{2}(\hat{i} - 2\hat{j} + \hat{k})$

Ans. D

Sol. $[\vec{p}\ \vec{q}\ \vec{r}]\vec{x} = (\vec{p}\times\vec{q})\times\vec{r}$

$\Rightarrow \quad \left(-\frac{8}{3}\right)\vec{x} = (\vec{p}\cdot\vec{r})\vec{q} - (\vec{q}\cdot\vec{r})\vec{p}$

or $\quad \left(-\frac{8}{3}\right)\vec{x} = 2\vec{q} - \frac{2\vec{p}}{3} \quad \left(\vec{q}\cdot\vec{r} = \frac{2}{3} \text{ Verify your self}\right)$

$\therefore \quad \vec{x} = -\frac{3}{8}\cdot\frac{2}{3}(3\vec{q} - \vec{p}) = -\frac{1}{4}(3\hat{i} - 3\hat{j} + 3\hat{k} - \hat{i} - \hat{j} - \hat{k}) = -\frac{1}{4}(2\vec{i} - 4\vec{j} + 2\vec{k})$

$\therefore \quad \vec{x} = -\frac{1}{2}(\hat{i} - 2\hat{j} + \hat{k})$

39. If $\vec{y}$ is a vector satisfying $(1 + c)\,\vec{y} = \vec{p}\times(\vec{q}\times\vec{r})$ then the vectors $\vec{x}, \vec{y}, \vec{r}$:

(A) are collinear

(B) are coplanar

(C) represent the coterminus edges of a tetrahedron whose volume is | c | cubic units.

(D) represent the coterminus edges of a parallelepiped whose volume is | c | cubic units

यदि $\vec{y}$ एक सदिश है जो $(1 + c)\,\vec{y} = \vec{p}\times(\vec{q}\times\vec{r})$ को संतुष्ट करता है, तो सदिश $\vec{x}, \vec{y}, \vec{r}$ है :

(A) समरेखीय

(B) समतलीय

(C) एक चतुष्फलक की संगामी भुजाऐं दर्शाते है जिसका आयतन | c | घन इकाई है

(D) एक समान्तर षट्फलक की संगामी भुजाऐं दर्शाते है जिसका आयतन | c | घन इकाई है

Ans. C

Sol. As $\quad c = -\frac{1}{3}$ from (vii)

$\therefore \quad \left(1 - \frac{1}{3}\right)\vec{y} = (\vec{p}\cdot\vec{r})\vec{q} - (\vec{p}\cdot\vec{q})\vec{r}$

$\therefore \quad \frac{2}{3}\vec{y} = 2\vec{q} - (1)\vec{r} \qquad (\text{As } \vec{p}\cdot\vec{q} = 1 \text{ from .... (iv)})$

$\therefore \quad \vec{y} = \frac{3}{2}\left(2\hat{i} - 2\hat{j} + 2\hat{k} - \frac{2}{3}\hat{j} - \frac{4}{3}\hat{k}\right) = \frac{3}{2}\,\frac{(6\hat{i} - 8\hat{j} + 2\hat{k})}{3}$

$\vec{y} = 3\hat{i} - 4\hat{j} + \hat{k}$

$\therefore \quad [\vec{x}\ \vec{y}\ \vec{r}] = \begin{vmatrix} -\frac{1}{2} & 1 & -\frac{1}{2} \\ 3 & -4 & 1 \\ 0 & \frac{2}{3} & \frac{4}{3} \end{vmatrix} = -\frac{1}{2}\left(-\frac{16}{3} - \frac{2}{3}\right) - 1\,(4) - \frac{1}{2}\,(2)$

$\therefore \quad [\vec{x}\ \vec{y}\ \vec{r}\,] = 3 - 4 - 1 = -2$

$\therefore \quad \left|\frac{1}{6}[\vec{x}\ \vec{y}\ \vec{r}\,]\right| = \left|-\frac{1}{3}\right| = |c|$

$\Rightarrow$ $\vec{x}, \vec{y}, \vec{r}$ are the coterminus edges of a tetrahedron whose volume is $|c|$.

Paragraph for Question Nos. 40 to 42

Suppose the three vectors $\vec{a}, \vec{b}, \vec{c}$ on a plane satisfy the condition that $|\vec{a}| = |\vec{b}| = |\vec{c}| = |\vec{a} + \vec{b}| = 1$;

$\vec{c}$ is perpendicular to $\vec{a}$ and $\vec{b} \cdot \vec{c} > 0$, then :

माना कि किसी तल में तीन सदिश $\vec{a}, \vec{b}, \vec{c}$ है जो कि दिये हुये प्रतिबन्धों को सन्तुष्ट करते है $|\vec{a}| = |\vec{b}| = |\vec{c}| = |\vec{a} + \vec{b}| = 1$;

यदि $\vec{c}$ सदिश $\vec{a}$ के लम्बवत् है तथा $\vec{b} \cdot \vec{c} > 0$, तो :

40. Find the angle formed by $2\vec{a} + \vec{b}$ and $\vec{b}$.

$2\vec{a} + \vec{b}$ एवं $\vec{b}$ के बीच का कोण बताइये।

Ans. $\frac{\pi}{2}$

Sol. $\left|\bar{a} + \bar{b}\right|^2 = a^2 + b^2 + 2ab\cos\theta = 1$

$\Rightarrow \cos\theta = -\frac{1}{2}$ and $\bar{a} \cdot \bar{b} = -\frac{1}{2}$

$$\cos\theta = \frac{\bar{b} \cdot \left(2\bar{a} + \bar{b}\right)}{\left|\bar{b}\right|\left|2\bar{a} + \bar{b}\right|} = \frac{-1+1}{\left|\bar{b}\right|\left|2\bar{a} + \bar{b}\right|} = 0$$

So, $\theta = \frac{\pi}{2}$

41. If the vector $\vec{c}$ is expressed as a linear combination $\lambda\vec{a} + \mu\vec{b}$ then find the ordered pair (λ, μ).

यदि सदिश $\vec{c}$ को एक घात संचय के रूप में यदि $\lambda\vec{a} + \mu\vec{b}$ से व्यक्त किया जाय तो (λ, μ) का मान बताइये।

Ans. $\left(\frac{1}{\sqrt{3}}, \frac{2}{\sqrt{3}}\right)$

Sol. $\left|\bar{a}+\bar{b}\right|^2 = a^2+b^2+2ab\cos\theta = 1$

$\Rightarrow \cos\theta = -\frac{1}{2}$ and $\bar{a}\cdot\bar{b} = -\frac{1}{2}$

$\bar{c} = \lambda\bar{a}+\mu\bar{b}$

$\bar{c}\cdot\bar{c} = \lambda a^2 + \mu\left(\bar{a}\cdot\bar{b}\right)$

$\Rightarrow 0 = \lambda - \frac{\mu}{2}$ $\qquad\qquad$ $\Rightarrow m = 2\lambda$

and $|\bar{c}| = 1$

$\Rightarrow \lambda^2a^2 + \mu^2b^2 + 2\lambda\mu\left(\bar{a}\cdot\bar{b}\right) = 1$

$\lambda^2 + 4\lambda^2 - 2\lambda^2 = 1$

$\Rightarrow 3\lambda^2 = 1$

$\Rightarrow \lambda = \frac{1}{\sqrt{3}}$ $\qquad\qquad$ So, $\mu = \frac{2}{\sqrt{3}}$

Note : $\bar{b}\cdot\bar{c} > 0$

$\Rightarrow \lambda\left(\bar{a}\cdot\bar{b}\right)+\mu b^2 > 0$

$\Rightarrow \mu - \frac{\lambda}{2} > 0$

$\Rightarrow \mu > \frac{\lambda}{2}$

42. For real numbers x, y the vector $\vec{p} = x\vec{a} + y\vec{c}$ satisfies the condition $0 \le \vec{p}\cdot\vec{a} \le 1$ and $0 \le \vec{p}\cdot\vec{b} \le 1$. Find the maximum value of $\vec{p}\cdot\vec{c}$.

वास्तविक संख्या x एवं y के लिये यदि सदिश $\vec{p} = x\vec{a} + y\vec{c}$ यदि शर्त $0 \le \vec{p}\cdot\vec{a} \le 1$ एवं $0 \le \vec{p}\cdot\vec{b} \le 1$ को संतुष्ट कर रहा है तो $\vec{p}\cdot\vec{c}$ अधिकतम मान बताओं।

Ans. $\sqrt{3}$

Sol. $\bar{p} = x\bar{a} + y\bar{c}$

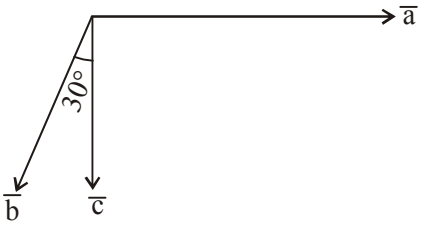
$\bar{p}\cdot\bar{a}=x \qquad \Rightarrow 0\le x\le 1$

$\bar{p}\cdot\bar{b}-x\left(\frac{-1}{2}\right)+y\left(\bar{b}\cdot\bar{c}\right) \qquad \Rightarrow 0\le -\frac{x}{2}+y\left(\bar{b}\cdot\bar{c}\right)\le 1 \qquad$ _____(i)

and $\bar{p}\cdot\bar{c}=y$

$\bar{a},\bar{b},\bar{c}$ are in same plane

$\bar{a}\cdot\bar{b}=-\frac{1}{2} \qquad \hat{a}\wedge\bar{b}=120°$



and $\bar{a}\perp\bar{c}$

From equation (i)

$$y=\frac{1+\frac{x}{2}}{\bar{b}\cdot\bar{c}}\le\frac{(2+x)}{2\left(\bar{b}\cdot\bar{c}\right)}$$

$$\Rightarrow y_{max}=\frac{2+1}{2\left(\bar{b}\cdot\bar{c}\right)} \qquad \{\because\ 0\le x\le 1\}$$

$$=\frac{3}{2\cdot\frac{\sqrt{3}}{2}}=\sqrt{3}$$

43. Match the statements/expressions given in Column I with the values given in Column II :

Column I	**Column II**
A. Root(s) of the equation $2\sin^2\theta+\sin^2 2\theta=2$	p. $\frac{\pi}{6}$
B. Points of discontinuity of the function $f(x)=\left[\frac{6x}{\pi}\right]\cos\left[\frac{3x}{\pi}\right]$, where [y] denotes the largest integer less than or equal to y	q. $\frac{\pi}{4}$
C. Volume of the parallelopiped with its edges represented by the vectors $\hat{i}+\hat{j},\ \hat{i}+2\hat{j}$ and $\hat{i}+\hat{j}+\pi\hat{k}$	r. $\frac{\pi}{3}$
D. Angle between vectors $\vec{a}$ and $\vec{b}$ where $\vec{a},\vec{b}$ and $\vec{c}$ are unit vectors satisfying $\vec{a}+\vec{b}+\sqrt{3}\,\vec{c}=\vec{0}$	s. $\frac{\pi}{2}$
	t. π

कॉलम–I में दिए गये वक्तयों/व्यंजको का कॉलम–II में दिये गये मानों से सुमेलित कीजिए :

Column I	**Column II**
A. समीकरण $2\sin^2\theta + \sin^2 2\theta = 2$ के मूल है	p. $\frac{\pi}{6}$
B. फलन $f(x) = \left[\frac{6x}{\pi}\right]\cos\left[\frac{3x}{\pi}\right]$ के असतत्ता बिन्दु [y] उस अधिकतम पूर्णांक को दर्शाता है जो कि y के बराबर या उससे कम हो	q. $\frac{\pi}{4}$
C. सदिशो $\hat{i}+\hat{j}, \hat{i}+2\hat{j}$ तथा $\hat{i}+\hat{j}+\pi\hat{k}$ से निरूपित कोरों वाले समान्तर षट्फलक का आयतन	r. $\frac{\pi}{3}$
D. सदिशो $\vec{a}$ एवं $\vec{b}$ के बीच को कोण, जबकि $\vec{a},\vec{b}$ व $\vec{c}$, $\vec{a}+\vec{b}+\sqrt{3}\vec{c}=\vec{0}$ को संतुष्ट करने वाले मात्रक सदिश है	s. $\frac{\pi}{2}$
	t. π

Ans. (A) Q, S ; (B) P, R, S, T ; (C) T ; (D) R

Sol. (A) $\sin^2 2\theta = 2 - 2\sin^2\theta = 2\cos^2\theta$

$4\sin^2\theta\cos^2\theta = 2\cos^2\theta$

$\cos\theta = 0$ OR $\sin^2\theta = \frac{1}{2}$

(B) $\frac{6x}{\pi} = I_1 \quad \Rightarrow x = \frac{\pi I_1}{6}$

and $\frac{3x}{\pi} = I_2 \quad \Rightarrow x = \frac{I_2\pi}{3}$

$x = \frac{\pi}{6}, \frac{\pi}{3}, \frac{\pi}{2}, \pi \qquad x \neq \frac{\pi}{4}$

(C) $\text{Volume} = \left[\hat{i}+\hat{j}, \hat{i}+2\hat{j}, \hat{i}+\hat{j}+\pi\hat{k}\right] = \pi$

(D) $\bar{a}+\bar{b} = -\sqrt{3}\bar{c}$

$a^2 + b^2 + 2ab + \cos\theta = 3c^2$

$\Rightarrow \cos\theta = \frac{1}{2} \qquad \Rightarrow \theta = \frac{\pi}{3}$

44. Let $\hat{a}$, $\hat{b}$ and $\hat{c}$ are three unit vectors mutually inclined to each other at an angle θ. $\hat{d}$ is also a unit vector such that $\hat{d}\cdot\hat{a} = \hat{d}\cdot\hat{b} = \hat{d}\cdot\hat{c} = \cos\alpha$. Then match the entries for θ given in column-I with their corresponding values for α given in column-II.

यदि $\hat{a}$, $\hat{b}$ तथा $\hat{c}$ तीन इकाई सदिश इस प्रकार है कि वे परस्पर एक दूसरे से θ कोण बनाते है यदि $\hat{d}$ की एक इकाई सदिश इस प्रकार से है कि $\hat{d}\cdot\hat{a} = \hat{d}\cdot\hat{b} = \hat{d}\cdot\hat{c} = \cos\alpha$,तो स्तम्भ–I में θ को स्तम्भ–II में α से मिलान कीजिए।

Column-I (θ)		**Column-II** (α)	
A.	$\frac{\pi}{2}$	p.	$\frac{\pi}{4}$
B.	$\frac{\pi}{3}$	q.	$\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$
C.	$\cos^{-1}\left(\frac{1}{4}\right)$	r.	$\cos^{-1}\sqrt{\frac{2}{3}}$
D.	$\cos^{-1}\left(\frac{5}{8}\right)$	s.	$\frac{\pi}{6}$

Ans. (A) Q (B) R (C) P (D) R

Sol. $\hat{d} = k\left(\hat{a}+\hat{b}+\hat{c}\right)$ _____(i)

$\hat{d}\cdot\hat{a} = k\left(\hat{a}\cdot\hat{a}+\hat{a}\cdot\hat{b}+\hat{a}\cdot\hat{c}\right)$

$\Rightarrow \cos\alpha = k(1 + 2\cos\theta)$ _____(ii)

dot with $\hat{d}$

$\hat{d}\cdot\hat{d} = k\left(\hat{d}\cdot\hat{a}+\hat{d}\cdot\hat{b}+\hat{d}\cdot\hat{c}\right)$

$1 = k(3\cos\alpha)$ _____(iii)

(ii) + (iii)

$|3\cos^2\alpha = 1 + 2\cos\theta|$

45. Match the statements in **Column-I** with those in **Column-II.** **[IIT-JEE-2010,]**

Column-I		**Column-II**	
(A)	A line from the origin meets the lines	(p)	– 4

$\frac{x-2}{1}=\frac{y-1}{-2}=\frac{z+1}{1}$ and $\frac{x-\frac{8}{3}}{2}=\frac{y+3}{-1}=\frac{z-1}{1}$

at P and Q respectively. If length PQ = d, then d^2 is

(B) The values of x satisfying (q) 0

$\tan^{-1}(x+3)-\tan^{-1}(x-3)=\sin^{-1}\left(\frac{3}{5}\right)$ are

(C) Non-zero vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ satisfy $\vec{a}\cdot\vec{b}=0$, (r) 4

$(\vec{b}-\vec{a}).(\vec{b}+\vec{c})=0$ and $2|\vec{b}+\vec{c}|=|\vec{b}-\vec{a}|$. If $\vec{a}=\mu\vec{b}+4\vec{c}$ then possible value of μ are

(D) Let f be the function on $[-\pi, \pi]$ given by (s) 5

$f(0)=9$ and $f(x)=\frac{\sin\left(\frac{9x}{2}\right)}{\sin\left(\frac{x}{2}\right)}$ for $x\neq 0$. The value

of $\frac{2}{\pi}\int_{-\pi}^{\pi} f(x)dx$ is

(t) 6

कॉलम **-I** में दिए गए वक्तव्यों का कॉलम **-II** में दिए वक्तव्यों से सुमेल करें।

कॉलम -I **कॉलम -II**

(A) मूल बिन्दु से खींची गयी एक सरल रेखा, सरल रेखाओं (p) – 4

$\frac{x-2}{1}=\frac{y-1}{-2}=\frac{z+1}{1}$ तथा $\frac{x-\frac{8}{3}}{2}=\frac{y+3}{-1}=\frac{z-1}{1}$

को क्रमशः P एवं Q पर काटती है। यदि लम्बाई PQ = d हो तो d^2 है–

(B) $\tan^{-1}(x+3)-\tan^{-1}(x-3)=\sin^{-1}\left(\frac{3}{5}\right)$ को संतुष्ट करने वाले (q) 0

x के मान हैं–

(C) अशुन्य सदिश $\vec{a}$, $\vec{b}$ तथा $\vec{c}$, $\vec{a}\cdot\vec{b}=0$, (r) 4

$(\vec{b}-\vec{a}).(\vec{b}+\vec{c})=0$ तथा $2|\vec{b}+\vec{c}|=|\vec{b}-\vec{a}|$ को संतुष्ट करते हैं।

यदि $\vec{a}=\mu\vec{b}+4\vec{c}$ तो μ का/के के संभावित मान है/हैं–

(D) मानाकि $[-\pi, \pi]$ पर फलन f; $f(0)=9$ तथा $f(x)=\frac{\sin\left(\frac{9x}{2}\right)}{\sin\left(\frac{x}{2}\right)}$ (s) 5

; $x\neq 0$ द्वारा परिभाषित है। तो $\frac{2}{\pi}\int_{-\pi}^{\pi} f(x)dx$ का मान है–

(t) 6

Ans. **(A) → (t), (B) → (p, r), (C) → (q,s) (D) → (r)**

Sol. **(A)** Let the line through origin is $\frac{x}{\lambda}=\frac{y}{\mu}=\frac{z}{1}$

$\Rightarrow x = \lambda z, y = \mu z$ .....(1)

To find point of intersection of line (1) and line $\frac{x-2}{1} = \frac{y-1}{-2} = \frac{z+1}{1}$ .....(2)

we have $\frac{\lambda z - 2}{1} = \frac{\mu z - 1}{-2} = z + 1$

$\Rightarrow z = \frac{3}{\lambda - 1} = \frac{-1}{\mu + 2}$

$\Rightarrow \lambda + 3\mu + 5 = 0$ .....(3)

To find point of intersection of line (1) and line $\frac{x - \frac{8}{3}}{2} = \frac{y+3}{-1} = \frac{z-1}{1}$ .....(4)

we have $\frac{\lambda z - \frac{8}{3}}{2} = \frac{\mu z + 3}{-1} = \frac{z-1}{1}$

$\Rightarrow z = \frac{2}{3(\lambda - 2)} = \frac{-2}{\mu + 1}$

$\Rightarrow 3\lambda + \mu = 5$ .....(5)

Solving (3) and (5), $\lambda = \frac{5}{2}$ and $\mu = -\frac{5}{2}$

$\therefore$ $z = 2, x = 5, y = -5$ for point P

and $z = \frac{4}{3}$, $x = \frac{10}{3}$, $y = -\frac{10}{3}$ for point Q

$\therefore$ $PQ^2 = \frac{4}{9} + \frac{25}{9} + \frac{25}{9} = 6$

(B) $\tan^{-1}(x+3) - \tan^{-1}(x-3) = \sin^{-1}(3/5)$

$\Rightarrow \tan^{-1}\left(\frac{x+3-x+3}{1+x^2-9}\right) = \tan^{-1}(3/4)$

$\Rightarrow \frac{6}{x^2-8} = \frac{3}{4} \Rightarrow x^2 = 16$

$\therefore$ $x = \pm 4$

(C) Since $\vec{a} \cdot \vec{b} = 0$

$\therefore$ Let $\vec{b} = \lambda_1 \hat{i}$, $\vec{a} = \lambda_2 \hat{j}$

Now $2|\vec{b} + \vec{c}| = |\vec{b} - \vec{a}|$ & $\vec{a} = \mu\vec{b} + 4\vec{c}$ $\Rightarrow$ $2\left|\lambda_1\hat{i} + \frac{\lambda_2\hat{j} - \lambda_1\mu\vec{b}}{4}\right| = |\lambda_1\hat{i} - \lambda_2\hat{j}|$

$|\lambda_1(4-\mu)\hat{i} + \lambda_2\hat{j}| = 2|\lambda_1\hat{i} + \lambda_2\hat{j}|$

squaring

$\lambda_1^2(4-\mu)^2 + \lambda_2^2 = 4\lambda_1^2 + 4\lambda_2^2$ $\Rightarrow$ $3\lambda_2^2 = (12 + \mu^2 - 8\mu)\lambda_1^2$ .....(1)

Also $(\vec{b} - \vec{a}).(\vec{b} + \vec{c}) = 0$

$\Rightarrow \quad (\lambda_1\hat{i}-\lambda_2\hat{j})\cdot\left(\lambda_1\hat{i}+\frac{\lambda_2\hat{j}-\lambda_1\mu\hat{i}}{4}\right)=0 \quad \Rightarrow \quad \frac{\lambda_1^2(4-\mu)-\lambda_2^2}{4}=0$

$\Rightarrow \quad \lambda_2^2=\lambda_1^2(4-\mu)$ .....(2)

from (1) & (2)

$12+\mu^2-8\mu=12-3\mu$

$\Rightarrow \quad \mu^2-5\mu=0 \quad \Rightarrow \quad \mu=0,5\ (\mu=5 \text{ reject})$

(D) $I=\frac{2}{\pi}\int_{-\pi}^{\pi}\frac{\sin\frac{9x}{2}}{\sin\frac{x}{2}}dx=\frac{4}{\pi}\int_0^{\pi}\frac{\sin\frac{9x}{2}\cos\frac{x}{2}}{\sin\frac{x}{2}\cos\frac{x}{2}}dx=\frac{8}{\pi}\int_0^{\pi}\frac{\sin\frac{9x}{2}\cos\frac{x}{2}}{\sin x}dx$

$I=\frac{4}{\pi}\int_0^{\pi}\frac{\sin 5x+\sin 4x}{\sin x}dx$ .....(i)

(using $\int_0^b f(x)dx=\int_0^b f(a+b-x)dx$)

$=\frac{4}{\pi}\int_0^{\pi}\frac{\sin 5x-\sin 4x}{\sin x}dx$ .....(ii)

Add (i) and (ii)

$I=\frac{4}{\pi}\int_0^{\pi}\frac{\sin 5x}{\sin x}dx$

Consider

$I_k-I_{k-2}=\frac{4}{\pi}\int_0^{\pi}\frac{\sin kx-\sin(k-2)x}{\sin x}dx=\frac{8}{\pi}\int_0^{\pi}\frac{\cos(k-1)x\sin x}{\sin x}dx$

$I_k=I_{k-2}$

so $I_5=I_3 \quad \Rightarrow \quad I_5=I_1=\frac{4}{\pi}\int_0^{\pi}dx=4$

Aliter

Let $I=\frac{2}{\pi}\int_{-\pi}^{\pi}\frac{\sin(9x/2)}{\sin(x/2)}dx$

$I=\frac{4}{\pi}\int_0^{\pi}\frac{\sin(9x/2)}{\sin(x/2)}dx$ .....(1) ($\because$ f(x) is even function)

$I=\frac{4}{\pi}\int_0^{\pi}\frac{\cos(9x/2)}{\cos(x/2)}dx$ .....(2)

(using $\int_0^b f(x)dx=\int_0^b f(a+b-x)dx$)

Add (1) & (2)

$$I = \frac{4}{\pi}\int_0^{\pi} \frac{\sin 5x}{2\sin(x/2)\cos(x/2)}\,dx = \frac{4}{\pi}\int_0^{\pi} \frac{\sin 5x}{\sin x}\,dx$$

$$\Rightarrow I = \frac{8}{\pi}\int_0^{\pi/2} \frac{\sin 5x}{\sin x}\,dx$$

$$\Rightarrow I = \frac{8}{\pi}\int_0^{\pi/2} \left(\frac{16\sin^5 x - 20\sin^3 x + 5\sin x}{\sin x}\right)dx$$

$$\Rightarrow I = \frac{8}{\pi}\int_0^{\pi/2} (16\sin^4 x - 20\sin^2 x + 5)\,dx$$

$$\Rightarrow I = \frac{8}{\pi}\left[16x\frac{3\times1\times\pi}{4\times2\times2} - 20\times\frac{1\times\pi}{2\times2} + \frac{5\pi}{2}\right]$$

$$\Rightarrow I = \frac{8}{\pi}\left[3\pi - 5\pi + \frac{5\pi}{2}\right]$$

$$\Rightarrow I = 4$$

Hindi (A) माना मूल बिन्दु से जाने वाली रेखा $\frac{x}{\lambda} = \frac{y}{\mu} = \frac{z}{1}$

$\Rightarrow x = \lambda z, y = \mu z$ .....(1)

रेखा (1) और रेखा $\frac{x-2}{1} = \frac{y-1}{-2} = \frac{z+1}{1}$ के प्रतिच्छेदन बिन्दु के लिए .....(2)

$$\frac{\lambda z - 2}{1} = \frac{\mu z - 1}{-2} = z + 1$$

$$\Rightarrow z = \frac{3}{\lambda - 1} = \frac{-1}{\mu + 2}$$

$\Rightarrow \lambda + 3\mu + 5 = 0$ .....(3)

रेखा (1) और रेखा $\frac{x - \frac{8}{3}}{2} = \frac{y+3}{-1} = \frac{z-1}{1}$ के प्रतिच्छेदन के लिए .....(4)

$$\frac{\lambda z - \frac{8}{3}}{2} = \frac{\mu z + 3}{-1} = \frac{z-1}{1}$$

$$\Rightarrow z = \frac{2}{3(\lambda - 2)} = \frac{-2}{\mu + 1}$$

$\Rightarrow 3\lambda + \mu = 5$ .....(5)

(3) और (5) से , $\lambda = \frac{5}{2}$ और $\mu = -\frac{5}{2}$

$\therefore$ बिन्दु P के लिए $z = 2, x = 5, y = -5$

और बिन्दु Q के लिए $z = \frac{4}{3}, x = \frac{10}{3}, y = -\frac{10}{3}$

$\therefore \quad PQ^2 = \frac{4}{9} + \frac{25}{9} + \frac{25}{9} = 6$

(B) $\tan^{-1}(x+3) - \tan^{-1}(x-3) = \sin^{-1}(3/5)$

$\Rightarrow \tan^{-1}\left(\frac{x+3-x+3}{1+x^2-9}\right) = \tan^{-1}(3/4)$

$\Rightarrow \frac{6}{x^2-8} = \frac{3}{4} \Rightarrow x^2 = 16$

$\therefore x = \pm 4$

(C) चूंकि $\vec{a}\cdot\vec{b} = 0$

$\therefore$ माना $\vec{b} = \lambda_1 \hat{i}$, $\vec{a} = \lambda_2 \hat{j}$

अब $2|\vec{b} + \vec{c}| = |\vec{b} - \vec{a}|$ और $\vec{a} = \mu\vec{b} + 4\vec{c}$

$\Rightarrow \quad 2\left|\lambda_1\hat{i} + \frac{\lambda_2\hat{j} - \lambda_1\mu\vec{b}}{4}\right| = |\lambda_1\hat{i} - \lambda_2\hat{j}|$

$|\lambda_1(4-\mu)\hat{i} + \lambda_2\hat{j}| = 2|\lambda_1\hat{i} + \lambda_2\hat{j}|$

वर्ग करने पर

$\lambda_1^2(4-\mu)^2 + \lambda_2^2 = 4\lambda_1^2 + 4\lambda_2^2$

$\Rightarrow \quad 3\lambda_2^2 = (12 + \mu^2 - 8\mu)\lambda_1^2$ .....(1)

तथा $(\vec{b} - \vec{a})\cdot(\vec{b} + \vec{c}) = 0$

$\Rightarrow \quad (\lambda_1\hat{i} - \lambda_2\hat{j})\cdot\left(\lambda_1\hat{i} + \frac{\lambda_2\hat{j} - \lambda_1\mu\hat{i}}{4}\right) = 0$

$\Rightarrow \quad \frac{\lambda_1^2(4-\mu) - \lambda_2^2}{4} = 0$

$\Rightarrow \quad \lambda_2^2 = \lambda_1^2(4-\mu)$ .....(2)

(1) व (2) से

$12 + \mu^2 - 8\mu = 12 - 3\mu$

$\Rightarrow \quad \mu^2 - 5\mu = 0 \quad \Rightarrow \quad \mu = 0, 5$ ($\mu = 5$ reject)

(D) $I = \frac{2}{\pi}\int_{-\pi}^{\pi} \frac{\sin\frac{9x}{2}}{\sin\frac{x}{2}}\,dx = \frac{4}{\pi}\int_0^{\pi} \frac{\sin\frac{9x}{2}\cos\frac{x}{2}}{\sin\frac{x}{2}\cos\frac{x}{2}}\,dx = \frac{8}{\pi}\int_0^{\pi} \frac{\sin\frac{9x}{2}\cos\frac{x}{2}}{\sin x}\,dx$

$$I = \frac{4}{\pi}\int_0^{\pi} \frac{\sin 5x + \sin 4x}{\sin x}\, dx \text{.....(i)}$$

$\left(\int_0^b f(x)dx = \int_0^b f(a+b-x)dx \text{ के उपयोग से}\right)$

$$= \frac{4}{\pi}\int_0^{\pi} \frac{\sin 5x - \sin 4x}{\sin x}\, dx \text{.....(ii)}$$

(i) और (ii) को जोड़ने पर

$$I = \frac{4}{\pi}\int_0^{\pi} \frac{\sin 5x}{\sin x}\, dx$$

मानाकि

$$I_k - I_{k-2} = \frac{4}{\pi}\int_0^{\pi} \frac{\sin kx - \sin(k-2)x}{\sin x}dx = \frac{8}{\pi}\int_0^{\pi} \frac{\cos(k-1)x \sin x}{\sin x}dx$$

$$I_k = I_{k-2}$$

अतः $\quad I_5 = I_3 \Rightarrow \quad I_5 = I_1 = \frac{4}{\pi}\int_0^{\pi} dx = 4$

वैकल्पिक

माना $I = \frac{2}{\pi}\int_{-\pi}^{\pi} \frac{\sin(9x/2)}{\sin(x/2)}\, dx$

$$I = \frac{4}{\pi}\int_0^{\pi} \frac{\sin(9x/2)}{\sin(x/2)}dx \quad \text{.....(1)} \qquad (\because\ f(x) \text{ सम फलन है})$$

$$I = \frac{4}{\pi}\int_0^{\pi} \frac{\cos(9x/2)}{\cos(x/2)}dx \quad \text{.....(2)}$$

$\left(\int_0^b f(x)dx = \int_0^b f(a+b-x)dx \text{ के उपयोग से}\right)$

(1) व (2) को जोड़ने पर

$$I = \frac{4}{\pi}\int_0^{\pi} \frac{\sin 5x}{2\sin(x/2)\cos(x/2)}dx = \frac{4}{\pi}\int_0^{\pi} \frac{\sin 5x}{\sin x}dx$$

$$\Rightarrow I = \frac{8}{\pi}\int_0^{\pi/2} \frac{\sin 5x}{\sin x}dx$$

$$\Rightarrow\ I = \frac{8}{\pi}\int_0^{\pi/2} \left(\frac{16\sin^5 x - 20\sin^3 x + 5\sin x}{\sin x}\right)dx$$

$$\Rightarrow I = \frac{8}{\pi}\int_0^{\pi/2}(16\sin^4 x - 20\sin^2 x + 5)\,dx$$

$$\Rightarrow I = \frac{8}{\pi}\left[16x\frac{3\times1\times\pi}{4\times2\times2} - 20\times\frac{1\times\pi}{2\times2} + \frac{5\pi}{2}\right]$$

$$\Rightarrow I = \frac{8}{\pi}\left[3\pi - 5\pi + \frac{5\pi}{2}\right]$$

$$\Rightarrow I = 4$$

46. Match the following column :

Column-I	**Column-II**
A. $\left\|\int_0^1 (y^2-1)\,dy\right\| + \left\|\int_0^1 (1-y^2)\,dy\right\|$	p. 2
B. A point (α, β, γ) lies on the plane $x + y + z = 2$. The value of γ such that the vector $\vec{a} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ satisfies $\hat{k}\times(\hat{k}\times\vec{a}) = \vec{0}$	q. $\left\|\int_{-1}^0 \sqrt{1+x}\,dx\right\| + \left\|\int_0^1 \sqrt{1-x}\,dx\right\|$
C. In a triangle ABC, if $\cos A\cos B + \sin A\sin B\sin C = 1$, then the value of $\sin C$ is	r. $\frac{4}{3}$
D. The set of values of a for which the lines $x + y = \|a\|$, $ax - y = 4$ intersect in the region $x > 0, y > 0$, is the interval (a_0, ∞). Then the value of a_0 is	s. 1

निम्न column को सुमेलित कीजिए :

Column-I	**Column-II**
A. $\left\|\int_0^1 (y^2-1)\,dy\right\| + \left\|\int_0^1 (1-y^2)\,dy\right\|$	p. 2
B. एक बिन्दु (α, β, γ) तल $x + y + z = 2$ पर स्थित है γ का मान होगा इस प्रकार कि $\vec{a} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$, $\hat{k}\times(\hat{k}\times\vec{a}) = \vec{0}$ को सतुष्ट करता है	q. $\left\|\int_{-1}^0 \sqrt{1+x}\,dx\right\| + \left\|\int_0^1 \sqrt{1-x}\,dx\right\|$
C. ΔABC में यदि $\cos A\cos B + \sin A\sin B\sin C = 1$, तो $\sin C$ का मान होगा	r. $\frac{4}{3}$
D. a के मानों का समुच्चय जिसके लिये रेखाऐं $x + y = \|a\|$, $ax - y = 4$	s. 1

क्षेत्र $x > 0, y > 0$ में प्रतिछेद करती हैं, अन्तराल (a_0, ∞)

है तो a_0 का मान होगा

Ans. (A) Q, R (B) P (C) S (D) P

Sol. (A) Solving the two equations, say i.e.,

$x + y = |a|$ and $ax - y = 1$, we get

$x = \frac{|a| + 1}{a + 1} > 0$ and $y = \frac{|a| - 1}{a + 1} > 0$

when $a + 1 > 0$; we get $a > 1$

$\therefore \quad a_0 = 1$

(B) We have, $\vec{a} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$

$\Rightarrow \vec{a}.\hat{k} = \gamma$

Now, $\hat{k} \times (\hat{k} \times \vec{a}) = (\hat{k}.\vec{a})\hat{k} - (\hat{k}.\hat{k})\vec{a}$

$= \gamma\hat{k} - (\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k})$

$\Rightarrow \alpha\hat{i} + \beta\hat{j} = 0$

$\Rightarrow \alpha = \beta = 0$

Also $\alpha + \beta + \gamma = 2$

$\Rightarrow \quad \gamma = 2.$

(C) $\left|\int_0^1 \left(1 - y^2\right) dy\right| + \left|\int_0^1 \left(y^2 - 1\right) dy\right|$

$= 2\int_0^1 (1 - y^2) dy = \frac{4}{3}$

Also $\left|\int_0^1 \sqrt{1 - x}\, dx\right| + \left|\int_{-1}^0 \sqrt{1 + x}\, dx\right|$

$= 2\int_0^1 \sqrt{1 - x}\, dx = \frac{4}{3}$

(D) $\sin A \sin B \sin C + \cos A \cos B$

$\leq \sin A \sin B + \cos A \cos B = \cos(A - B)$

$\Rightarrow \quad \cos(A - B) \geq 1$

$\Rightarrow \quad \cos(A - B) = 1$

$\Rightarrow \quad \sin C = 1.$

47. Match the following column :

Column-I	**Column-II**
A. Let P be the plane passing through the point (2, 1, – 1) and	p. $\frac{\sqrt{5}}{3}$

perpendicular to the line of intersection of the planes $2x + y - z = 3$ and $x + 2y + z = 2$. Then the distance from the point $\left(\sqrt{3}, 2, 2\right)$ to the plane P is

B. If $\lim\limits_{n\to\infty} \sum\limits_{i=1}^{n} \tan^{-1}\left(\frac{1}{2i^2}\right) = t$, then tan t is — q. 1

C. The sides a, b, c of a triangle ABC are in A.P. If $\cos\theta_1 = \frac{a}{b+c}$, $\cos\theta_2 = \frac{b}{a+c}$, $\cos\theta_3 = \frac{c}{a+b}$, then $\tan^2\frac{\theta_1}{2} + \tan^2\frac{\theta_3}{2} =$ — r. $\frac{2}{3}$

D. Let L be the line passing through the point (0, 1, 0) and perpendicular to the plane $x + 2y + 2z = 0$. Then the distance from the point (0, 0, 0) to the line L is — s. 0

निम्न column को सुमेलित कीजिए :

Column-I	**Column-II**
A. माना P एक तल है जो बिन्दु (2, 1, – 1) से गुजरता है एवं तल $2x + y - z = 3$ तथा $x + 2y + z = 2$ की प्रतिच्छेदी रेखा से गुजरता है तो बिन्दु $\left(\sqrt{3}, 2, 2\right)$ की तल P से दूरी बराबर है	p. $\frac{\sqrt{5}}{3}$
B. यदि $\lim\limits_{n\to\infty} \sum\limits_{i=1}^{n} \tan^{-1}\left(\frac{1}{2i^2}\right) = t$, तो tan t का मान होगा	q. 1
C. यदि त्रिभुज ABC की भुजाऐं a, b, c सं. श्रे. में हो तथा $\cos\theta_1 = \frac{a}{b+c}$, $\cos\theta_2 = \frac{b}{a+c}$, $\cos\theta_3 = \frac{c}{a+b}$, तो $\tan^2\frac{\theta_1}{2} + \tan^2\frac{\theta_3}{2} =$	r. $\frac{2}{3}$
D. माना L एक रेखा है जो बिन्दु (0, 1, 0) से गुजरती है तथा तल $x + 2y + 2z = 0$ के लम्बवत् है तो बिन्दु (0, 0, 0) की रेखा L से दूरी होगी	s. 0

Ans. (A) Q (B) Q (C) R (D) P

Sol. (B) $\sum\limits_{i=1}^{\infty} \tan^{-1}\left(\frac{1}{2i^2}\right) = t$.

$$\Rightarrow \sum_{i=1}^{\infty} \tan^{-1}\left(\frac{2}{4i^2 - 1 + 1}\right)$$

$$= \sum_{i=1}^{\infty} \tan^{-1}\left\{\frac{(2i+1)-(2i-1)}{1+(2i-1)(2i+1)}\right\}$$

$= (\tan^{-1}3 - \tan^{-1}1)+(\tan^{-1}5 - \tan^{-1}3) +....$
$+ \{(\tan^{-1}(2n+1) - \tan^{-1}(2n-1))$

$\therefore \quad t = \lim_{n\to\infty} (\tan^{-1}(2n+1) - \tan^{-1}1)$

$= \lim_{n\to\infty} \tan^{-1}\left(\frac{2n}{1+2n+1}\right) = \frac{\pi}{4}$

$\therefore \quad \tan t = 1.$

(C) We have, $\cos\theta_1 = \dfrac{1-\tan^2\frac{\theta_1}{2}}{1+\tan^2\frac{\theta_1}{2}} = \dfrac{a}{b+c}$

$\Rightarrow \quad \tan^2\left(\dfrac{\theta_1}{2}\right) = \dfrac{b+c-a}{b+c+a}$

Also, $\cos\theta_3 = \dfrac{1-\tan^2\frac{\theta_3}{2}}{1+\tan^2\frac{\theta_3}{2}} = \dfrac{c}{a+b}$

$\Rightarrow \quad \tan^2\dfrac{\theta_3}{2} = \dfrac{a+b-c}{a+b+c}$

$\Rightarrow \quad \tan^2\dfrac{\theta_1}{2} + \tan^2\dfrac{\theta_3}{2}$

$= \dfrac{2b}{a+b+c} = \dfrac{2b}{3b} = \dfrac{2}{3}$

{as, a, b, c are in AP $\Rightarrow 2b = a + c$}

(D) Line through (0, 1, 0) and perpendicular to plane $x + 2y + 2z = 0$ is given by

$\dfrac{x-0}{1} = \dfrac{y-1}{2} = \dfrac{z-0}{2} = r$

$\therefore$ P(r, 2r + 1, 2r) be the foot of perpendicular on the straight line then

$r \cdot 1 + (2r + 1) \cdot 2 + (2r) \cdot 2 = 0$

$\Rightarrow \quad r = -\dfrac{2}{9}$

$\therefore \quad P\left(-\dfrac{2}{9}, \dfrac{5}{9}, -\dfrac{4}{9}\right)$

$\therefore$ Required perpendicular distance

$$= \sqrt{\frac{4+25+16}{81}} = \frac{\sqrt{5}}{3} \text{ unit.}$$